



PhD Dissertation

The Power of Input Perturbation for Data Privacy

Revised Version for PhD Defense, Sep 28 2026

Lukas Retschmeier

Advisor: Rasmus Pagh

Co-Advisor: Martin Aumüller

Submitted: July 7, 2027

This thesis has been submitted to the PhD School of The Faculty of Science, University of Copenhagen

THE POWER OF INPUT PERTURBATION FOR DATA PRIVACY

LUKAS RETSCHMEIER



PhD Dissertation

Department of Computer Science

Faculty of Science

University of Copenhagen

July 2026

¹² Lukas Retschmeier: *The Power of Input Perturbation for Data Privacy*,
¹³ PhD Dissertation, © July 2026, Revision based on committee feedback

14
15
16

Dedicated to my family,
for their love and unwavering support,
on every path I chose.

Differential privacy protects individuals by releasing statistics that have been perturbed with carefully calibrated random noise. The thesis studies *input-perturbation* mechanisms, which first perturb the input, then run an arbitrary non-private algorithm on the perturbed input, and only then release its output. Canonical Additive Noise Mechanisms such as the Laplace or Gaussian mechanisms are special cases where the final step is the identity map. We ask when these approaches achieve optimal privacy-utility trade-offs, and develop matching error upper and lower bounds in three different settings. The main contributions of this thesis are:

- Private Graph Optimization.** We establish nearly tight upper and lower bounds in the edge-weight model of Sealfon [Sea16], where a graph topology $G = (V, E)$ with n vertices and m edges is public and only the edge weights $\mathbf{w} \in \mathbb{R}^m$ contain sensitive information. Under ℓ_∞ neighboring weights, we show that privately releasing the edges of a Minimum-Weight Spanning Tree reduces to adding Gumbel noise to each edge *once* before running any non-private **MST** algorithm $\mathcal{A}$ on top: the output distribution exactly matches a privatized Kruskal algorithm that iteratively calls the exponential mechanism. This runs in time $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, where $t_{\mathcal{A}}(m, n)$ is the running time of $\mathcal{A}$, improving on the previous $\mathcal{O}(m \cdot n)$. Through a connection to private top- k selection, we prove a matching lower bound, showing the $\tilde{\Theta}(n^{3/2})$ expected additive error is optimal up to log factors. Under ℓ_1 neighbors, we give novel reconstruction-based lower bounds: for Minimum-Weight Spanning Tree and Minimum-Weight (Perfect) Matching we close a remaining logarithmic gap, matching the Laplace mechanism and proving that input perturbation is optimal. For **HC!** under Dasgupta's cost function we give the first approximate DP lower bound. Our bound is parameterized by the balanced min-cut. We show that the pure DP lower bound given in [Den+25] also holds for approximate DP as well as for sparse graphs with high connectivity, such as $\Theta(1)$ -degree expanders.
- Private Lossless Multiple Release.** We introduce the framework of Private Lossless Multiple Release, providing a general characterization of when the same statistic can be released under multiple privacy levels, in any order, and without *any* privacy loss beyond that of the least private release. We show that this is possible for every additive noise mechanism whose noise distribution satisfies a decomposition property we call *convolution preorder*, which is satisfied by, e.g., the Gaussian, Laplace, and Poisson distributions. Furthermore, we initiate the study of non-additive mechanisms like *Randomized Response* via a Markov-chain formulation and provide a counterexample where lossless release is provably impossible.
- Sparse Gaussian Histograms with Correlated Noise.** We improve the private release of sparse histograms by exploiting the sensitivity structure induced by the add/remove neighboring relationship. Combining insights about correlating noise from [Leb25] together with the exact analysis of the (Sparse) Gaussian mechanism [Wil+24; BW18], we obtain a utility improvement of up to a factor of two. We extend these techniques to aggregate and top- k queries. Furthermore, our experiments confirm these gains.

61 Differentiel privathed beskytter individer ved at offentliggøre statistikker, der er blevet
62 forstyrret med nøje kalibreret tilfældig støj. Denne afhandling studerer *input-perturbation*-
63 mekanismer, som først tilføjer støj til inputtet, derefter kører en vilkårlig ikke-privat algo-
64 ritme på det perturberede input og først derefter offentliggør algoritmens output. Kanoniske
65 *additive støjmekanismer*, såsom Laplace- eller Gauss-mekanismen, er specialtilfælde, hvor
66 det sidste trin er identitetsafbildningen. Vi undersøger, hvornår disse tilgange opnår opti-
67 male afvejninger af privathed mod nytte i tre forskellige problemstillinger. Afhandlingens
68 hovedbidrag er:

- 69 • **Privat grafoptimering.** Vi etablerer næsten tætte øvre og nedre grænser i modellen
70 fra Sealfon [Sea16], hvor en graftopologi $G = (V, E)$ med n knuder og m kanter er
71 offentlig, og kun kantvægtene $\mathbf{w} \in \mathbb{R}^m$ indeholder sensitiv information.

72 Under ℓ_∞ -naborelationer mellem vægte viser vi, at privat offentliggørelse af et mini-
73 mum udspændende træ reduceres til at tilføje støj fra en Gumbel-fordeling til hver
74 kant *én gang*, før en vilkårlig ikke-privat **MST**-algoritme $\mathcal{A}$ køres. Outputfordelingen
75 stemmer præcist overens med en privatiseret Kruskal-algoritme, som iterativt kalder
76 den eksponentielle mekanisme. Dette kører i tid $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, hvor $t_{\mathcal{A}}(m, n)$ er
77 køretiden for $\mathcal{A}$, og forbedrer den tidligere $\mathcal{O}(m \cdot n)$. Vi etablerer en forbindelse
78 til privat top- k -udvælgelse og beviser en matchende nedre grænse, som viser, at
79 den forventede additive fejl $\tilde{\Theta}(n^{3/2})$ er næsten optimal. Under ℓ_1 -naboer giver vi
80 nye rekonstruktionsbaserede nedre grænser: For Minimum-Weight Spanning Tree
81 og Minimum-Weight (Perfect) Matching lukker vi et tilbageværende logaritmisk
82 gab, matcher Laplace-mekanismen og beviser, at inputperturbation er optimal. For
83 **HC!** under Dasguptas omkostningsfunktion giver vi den første nedre grænse for
84 approksimativ DP parameteriseret ved det balancerede min-cut. Vi viser, at den
85 nedre grænse for ren DP givet i [Den+25] også gælder for approksimativ DP og
86 sparse grafer med høj konnektivitet, såsom tilfældige expandere af grad $\Theta(1)$.

- 87 • **Privat tabsfri multipel offentliggørelse.** Vi introducerer frameworket *Private Lossless*
88 *Multiple Release*, som giver en generel karakterisering af, hvornår den samme statistik
89 kan offentliggøres under flere privathedsniveauer, i vilkårlig rækkefølge og uden
90 *noget* privathedstab ud over tabet fra den mindst private offentliggørelse. Vi viser, at
91 dette er muligt for enhver mekanisme med additiv støj, hvis støjfordeling opfylder
92 en dekompositionsegenskab, som vi kalder *convolution preorder*, og som blandt andet
93 opfyldes af Gaussian-, Laplace- og Poissonfordelingen. Desuden påbegynder vi
94 studiet af ikke-additive mekanismer såsom *Randomized Response* via en Markovkæde-
95 formulering og giver et modeksempel, hvor tabsfri offentliggørelse beviseligt er
96 umulig.

- 97 • **Sparse Gaussian-histogrammer med korreleret støj.** Vi forbedrer den private of-
98 fentliggørelse af sparse histogrammer ved at udnytte sensitivitsstrukturen induceret
99 af add/remove-naborelationen. Ved at kombinere indsigter om korreleret støj fra
100 [Leb25] med den eksakte analyse af den (Sparse) Gaussian-mekanisme [Wil+24;
101 BW18], opnår vi en nytteforbedring på op til en faktor to. Vi udvider disse teknikker
102 til aggregat- og top- k -forespørgsler. Desuden bekræfter vores eksperimenter disse
103 forbedringer empirisk.

Differential Privacy (DP) schützt Individuen, indem Statistiken vor dem Veröffentlichen durch zufälliges Rauschen verändert werden. Dabei lässt sich der entstehende Fehler gegen ein formales Maß an Privatsphäre abwägen. Intuitiv führt ein größerer Rauschfaktor zu mehr Privatsphäre, wohingegen ein Kleinerer mehr Rückschlüsse auf die Personen zulässt, die Teil des Datensatzes sind.

Diese Doktorarbeit untersucht Algorithmen, die zunächst nur die Eingabedaten verändern, anschließend einen beliebigen, nicht privaten Algorithmus auf der verrauschten Eingabe ausführen und erst danach die Ergebnisse veröffentlichen. *Additive Rauschmechanismen*, wie der Laplace- oder Gauß-Mechanismus, sind Spezialfälle, in denen der letzte Schritt die Identitätsabbildung ist. Wir untersuchen, wann diese Ansätze eine optimale Abwägung zwischen Privatsphäre und Fehler erreichen. Die Hauptbeiträge dieser Dissertation sind:

- **Netzwerkoptimierungsprobleme.** Wir etablieren nahezu optimale Fehlerschranken im Kantengewichtsmodell von Sealfon [Seal16]. Hierbei ist eine Graphtopologie $G = (V, E)$ mit n Knoten und m Kanten öffentlich und die Kantengewichte $\mathbf{w} \in \mathbb{R}^m$ enthalten sensible Informationen. Für ℓ_∞ -benachbarte Kantengewichte zeigen wir, dass die private Veröffentlichung der Kanten eines minimalen Spannbaumes (MST) darauf reduziert werden kann, *einmal* Zufallsvariablen einer Gumbelverteilung zu jeder Kante hinzuzufügen und anschließend einen beliebigen nicht privaten MST-Algorithmus darauf auszuführen. Der resultierende Algorithmus hat eine Laufzeit von $\mathcal{O}(m + t_A(m, n))$, wobei $t_A(m, n)$ die Laufzeit von $\mathcal{A}$ bezeichnet. Dies verbessert den früheren Algorithmus mit Laufzeit $\mathcal{O}(m \cdot n)$. Über eine Verbindung zu privater Top- k -Selektion beweisen wir eine untere Schranke und zeigen, dass der erwartete additive Fehler von $\tilde{\mathcal{O}}(n^{3/2})$ bis auf logarithmische Faktoren optimal ist. Für ℓ_1 -Nachbarn geben wir neue untere Schranken an, die auf Datenrekonstruktionen basieren. Für MST und das Problem eines perfekten Matchings mit minimalem Gewicht (MWPM) schließen wir eine verbleibende logarithmische Lücke in der oberen und unteren Schranke. Für hierarchisches Clustering unter Dasguptas Kostenfunktion [STOC 2016] geben wir die erste untere Schranke für (ϵ, δ) -DP an. Unsere Schranke ist durch den minimalen balancierten Schnitt parametrisiert. Wir zeigen, dass die untere Schranke für ϵ -DP aus [Den+25] auch für approximatives DP sowie für dünne Graphen mit hoher Konnektivität gilt, etwa für Expander mit konstantem Grad.
- **Private Verlustfreie Mehrfachveröffentlichung.** Wir führen das Framework der *Verlustfreien Mehrfachveröffentlichung* (Private Lossless Multiple Release) ein und geben eine allgemeine Charakterisierung dafür, wann dieselbe Statistik unter mehreren Privatsphärenniveaus, in beliebiger Reihenfolge und ohne *jedlichen* zusätzlichen Verlust veröffentlicht werden kann. Wir zeigen, dass dies für jeden additiven Rauschmechanismus möglich ist, dessen Rauschverteilung eine Zerlegungseigenschaft erfüllt. Wir bezeichnen diese Eigenschaft *convolution preorder* und zeigen, dass sie etwa von der Gauß-, Laplace- und der Poisson-Verteilung erfüllt wird. Darüber hinaus initiieren wir die Untersuchung nicht-additiver Mechanismen wie *Randomized Response* mittels Markovketten und geben ein Gegenbeispiel an, in dem PLMR nachweislich unmöglich ist.
- **Dünne Histogramme mit korreliertem Rauschen.** Wir verbessern die private Veröffentlichung dünner Histogramme, indem wir die Sensitivitätsstruktur ausnutzen, die durch eine Nachbarschaftsrelation induziert wird, die nur Hinzufügen und Entfernen zulässt. Indem wir Beobachtungen aus Lebeda [Leb25] mit der genauen Analyse des Gaußschen Mechanismus für dünnbesetzte Histogramme kombinieren [Wil+24; BW18], erzielen wir eine konstante Verbesserung des maximalen Fehlers um bis zu einen Faktor von zwei. Des Weiteren bestätigen Experimente diese theoretischen Ergebnisse empirisch.

157 It is difficult to express how deeply grateful I am to have had Rasmus as my
158 supervisor. When I began my PhD, I knew almost nothing about randomized
159 algorithms, yet he believed in my potential and decided to work with me. Beyond
160 being an outstanding researcher, his patience and guidance as a mentor really
161 helped me stay on track. This thesis would not have been possible without him.
162 “Kudos” also goes to my co-supervisor Martin for his encouragement, motivation,
163 and steady support. I also want to thank Inge Li Gørtz, Haim Kaplan, and Mikkel
164 Abrahamsen for serving on my thesis committee.

165 I am also grateful to all my other coauthors, especially Anders, Boel, Christian,
166 Hanwen, Hao, Jacob, and Joel, for the many ideas, discussions, and collaborations
167 that shaped this thesis.

168 I want to thank Sofya for hosting me at Boston University. I have rarely met
169 people as welcoming and caring as Sofya and Adam, and their hospitality made
170 my research stay truly enjoyable. Thanks also to Rathin, Themis, Anatoly, and
171 Duy for immediately making me feel at home in Boston, for many intriguing
172 discussions, and for becoming friends along the way.

173 My sincere thanks go to Anne for holding the pack together. Her energy and
174 boundless creativity really make BARC a special place that is more than just an
175 academic environment.

176 Furthermore, I want to thank the whole Providentia group, [again](#) in randomized
177 order: Hao, Fabrizio, Johanna, Joel, Boel, Carolin, Amartya, Christian, Sia, Quentin,
178 and Anders, and all the guests we had in the past years. I am grateful for the
179 many discussions, but also the moments we shared along the way. Thank you to
180 all the 4B people at ITU, who by now have grown too numerous to even start
181 enumerating.

182 I owe particular thanks to Christian, who, during some late hours at a retreat
183 in Helsingør, helped lay the foundations for the following three years by telling
184 me about differential privacy. He is also a strong advocate for the monograph
185 by Steinke [Ste22]. Although I did not fully appreciate it at the beginning, this text
186 shaped much of my understanding of differential privacy and strongly influenced
187 the presentation in the first chapter of this thesis.

188 Thanks to Sia and Quentin for providing feedback early on and to Marco for
189 taking a look at the final draft.

190 Thank you, Maren, for your patience and for always being willing to listen to
191 the problems I am working on, even when their connection to the real world is
192 not always crystal clear at first glance. Sharing these thoughts with you helped
193 preserve a life outside my research bubble, and for that I am deeply grateful.

194 Finally, I want to thank my family. Without their support, I would never have
195 made it this far.

198 I SYNOPSIS

199	1	INTRODUCTION	1
200	2	DIFFERENTIAL PRIVACY	5
201	2.1	Three different angles on Differential Privacy	5
202	2.2	Classic Mechanisms	8
203	2.3	Properties of Differential Privacy	10
204	2.4	Differential Privacy on Graphs	12
205	3	CONTRIBUTIONS AND OVERVIEW	15

206 II INCLUDED PAPERS

207	4	OPTIMAL BOUNDS VIA INPUT PERTURBATION	22
208	4.1	Preliminaries	24
209	4.2	Reduction from Private MST to Non-Private MST	25
210	4.3	A Lower Bound for Approximate DP with ℓ_∞	32
211	4.4	Related Work	36
212	4.5	Empirical Evaluation	38
213	4.6	Conclusion and Open Problems	39
214	4.7	Appendix	40
215	5	OPTIMAL BOUNDS VIA INPUT PERTURBATION	48
216	5.1	Preliminaries	53
217	5.2	Private Minimum-Weight Spanning Trees	55
218	5.3	Minimum-Weight Perfect Matchings	59
219	5.4	Hierarchical Clustering under Dasgupta's Cost Function	63
220	5.5	Private Algorithms with Tight Error	68
221	5.6	Conclusion and Future Work	69
222	5.7	Appendix: Reconstruction-Lower Bounds	69
223	6	LOSSLESS PRIVATE MULTIPLE RELEASE	77
224	6.1	Threat Model and Goals	80
225	6.2	Gaussian Lossless Multiple Release	81
226	6.3	Extending to Independent Additive Noise	84
227	6.4	Applications	87
228	6.5	Conclusion and Open Questions	90
229	6.6	Appendix	91
230	6.7	Extension: Non-Additive Noise Mechanisms	106
231	7	THE CORRELATED SPARSE GAUSSIAN HISTOGRAM MECHANISM	112
232	7.1	Preliminaries and Background	114
233	7.2	Algorithmic Framework	118
234	7.3	Top-k Counting Queries	123
235	7.4	Numerical Evaluations	124
236	7.5	Extensions	125
237	7.6	From Theory towards Practice: Discrete Gaussian Noise	126
238	7.7	Conclusion and Open Problems	128
239	7.8	Appendix: Proof of Theorem 7.9 (Tighter Analysis)	129

240 III BACKGROUND

241	A	PROBABILITY THEORY	135
-----	---	--------------------	-----

242

Part I

243

SYNOPSIS

244

247
248 INTRODUCTION

249 Data often reveals more about individuals than it appears to at first.
250 In 2013, a Freedom of Information request forced the public release of all 173
251 million New York City taxi trips from that year. To protect privacy, each medal-
252 lion number was replaced by a seemingly random identifier [Pan14; Who14]. It
253 turned out that this *pseudonymization* was too weak: Trotter [Tro14] quickly recov-
254 ered the original identifiers. Combined with side information such as paparazzi
255 photographs of celebrities entering a cab, individual trips could be recovered —
256 including destinations, fares, and even tip amounts [Toc14]. Even the drivers were
257 exposed: their routes, working patterns, and income could be inferred. Similar
258 failures had already exposed Netflix viewers by linking movie preferences to public
259 IMDb accounts [NSo8], and AOL users through the partial release of search query
260 logs [Arro6].

261 These early *deanonymization attacks* required substantial manual effort: auxiliary
262 datasets had to be located, linked, and interpreted by hand. More recently, Lermen,
263 Paleka, Swanson, Aerni, Carlini, and Tramèr [Ler+26] demonstrated that large
264 language models can be very effective at these deanonymization tasks when
265 equipped with agentic abilities such as web search and reasoning capabilities. The
266 authors showed that given only an anonymized *Hacker News* profile, such an agent
267 can autonomously find the corresponding *LinkedIn* account in roughly 70% of the
268 cases. Their experiments further suggest that scaling to larger candidate pools
269 may be possible with only a minuscule loss in recall. Concurrent work reports
270 similar results in other domains [NSN24; Li26], suggesting that deanonymization
271 increasingly requires less manual effort.

272 **STATISTICAL RELEASES** Since removing identifiers is insufficient, a natural
273 response is to release an aggregated statistic, which is a summary over many
274 records. Examples include the average fare of taxi trips between two neighborhoods
275 or the count of trips leaving a specific area late at night. At first sight, such
276 summaries seem much safer than simply releasing the raw data. To see that this
277 is not the case, suppose, for instance, that we want to know whether a particular
278 person took a taxi between two neighborhoods. If the released count is zero, then it
279 rules out that possibility immediately. Thus, even an aggregated value can disclose
280 information about individual records.

281 A statistic is usually supposed to describe an overall population. For instance, if
282 we want to know which streets to avoid on a busy Monday morning, it is not a
283 single person, assuming no accidents, who causes the traffic jam, but rather that
284 many people use this street at the same time. Furthermore, its value should only
285 change marginally whether a particular record is present. We call this maximal
286 possible change the *sensitivity* of a statistical release. For example, a single trip
287 between two places only changes the total count by one. Among the thousands
288 of trips that may occur every day, the contribution of a single trip is negligible,
289 and in almost all cases, it would be sufficient to only get a rough estimate. This
290 motivates the idea of carefully altering the statistic. Intuitively, just from observing

the release, it should be provably difficult to infer anything about whether any single individual’s data has been used to compute it. This is precisely the intuition captured by the robust definition of *privacy* we use in this thesis.

One way to achieve this goal is to randomly perturb the release. Instead of publishing the exact value of a statistic, we publish a randomized version. This introduces uncertainty, usually called *noise*, which masks the effect that any single individual can have on the output. Intuitively, if the impact of a single record is small, we will get away with only a small amount of noise that covers its impact. This intuition was formalized as Differential Privacy (DP) by Dwork, McSherry, Nissim, and Smith [Dwo+06b]. DP requires that the released statistic of two datasets that differ in only one individual’s record should look very similar. We defer the precise definitions to Chapter 2. Releases under DP have some desirable properties. For example, *post-processing* ensures that any function applied to a release (independent of the dataset) preserves privacy and *composition* theorems allow us to quantify how much privacy is lost if multiple mechanisms are run on the same dataset.

ADDITIVE NOISE MECHANISMS Among the many mechanisms that satisfy DP, Additive Noise Mechanisms (ANMs) play a particularly important role. Given a statistical query $f : \mathcal{X} \rightarrow \mathbb{R}^d$, an ANM releases $\mathcal{A}(x) = f(x) + Z$ where Z is a random data-independent noise vector. Canonical examples are the *Laplace* and *Gaussian* mechanisms where Z is drawn from the corresponding distribution. The noise is calibrated to the sensitivity of f , i.e., the largest possible change of f under neighboring inputs. This makes ANMs a very general way to provide formal privacy guarantees. As long as the sensitivity of f can be bounded, we can directly construct a private mechanism.

In this direct form, however, additive noise uses only a worst-case sensitivity bound and ignores much of the structure of the task. Many private release problems, such as *Private Selection*, do not ask for a numerical vector at all, but for example for an object that maximizes some utility score. In such cases, adding noise to every score and releasing the entire perturbed vector leaks more information than required. Even when the desired output is numerical, the query might impose additional structure we can exploit. For example, the *Sparse Vector Technique* allows releasing private answers to only those queries that exceed some defined threshold without “paying” privacy cost for those queries whose answers are not released. This is strictly better than the naive ANM-based approach that simply adds noisy answers to all the queries.

THE POWER OF INPUT PERTURBATION In this thesis we study more general *input perturbation* mechanisms: add data-independent noise and then apply a private operation on top of it. The ANMs from above are special cases where this last step is the identity map. Otherwise, the noisy internal quantities are not directly released, but only an output that has been computed on top of them. This final step is required to obtain the privacy guarantees we are looking for. The *Exponential Mechanism*, which solves the *Private Selection* problem, can be implemented by first adding appropriate noise to each score and then outputting the element with the maximum perturbed score. Likewise, the *Sparse Vector Technique* can be implemented by adding noise once to the threshold and once to each query, releasing those answers that exceed the threshold.

These examples show that perturbing an internal representation before post-processing can be substantially more powerful than releasing a noisy statistic directly. This motivates the driving question of this thesis.

When is *Input Perturbation* optimal for a differentially private release?

We investigate this question in three fundamentally different private release settings. We defer an overview of the exact results to Chapter 3.

NETWORK DATA. We first study the private release of approximate solutions to fundamental graph optimization problems. We assume a public graph topology $G = (V, E)$ with n vertices and m edges together with edge weights $\mathbf{w} \in \mathbb{R}^m$ that carry sensitive information. Two graphs over the same topology are neighboring if their corresponding weight vectors differ by at most some constant in ℓ_1 or ℓ_∞ distance. This model, introduced by Sealfon [Sea16], arises naturally when edge weights represent aggregated statistics, for example, traffic intensities collected from the taxi drivers in New York.

We first study the problem of releasing an approximate Minimum-Weight Spanning Tree (**MST**) under privacy constraints. A spanning tree $T \subseteq E$ is an acyclic subset of the edges of size $n - 1$ making the graph connected. We want to release a spanning tree T that minimizes the sum of the weights $\sum_{t \in T} w(t)$ and consider the additive error $|w(T) - w(T^*)|$ relative to the optimal solution T^* .

Our results show that input-perturbation based mechanisms are surprisingly powerful for this problem. Under ℓ_1 neighbors, simply privatizing the weights with independent Laplace noise gives an additive error of $\mathcal{O}(\frac{m}{n} \log(m/n))$ (for fixed privacy parameters). We start by showing a reconstruction-based lower bound proving that this error is asymptotically optimal. Similar to existing (worst-case) lower bounds [Sea16], we encode a random dataset into a fixed dense graph. Releasing the **MST** with too high accuracy would then immediately contradict privacy.

Under ℓ_∞ , previous works [Pin+18; MMS21] already showed that private variants of standard textbook algorithms achieve asymptotically better utility than the naive approach of simply adding noise to each edge. Intuitively, these better bounds can be achieved by exploiting a tighter composition theorem and injecting the noise during each round of the algorithm directly. However, these algorithms require fresh noise in each round and are therefore computationally expensive. In a first preprint [PR24] we used a simulation argument to improve the running time of a private version of Prim–Jarník’s **MST** algorithm down from $\mathcal{O}(m \cdot n)$ to $\mathcal{O}(m + n^{3/2} \log(n))$. In follow-up work, we realized that there is an even simpler algorithm [Pag+25]. We prove that an optimal algorithm can be obtained by adding noise from a Gumbel distribution to each weight exactly *once* and then only releasing the output of any **MST** algorithm. We show that this output distribution exactly matches that of a private version of Kruskal’s algorithm. Note that similar to the *Exponential Mechanism*, releasing the perturbed weights directly does not give the desired privacy guarantees. It is only through the algorithm that the actual privacy guarantees are established. We complement this upper bound with a corresponding lower bound showing that the resulting asymptotic error is optimal up to a logarithmic factor.

Furthermore, we show that several of these bounds also hold for the problem of privately releasing a Minimum-Weight (Perfect) Matching.

384 **MULTIPLE PRIVACY LEVELS.** Our second setting studies the problem of re-
 385 leasing the same statistic under different privacy levels simultaneously. A naive
 386 approach is to generate both releases independently, but a standard composition
 387 result implies that the overall privacy budget accumulates across those releases. We
 388 show how to bypass this for a large family of **ANMs**, so that the overall guarantee is
 389 solely determined by the least private release. Intuitively, we add a small amount of
 390 additional noise on top of previous releases, which does not cause any additional
 391 privacy costs as it only post-processes some previous output. We show that this
 392 way of correlating the noise works for all noise distributions that satisfy a very
 393 general decomposition property, which we call *convolution preorder*. This works,
 394 among others, for *Laplace*, *Gaussian* and *Poisson* noise. Our algorithm works even
 395 in a streaming setting, where the required privacy levels arrive one at a time and
 396 in arbitrary order.

397 **SPARSE GAUSSIAN HISTOGRAMS** Finally, we study the *Gaussian mechanism* un-
 398 der the add-remove neighboring relationship in a sparse setting. We combine some
 399 recent results [Leb25] with the exact analysis of the Gaussian mechanism [Wil+24]
 400 in the sparse setting. For example, consider releasing how often each term has
 401 been searched for in a search engine. Since only a small fraction of all possible
 402 terms ever appears, the resulting histogram is *sparse*, and we only want to release
 403 its few non-zero counts rather than a value for every possible term. We show how
 404 a small amount of correlated noise can help to improve the utility by a constant
 405 factor of up to two. Again, the algorithm is based on a simple perturbation of the
 406 input using additive noise.

407

408 **Thesis Organization.** We introduce Differential Privacy (**DP**) in more detail in
 409 Chapter 2 and then give a technical summary of the exact papers included in this
 410 thesis in Chapter 3. Many standard probability distributions are frequently used in
 411 all chapters. We defer their definitions to Part iii. The rest of the thesis includes
 412 the corresponding papers chapter-wise with small modifications to the published
 413 versions to reduce redundancy.

414 But if the difference between the heap [of sand grains] and the heap-minus-one is miniscule,
 415 how can you ever transform a heap into a non-heap?
 416 — “Sorites Paradox”, R. F. Kuang, *Katabasis* [Kua25]

2

417 DIFFERENTIAL PRIVACY 418

419 Introduced in seminal work by Dwork, McSherry, Nissim, and Smith [Dwo+06b],
 420 Differential Privacy (DP) gives a framework for providing privacy guarantees
 421 by limiting how much a data release may reveal about any individual. DP has
 422 been deployed in real-world settings such as the release of *Wikipedia usage* statis-
 423 tics [Ade+23], *Israeli birth data* [HC25], *product analytics in Brave* [Sha+25], and
 424 probably most notably, the *2020 US Census* [Abo+22]. A community-maintained
 425 list of deployments has recently been established by Howarth [How25].

426 2.1 THREE DIFFERENT ANGLES ON DIFFERENTIAL PRIVACY

427 We will approach a formal definition of DP from three equivalent perspectives. But
 428 first, we need some definitions. Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be a randomized algorithm that
 429 releases a statistic computed on a dataset. For simplicity, we assume that $\mathcal{Y}$ is a
 430 discrete output space, but all statements can be extended to a continuous domain.
 431 We write $\mathbf{x} \sim \mathbf{x}'$ to indicate that two datasets are *neighboring*; the treatment of
 432 possible neighboring relationships is deferred to Section 2.2. For now, it suffices to
 433 think of $\mathbf{x}'$ as $\mathbf{x}$ but with a single record added or removed.

434 **HYPOTHESIS TESTING** One way to view Differential Privacy is from the per-
 435 spective of an adversary who, given only a released statistic, must decide whether
 436 a particular record was part of the underlying dataset. If she can answer correctly
 437 with high confidence, the release leaks information about that record. If she can
 438 do no better than random guessing, the release reveals nothing about it. This is
 439 precisely the binary hypothesis test setting [KOV15; WZ10; Ste22; DRS19]: the
 440 adversary must decide whether the observed output came from a dataset where
 441 one record was present, or from one in which it wasn’t.

442 Suppose we release $Y \leftarrow \mathcal{A}(\mathbf{x})$ and want to hide whether some record was
 443 present in $\mathbf{x}$. Formally, an adversary who only sees the output Y faces a binary
 444 hypothesis test where the null hypothesis $\mathbf{H}_0$ states that Y was drawn from $\mathcal{A}(\mathbf{x})$.
 445 We test it against the alternative $\mathbf{H}_1$ that Y was drawn from $\mathcal{A}(\mathbf{x}')$ for a neighboring
 446 dataset $\mathbf{x}'$ in which the record is absent, i.e.,

$$\mathbf{H}_0 : Y \leftarrow \mathcal{A}(\mathbf{x}) \quad \mathbf{H}_1 : Y \leftarrow \mathcal{A}(\mathbf{x}').$$

447 A test can be specified by a *rejection region*: we reject $\mathbf{H}_0$ if the observed outcome
 448 lies in that region and accept it otherwise. For a rejection region S , the *type-I* error
 449 $\alpha(S)$ is the probability of falsely rejecting $\mathbf{H}_0$, and the *type-II* error $\beta(S)$ that of
 450 failing to reject it when $\mathbf{H}_1$ holds. The adversary’s goal is to find an S for which
 451 both $\alpha(S)$ and $\beta(S)$ are as small as possible. Differential Privacy (DP) is now the
 452 requirement that no test can reliably distinguish $\mathbf{H}_0$ from $\mathbf{H}_1$, that is, for every
 453 rejection region $S \subseteq \mathcal{Y}$ we require that at least one of $\alpha(S)$ or $\beta(S)$ is large.

454 **PRIVACY LOSS DISTRIBUTIONS** A classic result by Neyman and Pearson [NP33]
 455 from 1933 states that (in the binary hypothesis test setting) for any target type-
 456 I error level, the test that minimizes the type-II error is the one that thresholds
 457 $\log(\mathbf{P}[\mathcal{A}(\mathbf{x}) = y] / \mathbf{P}[\mathcal{A}(\mathbf{x}') = y])$, also known as the log-likelihood ratio. This identity
 458 is closely related to the concept of *Privacy Loss*, which leads us to the next way to
 459 characterize privacy. This perspective is often used to give tighter guarantees for
 460 the composition of multiple private mechanisms.

Definition 2.1 (Privacy Loss). *For a pair of distributions $\mathcal{A}(\mathbf{x})$ and $\mathcal{A}(\mathbf{x}')$ we define the privacy loss as*

$$\mathcal{L}_{\mathcal{A}(\mathbf{x})\|\mathcal{A}(\mathbf{x}')}(\mathbf{y}) := \begin{cases} -\infty & \text{if only } \mathbf{P}[\mathcal{A}(\mathbf{x}) = y] = 0, \\ \log\left(\frac{\mathbf{P}[\mathcal{A}(\mathbf{x}) = y]}{\mathbf{P}[\mathcal{A}(\mathbf{x}') = y]}\right) & \text{if both are positive,} \\ +\infty & \text{if only } \mathbf{P}[\mathcal{A}(\mathbf{x}') = y] = 0. \end{cases}$$

The loss is undefined if both $\mathbf{P}[\mathcal{A}(\mathbf{x}) = y] = \mathbf{P}[\mathcal{A}(\mathbf{x}') = y] = 0$.

461 The privacy loss measures how much an observation favors $\mathcal{A}(\mathbf{x})$ over $\mathcal{A}(\mathbf{x}')$: it is
 462 positive when y is more likely under $\mathcal{A}(\mathbf{x})$, and negative when it favors $\mathcal{A}(\mathbf{x}')$. A
 463 loss close to zero indicates that distinguishing $\mathbf{x}$ and $\mathbf{x}'$ is hard, while higher values
 464 tell us the opposite. Note that the case of infinite privacy loss occurs when one of
 465 the probabilities is zero. In that case, an attacker can immediately tell where the
 466 data must have come from. As we will see in Chapter 7, these kinds of events can
 467 happen, for instance, in algorithms that preserve sparsity by a thresholding step.
 468 The *Privacy Loss Random Variable* can now be defined as the loss we get when we
 469 sample from $\mathcal{A}(\mathbf{x})$:

Definition 2.2 (Privacy Loss Random Variable). *For a pair of distributions $\mathcal{A}(\mathbf{x})$ and $\mathcal{A}(\mathbf{x}')$, the privacy loss random variable $Z \leftarrow \mathcal{L}_{\mathcal{A}(\mathbf{x})\|\mathcal{A}(\mathbf{x}')}$ is obtained by drawing $Y \sim \mathcal{A}(\mathbf{x})$ and setting $Z := \mathcal{L}_{\mathcal{A}(\mathbf{x})\|\mathcal{A}(\mathbf{x}')}(\mathbf{Y})$.*

470 Intuitively, privacy requires that, when we release statistics computed on any
 471 pair of neighboring datasets $\mathbf{x} \sim \mathbf{x}'$, the *Privacy Loss Random Variable* is not too large.
 472 We would expect strong guarantees if we can bound the loss by some constant
 473 $\varepsilon > 0$, but even a probabilistic bound on the loss should give us “something”. The
 474 privacy loss distribution is closely related to the textbook definition of **DP**.
 find a survey of different notions in [Des20]

475 **THE CLASSIC DEFINITION OF DIFFERENTIAL PRIVACY.** We now introduce the
 476 standard textbook definition of (ε, δ) -**DP**. The parameter ε bounds the likelihood
 477 ratio multiplicatively and $\delta > 0$ accounts for a small additive error (in the sense of
 478 total variation distance).

Definition 2.3 $((\varepsilon, \delta)$ -indistinguishability). *We say that two distributions $\mathcal{A}(\mathbf{x})$*

and $\mathcal{A}(\mathbf{x}')$ over $\mathcal{Y}$ are (ϵ, δ) -indistinguishable (denoted $\mathcal{A}(\mathbf{x}) \approx_{\epsilon, \delta} \mathcal{A}(\mathbf{x}')$) if for all $S \subseteq \mathcal{Y}$,

$$\begin{aligned} \mathbf{P}_{Y \leftarrow \mathcal{A}(\mathbf{x})}[Y \in S] &\leq e^\epsilon \cdot \mathbf{P}_{Y \leftarrow \mathcal{A}(\mathbf{x}')}[Y \in S] + \delta, \text{ and} \\ \mathbf{P}_{Y \leftarrow \mathcal{A}(\mathbf{x}')}[Y \in S] &\leq e^\epsilon \cdot \mathbf{P}_{Y \leftarrow \mathcal{A}(\mathbf{x})}[Y \in S] + \delta. \end{aligned}$$

480

481 **DP** is satisfied if the output distributions for every pair of neighboring datasets are
482 (ϵ, δ) -indistinguishable.

Definition 2.4 ([Dwo+o6b] (ϵ, δ) -differential privacy). Given $\epsilon \in \mathbb{R}$ and $\delta \in [0, 1]$, a randomized mechanism $\mathcal{A} : \mathcal{U}^* \rightarrow \mathcal{Y}$ satisfies (ϵ, δ) -DP if for every pair of neighboring datasets $\mathbf{x} \sim \mathbf{x}'$, $\mathcal{A}(\mathbf{x})$ is (ϵ, δ) -indistinguishable from $\mathcal{A}(\mathbf{x}')$, i.e.,

$$\mathcal{A}(\mathbf{x}) \approx_{\epsilon, \delta} \mathcal{A}(\mathbf{x}').$$

483 We refer to the case $\delta = 0$ as *pure DP*. It is easy to see that pure DP immediately
484 recovers the requirement that the privacy loss $\mathcal{L}_{\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')}$ is bounded by ϵ in
485 absolute value for every pair of neighboring datasets. We refer to the other case
486 where $\delta > 0$ as *approximate DP*. It turns out that interpreting δ directly is non-trivial
487 but thinking in terms of the privacy loss gives some intuition. One can show that
488 if the probability that the privacy loss exceeds some fixed ϵ can be bounded by δ ,
489 then the mechanism satisfies (ϵ, δ) -DP, but the reverse direction is not true. We will
490 give the exact characterization in Lemma 2.1.

491 We usually think of $\delta = n^{-\Omega(1)}$ where n is the size of the dataset. Otherwise, there
492 are algorithms satisfying strong privacy guarantees that we would not consider
493 private. Consider, for example, the following “Name and Shame” algorithm that
494 releases each record with probability δ independently. It can be shown that this
495 algorithm satisfies $(0, \delta)$ -DP [Sm12]. So, unless δ is much smaller than $1/n$, we have
496 a significant chance of exactly releasing the private records, something we would
497 definitely not consider private. All of these three perspectives — the hypothesis test,

Meiser [Mei18] gives
more intuition about
interpreting δ .

498 the privacy loss, and the standard (ϵ, δ) -formulation — are closely related. In fact,
499 the following lemma states that all of these characterizations of DP are equivalent.

Lemma 2.1 ([KOV15; WZ10]). Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be any randomized algorithm.

500

The following statements are equivalent for any neighboring relationship $\sim$ and fixed privacy parameters ϵ, δ .

1. (classic definition) $\mathcal{A}(\mathbf{x}) \approx_{\epsilon, \delta} \mathcal{A}(\mathbf{x}')$ for each pair of neighboring datasets $\mathbf{x} \sim \mathbf{x}'$.
2. (privacy loss)

$$\delta \geq \mathbf{P}_{Z \leftarrow \mathcal{L}_{\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')}} [Z > \epsilon] - e^\epsilon \cdot \mathbf{P}_{Z \leftarrow \mathcal{L}_{\mathcal{A}(\mathbf{x}') \parallel \mathcal{A}(\mathbf{x})}} [-Z > \epsilon] .$$

3. (binary hypothesis test) For every rejection region $S \subseteq \mathcal{Y}$, we have

$$\alpha(S) + e^\epsilon \beta(S) \geq 1 - \delta \quad \text{and} \quad e^\epsilon \alpha(S) + \beta(S) \geq 1 - \delta ,$$

$$\text{where } \alpha(S) := \mathbf{P}_{Y \sim \mathcal{A}(\mathbf{x})} [Y \in S] \text{ and } \beta(S) := \mathbf{P}_{Y \sim \mathcal{A}(\mathbf{x}')} [Y \notin S].$$

501

502 2.2 CLASSIC MECHANISMS

503 Arguable the most widely used mechanisms are the *Gaussian* and the *Laplace*
 504 mechanisms where the amount of noise is calibrated to the *sensitivity* of the
 505 released statistic.

506 **SENSITIVITY** Suppose we want to release some statistical query $f : \mathcal{X} \rightarrow \mathbb{R}^d$
 507 computed from some private dataset $\mathbf{x}$. The *sensitivity* Δ_p of f captures how much
 508 $f(\mathbf{x})$ can change if instead, the dataset were some neighboring $\mathbf{x}'$.

Definition 2.5 (ℓ_p Sensitivity). The ℓ_p sensitivity of a function $f : \mathcal{X} \rightarrow \mathbb{R}^d$ is

$$\Delta_p := \sup_{\mathbf{x} \sim \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_p ,$$

where the supremum is taken over all neighboring datasets $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$.

509 The concrete adjacency notion can have a big influence on the *sensitivity*. Con-
 510 sider, for example, the setting where two vectors $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^d$ are neighboring if they
 511 are close in ℓ_1 distance, i.e., $\|\mathbf{x} - \mathbf{x}'\|_1 \leq 1$ implies $\Delta_1 = \Delta_2 = 1$. On the other hand,
 512 if we define the neighboring relationship as $\|\mathbf{x} - \mathbf{x}'\|_\infty \leq 1$ where each coordinate
 513 can change by one, the sensitivity blows up to $\Delta_2 = \sqrt{d}$ and even $\Delta_1 \leq d$. This
 514 difference will become important in Chapter 4 and 5 where we show that optimal
 515 algorithms can be obtained by Additive Noise Mechanism.

516 Many algorithms in this thesis are based on *additive* noise mechanisms, which
 517 add noise from some distribution to the computed statistic $f(\mathbf{x})$ before releasing
 518 it. The sensitivity directly influences how the noise must be “calibrated” such
 519 that the output distribution satisfies privacy guarantees. This is one of the key
 520 contributions shown by Dwork, McSherry, Nissim, and Smith [Dwo+06b].

Definition 2.6 (Additive Noise Mechanism (ANM)). For some positive integer d ,

521

a randomized mechanism $\mathcal{A} : \mathcal{X} \rightarrow \mathbb{R}^d$ is an additive noise mechanism if

$$\mathcal{A}(\mathbf{x}) = f(\mathbf{x}) + Z, \quad Z \sim \mathcal{D},$$

where $f : \mathcal{X} \rightarrow \mathbb{R}^d$ is some statistical query and $\mathcal{D}$ an arbitrary (data-independent) probability distribution over $\mathbb{R}^d$.

We will mostly assume that $\mathcal{D}$ independently adds noise to each $f(\mathbf{x})_i$ but sometimes, like in Chapter 7, the noise might be correlated across different coordinates. The two classic instantiations of (independent) ANMs are the Laplace and Gaussian mechanisms.

more on correlated noise in Chapter 7

Definition 2.7 ([Dwo+06b] Laplace mechanism). Let Δ_1 be the ℓ_1 sensitivity of f . The Laplace mechanism is the ANM where $Z = (Z_1, \dots, Z_d)$ with each $Z_i \sim \text{Lap}(\Delta_1/\epsilon)$ independently. It satisfies ϵ -DP.

Definition 2.8 ([DR14] Gaussian mechanism). Let Δ_2 be the ℓ_2 sensitivity of f . The Gaussian mechanism is the ANM where $Z = (Z_1, \dots, Z_d)$ with each $Z_i \sim \mathcal{N}(0, \sigma^2)$ independently.

The scaling of the noise can be exactly characterized. [BW18] showed that the Gaussian mechanism satisfies (ϵ, δ) -DP, if and only if

$$\Phi\left(\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right) - e^\epsilon \Phi\left(-\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right) \leq \delta, \quad (1)$$

where Φ denotes the CDF of the normal distribution. One can derive Definition 2.8 by observing that the Privacy Loss Random Variable of the Gaussian mechanism is also a Gaussian and then use the second characterization in Lemma 2.1. Although not giving the tightest guarantees, most of the times, one can think of $\sigma \geq \Delta_2 \sqrt{2 \ln(1.25/\delta)} \cdot \frac{1}{\epsilon}$, which is sufficient to achieve (ϵ, δ) -DP.

NEIGHBORING RELATIONSHIP Until now, we only considered datasets where we wanted to protect the membership of a record in the dataset and neighboring (or adjacent) datasets were defined to be exactly those that differ by the presence or absence of a single record. This setting is known as the *add/remove* neighboring relationship. Differential Privacy itself is oblivious to this notion. This freedom is meaningful because the choice of adjacency gives us the flexibility to exactly decide what to protect.

Another popular notion is that of *replacement* where the dataset size stays the same but a single record itself is changed. In this case, we no longer hide whether an individual contributed to the statistic at all, but rather what their record exactly contained. Note that a single replacement can be simulated by combining an add followed by a remove operation. A large part of this thesis is dedicated to studying differentially private releases when the input is a graph. Graphs admit several natural neighboring relations, like protecting the existence of edges (*edge-DP*), nodes (*node-DP*) or associated attributes such as edge weights (*edge-weight-DP*).

find an introduction to Differential Privacy in Section 2.4

VARIANTS There are several other variants of DP that behave better in certain settings like composition or subsampling. We refer to the survey by Desfontaines and Pejó [DP20] for a comprehensive overview and how these notions relate to

each other. For example, f -DP formalizes the binary hypothesis testing intuition previously described directly by describing privacy in terms of the best possible trade-off function between type-I and type-II errors [DRS19]. *Gaussian-DP* (μ -GDP) is a natural instantiation that requires that distinguishing the mechanism's output is at least as hard as distinguishing the two Gaussians $\mathcal{N}(0, 1)$ and $\mathcal{N}(\mu, 1)$. ρ -zero concentrated DP on the other hand, is closely related to the privacy loss distribution and provides a simple way to exactly characterize privacy guarantees under composition.

Definition 2.9 ([BS16], ρ -zero concentrated DP (ρ -zCDP)). *Let $\rho > 0$. An algorithm $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ satisfies ρ -zCDP, if for all $\alpha > 1$ and all pairs of neighboring inputs $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$, it holds that*

$$D_\alpha(\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')) \leq \rho\alpha,$$

where $D_\alpha(\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')) = (1 - \alpha)^{-1} \cdot \ln \left(\mathbb{E}_{x \leftarrow \mathcal{A}(\mathbf{x})} \left[\left(\mathcal{A}(\mathbf{x}) / \mathcal{A}(\mathbf{x}') \right)^{\alpha-1} \right] \right)$ denotes the α -Rényi divergence between the two output distributions of $\mathcal{A}(\mathbf{x})$ and $\mathcal{A}(\mathbf{x}')$.

The advantage is that if two mechanisms are ρ_1 and ρ_2 DP respectively, then the composition of those algorithms is exactly $(\rho_1 + \rho_2)$ -DP. This characterization often gives better bounds than the advanced composition theorems applied to two approximate DP mechanisms. Therefore, we need a way to convert between ρ -zCDP and (ϵ, δ) -DP.

Fact 2.2 ([BS16] Conversion ρ -zCDP $\Leftrightarrow$ approximate DP).

1. If $\mathcal{M}$ satisfies ρ -zCDP, then $\mathcal{M}$ is $(\rho + 2\sqrt{\rho \log(1/\delta)}, \delta)$ -DP for all $\delta > 0$.
2. Conversely, if $\mathcal{M}$ satisfies $(\epsilon, 0)$ -DP, then $\mathcal{M}$ satisfies $(\epsilon^2/2)$ -zCDP.

2.3 PROPERTIES OF DIFFERENTIAL PRIVACY

As already mentioned, DP satisfies many desirable properties. First, it prohibits *blatant non-privacy* and formally rules out any kind of linear reconstruction attack. It is immune to *post-processing*, meaning that applying any data-independent function to a private output cannot weaken the privacy guarantees. Furthermore, it provides nice composition properties and can easily extend to providing guarantees for groups.

COMPOSITION What happens with the overall privacy guarantee of a mechanism that is built from several private subroutines that all have access to the same dataset? This question is answered by another key property *composition*. Composition theorems then provide a way to bound the overall privacy cost directly from the guarantees of the individual building blocks. This offers the advantage of allowing modularization: complex mechanisms can be assembled from known private primitives, whose privacy guarantees are already known.

Proposition 2.3 (Basic Composition). *Let $\mathcal{A}_1, \dots, \mathcal{A}_k : \mathcal{X} \rightarrow \mathcal{Y}$ be randomized algorithms, each of them (ϵ_i, δ_i) -DP. Then the composed mechanism $\mathcal{A}(x) : \mathcal{X} \rightarrow \mathcal{Y}^k$ with $\mathcal{A} := (\mathcal{A}_1(\mathbf{x}), \dots, \mathcal{A}_k(\mathbf{x}))$ is (ϵ', δ') -DP where*

$$\epsilon' := \sum_{i \in [k]} \epsilon_i \quad \text{and} \quad \delta' := \sum_{i \in [k]} \delta_i. \quad (2)$$

In the case of pure-DP, this bound is very clean and even optimal [Ste22], but it turns out that for approximate DP, we can do significantly better.

Lemma 2.4 (Advanced Composition). *Let $\mathcal{A}_1, \dots, \mathcal{A}_k : \mathcal{U}' \rightarrow \mathcal{Y}$ be randomized algorithms, each of them (ϵ_i, δ_i) -DP. Then $\mathcal{A}(\mathbf{x}) := (\mathcal{A}_1(\mathbf{x}), \dots, \mathcal{A}_k(\mathbf{x}))$ is (ϵ', δ') -DP for all $\delta > 0$ where*

$$\epsilon' := \frac{1}{2} \cdot \sum_{j=1}^k \epsilon_j^2 + \sqrt{2 \log(1/\delta) \sum_{j=1}^k \epsilon_j^2} \quad \text{and} \quad \delta' = \sum_{j=1}^k \delta_j.$$

As mentioned, ρ -zCDP provides a particularly clean way of composition.

Fact 2.5 ([BS16] Composition). *If $\mathcal{A}_1$ and $\mathcal{A}_2$ satisfy ρ_1 -zCDP and ρ_2 -zCDP, respectively, then $\mathcal{A} = (\mathcal{M}_1, \mathcal{M}_2)$ satisfies $(\rho_1 + \rho_2)$ -zCDP.*

This concept will play a major role in Section 6.3 where we want to create multiple releases with different privacy levels simultaneously and avoid the loss imposed by composition.

POST-PROCESSING Naturally, privacy must be preserved, regardless of what (data-independent) function we apply to the output. This really is a basic requirement: an adversary without access to the dataset should not be able to weaken the privacy guarantee just by *post-processing* the output.

Proposition 2.6 ([DR14] Post-Processing). *Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be a randomized algorithm that is (ϵ, δ) -differentially private. Let $f : \mathcal{Y} \rightarrow \mathcal{Y}'$ be any (potentially randomized) data-independent function. Then the composition $f \circ \mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}'$ is (ϵ, δ) -differentially private.*

Many of our graph results are obtained by releasing a privatized version of the weight vector. Together with the public topology, we can compute combinatorial objects like an approximate MST by post-processing these perturbed weights.

GROUP PRIVACY Occasionally, we do not only want to protect the privacy of a single individual, but rather a whole group of individuals. One can also view this requirement as the similarity between the output for two inputs that are at most " k -hops away", where $k \in \mathbb{N}^+$. Of course, an increasing group size degrades the overall privacy guarantees, but the good news is that DP can already handle this privacy loss.

Fact 2.7 ([Vad17] Group Privacy). *Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be any (ϵ, δ) -DP mechanism*

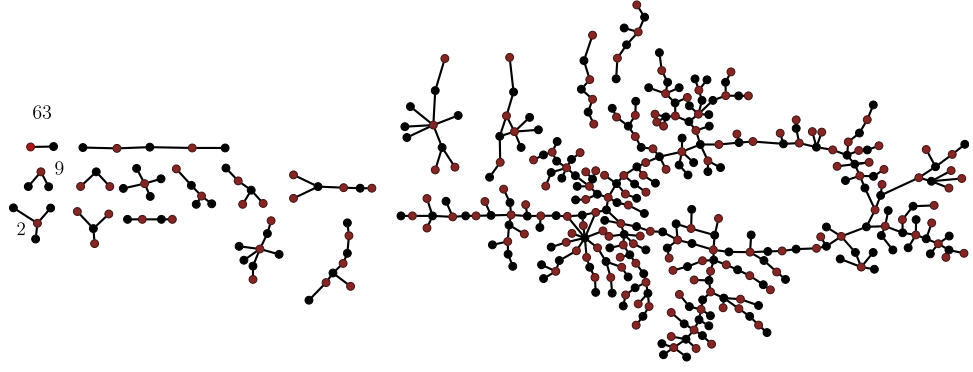


Figure 2.1: Sexual relationships at a US high school [BMS04] from the Add Health longitudinal study [Nat]. (Replicated with explicit permission of University of Chicago Press). The color indicates the reported gender.

with $\varepsilon < e$. Given $k \in \mathbb{N}^+$ and $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$, if there exist $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k-1)}$ such that $\mathbf{x}^{(i-1)} \sim \mathbf{x}^{(i)}$ for each $i \in [k]$ (where we define $\mathbf{x}^{(0)} = \mathbf{x}$ and $\mathbf{x}^{(k)} = \mathbf{x}'$), then for each subset $S \subseteq \mathcal{Y}$, we have:

$$\mathbf{P}[\mathcal{A}(\mathbf{x}) \in S] \leq e^{k\varepsilon} \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}') \in S] + \frac{e^{k\varepsilon} - 1}{e^\varepsilon - 1} \cdot \delta. \quad (3)$$

606

Proof. By simply applying the definition of (ε, δ) -DP k times, we have for all $S \subseteq \mathcal{Y}$

607

$$\begin{aligned} \mathbf{P}[\mathcal{A}(\mathbf{x}) \in S] &\leq e^\varepsilon \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}^{(1)}) \in S] + \delta \\ &\leq \dots \leq e^{k\varepsilon} \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}') \in S] + \delta \sum_{i=1}^k (e^\varepsilon)^i \\ &\leq e^{k\varepsilon} \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}') \in S] + \frac{e^{k\varepsilon} - 1}{e^\varepsilon - 1} \cdot \delta. \end{aligned}$$

The last line follows from a geometric series. □

608

In Chapter 5, we use the simpler $\delta \sum_{i=1}^k (e^\varepsilon)^i \leq \varepsilon k e^{\varepsilon k} \delta$.

609

2.4 DIFFERENTIAL PRIVACY ON GRAPHS

610

Graphs are rich theoretical objects [Die12] that can be used to model complex relationships. They have found applications across diverse domains such as social networks [Gra73], fraud detection [Web+18], or protein structure prediction [Jum+21]. Sensitive data also arises in these contexts, often encoded in the graph structure.

611

612

613

614

615

616

617

618

619

620

621

622

623

624

Consider, for example, the graph depicted in Figure 2.1. The vertices represent students at a US high school extracted from interview data. The edges indicate romantic or sexual partnerships that the students reported. This data is extremely sensitive and reidentifying even a single edge can reveal a student's sexual orientation or, via the chain of contacts, place them at inferred risk of sexually transmitted infection. Even years later, reidentification can have severe consequences. Similar to the reidentification attacks on tabular data as mentioned in the introduction, it has been repeatedly shown that such attacks are possible [Hay+08; NS09; JMo9]. Protecting against such attacks motivates the use of Differential Privacy (DP) for graph-structured data.

NEIGHBORING RELATIONSHIPS As before, privacy guarantees can be given respecting some underlying neighboring relationship that dictates what exactly we want to protect. In the setting above, it is natural to protect the relationships itself, while we are content with revealing that someone participated in the study. Introduced in [NRS07], this setting has become known as *edge-DP* where neighboring graphs differ by the addition or removal of a single edge. Kasiviswanathan, Nissim, Raskhodnikova, and Smith [Kas+13] and Blocki, Blum, Datta, and Sheffet [Blo+13] independently initiated the study of graphs under the stronger notion of *node-DP*. Here, neighboring graphs are obtained by adding or removing a node, including all of its connections. In the above context, this would also protect, whether someone has taken part in the study or not.

This means that she has full access to the private dataset and it suffices to give privacy guarantees for the final release. A more stringent notion is that of local DP that can be traced back to [War65]. The idea is that privacy should already be satisfied, as soon as data leaves some individual. In other words, this requires that the interviewee already lies about some of her answers. In the literature, mostly *local edge-DP* and, most recently, *local node-DP* [Ras+26] are considered. We refer to the recent surveys due to [DNK26; Mue+22; Li+23; BMM24] for more background.

EDGE-WEIGHT DIFFERENTIAL PRIVACY In this thesis, we study a slightly different model due to Sealfon [Sea16] that aims to protect information that is encoded in the weights of the edges. This models settings where the network topology is publicly known, while the data carried by its connections is sensitive. Consider again the example of the New York road network and releasing the count of taxi trips between two locations. The road network is known to everyone and definitely not considered private while the counts itself encode private information. In many settings, the edge-weights are computed by aggregating the data of an underlying private dataset. For example, a Chow-Liu tree [CL68] can be used to privately generate synthetic data [MMS21]. Here, a graph encodes the pairwise mutual information between all (aggregated) features of some underlying private dataset. This setting is known as the notion of *Edge-Weight Differential Privacy* as defined next.

Definition 2.10 ([Sea16] Edge-Weight DP). For some constant $\Delta > 0$ and two weight vectors $\mathbf{w}, \mathbf{w}' \in \mathbb{R}^E$ are neighboring, denoted $\mathbf{w} \sim \mathbf{w}'$, if and only if $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \Delta$. Two graphs $G = (V, E, \mathbf{w})$ and $G' = (V', E', \mathbf{w}')$ are neighboring if and only if $V = V'$, $E = E'$, and $\mathbf{w} \sim \mathbf{w}'$.

Analogously, two weight vectors are ℓ_∞ -neighboring if $\max_{e \in E} |w_e - w'_e| \leq \Delta$. Throughout the thesis, we usually consider $\Delta = 1$, but there are settings where this is a bit more intricate. For example, in the Chow-Liu setting above, $\Delta = \mathcal{O}(\log(s)/s)$ is the maximal amount the value of mutual information between two binary features can change if the underlying dataset had s users. In all our error bounds, Δ is simply an additional multiplicative factor. We sometimes overload notation and simply note $G \sim G'$ to be two neighboring graphs, implicitly meaning that the topology is the same and their weights are neighboring. Observe that the *edge-DP* setting above can be simulated by using a complete graph, where the existence of an edge is simply encoded by binary weight. Graphs would then be neighboring if their weight vectors $\mathbf{w}$ and $\mathbf{w}'$ are close in Hamming distance.

GRAPH OPTIMIZATION PROBLEMS There is a plethora of graph analysis tasks in the edge-weight model and we refer to [Mue+22] for an overview. We mainly consider problems where the output itself is also a combinatorial object that minimizes or maximizes some cost function.

The Minimum-Weight Spanning Tree problem is one of those with many applications. A *MST* can, for example, be used as a sparse representation of low-cost routes in road networks like the New York taxi cab setting. Other applications were found as a subroute in marginal selection in synthetic data generation [MMS21] and in the context of clustering algorithms [Pin18; Pin+18; Bat+17; Jay+24; LRNo9].

We also consider the problem of releasing a Minimum-Weight (Perfect) Matching analogously. Finally, in Chapter 5 we consider the problem of privately releasing a *dendrogram* which is a tree-based data structure encoding a hierarchical clustering. Releasing hierarchical has many applications, e.g. in data visualization [SSo2; HHEo4] or mining [Bro+92]. We consider the rather recently proposed cost function due to Dasgupta [Das16]. Our main goal is to show that this problem is very hard under privacy constraints, even if the graph is very sparse.

That cost function recently won a Test of Time Award at STOC [ACM26].

CONTRIBUTIONS AND OVERVIEW

We now give a summary of the results in this thesis. The thesis is written as a “synopsis with manuscripts of papers or already published papers attached” in accordance with the *general rules and guidelines for the PhD program* at the Faculty of Science at the University of Copenhagen. This thesis is based on the following published results.

P1 Optimal Bounds for Private MSTs via Input Perturbation [Pag+25]

with Rasmus Pagh, Hao Wu, and Hanwen Zhang

Principles of Database Systems (PODS) 2025

Also presented as a research highlight at FORC 2025

P2 Private Lossless Multiple Release [And+25]

with Joel Daniel Andersson, Boel Nelson, and Rasmus Pagh

International Conference on Machine Learning (ICML) 2025

P3 The Correlated Gaussian Sparse Histogram Mechanism [LR25],

with Christian Janos Lebeda

Foundations of Responsible Computing (FORC) 2025

P4 Lower Bounds for Private Graph Optimization Problems using Reconstruction Attacks [IPR27]

with Jacob Imola and Rasmus Pagh

Principles of Database Systems (PODS) 2027

Another paper called *Faster Private Minimum Spanning Trees*, had been accepted for poster presentation at the annual privacy workshop *Theory and Practice of Differential Privacy* (TPDP) but had not been published at a conference because **P1** provides strictly stronger results. Therefore, this work is excluded, but we briefly describe the main ideas below.

P4 has been accepted between the thesis submission and the defense. Furthermore, some of the results overlap with a different manuscript called *Optimal Bounds for Private Graph Problems* together with Anders Aamand and Rasmus Pagh that was presented at TPDP 2025. All published work was additionally presented at one of the TPDP workshops.

Joint work with Geis, Leonhardt, Meintrup, Meyer and Penschuck [Gei+26] was accepted at ALENEX 2026; another work, extending the author’s master’s thesis together with Paloma Lima is under submission. These results fall outside the scope of this thesis and are therefore excluded.

SUMMARY OF THE RESULTS

We call a private mechanism $\mathcal{M}$ *input-perturbation based* if it is a composition $\mathcal{M} = \mathcal{A} \circ \mathcal{P}$ of a randomized perturbation $\mathcal{P}$ of the input and an arbitrary non-private algorithm $\mathcal{A}$. In other words, on input $\mathbf{x}$, it first perturbs the input $\tilde{\mathbf{x}} \leftarrow \mathcal{P}(\mathbf{x})$, runs $\mathcal{A}$ on it, and only then releases the final output $\mathcal{A}(\tilde{\mathbf{x}})$. Additive Noise Mechanisms are exactly those mechanisms where $\mathcal{A}$ is the identity map.

In this thesis, we investigate when input perturbation gives optimal private mechanisms. We study the question in the following three settings: private graph optimization problems, multiple releases, and sparse histograms.

(T2: FASTER PRIVATE MINIMUM SPANNING TREES) We start by considering the problem of releasing an **MST** under approximate **DP** and the ℓ_∞ neighboring relationship. The error α of a released tree $T \subseteq E$ is measured in an (expected) additive sense with respect to the underlying private weights:

Recall that ℓ_∞ allows
all weights to change
by at most one.

Definition 3.1. For a graph $G = (V, E, \mathbf{w})$, we define the error of some spanning tree $T \leftarrow \mathcal{M}(G)$ drawn from some **MST** algorithm $\mathcal{M}$ as

$$\alpha := \mathbb{E}_{T \leftarrow \mathcal{M}(G)} \left[\sum_{e \in T} w_e \right] - \sum_{e \in T^*} w_e, \quad (1)$$

where T^* is the optimal **MST** in G .

We consider the maximization version, which is equivalent to the minimization version after negating all weights. This work is not included in the thesis because **P1** provides strictly stronger results. Nevertheless, we quickly sketch the main ideas here. A privatized version of Prim–Jarník’s algorithm for finding an **MST** starts at some vertex and keeps a frontier of edges that can be added to the solution without creating a cycle. The cut property ensures that the minimum-weight edge crossing any cut must be part of an optimal **MST**. Pinot, Morvan, Yger, Gouy-Pailler, and Atif [Pin+18] proved that this selection process can be replaced by a private selection mechanism, in their case the exponential mechanism, to achieve privacy guarantees. The problem with this approach is that it needs to generate $\mathcal{O}(m)$ noise terms in each of the $\mathcal{O}(n)$ steps of the algorithm. This naively results in a running time of $\mathcal{O}(m \cdot n)$. We provide an efficient implementation of *Report-Noisy-Max* where the set of possible edges is constantly updated. *Report-Noisy-Max* adds independent exponential noise to each weight and then releases the edge with the maximum noisy weight. By the composition of ρ -zCDP over the $\mathcal{O}(n)$ rounds of Prim–Jarník’s algorithm, we need to scale the noise proportionally to $\sqrt{n}$.

For a fixed sensitivity and privacy level, the main idea is to discretize and group all weights into buckets of constant width. We then only need to consider $\mathcal{O}(\sqrt{n} \log m)$ of the largest groups by a concentration bound of the exponential distribution. The remaining edges that are further away from the maximum are extremely unlikely to get enough noise to become the maximum. We provide a simulation argument that preserves these low-probability events exactly. Furthermore, we show that a single sample is sufficient within each group by another simulation argument—essentially, one only needs the maximum noise within each group, and by a symmetry argument, we can just assign it to some uniformly drawn representative within the group. We provide a special priority queue that keeps the maximum in sublinear time $\mathcal{O}(m^{1/4})$ and supports constant-time insertions and deletions. In this way, we achieve a running time of $\mathcal{O}(m + n \cdot (m^{1/4} + \sqrt{n} \log m)) = \mathcal{O}(m + n^{3/2} \log m)$: the $\mathcal{O}(m)$ terms accounts for adding and removing elements to the frontier over the whole execution (each of the m edges is inserted and removed once in constant time). Furthermore, in each of the $\mathcal{O}(n)$ rounds, we spend $\mathcal{O}(m^{1/4})$ to find the maximum edge in the current set of candidates (frontier) but this becomes asymptotically negligible. The remaining factor of $\mathcal{O}(\sqrt{n} \cdot \log m)$ comes exactly from the running time of our private selection mechanism..

For dense graphs with $m = \Theta(n^2)$, our algorithm runs in time linear in the size of the input. Furthermore, the algorithm matches the previous error bound of $\mathcal{O}(n^{3/2} \log n)$ up to a constant factor.

PAPER P1 In Chapter 4 we improve these results. We provide an optimal algorithm for privately releasing an **MST** under the ℓ_∞ neighboring relationship and approximate **DP**. We improve the previous work by providing a simple input-perturbation based algorithm with error $\alpha \in \mathcal{O}\left(\frac{1}{\epsilon} \cdot n^{3/2} \cdot \log n \cdot \sqrt{\log(1/\delta)}\right)$. The main idea is to add carefully calibrated Gumbel noise to each edge and then release the output of *any* chosen **MST** algorithm $\mathcal{A}$. Therefore, the overall running time is given by $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, where $t_{\mathcal{A}}(m, n)$ is the running time of $\mathcal{A}$ on a graph with m edges and n vertices. This improves the previous $\mathcal{O}(m \cdot n)$ time algorithm due to [Pin+18] and our subsequent work **T2**, which gives $\mathcal{O}(m + n^{3/2} \log n)$.

We first observe that the **MST** is unique after adding noise with high probability. Hence, every **MST** algorithm returns the same tree as the one directly found by Kruskal’s algorithm. On the other hand, a privatized version of Kruskal’s algorithm can be obtained by iteratively adding an approximately heaviest edge via the *Exponential Mechanism* and then discarding edges that would create a cycle in subsequent steps. We show that the output distributions of those two mechanisms match exactly.

This is closely related to a known *one-shot* mechanism for private top- k selection, where Gumbel noise is likewise used to simulate the iterative application of the *Exponential Mechanism* (the “peeling mechanism”) [DR19]. In some sense, this is precisely what happens when Kruskal’s algorithm is run on the graph with the perturbed weights. Our setting is more involved because the edges with the largest noisy weights need not form a valid spanning tree, as some of them may close a cycle. Nevertheless, we show by a careful analysis that this topological constraint is not an obstacle at all. We show that the one-shot approach still exactly simulates the *Exponential Mechanism* even when in each step, some of the possible future candidates are removed. We complement our upper bound with a lower bound of $\mathcal{O}\left(\frac{1}{\epsilon} \cdot n^{3/2} \cdot \ln n\right)$ for $\delta \leq \mathcal{O}(1/\sqrt{n})$. This proves that the error is optimal up to a logarithmic factor. We confirm our theoretical findings with some carefully designed experiments.

PAPER T1 In Chapter 5, we provide new *reconstruction-attack* based lower bounds for private graph optimization problems. Recently, Hladík and Tetek [HT24] found a new lower bound for private **MST** under pure **DP**. Their packing-based lower bound shows that an error of $\Omega(\frac{1}{\epsilon} \cdot n \cdot \log n)$ under ℓ_1 and $\Omega(\frac{1}{\epsilon} \cdot n^2 \cdot \ln n)$ under ℓ_∞ is necessary for the complete graph. This worst-case lower bound asymptotically matches the error given by the Laplace mechanism. The authors explicitly asked for lower bounds for approximate **DP** as well. In **P1**, we already provided a nearly-optimal lower bound under ℓ_∞ .

In the manuscript **T1**, we focus on the ℓ_1 neighboring relationship. One way to get lower bounds for approximate **DP** is via reconstruction attacks from the perspective of an adversary. The overall idea is that any private mechanism cannot be *blatantly non-private*, i.e., it cannot leak a large fraction of its input (assuming sufficiently high entropy in the dataset). This motivates the idea of encoding the private dataset into the solution of the graph problem for which one wants to obtain a lower bound. Any solution that is too accurate would then allow reconstruction of the private input, which in turn contradicts privacy.

For **MST** and Minimum-Weight (Perfect) Matching (**MWPM**), we provide a novel reconstruction-attack based lower bound that shows that an error of $\Omega(\frac{1}{\epsilon} \cdot \frac{m}{n} \cdot \log(m/n))$ is optimal (assuming δ is polynomially small). Consequently, any private **MST** algorithm that is too accurate would violate fundamental reconstruction lower bounds of **DP**.

This exactly matches the upper bound obtained by simply adding Laplace noise to each edge and closes a remaining logarithmic gap from [Sea16]. Observe that this implies that input perturbation is asymptotically optimal for private **MST**. With minor modifications, we also show that this lower bound works for the problem of privately releasing a **MWPM** and additionally, we extend the packing-based argument in [HT24] under pure **DP** to Minimum-Weight (Perfect) Matching.

We then turn our attention to a different problem, namely privately releasing a hierarchical clustering under Dasgupta’s cost function. Recently, Deng, Gao, Upadhyay, Wang, and Zhou [Den+25] showed a packing-based lower bound of $\Omega(n^2/\epsilon)$ on the complete graph topology. We give a reconstruction-based lower bound of $\Omega(\frac{1}{\epsilon} \cdot n \cdot \frac{\phi}{d_{\max}})$ that is parameterized by the balanced min-cut ϕ and the maximum degree $d_{\max}$ of the graph topology. It holds in expectation for a reasonable regime of the privacy parameters. This immediately recovers the previous bound [Den+25] on the complete graph (where $\phi = \Theta(n^2)$ and $d_{\max} = \Theta(n)$). Moreover, observe that, e.g., for d -regular expander graphs $\phi = \Omega(nd)$ and hence the quadratic lower bound still holds for extremely sparse graphs.

Again, the main idea is to encode a random binary vector into a random subset of the edges. The reconstruction attack then works by using new insights into the connection between the released clustering tree and an induced balanced min-cut. At a high level, we can reconstruct a large fraction of the edges that cross this cut if the private HC algorithm was too accurate. Finally, it matches the known (exponential-time) upper bound of $\mathcal{O}(\frac{1}{\epsilon} \cdot n^2 \cdot \log n)$ up to a logarithmic factor.

P2: PRIVATE MULTIPLE LOSSLESS RELEASES In Chapter 6, we study the setting where the same statistic is released under different privacy levels. This resembles natural settings where some parties are trusted more than others. As a simple baseline, one could just create these releases independently but this causes privacy costs in terms of composition. Our ultimate goal is to find cases where we can bypass this, so that the full sequence is jointly as private as the least private release within this sequence.

We formalize this intuition by introducing the notion of *lossless multiple release*. Informally, we say that a mechanism satisfies *lossless multiple release* if for any set of outputs, the joint distribution can be viewed as post-processing of the least private release in that subset. Additionally, we want each of those releases to be distributed as a direct release with its corresponding privacy level. A simple example where this works is the Gaussian mechanism: adding fresh Gaussian noise to an existing release yields a more private release that is obtained by post-processing.

We generalize this observation of “going from one noise term to another” and introduce a distributional property that we call *convolution preorder*. For any (possibly infinite) set of privacy parameters S , we say that a family of distributions $\mathcal{D}(\rho)$ parameterized by $\rho \in S$ admits *convolution preorder*, if there exists another distribution $\mathcal{C}(\rho_1, \rho_2)$ such that the convolution $\mathcal{D}(\rho_2) * \mathcal{C}(\rho_1, \rho_2)$ is distributed as $\mathcal{D}(\rho_1)$ for all $\rho_1 < \rho_2$. Note that the ρ ’s naturally correspond to some privacy level where a higher value means less privacy. Using this, one can not only create a more private release, but (given access to an additional independent release), one can also sample a less private release by combining the noise in a particular way

privacy level here is
a very general
concept and not tied
to any particular
definition.

known as inverse variance weighting. These settings are known as the *gradual release* and the *privacy relaxation*. Our work goes one step further and allows for *arbitrary release orders*.

Our main result proves a meta theorem stating that Private Lossless Multiple Release is possible for every Independent Additive Noise Mechanism that uses a noise distribution satisfying *convolution preorder*. Furthermore, we show that many distributions such as Gaussian, Laplace (and the discrete counterpart) and Poisson, satisfy this property.

In unpublished follow-up work, we initiate the study of non-additive noise mechanisms. We derive a Markov chain formulation that is equivalent to our lossless multiple release definition. We then use this to enable arbitrary releases for k -ary *Randomized Response*, extending and simplifying previous work [Pan24]. Finally, we give a numerical counterexample where Private Lossless Multiple Release is not possible, answering a question we raised in the published paper.

P3: THE CORRELATED SPARSE GAUSSIAN HISTOGRAM MECHANISM In Chapter 7, we show how a single correlated noise sample can improve the utility when releasing a sparse histogram under DP. Recently, Lebeda [Leb25] has observed that the release of d counting queries with the Gaussian mechanism can be improved in the case of the *add/remove* relationship. A very good overview about the geometric intuition was written by Desfontaines [Des24]. His result shows that instead of scaling with $\sqrt{d}$, the standard deviation scales as $\frac{1}{2}(\sqrt{d} + 1)$, which improves the utility by up to a factor of two. This is because, in this particular neighboring relationship, it is impossible for some counts to increase while others decrease. Their main insight is that using a single correlated noise sample can exploit this structural constraint.

In that paper, we extend this approach to *sparse* histograms. One way to preserve the sparsity of a histogram is to only add noise to the non-zero entries. However, this can result in an event of infinite privacy loss when a zero changes to a one in neighboring datasets (or vice versa). In that case, an adversary can immediately infer what the underlying dataset was. One solution is to introduce a threshold $\tau > 0$ and only release those values whose noisy value is above τ .

This mechanism can still have two kinds of bad events: some privacy loss δ_{gauss} that is caused by the Gaussian mechanism and some loss δ_{inf} when the noise of a neighboring non-zero count exceeds the threshold.

An observation due to [GAT20] is that if the Gaussian mechanism satisfies $(\epsilon, \delta_{\text{gauss}})$ -DP, adding the thresholding technique gives a mechanism that is $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -DP. We use this result together with the exact analysis of the Gaussian mechanism due to Wilkins, Kifer, Zhang, and Karrer [Wil+24] to prove that a correlated sample can also give an advantage in this setting. Finally, we give an even tighter analysis using ideas of Wilkins, Kifer, Zhang, and Karrer [Wil+24], making use of the fact that each coordinate can contribute to either δ_{inf} or δ_{gauss} , but never to both at the same time. We complement these results by extending them to other settings such as aggregate queries and top- k counting queries, by giving practical considerations for discrete noise, and by running experiments that verify the utility gain of our approach.

909

Part II

910

INCLUDED PAPERS

911

915 *published in Principles of Database Systems (PODS) 2025*

916 *Joint work with Rasmus Pagh, Hao Wu and Hanwen Zhang*
 917

918 We study the problem of privately releasing an approximate minimum span-
 919 ning tree (MST). Given a graph $G = (V, E, \mathbf{w})$ where V is a set of n vertices,
 920 E is a set of m undirected edges, and $\mathbf{w} \in \mathbb{R}^{|E|}$ is an edge-weight vector, our
 921 goal is to publish an approximate Minimum-Weight Spanning Tree (MST)
 922 under *edge-weight* differential privacy, as introduced by Sealfon in PODS 2016,
 923 where V and E are considered public and the weight vector is private.
 924 We use the ℓ_∞ -distance on weights: for a sensitivity parameter Δ_∞ , graphs
 925 $G = (V, E, \mathbf{w})$ and $G' = (V, E, \mathbf{w}')$ are neighboring if $\|\mathbf{w} - \mathbf{w}'\|_\infty \leq \Delta_\infty$.

926 Existing private MST algorithms face a trade-off, sacrificing either compu-
 927 tational efficiency or accuracy. We show that it is possible to get the best
 928 of both worlds: With a suitable random perturbation of the input that does
 929 *not* suffice to make the weight vector private, the result of *any* non-private
 930 MST algorithm will be private and achieves a state-of-the-art error guarantee.
 931 Furthermore, by establishing a connection to *Private Top-k Selection* [Steinke
 932 and Ullman, FOCS '17], we give the first privacy-utility trade-off lower bound
 933 for MST under approximate differential privacy, demonstrating that the error
 934 magnitude, $\tilde{O}(n^{3/2})$, is optimal up to logarithmic factors. That is, our approach
 935 matches the time complexity of any non-private MST algorithm and at the
 936 same time achieves optimal error. We complement our theoretical treatment
 937 with experiments that confirm the practicality of our approach.

938 In this section, we consider the problem of privately releasing an MST for a
 939 graph G where both the set of vertices $V = \{1, \dots, n\}$ and edges $E = \{1, \dots, m\}$
 940 are public, but we keep the weight vector $\mathbf{w} = (w_1, \dots, w_m) \in \mathbb{R}^{|E|}$ private. We
 941 consider the ℓ_∞ neighborhood relation, where all weights can differ by at most
 942 Δ_∞ and want to protect the information encoded on the weights, which is known
 943 as *edge-weight differential privacy* [Sea16]. An application example is to model
 944 passenger data on a public city traffic network. Another setting comes from
 945 graphical modeling: consider a dataset $D = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(d)})$ of size d where each
 946 vector $\mathbf{x}^{(i)} \in \{0, 1\}^n$ represents a list of sensitive binary attributes. The *Chow-*
 947 *Liu Tree* [CL68] is the minimum spanning tree derived from the negated *mutual*
 948 *information matrix* encoding all pairwise mutual information of the attributes.
 949 Changing one $\mathbf{x}$ in D could simultaneously alter *all* weights.

950 Currently, there are two different algorithmic approaches: *input-privatization* [Sea16]
 951 and *in-place* [Pin+18; MMS21]. The former method adds noise (e.g., Laplace or
 952 Gaussian) to each edge weight and publishes the entire noisy weight vector. This
 953 allows anyone to compute the MST while privacy is ensured by the *post-processing*
 954 property of differential privacy. While this approach allows the design of an
 955 (expected) linear-time algorithm, it provides only a worst-case additive error of
 956 $\tilde{O}(n^2)$.

957 The latter approach, which adds noise during the execution of an MST algorithm,
 958 achieves a better error bound of $\tilde{O}(n^{3/2})$, but its best-known running time is
 959 $\tilde{O}(n^{3/2} + m)$ for a fixed privacy parameter [PR24]. Thus, there is a gap, and whether

input-privatization
 is equivalent to
 input-perturbation
 with the identity map.

Reference	Additive Error	Time	Technique
[Sea16]	$O\left((1/\varepsilon) \cdot n\sqrt{m} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$\text{MST} + O(m)$	Input privatization
[Sea16]	$\Omega(n)$	–	Lower bound
[Pin18]	$O\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$O(nm)$	In-place noise
[PR24]	$O\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$O\left(m + n^{3/2} \log n\right)$	In-place noise
New	$O\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$\text{MST} + O(m)$	Input perturbation
New	$\Omega\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n)\right)$	–	Lower bound

Table 4.1: Results on (ε, δ) -DP MST with ℓ_∞ neighboring relationship on edge weights, assuming sensitivity $\Delta_\infty = 1$. Sealfon [Sea16] and Pinot [Pin18] originally provided upper bounds for pure DP, but these can be adapted for approximate DP. Sealfon considered the ℓ_1 neighboring relationship, which leads to the same lower bound and a weaker upper bound for ℓ_∞ . The unpublished manuscript [PR24] was written by a subset of the authors. The “MST” in the **Time** column refers to the running time of any non-private MST algorithm.

it is possible to achieve the “best of both worlds” has remained unknown. We answer these two open questions in the affirmative:

Question 1: Can we design an (expected) linear-time private MST algorithm that matches the error guarantee of $\tilde{O}(n^{3/2})$ of the *in-place* approach?

Question 2: Is the error asymptotically optimal?

OUR CONTRIBUTION We introduce the *first* algorithmic framework that reduces the private MST problem (under the ℓ_∞ neighboring relationship) to the non-private one, achieving an additive error of $\tilde{O}(n^{3/2})$. Compatible with any non-private MST algorithm, it enables the *first* (expected) linear-time private MST solution. We also establish the *first* $\tilde{\Omega}(n^{3/2})$ error lower bound, proving the asymptotic optimality of our framework. Formally, we show the following upper and lower bounds:

Theorem 4.1 (Upper Bound). *Let $G = (V, E, \mathbf{w})$ be a graph with n vertices and m edges, and let $\varepsilon, \delta > 0$. Consider an arbitrary (non-private) algorithm that computes an MST of G within time $t(n, m)$, independent of the weight vector. Then there exists an (ε, δ) -differentially private mechanism $\mathcal{M}$ that releases a random, approximately minimum spanning tree $\mathcal{M}(G)$, such that, if T^* is a minimum spanning tree:*

- *the weight difference satisfies*

$$\mathbb{E} \left[\sum_{e \in \mathcal{M}(G)} w_e \right] - \sum_{e \in T^*} w_e \in O\left(\frac{1}{\varepsilon} \cdot n^{3/2} \cdot (\log n) \cdot \sqrt{\log(1/\delta)}\right), \text{ and}$$

- *the running time is $t(n, m) + \mathcal{O}(m)$.*

For example, the randomized linear time algorithm by Karger, Klein, and Tarjan [KKT95] or the best-known near-linear time deterministic algorithm due to Chazelle [Cha00] can be used in Theorem 4.1. Our approach matches all known upper bounds, meanwhile it is very simple to implement. See Table 4.1 for a comprehensive summary. Moreover, the result can be easily extended to finding a maximum weight independent set in any matroid with rank n and m elements, where $t(n, m)$ is the running time for any algorithm that computes the maximum weight independent set in such a matroid.

The key technique is a novel approach (formally introduced in Section 4.2.3) to iteratively sample k items from a larger candidate set with probabilities proportional to their weights. After each selection, the chosen item and a subset of the remaining candidates—determined by the sequence of previously selected items—are removed before the next selection. Our approach generates noisy weights for all items at the beginning of the sampling process, which are reused throughout the iterative selection procedure, thereby eliminating the need to regenerate fresh randomness at each step. As discussed in Section 4.4, this technique extends a substantial body of research on top- k sampling [Yel77; Ohl90; Ros97; Coh97].

Lower Bound. Our lower bound partially resolves an open question of Hladík and Tetek [HT24], establishing a tight bound for worst-case instances under approximate differential privacy:

Theorem 4.2 (Lower Bound). *Let $\varepsilon \leq 1$ and $\delta \in O(1/\sqrt{n})$, and let $\mathcal{M}$ be an (ε, δ) -DP MST algorithm. There exists an input graph $G = (V, E, \mathbf{w})$ with n vertices such that, if T^* is the minimum spanning tree of G , the (random) spanning tree $\mathcal{M}(G)$ released by $\mathcal{M}$ satisfies:*

$$\mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w_e] - \sum_{e \in T^*} w_e \in \Omega\left(\frac{1}{\varepsilon} \cdot n^{3/2} \cdot \ln n\right). \quad (1)$$

The proof technique is based on a lower bound technique for top- k selection (reporting k elements of approximately maximum value) in [SU17]. First, observe that maximization and minimization are equivalent for spanning trees by simply flipping the sign of all edge weights. A maximum spanning tree can be seen as a top- $(n-1)$ selection problem (selecting $n-1$ edges of the approximately largest value) with an additional constraint that the edge set is acyclic. Thus, it is natural to guess that the hard instances of Steinke and Ullman [SU17], embedded into the weights of a complete graph, yield a hard instance for the maximum spanning tree as well. By carefully adapting the proof in [SU17], we show that this is indeed the case. In particular, leveraging a result from the Erdős–Rényi random graph model [ER59], we show the existence of a spanning tree with weight similar to the top- $(n-1)$ edge weights.

Organization. The rest of the section is organized as follows. Section 4.1 introduces the problem formally and reviews the necessary preliminaries. Section 4.2 describes the reduction from normal non-private MST algorithm to private MST algorithm that achieve the desired utility and running time. Section 4.3 establishes the lower bound for the problem. Section 4.4 presents a detailed comparison of our techniques with previous ones and explores the broader background of the problem. Experiments in Section 4.5 show the practicality of our approach.

4.1 PRELIMINARIES

We consider a graph $G = (V, E, \mathbf{w})$ where the set of vertices $V = \{1, \dots, n\}$ and edges $E = \{1, \dots, m\}$ are public, and the weight vector $\mathbf{w} = (w_1, \dots, w_m) \in \mathbb{R}^m$ is private. For each edge $e \in E$, let w_e denote its weight. The cost of a subset $T \subseteq E$ is defined as $w(T) \doteq \sum_{e \in T} w_e$. A *spanning tree* is an acyclic subset $T \subseteq E$ that makes the graph connected. Denote $\mathcal{T}(G)$ the collection of all possible spanning trees in G . A *minimum (cost) spanning tree* (MST) is a spanning tree T^* which minimizes $w(T^*)$. We sometimes hide logarithmic factors using tilde notation for $\tilde{O}$, $\tilde{\Theta}$ and $\tilde{\Omega}$.

This section makes use several probability distributions and we refer to Part [iii](#) for their definitions.

4.1.1 Differential Privacy

Given an input graph $G = (V, E, \mathbf{w})$, the private **MST** problem aims to find a spanning tree T while preserving the privacy of the weight vector $\mathbf{w}$. In this section, we consider the ℓ_∞ neighboring relationship, restated here for convenience.

Definition 4.1 (ℓ_∞ -neighboring inputs). *Given $\Delta_\infty > 0$, two weight vectors $\mathbf{w}, \mathbf{w}' \subseteq \mathbb{R}^E$ are neighboring, denoted $\mathbf{w} \sim \mathbf{w}'$, if and only if $\|\mathbf{w} - \mathbf{w}'\|_\infty = \max_{e \in E} |w_e - w'_e| \leq \Delta_\infty$. Two graphs $G = (V, E, \mathbf{w})$ and $G' = (V', E', \mathbf{w}')$ are neighboring, denoted $G \sim G'$, if and only if $V = V'$, $E = E'$, and $\mathbf{w} \sim \mathbf{w}'$.*

For simplicity, we assume throughout the paper that $\Delta_\infty = 1$ and all bounds presented can be generalized by scaling a factor of Δ_∞ .

We will use ρ -zero concentrated DP (ρ -zCDP) as defined in Definition [2.9](#).

Exponential Mechanism. The exponential mechanism [\[MT07\]](#) can be used to differentially private release discrete outputs. It operates over an input space $\mathcal{X}$ (with neighboring datasets defined by a relation $\sim$) and a finite output space $\mathcal{Y}$. The mechanism $\mathcal{M}_{\text{EXP}} : \mathcal{X} \rightarrow \mathcal{Y}$ assigns the following probabilities for a given dataset $x \in \mathcal{X}$:

$$\mathbb{P}[\mathcal{M}_{\text{EXP}}(x) = y] \propto \exp(-\varepsilon \cdot \mathcal{E}(x, y) / (2 \cdot \Delta_{\text{EXP}})), \quad \forall y \in \mathcal{Y} \quad (2)$$

where $\mathcal{E} : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is the *loss function* quantifying the cost of selecting y given input x , and Δ_{EXP} is the *sensitivity* of $\mathcal{E}$, i.e., the maximum change in the loss function for neighboring datasets:

$$\Delta_{\text{EXP}} \doteq \max_{x \sim x', y \in \mathcal{Y}} |\mathcal{E}(x, y) - \mathcal{E}(x', y)|.$$

Fact 4.3 ([\[MT07\]](#) Properties of $\mathcal{M}_{\text{EXP}}$). *The exponential mechanism $\mathcal{M}_{\text{EXP}}$ satisfies ε -differential privacy. Moreover, for each $\beta \in (0, 1)$, and $\tau \doteq \frac{2 \cdot \Delta_{\text{EXP}}}{\varepsilon} \cdot \ln \frac{|\mathcal{Y}|}{\beta}$, it holds that:*

$$\mathbb{P}[\mathcal{E}(x, \mathcal{M}_{\text{EXP}}(x)) \geq \min_{y \in \mathcal{Y}} \mathcal{E}(x, y) + \tau] \leq \beta, \quad \forall x \in \mathcal{X}. \quad (3)$$

4.2 REDUCTION FROM PRIVATE MST TO NON-PRIVATE MST

In this section, we present a novel algorithmic framework (Algorithm [1](#)) that enables a reduction from the private **MST** problem to the non-private one, thereby proving Theorem [4.1](#). A key feature of this framework is its independence from the chosen **MST** algorithm. Since the graph's topology is public, the most suitable algorithm can always be selected in advance. Notably, this framework facilitates an (expected) linear-time private **MST** algorithm.

Overview. Figure [4.1](#) outlines the roadmap of this section. In Section [4.2.1](#), we introduce a simple algorithmic framework that fulfills Theorem [4.1](#). The framework adds certain noise to each edge weight, then applies an arbitrary non-private

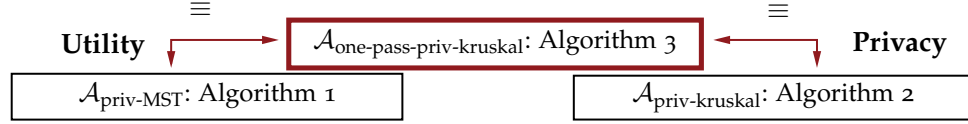


Figure 4.1: Roadmap of Section 4.2. The figure outlines the proof structure: the utility guarantee is established for Algorithm 1, and the privacy guarantee is proven for Algorithm 2. A "bridging" algorithm (Algorithm 3) is introduced to demonstrate the equivalence of Algorithms 1 and 2 by showing they share the same output distribution.

1045 **MST** algorithm to publish only the set of **MST** edges with respect to the noisy
 1046 weights. Proving its time complexity and utility is straightforward, but its privacy
 1047 guarantee requires more effort. Instead of proving privacy directly, we establish
 1048 it through two equivalent algorithms. Section 4.2.2 describes a private version
 1049 of Kruskal's algorithm [Kru56] (Algorithm 2), which iteratively and privately
 1050 selects the approximate **MST** edges. Its privacy analysis follows directly from the
 1051 composition property of DP, though its utility analysis is more intricate. Finally,
 1052 in Section 4.2.3, we present a "bridging" algorithm (Algorithm 3) that has the
 1053 same output distribution as both Algorithm 1 and Algorithm 2, establishing their
 1054 equivalence. This allows us to transfer privacy and utility guarantees between the
 1055 algorithms, simplifying the proofs.

1056 A detailed comparison of our algorithmic techniques with prior work, along
 1057 with a broader discussion of the problem's background, is deferred to Section 4.4.

1058 4.2.1 Reduction

1059 The framework is outlined in Algorithm 1. It is straightforward to implement and
 1060 requires only 3 lines of code using standard libraries. Given a privacy parameter ρ
 1061 and an input graph $G = (V, E, \mathbf{w})$, it computes for each edge $e \in E$ a noisy weight
 1062 $\tilde{w}_e \doteq w_e + (2/\epsilon') \cdot \ln \mathbb{Exp}(1)$, where $\mathbb{Exp}(1)$ denotes a random variable following
 1063 the exponential distribution (Definition A.4). Finally, it computes the **MST** using an
 1064 existing (non-private) **MST** algorithm $\mathcal{A}_{MST}$ on the noisy weights $\tilde{\mathbf{w}} = (\tilde{w}_1, \dots, \tilde{w}_m)$.

Algorithm 1 $\mathcal{A}_{\text{priv-MST}}$: Private **MST** Framework

Require: $G = (V, E, \mathbf{w})$, any **MST** algorithm $\mathcal{A}_{MST}$, privacy parameters (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2 \cdot \rho / (n-1)}$;
 - 2: $\tilde{w}_e \leftarrow w_e + (2/\epsilon') \cdot \ln(\mathbb{Exp}(1))$ for all $e \in E$.
 - 3: **return** $\mathcal{A}_{MST}(V, E, \tilde{\mathbf{w}})$.
-

1065 **EXTENSION TO FINDING A MAXIMUM WEIGHT INDEPENDENT SET IN MA-**
 1066 **TROIDS** The framework extends to general matroids by perturbing each ele-
 1067 ment's weight with similar noise and computing the maximum weight independent
 1068 set using any suitable algorithm on the noisy weights, see Section 4.7.1.

1069 The properties of Algorithm 1 are outlined in Theorem 4.1.

1070 *Proof of Theorem 4.1.* It remains to analyze its running time, privacy and utility
 1071 guarantees.

1072 **Running Time.** Adding noise to all edge weights requires only $O(m)$ time. Con-
 1073 sequently, the overall running time of Algorithm 1 is given by $O(m + t_{\mathcal{A}_{MST}}(n, m))$,
 1074 where $t_{\mathcal{A}_{MST}}(n, m)$ denotes the running time of the chosen **MST** algorithm $\mathcal{A}_{MST}$
 1075 for a graph with n vertices and m edges. For example, with the celebrated **MST**

algorithm by Karger, Klein, and Tarjan [KKT95], Algorithm 1 achieves an expected linear running time, where $t_{A_{MST}}(n, m) \in O(n + m)$. Our reduction is compatible with any **MST** algorithm, even in parallel or distributed settings.

Utility Guarantee. We analyze the utility of Algorithm 1 before establishing the privacy guarantee, as the former is more straightforward. We need the following fact, whose proof is in Appendix 4.7.3.

Fact 4.4. Let $Z_i \sim \text{ln Exp}(1)$ for $i \in [m]$ be independent. Then,

$$\mathbb{E} \left[\max_{i \in [m]} |Z_i| \right] \in O(\ln m).$$

Let T^* and T denote the MSTs with respect to the original edge weights $\mathbf{w}$ and the noisy edge weights $\tilde{\mathbf{w}}$, respectively. Note that T is precisely the spanning tree returned by Algorithm 1. We aim to prove that T satisfies the utility guarantee specified in Theorem 4.1. By Fact 4.4 and noting that both T and T^* contain fewer than n edges, it follows that

$$\mathbb{E} [\max_{e \in E} |\tilde{w}_e - w_e|] \in O(1/\epsilon' \cdot \ln m) \implies \frac{\mathbb{E} [|w(T) - \tilde{w}(T)|]}{\mathbb{E} [|w(T^*) - \tilde{w}(T^*)|]} \in O(n/\epsilon' \cdot \ln m). \quad (4)$$

As T is the **MST** on the noisy weights $\tilde{\mathbf{w}}$, we have $\tilde{w}(T) \leq \tilde{w}(T^*)$. Therefore,

$$\begin{aligned} \mathbb{E} [w(T) - w(T^*)] &\leq \mathbb{E} [w(T) - \tilde{w}(T) + \tilde{w}(T^*) - w(T^*)] \\ &\leq \mathbb{E} [|w(T) - \tilde{w}(T)|] + \mathbb{E} [|\tilde{w}(T^*) - w(T^*)|] \in O(n/\epsilon' \cdot \ln m). \end{aligned} \quad (5)$$

Finally, to obtain the utility bound in Theorem 4.1, based on Algorithm 1, Line 1, we observe that

$$\begin{aligned} \frac{1}{\epsilon'} &= \sqrt{\frac{n-1}{2\rho}} \in O\left(\frac{1}{\epsilon} \sqrt{n \log \frac{1}{\delta}}\right), \text{ since,} \\ \frac{1}{\sqrt{\rho}} &= \frac{1}{\sqrt{\epsilon + \log \frac{1}{\delta}} - \sqrt{\log \frac{1}{\delta}}} = \frac{\sqrt{\epsilon + \log \frac{1}{\delta}} + \sqrt{\log \frac{1}{\delta}}}{\epsilon} \in O\left(\frac{1}{\epsilon} \sqrt{\log \frac{1}{\delta}}\right). \end{aligned}$$

Substituting the expression for $1/\epsilon'$ into Equation (5) completes the proof.

Privacy Guarantee. As outlined in the *roadmap* and illustrated in Figure 4.1, directly proving the privacy guarantees is challenging. To address this, we introduce Algorithm 2 in Section 4.2.2, whose privacy guarantee can be easily established, and Algorithm 3 in Section 4.2.3, which shares the same output distribution as Algorithms 1 and 2, transferring the privacy guarantee to the former. $\square$

4.2.2 Private Kruskal Algorithm

Algorithm 2 presents a privatized version of Kruskal's algorithm. Compared to the non-private one [Kru56], at each iteration (Line 3-6), instead of selecting the edge with minimum weight, it selects an approximately minimum-weight edge by the exponential mechanism [MT07] which samples an edge e with probability proportional to $e^{-(\epsilon'/2) \cdot w_e}$.

Algorithm 2 $\mathcal{A}_{\text{private-Kruskal}}$: Private Kruskal's Algorithm**Require:** Graph $G = (V, E, \mathbf{w})$, privacy parameter (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2} \cdot \rho / (n - 1)$;
- 2: Initialize $T \leftarrow \emptyset, F \leftarrow E$.
- 3: **for** $i \leftarrow 1$ to $|V| - 1$ **do**
- 4: Sample $e \in F$ with probability $\propto e^{-\frac{\epsilon'}{2} \cdot w_e}$.
- 5: $T \leftarrow T \cup \{e\}$.
- 6: $F \leftarrow F \setminus \{e' \in F : \{e'\} \cup T \text{ contains a cycle}\}$.
- 7: **return** T .

Theorem 4.5 (Properties of Algorithm 2). *Given an input graph $G = (V, E, \mathbf{w})$ with n vertices and m edges, Algorithm 2 runs in $\mathcal{O}(m + n \log n)$ time, is (ϵ, δ) -DP, and returns an approximately minimum spanning tree $\mathcal{M}(G)$ such that, if T^* is a minimum spanning tree:*

$$\mathbb{E} [\sum_{e \in \mathcal{M}(G)} w_e] - \sum_{e \in T^*} w_e \in O\left(\frac{1}{\epsilon} \cdot n^{3/2} \cdot (\log n) \cdot \sqrt{\log(1/\delta)}\right). \quad (6)$$

1103 *Proof of Theorem 4.5.* We analyze the running time, privacy and utility guarantees
1104 of Algorithm 2.

1105 **Running Time.** Algorithm 2 can be implemented in $\mathcal{O}(m + n \log n)$ time using a
1106 designated data structure. As it is not our primary focus, we refer for the details to
1107 Section 4.7.5. It remains to analyze the privacy-utility trade-offs of Algorithm 2.

1108 **Privacy Guarantee.** We leverage ρ -zCDP for its tighter composition bounds, so the
1109 proof involves conversion from DP to ρ -zCDP and back. We need to show that
1110 Algorithm 2 guarantees (ϵ, δ) -DP.

- 1111 • Based on the privacy guarantee of the exponential mechanism (Fact 4.3), each
1112 round of Algorithm 2 is ϵ' -DP. Using the conversion from DP to ρ -zCDP (Fact 2.2)
1113 and the initialization of ϵ' (Line 1), this corresponds to ρ' -zCDP, where $\rho' =$
1114 $(\epsilon')^2/2 = \rho/(n - 1)$.
- 1115 • Since the algorithm runs $n - 1$ rounds, by the composition property (Fact 2.5),
1116 Algorithm 2 satisfies $(n - 1) \cdot \rho' = \rho$ -zCDP.
- 1117 • Applying Fact 2.2 in reverse (from ρ -zCDP to DP), we conclude that the algorithm
1118 satisfies $(\rho + 2 \cdot \sqrt{\rho \log(1/\delta)}, \delta)$ -DP. Substituting

$$\rho \leftarrow \left(\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)} \right)^2 \text{ yields, } \rho + 2 \cdot \sqrt{\rho \log(1/\delta)} = \epsilon.$$

1119 **Utility Guarantee.** The utility analysis is more involved, so we defer it to the
1120 next section, where we introduce a simplified implementation of Algorithm 2 and
1121 establish its equivalence to Algorithm 1. This allows the straightforward utility
1122 analysis of Algorithm 1 to transfer directly to Algorithm 2, demonstrating the
1123 benefit of proving the algorithms' equivalence.

1124 It is worth noting that the utility guarantee of Algorithm 2 can be proven directly,
1125 although it is much more complicated and requires a much deeper insight into
1126 Kruskal's algorithm. A proof is provided in Section 4.7.4 for reference.

1127

□

4.2.3 One-Pass Private Kruskal

Finally, we introduce the last private **MST** algorithm. It achieves the same running time as Algorithm 2, but features a significantly simplified structure, allowing it to be implemented using existing libraries for non-private Kruskal's algorithm. More importantly, it serves as a bridge to establish the equivalence between Algorithm 1 and Algorithm 2.

The algorithm is outlined in Algorithm 3. It begins by generating noisy edge weights using $\tilde{w}_e \doteq \text{Exp}(1) / e^{-(\epsilon'/2)w_e}$ and then applies the non-private Kruskal's algorithm to the noisy weights.

Algorithm 3 $\mathcal{A}_{\text{one-pass-private-Kruskal}}$

Require: Graph $G = (V, E, \mathbf{w})$, privacy parameter (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2 \cdot \rho / (n-1)}$;
 - 2: Initialize $T \leftarrow \emptyset$, $F \leftarrow E$.
 - 3: $\tilde{w}_e \leftarrow \text{Exp}(1) / e^{-\frac{\epsilon'}{2} \cdot w_e}$. $\triangleright \text{Exp}(1)$ represents exponential noise
 - 4: **for** $i \leftarrow 1$ to $|V|-1$ **do**
 - 5: $e \leftarrow \arg \min_{e \in F} \tilde{w}_e$.
 - 6: $T \leftarrow T \cup \{e\}$.
 - 7: $F \leftarrow F \setminus \{e' \in F : \{e'\} \cup T \text{ contains a cycle}\}$.
 - 8: **return** T .
-

Theorem 4.6. Algorithm 3 shares the properties of Algorithm 2 as stated in Theorem 4.5.

Proof of Theorem 4.6. We analyze the runtime, utility, and privacy guarantees of Algorithm 3. In the process, we establish the equivalence among Algorithm 1, Algorithm 2, and Algorithm 3, i.e., they have the same output distributions.

Running Time. The running time is dominated by Kruskal's algorithm, which is $O(n + m \log n)$.

Utility Guarantee. The utility guarantee is established through the equivalence between Algorithm 1 and Algorithm 3, allowing the utility guarantee of the former to transfer to the latter.

First, observe that in line 5 of Algorithm 3, the returned edge satisfies

$$\arg \min_{e \in F} \tilde{w}_e = \arg \min_{e \in F} (w_e + \frac{2}{\epsilon'} \cdot \ln \text{Exp}(1)). \quad (7)$$

Hence, we can replace the noisy weight $\tilde{w}_e \doteq \text{Exp}(1) / e^{-\frac{\epsilon'}{2} \cdot w_e}$ in Algorithm 3 by $\tilde{w}_e \doteq w_e + \frac{2}{\epsilon'} \cdot \ln \text{Exp}(1)$. Second, since $\frac{2}{\epsilon'} \cdot \ln \text{Exp}(1)$ is a continuous random variable, the **MST** on the noisy graph is unique with probability 1. Consequently, any **MST** algorithm computes the same solution, thereby establishing the equivalence between Algorithm 1 and Algorithm 3.

Privacy Guarantee. The privacy guarantee is established through the equivalence between Algorithms 2 and 3, allowing the privacy guarantee of the former to transfer to the latter.

To establish the equivalence, we introduce a generalized model, *Probability Proportional to Sizes with Adaptive Candidate Removal* (Probability Proportional to Sizes with Adaptive Candidate Removal (**PPSACR**)), for top- k sampling, along with

a new sampling technique for this model. As will be shown, Algorithm 2 can be viewed as a special case of PPSACR, and Algorithm 3 directly implements the new sampling technique, thereby demonstrating its equivalence to Algorithm 2. It is worth noting that the equivalence between Algorithms 2 and 3 can also be established directly, without relying on PPSACR. However, we choose to present PPSACR here, as it is of independent interest and may have broader applications to other problems.

PPSACR MODEL: The model is described in Algorithm 4. Let U be a finite set of items, indexed from 1 to $|U|$. Each item $j \in U$ is associated with a weight $s(j) \in \mathbb{R}^+$. PPSACR iteratively samples k items. It maintains a candidate set $\mathcal{C}$, initialized as U , and a selected item list $\mathcal{I}$, which is initially empty. At each step, an item $j \in \mathcal{C}$ is sampled with probability proportional to $s(j)$ and appended to $\mathcal{I}$. Afterward, the sampled item j and an additional subset of items, denoted by $f(\mathcal{I})$, are removed from $\mathcal{C}$. Here, $f : (\bigcup_{i \in [k]} U^i) \rightarrow 2^U$ is a function that determines the additional items to be removed based on the sequence of selected items so far. Since the set $U \setminus \mathcal{C}$ at each step is fully determined by $\mathcal{I}$ and f , the function f , given $\mathcal{I}$ as input, effectively encodes the information about $U \setminus \mathcal{C}$.

Example 4.7. Algorithm 2 can be interpreted as a special case of this sampling model, where U represents the set of edges E , $s(e)$ is the exponential mechanism sampling weight $e^{-\lambda \cdot w_e^*}$ for each $e \in E$, k is $n - 1$ (the number of edges in a spanning tree), and f is the function that removes edges forming cycles based on the previously selected edges.

Algorithm 4 PPSACR

```

1:  $\mathcal{C} \leftarrow U, \mathcal{I} \leftarrow$  An empty list  $\triangleright$  possible candidates and sampled items so far.
2: while  $|\mathcal{I}| \leq k$  and  $\mathcal{C} \neq \emptyset$  do
3:   Sample  $j \in \mathcal{C}$  with probability proportional to  $s(j)$ .
4:   Add  $j$  to the back of  $\mathcal{I}$ .
5:    $\mathcal{C} \leftarrow \mathcal{C} \setminus (\{j\} \cup f(\mathcal{I}))$ .
6: return  $\mathcal{I}$ 

```

Theorem 4.8 (One-Shot-PPSACR, Algorithm 6). *Algorithm 4 can be equivalently implemented with the following modifications:*

1. Add a noisy weights generating step before line 1: $\tilde{s}(j) \leftarrow \text{Exp}(1) / s(j), \forall j \in U$.
2. Replace line 3 with $j \leftarrow \arg \min_{j' \in \mathcal{C}} \tilde{s}(j')$.

The modified algorithm is referred to as One-Shot-PPSACR.

Due to a cleaner exposition, the pseudo codes of the modified algorithm (Algorithm 6) is given in Section 4.7.2. Note that if Algorithm 2 is viewed as a special case of PPSACR, as described in Example 4.7, then Algorithm 3 directly implements the modifications proposed in Theorem 4.8. Therefore, Theorem 4.8 immediately establishes the equivalence between Algorithms 2 and 3.

To proceed, we require the following properties of the exponential distribution, the proofs of which are included in Appendix 4.7.3.

Fact 4.9 (Scaling). *If $X \sim \text{Exp}(1)$, then for all $\lambda > 0$, we have that $Y \doteq X/\lambda$ has distribution $\text{Exp}(\lambda)$.*

Fact 4.10 (Minimum). *If $X_1 \sim \text{Exp}(\lambda_1), \dots, X_d \sim \text{Exp}(\lambda_d)$, then*

$$\mathbf{P} \left[X_i = \min_{j \in [d]} X_j \right] = \frac{\lambda_i}{\sum_{j \in [d]} \lambda_j}.$$

Fact 4.11 (Memoryless property of exponential distribution). *If $X \sim \text{Exp}(\lambda)$, then*

$$\mathbf{P}[X \geq x + y \mid X \geq x] = \mathbf{P}[X \geq y], \forall x, y \geq 0.$$

Proof of Theorem 4.8. To simplify the discussion, we assume that k sampled items are returned by **PPSACR**. The proof for the case when less than k items are returned is similar. Let $\mathcal{J} \doteq (j_1, j_2, \dots, j_k) \in U^k$ be a feasible output sequence of **PPSACR**. We will prove that **One-Shot-PPSACR** outputs $\mathcal{J}$ with the same probability as **PPSACR**. **PPSACR**: For each $i \in [k]$, denote $\mathcal{C}_i$ the set of candidates to be sampled in Algorithm 4, Line 3 during the i -th iteration. The probability of selecting j_i is $s(j_i)/(\sum_{j \in \mathcal{C}_i} s(j))$. Thus, the probability that **PPSACR** outputs $\mathcal{J}$ is

$$\frac{s(j_1)}{\sum_{j \in \mathcal{C}_1} s(j)} \cdot \frac{s(j_2)}{\sum_{j \in \mathcal{C}_2} s(j)} \cdots \frac{s(j_k)}{\sum_{j \in \mathcal{C}_k} s(j)}. \quad (8)$$

One-Shot-PPSACR: For each $i \in [k]$, let $\mathcal{E}_i$ be the event that $\tilde{s}(j_i) = \min_{j \in \mathcal{C}_i} \tilde{s}(j)$, and define $\mathcal{E}_{1:i} \doteq \mathcal{E}_1 \wedge \dots \wedge \mathcal{E}_i$. First, by Fact 4.9, for each $j \in U$, $\tilde{s}(j) \doteq \text{Exp}(1)/s(j)$ follows distribution $\text{Exp}(s(j))$. Based on Fact 4.10, the minimum property of a collection of exponential random variables, it holds that

$$\mathbf{P}[\mathcal{E}_1] = \frac{s(j_1)}{\sum_{j \in \mathcal{C}_1} s(j)}. \quad (9)$$

Second, let $z_1 \leq z_2 \leq \dots \leq z_k$ be a feasible realization of $\tilde{s}(j_1), \dots, \tilde{s}(j_k)$. For each $i \in [2..k]$, define $\tilde{s}(j_{1:i-1}) \doteq (\tilde{s}(j_1), \dots, \tilde{s}(j_{i-1}))$ and $z_{1:i-1} = (z_1, \dots, z_{i-1})$. Conditioned on the events $\mathcal{E}_{1:i-1}$ and $\tilde{s}(j_{1:i-1}) = z_{1:i-1}$, we know that for each $j \in \mathcal{C}_i$, $\tilde{s}(j) \geq (\max z_{1:i-1}) = z_{i-1}$. Since the $\tilde{s}(j)$ are independent, by the memoryless property of the exponential distribution (Fact 4.11), $\tilde{s}(j) - z_{i-1}$ still follows the distribution $\text{Exp}(s(j))$. Together with the fact that the minimum is invariant under a constant shift, the event $\mathcal{E}_i$ is now equivalent to $\tilde{s}(j_i) - z_{i-1} = \min_{j \in \mathcal{C}_i} (\tilde{s}(j) - z_{i-1})$, applying Fact 4.10 again yields:

$$\begin{aligned} & \mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}, \tilde{s}(j_{1:i-1}) = z_{1:i-1}] \\ &= \mathbf{P} \left[\tilde{s}(j_i) - z_{i-1} = \min_{j \in \mathcal{C}_i} (\tilde{s}(j) - z_{i-1}) \mid \mathcal{E}_{1:i-1}, \tilde{s}(j_{1:i-1}) = z_{1:i-1} \right] \\ &= \frac{s(j_i)}{\sum_{j \in \mathcal{C}_i} s(j)}. \end{aligned}$$

1205 Taking expectation over $\tilde{s}(j_{1:i-1})$ gives

$$\mathbb{E}_{\tilde{s}(j_{1:i-1})} [\mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}, \tilde{s}(j_{1:i-1})]] = \mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}] = \frac{s(j_i)}{\sum_{j \in \mathcal{C}_i} s(j)}.$$

1206 Finally, based on chain rule of probability, $\mathcal{E}_{1:k}$ has exactly the same probability
1207 as Equation (8):

$$\mathbf{P}[\mathcal{E}_{1:k}] = \mathbf{P}[\mathcal{E}_1] \cdot \prod_{i=2}^k \mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}] = \frac{s(j_1)}{\sum_{j \in \mathcal{C}_1} s(j)} \cdot \frac{s(j_2)}{\sum_{j \in \mathcal{C}_2} s(j)} \cdots \frac{s(j_k)}{\sum_{j \in \mathcal{C}_k} s(j)}. \quad (10)$$

1208 □

1209 4.3 A LOWER BOUND FOR APPROXIMATE DP WITH ℓ_∞

1210 In this section, we prove the lower bound stated in Theorem 4.2. Our proof consists
1211 of two steps: in Section 4.3.1, we present a simple reduction from (ϵ, δ) -DP **MST**
1212 algorithms to $(1, \delta)$ -DP **MST** algorithms to facilitate the derivation of lower bounds;
1213 in Section 4.3.2, we present a lower bound for all $(1, \delta)$ -DP **MST** algorithms, which
1214 can be extended to (ϵ, δ) -DP lower bound using the previous reduction.

1215 4.3.1 Reduction from (ϵ, δ) -DP to $(1, \delta)$ -DP

Lemma 4.12. Assume that $\epsilon < 1$. Suppose that there is an $(\epsilon, \frac{\epsilon^\epsilon - 1}{\epsilon - 1} \cdot \delta)$ -DP **MST** algorithm $\mathcal{M}$ under the ℓ_∞ neighboring relationship, such that for every input graph $G = (V, E, \mathbf{w})$ (with the **MST** denoted by T^*), it holds that

$$\mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w_e] - \sum_{e \in T^*} w_e \in o\left((1/\epsilon) \cdot n^{3/2} \cdot \ln n\right). \quad (11)$$

Then there exists a $(1, \delta)$ -DP **MST** algorithm $\mathcal{M}'$, such that for every input graph $G' = (V', E', W')$ (with the **MST** denoted by T'), it holds that

$$\mathbb{E}_{\mathcal{M}'} \left[\sum_{e \in \mathcal{M}'(G')} w'_e \right] - \sum_{e \in T'} w'_e \in o\left(n^{3/2} \cdot \ln n\right). \quad (12)$$

1216 *Proof.* We demonstrate how to construct $\mathcal{M}'$ based on $\mathcal{M}$. Given a graph $G' =$
1217 (V', E', W') , we create a new graph $G = (V, E, W)$ as follows: $V = V'$, $E = E'$, and
1218 $W = (1/\epsilon) \cdot W'$. To simplify the discussion, we assume that $1/\epsilon \in \mathbb{N}^+$; otherwise,
1219 we replace $1/\epsilon$ with $\lceil 1/\epsilon \rceil$, which affects the privacy and utility guarantees only by
1220 a constant factor, as $\epsilon \leq 1$. Based on this construction, we have $T' = T^*$. Finally, we
1221 define $\mathcal{M}'$ as: $\mathcal{M}'(G') \doteq \mathcal{M}(G)$. Then based on group privacy property (Fact 2.7),
1222 $\mathcal{M}'$ is $\left(\epsilon \cdot \frac{1}{\epsilon}, \frac{\epsilon^{\frac{1}{\epsilon} \cdot \epsilon} - 1}{\epsilon^\epsilon - 1} \cdot \frac{\epsilon^\epsilon - 1}{\epsilon - 1} \cdot \delta\right) = (1, \delta)$ -DP. We conclude the proof by observing that

$$\begin{aligned} \mathbb{E}_{\mathcal{M}'} [\sum_{e \in \mathcal{M}'(G')} w'_e] &= \mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w'_e] = \mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} \epsilon \cdot w_e] \\ &= \epsilon \cdot \mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w_e] \\ &= \epsilon \cdot \left(\sum_{e \in T^*} w_e + o\left((1/\epsilon) \cdot n^{3/2} \cdot \ln n\right) \right) \\ &= \sum_{e \in T'} w'_e + o\left(n^{3/2} \cdot \ln n\right). \end{aligned} \quad (13)$$

1223 □

4.3.2 $(1, \delta)$ -DP Lower Bound

In this subsection, we show that no $(1, \delta)$ -DP algorithm $\mathcal{M}$ satisfies Equation (12) for $\delta = O(1/\sqrt{n})$, thereby proving Theorem 4.2. Our proof heavily relies on a technique originally developed by Steinke and Ullman [SU17] for establishing lower bounds for the private maximum top- k selection problem. To apply this technique, we negate the weights of the input graph. Instead of proving a lower bound for the minimum spanning tree problem, we demonstrate that any $(1, \delta)$ -DP maximum spanning tree algorithm $\mathcal{M}$ incurs an error of $\Omega(n^{3/2} \log n)$ for $\delta = O(1/\sqrt{n})$.

Theorem 4.13 (Theorem 3 in [SU17]). *Let $\beta, \gamma, \Delta, k > 0$ and $s, d \in \mathbb{N}^+$ be a fixed set of parameters. Let $P = (P_1, \dots, P_d)$ be independent samples from the beta distribution $\text{Beta}(\beta, \beta)$, and let $X \in \{0, 1\}^{s \times d}$ be a random dataset such that every $X_{i,j}$ is an independent sample from Bernoulli distribution with mean P_j for every $i \in [s]$ and $j \in [d]$.*

Let $\mathcal{A} : \{0, 1\}^{s \times d} \rightarrow \mathbb{R}^d$ be a $(1, \beta\gamma k/(s\Delta))$ -differentially private algorithm (where two datasets X and X' are neighboring if and only if they differ by at most one row).

Assume $\mathbb{E}_{P,X,\mathcal{A}} [\|\mathcal{A}(X)\|_2^2] = k$ and $\mathbf{P} [\|\mathcal{A}(X)\|_1 \leq \Delta] = 1$ and

$$\mathbb{E}_{P,X,\mathcal{A}} \left[\sum_{j \in [d]} \mathcal{A}(X)_j \cdot \left(P_j - \frac{1}{2} \right) \right] \geq \gamma k, \quad (14)$$

then $s \geq \beta\gamma\sqrt{k}$.

Informally speaking, from Theorem 4.13, any DP algorithm with a good utility requires a lot of samples. To translate the lower bound on samples to a lower bound on the utility for MST, our goal is to construct a hard distribution of graphs, where the weight vector simulates the random dataset X and the algorithm's utility ties to the same quantity in the theorem. The proof is by contradiction. When the weight vector, or equivalently, the number of samples, is fixed, Theorem 4.13 shows that any DP algorithm with a "good" utility requires more samples than this fixed number, leading to a contradiction. We will present the proof formally in the rest of this section.

Definition 4.2 (Hard Distribution). *Consider a complete graph G with n -vertices and let $m = n(n-1)/2$ be the number of edges. Let $\beta, \gamma > 0$ and $s \in \mathbb{N}^+$ be parameters to be determined later. For each edge e , we first sample P_e from $\text{Beta}(\beta, \beta)$ independently. Then let $X \in \{0, 1\}^{s \times m}$ be a random dataset such that $X_{i,e}$ is an independent sample from Bernoulli distribution with mean P_e , $\forall i \in [s], e \in [m]$. Finally, let $w_e \doteq \sum_{i \in [s]} X_{i,e}$ for each $e \in [m]$. Hence, w_e follows binomial distribution $\text{Bin}(s, P_e)$.*

Since a spanning tree algorithm $\mathcal{M}$ has a different input and output format from the algorithm in Theorem 4.13, a conversion is required.

Definition 4.3 (Conversion). *Let G be the random graph and X the random dataset*

generated as described in Definition 4.2. Given an algorithm $\mathcal{M}$ that takes G as input and outputs a set of spanning tree edges, define $\mathcal{A}_{\mathcal{M}}$ as the algorithm that, given input X , outputs an indicator vector $\mathcal{A}_{\mathcal{M}}(X) \in \{0, 1\}^{|E|}$, where $\mathcal{A}_{\mathcal{M}}(X)_e = 1$ if and only if $e \in \mathcal{M}(G)$.

It is straightforward to see that if $\mathcal{M}$ is $(1, \delta)$ -DP (where two input graphs are considered neighboring if the weights of each edge differ by at most 1), then $\mathcal{A}_{\mathcal{M}}$ is also $(1, \delta)$ -DP (where two input datasets are neighboring if they differ in at most one row). Therefore, under the hard distribution, the sample complexity lower bound in Theorem 4.13 for the algorithm $\mathcal{A}_{\mathcal{M}}$ transfer to $\mathcal{M}$.

Before presenting our main lower bound, we state a lemma that is critical to our proof.

Lemma 4.14. *For $s \geq 10$, $\beta = \frac{1}{2} \ln n$ and $n \geq 2 \times 10^7$, let G be a random graph sampled from the hard distribution in Definition 4.2. With probability at least 0.99, there exists a spanning tree of G where each edge has weight at least $3/4 \cdot s$.*

The complete proof of Lemma 4.14 is included in Appendix 4.7.6. Intuitively, the result follows from the properties of the beta distribution and the Erdős–Rényi random graph model [ER59]. Let H be the subgraph of G that includes all vertices and all edges e such that $w_e \geq \frac{3}{4} \cdot s$. Since the edge weights are independent and due to our choice of β , the graph H is an instance of the Erdős–Rényi model with edge sampling probability $p = \mathbf{P}[w_e \geq s \cdot 3/4] \in \Omega(\ln(n)/n)$. Finally, G has a spanning tree where all edges e satisfy $w_e \geq s \cdot 3/4$ if and only if H is connected, which occurs almost surely when $p \in \Omega(\ln(n)/n)$ [ER59].

Assume Lemma 4.14 holds, we directly have

$$\mathbb{E}_G \left[\sum_{e \in T^*} w_e \right] \geq 0.99 \cdot \frac{3}{4} \cdot s \cdot (n - 1). \quad (15)$$

where T^* be the maximum spanning tree in G . Next, we will present the main theorem in this section that captures the utility lower bound.

Theorem 4.15 (MST lower-bound). *Let $n \geq 9 \times 10^8$, $\beta = \frac{1}{2} \ln n$ and $s = \frac{1}{100} \sqrt{n} \ln n$, G be a random graph sampled from the hard distribution in Definition 4.2. Let $\gamma = 0.04$ and $\delta = \beta\gamma/s = 2/\sqrt{n}$. For any $(1, \delta)$ -differentially private maximum spanning tree algorithm $\mathcal{M}$, it holds that*

$$\mathbb{E}_{G, \mathcal{M}} \left[\sum_{e \in \mathcal{M}(G)} w_e \right] \leq \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] - \frac{n^{3/2} \ln n}{1000}, \quad (16)$$

where T_G^* be the maximum spanning tree given G .

Proof. The proof proceeds by contradiction. Assume there exists a $(1, \delta)$ -DP maximum spanning tree algorithm $\mathcal{M}$, s.t.

$$\mathbb{E}_{G, \mathcal{M}} \left[\sum_{e \in \mathcal{M}(G)} w_e \right] > \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] - \frac{n^{3/2} \ln n}{1000}, \text{ and}$$

let X be the random dataset generated when G is constructed, as described in Definition 4.2, and let $\mathcal{A}_\mathcal{M}$ denote the algorithm converted from $\mathcal{M}$, as described in Definition 4.3. We will prove that

$$\mathbb{E}_{P, X, \mathcal{A}_\mathcal{M}} \left[\sum_{e \in [m]} (\mathcal{A}_\mathcal{M}(X))_e (P_e - \frac{1}{2}) \right] > \gamma(n-1), \quad (17)$$

Since $\mathcal{M}$ outputs $(n-1)$ edges, we always have $\|\mathcal{A}_\mathcal{M}(X)\|_2^2 = \|\mathcal{A}_\mathcal{M}(X)\|_1 = n-1$. Hence, $\mathcal{A}_\mathcal{M}$ satisfies all conditions in Theorem 4.13, where we set $k = \Delta = (n-1)$ and $\delta = \beta\gamma k/(s\Delta) = \beta\gamma/s$. Therefore, Theorem 4.13 implies that $s \geq \beta\gamma\sqrt{n-1}$, leading to a contradiction since the assumption in Theorem 4.15 implies $s = 0.01\sqrt{n} \ln n < \beta\gamma\sqrt{n-1}$.

Proving Equation (17): Since $w_e \sim \text{Bin}(s, P_e)$, we have $\mathbb{E}[w_e] = s \cdot P_e$. Further, the following lemma holds, whose proof employs a standard technique using Hoeffding's inequality.

Lemma 4.16. $\mathbb{E}[\max_{e \in E} |w_e - s \cdot P_e|] \leq \sqrt{3s \ln n}$.

Proof. Since w_e is a sum of s independent random variables between $[0, 1]$, and has expectation $s \cdot P_e$, by Hoeffding's inequality, it holds that $\mathbf{P}[|w_e - s \cdot P_e| \geq t] \leq 2 \cdot \exp\left(-\frac{2t^2}{s}\right)$ for all $t \geq 0$.

By union bound, it holds that for all $t \geq 0$, $s \geq 1$ and $n \geq 100$:

$$\begin{aligned} \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t + \sqrt{s \cdot \ln n}\right] &\leq 2 \cdot \frac{n(n-1)}{2} \cdot \exp\left(-2 \cdot \left(t + \sqrt{s \cdot \ln n}\right)^2 / s\right) \\ &\leq n^2 \cdot \exp\left(-2 \cdot t^2 / s\right) \cdot \exp\left(-2 \cdot s \cdot (\ln n) / s\right) = \exp\left(-2 \cdot t^2 / s\right). \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}\left[\max_{e \in E} |w_e - s P_e|\right] &= \int_0^{\sqrt{s \ln n}} \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t\right] dt \\ &\quad + \int_{\sqrt{s \ln n}}^\infty \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t\right] dt \\ &\leq \sqrt{s \cdot \ln n} + \int_0^\infty \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t + \sqrt{s \cdot \ln n}\right] dt \\ &\leq \sqrt{s \cdot \ln n} + \int_0^\infty \exp\left(-2 \cdot t^2 / s\right) dt = \sqrt{s \cdot \ln n} + \frac{1}{2} \cdot \int_{-\infty}^\infty \exp\left(-\frac{t^2}{2 \cdot (s/4)}\right) dt \\ &= \sqrt{s \cdot \ln n} + \frac{\sqrt{2 \cdot \pi \cdot (s/4)}}{2} < \sqrt{3s \ln n}. \quad \square \end{aligned}$$

Combing Lemma 4.16 and that $\|\mathcal{A}_\mathcal{M}(X)\|_1 \equiv n-1$, we see

$$\begin{aligned} \mathbb{E}_{P, X, \mathcal{A}_\mathcal{M}} \left[\sum_{e \in [m]} (\mathcal{A}_\mathcal{M}(X))_e (s \cdot P_e - \frac{s}{2}) \right] &\geq \mathbb{E}_{P, X, \mathcal{A}_\mathcal{M}} \left[\sum_{e \in [m]} (\mathcal{A}_\mathcal{M}(X))_e w_e \right] \\ &\quad - (n-1) \cdot (\sqrt{3s \ln n} + \frac{s}{2}). \end{aligned}$$

By the construction of $\mathcal{A}_\mathcal{M}$ (Definition 4.3), along with the assumption that $\mathcal{M}$ violates Equation (16),

$$\mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e w_e \right] = \mathbb{E}_{G, M} \left[\sum_{e \in M(G)} w_e \right] > \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] \frac{n^{3/2} \ln n}{1000}.$$

1284 Hence, we can bound the expectation $\mu_E \doteq \mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e (s \cdot P_e - \frac{s}{2}) \right]$,

$$\begin{aligned} \mu_E &> \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] - \frac{n^{3/2} \ln n}{1000} - (n-1)(\sqrt{3s \ln n} + \frac{s}{2}) \\ &> 0.99 \cdot \frac{3}{4} s(n-1) - \frac{n^{3/2} \ln n}{1000} - (n-1)(\sqrt{3s \ln n} + \frac{s}{2}) \\ &= 0.2425s(n-1) - \frac{n^{3/2} \ln n}{1000} - (n-1)\sqrt{3s \ln n}. \end{aligned}$$

1285 Since $n \geq 9 \times 10^8$, $s = \frac{1}{100} \sqrt{n \ln n}$ and $\gamma = 0.04$, it follows that

$$\begin{aligned} \mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e (P_e - \frac{1}{2}) \right] &> 0.2424(n-1) - \left(\frac{(n-1)}{s} \sqrt{3s \ln n} + \frac{n^{3/2} \ln n}{1000} \right) \\ &> \gamma(n-1). \end{aligned}$$

1286

□

1287 4.4 RELATED WORK

1288 Releasing various graph statistics under differential privacy constraints is a funda-
1289 mental and well-studied task. For more background, see [Li+23; Mue+22].

1290 4.4.1 Top- k Selection.

1291 The private **MST** problem is closely related to the private top- k selection problem.
1292 Formulated in the framework of this paper, the problem involves m items with
1293 weights $w_1, \dots, w_m$, and the goal of private top- k selection is to privately sample k
1294 items with approximately maximum or minimum weights. Notably, maximization
1295 and minimization are equivalent by flipping the sign of all weights. The private
1296 **MST** problem can be seen as a variant of top- $(n-1)$ selection with the additional
1297 topological constraint that the selected items must form a spanning tree.

1298 Durfee and Rogers [DR19] introduced the first linear-time algorithm for private
1299 top- k selection with asymptotically optimal privacy-utility trade-off under approxi-
1300 mate DP. They showed that iteratively applying the exponential mechanism [MT07]
1301 to select k items with an approximately maximum weight, is equivalent to adding
1302 random noise following the Gumbel distribution (see Definition A.5) to each item
1303 weight and then returning the k items with maximum noisy weights. This technique
1304 has a rich research history in the context of the non-private top- k problem [Yel77;
1305 Ohl90; Ros97; Coh97], appearing under various names. For the top-1 maximum
1306 selection problem, Yellott [Yel77] showed very early that adding Gumbel noise to
1307 the $(\ln w_i)$'s and selecting the maximum allows sampling an item with probability
1308 proportional to their weights. Rosén [Ros97] studied the top- k maximum selection
1309 problem under the name of "order sampling" and proposed a method to sample k
1310 items with probability proportional to their weights, by generating noisy scores
1311 $\text{Exp}(w_i) = \text{Exp}(1)/w_i$ for $i \in [m]$ and selecting k items with the minimum scores.

Reference	PN	NH	Error	Time	Technique
[Sea16]	ε -DP	ℓ_1	$O((1/\varepsilon) \cdot n \log n)$	$\text{MST} + O(m)$	Input privatization
[Pin18]	ε -DP	ℓ_1	$O((1/\varepsilon) \cdot n \log n)$	$O(nm)$	In-place noise
[HT24]	ε -DP	ℓ_1	$\Omega((1/\varepsilon) \cdot n \log n)$	–	Lower bound
[Sea16]	(ε, δ) -DP	ℓ_1	$O\left((1/\varepsilon) \cdot n \sqrt{(\log n) \cdot \log(1/\delta)}\right)$	$\text{MST} + O(m)$	Input privatization
[Pin18]	(ε, δ) -DP	ℓ_1	$O\left((1/\varepsilon) \cdot n \sqrt{(\log n) \cdot \log(1/\delta)}\right)$	$O(nm)$	In-place noise
[Sea16]	(ε, δ) -DP	ℓ_1	$\Omega(n)$	–	Lower bound
[Sea16]	ε -DP	ℓ_∞	$O((1/\varepsilon) \cdot nm \log n)$	$\text{MST} + O(m)$	Input privatization
[Pin18]	ε -DP	ℓ_∞	$O((1/\varepsilon) \cdot n^2 \log n)$	$O(nm)$	In-place noise
[HT24]	ε -DP	ℓ_∞	$\Omega((1/\varepsilon) \cdot n^2 \log n)$	–	Lower bound

Table 4.2: Extended landscape of results and complementing Table 4.1 with further known results for a graph with n vertices and m edges. **PN** and **NH** denote the Privacy Notation and the Neighboring Relationship respectively. The table compares the previous works [Sea16; Pin18; HT24] for both the ℓ_1 and ℓ_∞ neighborhood relation. “MST” in the **Time** column is the running time of any non-private MST algorithm.

This is equivalent to finding the k items maximizing $-\ln \mathbb{E} \exp(1) + \ln w_i$, where $-\ln \mathbb{E} \exp(1)$ follows exactly a Gumbel distribution (Fact 4.17).

Our sampling model, PPSACR, and the corresponding technique in Algorithm 4, extend this line of research [Coh97; Yel77; Ros97] by enabling the adaptive removal of candidate items after each sampling step, based on previously selected items.

Qiao, Su, and Zhang [QSZ21] proposed a linear-time private top- k selection algorithm that achieves an asymptotically optimal privacy-utility trade-off (up to a logarithmic factor) under approximate differential privacy. Their approach involves adding independent Laplace noise to each item weight and returning the set of top- k items with the smallest noisy weights. However, their current privacy-utility trade-off analysis heavily depends on the returned items corresponding to the true k smallest noisy weights. Extending their approach and analysis to the private MST problem remains open, as the selected items or edges may not necessarily form a spanning tree.

4.4.2 Releasing an MST under DP

We supplement results for approximate DP (Table 4.1) with results for pure DP in Table 4.2. Before our work there were two approaches to private MST: input privatization and in-place algorithms [Pin+18] that we review next.

IN-PLACE Both Prim-Jarník’s and Kruskal’s algorithm [Kru56; Pri57; Jar30] start with an empty set of edges and then iteratively grow it while guaranteeing that it is still a subset of an MST. In each step, they greedily select the lightest new edge between cuts that respect the edges already chosen. One can privatize them by injecting noise whenever a weight is accessed during the computation of an MST. We can replace the selection step using any differentially private selection mechanism, for instance, Report-Noisy-Max [DR14], Permute-and-Flip [MS20], or the Exponential Mechanism [MT07] and get overall privacy by composition. PAMST [Pin18] is based on the Prim-Jarník algorithm and gives the same asymptotic utility as our approach. Furthermore, a private version of Kruskal’s algorithm has been suggested by McKenna, Miklau, and Sheldon [MMS21] as a subroutine in privately generating synthetic data using a probabilistic graphical model, but its utility has never been discussed. We provide such an analysis in Section 4.2.2. Compared to

the *input privatization* in the next paragraph [Sea16], this approach gives strictly better utility under the ℓ_∞ neighboring relationship, assuming the graph is not too sparse. Unfortunately, the caveat is the running time because of the cost of the noisy selection. In an unpublished manuscript, Pagh and Retschmeier [2024] brought down the running time to $O(m + n^{3/2} \log(n) / \sqrt{\rho})$ by designing a special priority queue for Prim–Jarník’s algorithm, which simulates *Report-Noisy-Max* in sublinear time $O(\sqrt{n/\rho} \log n)$, where ρ can be chosen as in Algorithm 1 to ensure (ϵ, δ) -DP.

INPUT PRIVATIZATION One simple idea is to release a private synthetic graph by adding noise to *all* the edge weights and obtain privacy by post-processing. Sealfon [Sea16] was the first to analyze *input perturbation* using Laplace noise to achieve ϵ -DP under the ℓ_1 neighborhood and gave error of $O(n \log n)$ which is known to be asymptotically optimal [HT24]. However, under ϵ -DP with ℓ_∞ neighboring relationship, this technique gives an additive error of $O(nm \log n)$. For dense graphs, this can leave a gap of up to a factor of $O(n)$ to the known lower bound of $\Omega(n^2 \log n)$ under ϵ -DP. The advantage is that it allows flexibility in choosing any (non-private) **MST** algorithm e.g. the expected linear time algorithm by Karger, Klein, and Tarjan [KKT95], the deterministic near-linear time algorithm by Chazelle [Chao0], or the deterministic linear time for dense graphs by Fredman and Tarjan [FT87]. The overall running time is compounded by the $O(m)$ time it takes to add fresh noise to each edge and the running time of the chosen **MST** algorithm. Another advantage is that releasing a single private synthetic graph simultaneously allows the computation of other graph statistics, such as finding *shortest paths* or *minimum-weight (perfect) matchings* [Sea16].

LOWER BOUNDS Recent work by Hladík and Tetek [HT24] showed tight asymptotical worst-case bounds for ϵ -DP using a packing argument: $\Omega(n \log n / \epsilon)$ for the ℓ_1 neighborhood and $\Omega(n^2 \log n / \epsilon)$ for ℓ_∞ . They improved Sealfon’s ℓ_1 bound of $\Omega(n)$. Under the ℓ_∞ neighborhood, we show in Theorem 4.1 together with Theorem 4.2, that an expected error of $\tilde{\Theta}(n^{3/2})$ is asymptotically tight.

4.5 EMPIRICAL EVALUATION

To confirm our theoretical claims, we implemented PAMST [Pin18], Sealfon’s input privatization [Sea16], and our Algorithm 1 (instantiated with Prim–Jarník) in *Python* 3.9. The implementation relies on version 3.2.1 of the *NetworkX* library [HSSo8]. All experiments ran locally on a MacBook Pro with an Apple M2 Pro processor (10 Cores, up to 3.7GHz) and 16GB of RAM. We present two experiments that support our theoretical claims. The first experiment resembles a natural setting in the context of synthetic data generation, and the second explores the influence of the graph’s density. The experiments indicate that the output distribution of our algorithm indeed matches PAMST and, hence, outperforms the *input privatization* approach if the graph is not too sparse. The results are shown in Figure 4.2, and we refer for more details to Section 4.7.7.

MUTUAL INFORMATION A natural setting is finding the Chow-Liu tree [CL68], which is the **MST** on the graph encoding the negated mutual information matrix on all pairwise attributes. Let $X_1, \dots, X_n \in \{0, 1\}$ be random bits. We draw $X_1 \sim \text{Ber}(1/2)$, and then recursively X_i by flipping the bit X_{i-1} with probability $0 < p < 1/2$. This simulates a natural scenario where there is mutual information

Note that in the Chow-Liu setting, we want to find the maximum spanning tree, which is the same as the **MST** on the negated weights.

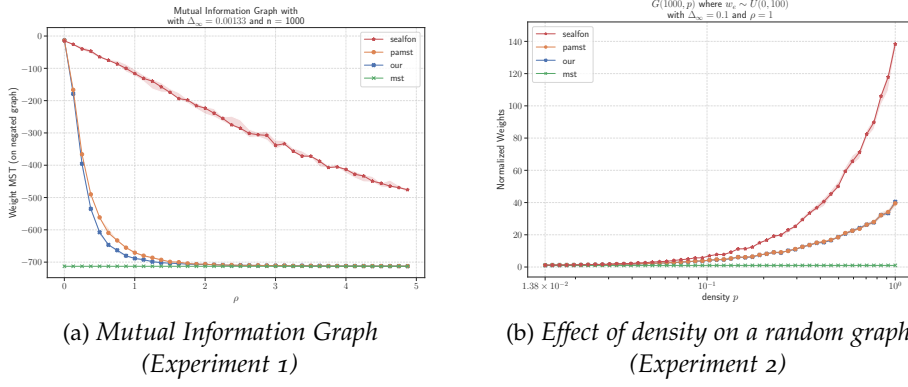


Figure 4.2: The results of our experiment. **a)** Shows the instance described in Experiment 1. Note that we have to negate all weights to find the maximum spanning tree on the mutual information graph. **b)** Shows the impact of the graph’s density on random graphs with $n = 1000$ vertices for a fixed privacy level $\rho = 1$: The figure shows the ratio between the real **MST** and the private one, where each edge weight is uniformly drawn from the interval $[0, 100]$. Because the noise scale of Sealfon’s input perturbation scales with the number of edges in the graph, we see a larger gap for denser graphs. Each data point shows the median of ten runs.

between X_i and X_{i-1} controlled by p . The **MST** is formed by the edges on the path $P = (X_1, \dots, X_n)$ (visualized in Figure 4.3). A potential underlying dataset of size d has the sensitivity of mutual information, which is $\Delta_{mi} = \log_2(d)/d$. As later shown in Section 4.7.7, for this process the pairwise mutual information between all X_i and X_j define a complete graph $G = (V, E)$ with weights

$$\forall i, j \in [n], i \neq j : w_{ij} = \frac{1}{2} (p_1 \log_2(p_1) + p_2 \log_2(p_2)) ,$$

where $p_1 = 1 + (1 - 2p)^k$ and $p_2 = 1 - (1 - 2p)^k$. In the experiment, we used $n = 1000$ with flip probability of $p = 0.05$. We set the underlying dataset size to $d = 10^5$, thus $\Delta_\infty = \Delta_{mi} \approx 0.00133$. The results are shown in Figure 4.2-a). We can see that the error of our approach closely resembles PAMST, outperforming the input privatization approach.

THE EFFECT OF THE DENSITY The second experiment follows the setup proposed in [Pin18]. We sample an Erdős–Rényi random graph $\mathcal{G}(n, p)$ where we draw the weights $w_e \sim U(0, 100)$ uniformly. The experiments shown in Figure 4.2-b) were run on graphs of size $n = 1000$ using $\Delta_\infty = 0.1$. As shown, input privatization does significantly worse as the graph’s density increases, whereas our approach resembles the output distribution of PAMST.

4.6 CONCLUSION AND OPEN PROBLEMS

Our work shows that a simple input perturbation yields a privacy guarantee for the output of any **MST** algorithm far better than what is implied by the post-processing property of DP. It is natural to wonder if something similar holds for other problems where the best existing in-place private algorithms add noise during the computation.

There remains a small gap of $O(\sqrt{\log(1/\delta)})$ between known upper and lower utility bounds. As discussed in Section 4.4.1, extending the Laplace noise-based

approach for top- k selection by Qiao, Su, and Zhang [QSZ21] to the private **MST** problem remains an open question.

ACKNOWLEDGEMENTS Pagh, Wu, Zhang, and Retschmeier carried out this work at Basic Algorithms Research Copenhagen (BARC), which was supported by the VILLUM Foundation grant 54451. *Providentia*, a Data Science Distinguished Investigator grant from the Novo Nordisk Fonden, supported Pagh, Retschmeier, and Wu. Hanwen Zhang is also partially supported by Starting Grant 1054-00032B from the Independent Research Fund Denmark under the Sapere Aude research career programme. We thank Edith Cohen for providing us with historical context for **PPSACR** and anonymous reviewers of a previous version of this paper for their valuable feedback.

4.7 APPENDIX

4.7.1 Algorithm for Maximum Weight Independent Set in a Matroid

A matroid $(U, \mathcal{I})$ is defined by a finite ground set U and a family of *independent sets* $\mathcal{I} \subseteq 2^U$. The family $\mathcal{I}$ satisfies the following three properties:

- $\emptyset \in \mathcal{I}$;
- $\forall S \in \mathcal{I}$ and $T \subseteq S, T \in \mathcal{I}$;
- $\forall S, T \in \mathcal{I}$ and $|S| < |T|$, there exists $t \in T \setminus S$ such that $S \cup \{t\} \in \mathcal{I}$.

As an example, let U be the set of edges in a graph and $\mathcal{I}$ be the family of all subsets of edges that form a forest. It's easy to verify that $(U, \mathcal{I})$ is indeed a matroid, more concretely a *graphic matroid*. Given a weight function $w : U \rightarrow \mathbb{R}^+$, the weight of an independent set S , $w(S)$, is defined as the sum of the weight of its elements. Therefore, **MST** problem is a special case of finding a maximum weight independent set in this matroid. Kruskal's [Kru56] algorithm was generalized to the problem of finding a maximum weight independent set in a matroid by Edmonds [Edm71]. The algorithm first sorts all the elements in the matroid by weight in decreasing order and then tries to insert elements one by one into the independent set if possible. Since the analysis for Algorithm 1 is based on Kruskal's algorithm, it naturally generalizes to finding a maximum weight independent set in general matroids. For matroids with rank n and m elements and since there is no upper bound for m on n , the expected error we have will be $O(n^{3/2} \log m / \sqrt{\rho})$.

Algorithm 5 Private Maximum-Weight-Independent-Set-in-Matroid Framework

Require: A matroid $(U, \mathcal{I})$ and the weights $\mathbf{w}$, algorithm $\mathcal{A}_{\text{MWIS}(\text{matroid})}$, privacy parameters (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2 \cdot \rho / (n-1)}$;
 - 2: $\tilde{w}_e \leftarrow w_e + (2/\epsilon') \cdot \ln(\text{Exp}(1))$ for all $e \in U$.
 - 3: **return** $\mathcal{A}_{\text{MWIS}(\text{matroid})}(U, \mathcal{I}, \tilde{\mathbf{w}})$.
-

1444 4.7.2 One-Shot-PPSACR

Algorithm 6 One-Shot-PPSACR

```

1:  $\tilde{s}(j) \leftrightarrow \mathbb{E} \exp(1) / s(j), \forall j \in U.$ 
2:  $\mathcal{C} \leftarrow U, \mathcal{I} \leftarrow$  An empty list.  $\triangleright$  possible candidates and sampled items so far
3: while  $|\mathcal{I}| \leq k$  and  $\mathcal{C} \neq \emptyset$  do
4:    $j \leftarrow \arg \min_{j' \in \mathcal{C}} \tilde{s}(j').$ 
5:   Add  $j$  to the back of  $\mathcal{I}$ 
6:    $\mathcal{C} \leftarrow \mathcal{C} \setminus (\{j\} \cup f(\mathcal{I})).$ 
7: return  $\mathcal{I}.$ 

```

1445 4.7.3 Further Background

1446 The distribution of the logarithm of an exponential random variable is closely
 1447 connected to the *Gumbel distribution*.

Fact 4.17. If $Z \sim \mathbb{E} \exp(1)$, then $-\ln Z \sim \text{Gumbel}(1)$.

1448 *Proof.* $\mathbf{P}[-\ln X \leq z] = \mathbf{P}[X \leq \exp(-z)] = \exp(-e^{-z})$ for all $z \in \mathbb{R}$. □

1449 **Proof of Fact 4.4.** For our proof, we require the following fact, which we will prove
 1450 later.

Fact 4.18. If $Z_i \sim \ln \mathbb{E} \exp(1)$, then for each $\beta \in (0, 1)$,

$$\mathbf{P}\left[Z_i > \ln \ln \frac{1}{\beta}\right] = \mathbf{P}\left[Z_i < \ln \ln \frac{1}{1-\beta}\right] = \beta.$$

1451 Since $\ln(1+x) \leq x, \forall x > -1$, it holds that

$$\ln \ln \frac{1}{1-\beta} = \ln(-\ln(1-\beta)) \geq \ln \beta = -\ln \frac{1}{\beta}.$$

1452 Then it follows that

$$\mathbf{P}\left[Z_i < -\ln \frac{1}{\beta}\right] \leq \mathbf{P}\left[Z_i > \ln \ln \frac{1}{\beta}\right] = \beta.$$

1453 Define $X \doteq \max_{i \in [m]} |Z_i|$. By a union bound, we have

$$\mathbf{P}\left[|X| \geq \ln \frac{2 \cdot m}{\beta}\right] \leq \beta, \quad \forall \beta \in (0, 1).$$

1454 Define $t \doteq \ln \frac{2 \cdot m}{\beta} \geq \ln(2 \cdot m)$. Then $\beta = (2 \cdot m) \cdot e^{-t}$. Therefore,

$$\begin{aligned} \mathbb{E}[|X|] &= \int_0^{\ln(2 \cdot m)} \mathbf{P}[|X| \geq t] dt + \int_{\ln(2 \cdot m)}^{\infty} \mathbf{P}[|X| \geq t] dt \\ &\leq \ln(2 \cdot m) + \int_{\ln(2 \cdot m)}^{\infty} (2 \cdot m) \cdot e^{-t} dt \leq \ln(2 \cdot m) + (2 \cdot m) \cdot e^{-\ln(2 \cdot m)} \\ &= \ln(2e \cdot m). \end{aligned}$$

1455 *Proof of Fact 4.18.* If $Y \sim \text{Exp}(1)$ where $\mathbf{P}[Y \leq y] = e^{-y}$ we have for each $\beta \in (0, 1)$,

$$\begin{aligned} \mathbf{P}\left[Y > \ln \frac{1}{\beta}\right] &= \exp\left(-\ln \frac{1}{\beta}\right) = \beta, \text{ and } \mathbf{P}\left[Y < \ln \frac{1}{1-\beta}\right] \\ &= 1 - \exp\left(-\ln \frac{1}{1-\beta}\right) = \beta. \end{aligned}$$

1456 It follows that

$$\mathbf{P}\left[\ln Y > \ln \ln \frac{1}{\beta}\right] = \mathbf{P}\left[\ln Y < \ln \ln \frac{1}{1-\beta}\right] = \beta.$$

1457

□

1458 **Proof of Fact 4.9.**

$$p_Y(y) = p_X(\lambda \cdot y) \cdot \frac{dX}{dY} = \lambda \cdot e^{-\lambda \cdot y}, \forall y \geq 0.$$

1459

□

Fact 4.19 (Order of Exponentials). *If $X \sim \text{Exp}(\lambda_1)$, and $Y \sim \text{Exp}(\lambda_2)$, then*

$$\mathbf{P}[X \leq Y] = \int_0^\infty \lambda_1 e^{-\lambda_1 x} \cdot \mathbf{P}[Y \geq x] dx = \int_0^\infty \lambda_1 e^{-\lambda_1 x} e^{-\lambda_2 x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

1460 **Proof of Fact 4.10.** Let $Y \doteq \min_{j \in [d] \setminus \{i\}} X_j$. Then

$$\mathbf{P}[Y \geq y] = \prod_{j \in [d] \setminus \{i\}} \mathbf{P}[X_j \geq y] = \exp\left(-y \sum_{j \in [d] \setminus \{i\}} \lambda_j\right). \quad (18)$$

1461 Therefore, $Y \sim \text{Exp}\left(\sum_{j \in [d] \setminus \{i\}} \lambda_j\right)$.

1462 Applying Fact 4.19 gives $\mathbf{P}\left[X_i = \min_{j \in [d]} X_j\right] = \frac{\lambda_i}{\sum_{j \in [d]} \lambda_j}$. □

1463 **Proof of Fact 4.11.** $\mathbf{P}[X \geq x + y \mid X \geq x] = \frac{\exp(x+y)}{\exp(x)} = \exp(y) = \mathbf{P}[X \geq y]$. □

1464 4.7.4 Utility Guarantee of Private Kruskal

1465 We prove the utility of private Kruskal's algorithm (Algorithm 2) directly. Suppose
 1466 we use Kruskal's algorithm to find the **MST** on the noisy and original graphs in
 1467 parallel. When we try to add the k -th edge in noisy graph, we are actually finding
 1468 the noisy minimum-weighted edge among all the cycle-free candidates. Also, there
 1469 must be a candidate from the first k added edges when we run Kruskal's algorithm
 1470 in the original graph, as all forests form a matroid. Therefore, the real minimum
 1471 weight is not greater than the payment for the k -th step on the original graph,
 1472 so the extra cost induced in this step only comes from selecting the noisy-min.
 1473 Accumulate the error over all steps, we show the desired utility guarantee of the
 1474 algorithm.

Proof. For each $i \in \{1, \dots, n-1\}$, let e_i be the edge added to T at the i^{th} iteration of Algorithm 2. Let $T_i \doteq \bigcup_{j \in [i]} \{e_j\}$ be the partial spanning forest after iteration i and for convenience, let $T_0 = \emptyset$. Further, initially set $F_0 = E$, and denote with F_i the set F immediately after the i^{th} iteration containing all edges that still could be added to T without creating a cycle. Assume that also run a non-private Kruskal's algorithm on the same graph with the true edge weights in parallel, to obtain the real MST $T_i^* = \bigcup_{j \in [i]} \{e_j^*\}$, where e_j^* is the edge added at iteration j .

To prove the utility guarantee, we want to show that for all steps, with probability $1 - \beta$ and for some universal constant $c \in \mathbb{R}^+$,

$$w_{e_i} \leq w_{e_i^*} + \frac{c}{\epsilon'} \cdot \ln \frac{n}{\beta}. \quad (19)$$

Note that the utility guarantee of the exponential mechanism gives us with probability $1 - \beta$

$$w_{e_i} \leq \min_{e \in F_{i-1}} w_e + \frac{c}{\epsilon'} \cdot \ln \frac{n}{\beta} \quad (20)$$

because e_i is exactly the noisy minimum of the set F_{i-1} chosen by the algorithm. Therefore, it suffices to show that

$$\min_{e \in F_{i-1}} w_e \leq w_{e_i^*}. \quad (21)$$

Note that $w_{e_{i-1}^*} \leq w_{e_i^*}$ by the way Kruskal adds new edges to T^* . Using this, we claim that F_{i-1} contains at least one edge which belongs to T_i^* and can be added to T_{i-1} without forming a cycle.

First, note that edges in T_{i-1} induce a forest. Let $\mathcal{T}$ be an arbitrary tree in this forest, and assume that it has t edges. Then, the number of edges $e = (u, v) \in T_i^*$, such that both $u, v \in \mathcal{T}$ is at most t without inducing a cycle.

There are $i-1$ edges in T_{i-1} , but i edges in T_i^* . Therefore, there must be *at least one* edge from T^* , which connects two trees in the forest induced by T_{i-1} . It follows that this edge belongs to F_{i-1} , proving Equation (21). Summing over all possible edges gives

$$\sum_{i \in [n-1]} w_{e_i} \leq \sum_{i \in [n-1]} w_{e_i^*} + \frac{c(n-1)}{\epsilon'} \ln \frac{n}{\beta}.$$

By a standard integration technique, this implies an expectation of $\mathcal{O}\left(\frac{n}{\epsilon'} \ln n\right)$. $\square$

4.7.5 Efficiently Implementing $\mathcal{A}_{\text{priv-kruskal}}$

This section describes how $\mathcal{A}_{\text{priv-kruskal}}$ can be efficiently implemented. Although this result is, of course, overshadowed by our main result, we believe it is interesting enough to be stated here.

The algorithm can be implemented in $O(n + m \log n)$ time. In the initialization phase, we construct a complete binary tree where every level, except possibly the last, is fully filled, and all nodes in the last level are as far left as possible—with m leaf nodes. Each leaf node represents an edge and is assigned a weight $e^{-\epsilon' w_e}$. Each internal node of the binary tree is assigned a weight equal to the sum of the weights of the leaf nodes in the subtree rooted at that node.

This binary tree is used to manage the nodes in F efficiently: it supports both the sampling operation (Algorithm 2, Line 4) and the update operation (Algorithm 2, Line 6) in $O(\log n)$ amortized time per edge.

1512 *Sampling.* We apply a top-down approach starting from the tree's root to sample
 1513 an edge (i.e., a leaf node). At each step, we move to the left child with probability
 1514 equal to the ratio of the left child's weight to the current node's weight, and move
 1515 to the right child otherwise. The sampling procedure terminates when the current
 1516 node is a leaf. It is straightforward to verify via an induction on the number of
 1517 levels that this method samples a leaf node with probability proportional to $e^{-\epsilon' w_e}$
 1518 and that the sampling process completes in $O(\log n)$ time.

1519 *Update.* We introduce two auxiliary data structures for this step. First, we main-
 1520 tain a union-find data structure [Tar75], which allows us to determine the connected
 1521 component (induced by the edges in T) to which a vertex belongs in $O(\alpha(n))$ amor-
 1522 tized time, where $\alpha(n)$ is the inverse Ackermann function. Additionally, for each
 1523 connected component, we maintain a linked list to track the vertices it contains.
 1524 When a new edge $e = (u, v)$ is added to the set T , let C_u and C_v denote the con-
 1525 nected components containing u and v , respectively. To merge C_u and C_v , we
 1526 update both the union-find structure and the linked list, which can be performed
 1527 in $O(\alpha(n) + 1)$ time. Next, we need to remove from F all edges connecting C_u and
 1528 C_v . To do so efficiently, we choose the smaller of C_u and C_v (in term of the number
 1529 of vertices)—assume, without loss of generality, it is C_u —and check all edges incident
 1530 to it. For each edge, if its other endpoint belongs to C_v , we remove it from F .

1531 We now analyze the number of times an edge is checked and the cost of its
 1532 removal from F . For the former, define the rank of a vertex as the number of
 1533 vertices in the connected component to which it belongs. Each time an edge is
 1534 checked, the rank of one of its endpoints at least doubles. Since a vertex can have a
 1535 rank of at most n , an edge can be checked at most $O(\log n)$ times.

1536 For the latter, removing an edge from F is handled by removing the correspond-
 1537 ing leaf node from the binary tree and then updating the weights of the nodes
 1538 along the leaf-to-root path. This operation can be completed in $O(\log n)$ time.

1539 4.7.6 Proof of Lemma 4.14

1540 Our proof relies on a standard result from the Erdős–Rényi model $\mathcal{G}(n, p)$, where
 1541 $\mathcal{G}$ is a random graph with n vertices such that each edge is independently included
 1542 with probability p . The following lemma is stated with explicit constants, as our
 1543 subsequent proof relies on them. Despite extensive searching, we could not find a
 1544 version of the lemma with explicit constants. For completeness, we provide a proof
 1545 of this lemma with explicit constants in the complete version [Pag+25].

Lemma 4.20 ([ER59] Connectivity). *For any real $\epsilon > 0$, if $p > \frac{(1+\epsilon)\ln n}{n}$, then a random graph in the Erdős–Rényi model $\mathcal{G}(n, p)$ with n vertices is almost surely connected. In particular, if $p \geq \frac{2\ln n}{n-1}$ and $n \geq 1700$, the probability that the random graph is not connected is at most $\frac{6}{n}$.*

1546 Recall that the random graph G considered in Lemma 4.14 is generated as
 1547 follows: G consists of n vertices and $m = \frac{n(n-1)}{2}$ edges. For each edge e , we first
 1548 sample $P_e \sim \text{Beta}(\beta, \beta)$ and then sample $w_e \sim \text{Bin}(s, P_e)$.

1549 Let H be the subgraph that includes all vertices in G and all edges e such that
 1550 $w_e \geq 3/4 \cdot s$. Since the edge weights are independent of each other, the graph H
 1551 is an instance of Erdős–Rényi model with $p = \mathbf{P}[w_e \geq 3/4 \cdot s]$. Note that G has a
 1552 spanning tree where all edges e have $w_e \geq 3/4 \cdot s$ if and only if H is connected.

Given Lemma 4.20, to prove Lemma 4.14, it just remain to show that for each edge e , we have

$$\mathbf{P} \left[w_e \geq \frac{3}{4} \cdot s \right] > \frac{2 \ln n}{n-1}. \quad (22)$$

Fact 4.21 (Steinke and Ullman [SU17]). *Let $\beta \geq 1$, and X be a random variable following beta distribution $\text{Beta}(\beta, \beta)$. Then for all $p \in [0, 1/2]$, it holds that*

$$\mathbf{P}[X \leq p] = \mathbf{P}[X \geq 1-p] \geq (4 \cdot p \cdot (1-p))^{\beta-1} \cdot \frac{p}{\beta}. \quad (23)$$

According to Fact 4.21, we have $\mathbf{P}[P_e \geq 0.9] \geq \frac{0.36^{\beta-1}}{10\beta}$ and since $\mathbb{E}[w_e] = s \cdot P_e$, by Hoeffding's inequality,

$$\mathbf{P}[w_e > s \cdot P_e - 0.15s \mid P_e \geq 0.9] < \exp(-0.045s). \quad (24)$$

Therefore,

$$\begin{aligned} \mathbf{P} \left[w_e \geq \frac{3}{4}s \right] &\geq \mathbf{P}[P_e \geq 0.9] \cdot \mathbf{P}[w_e \leq s \cdot P_e - 0.15s \mid P_e \geq 0.9] \\ &\geq \frac{0.36^{\beta-1}}{10\beta} \cdot (1 - \exp(-0.045s)). \end{aligned}$$

For $s \geq 10$ and $\beta = \frac{1}{2} \ln n$ and $n \geq 2 \times 10^7$, we have

$$\mathbf{P} \left[w_e \geq \frac{3}{4}s \right] \geq \frac{0.36^{\frac{1}{2} \ln n}}{1.8 \ln n} \cdot (1 - \exp(-0.45)) > \frac{0.36}{1.8 n^{0.52} \ln n} > \frac{2 \ln n}{n-1}. \quad (25)$$

□

4.7.7 Proofs for the Experiments

Assume $0 < p < \frac{1}{2}$. The mutual information $I(X; Y)$ between two discrete random variables X and Y quantifies the amount of information that X contains about Y (or vice versa).

Definition 4.4 (Mutual Information [Ash90]).

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \mathbf{P}[X = x, Y = y] \log_2 \left(\frac{\mathbf{P}[X = x, Y = y]}{\mathbf{P}[X = x] \mathbf{P}[Y = y]} \right). \quad (26)$$

We now prove the validity of the weight function given in Section 4.5.

Definition 4.5. We define the random variables $X_1, \dots, X_n \in \{0, 1\}$ recursively by

setting

$$X_1 \sim \text{Ber}\left(\frac{1}{2}\right) \quad \text{and} \quad X_i = \begin{cases} 1 - X_{i-1} & \text{with probability } p \\ X & \text{else} \end{cases}.$$

1567

1568 By definition, $\mathbf{P}[X_i = 0] = \mathbf{P}[X_i] = 1 = \frac{1}{2}$. Note that mutual information is min-
 1569 imized for $p \approx \frac{1}{2}$ and note that for all X_i : $\mathbf{P}[X_i = 1] = \mathbf{P}[X_i = 0] = \frac{1}{2}$. We need
 1570 another short lemma for the parity of a binomial random variable:

Lemma 4.22 (Parity of Binomials.). Denote $Z \sim \text{Bin}(n, p)$, then

$$\Pr[Z \text{ is even}] = 1 - \mathbf{P}[Z \text{ is odd}] = \frac{1}{2} + \frac{1}{2}(1 - 2p)^n. \quad (27)$$

1571 *Proof.* By direct computation and application of the binomial theorem in the last
 1572 step:

$$\begin{aligned} \mathbf{P}_{Z \sim \text{Bin}(n, p)}[Z \text{ even}] &= \sum_{k=0}^n \frac{(-1)^k + 1}{2} \mathbf{P}[Z = k] \\ &= \frac{1}{2} \left(\sum_{k=0}^n (-1)^k \mathbf{P}[Z = k] + \sum_{k=0}^n \mathbf{P}[Z = k] \right) \\ &= \frac{1}{2} \left(\sum_{k=0}^n ((-1)^k \mathbf{P}[Z = k]) + 1 \right) \\ &= \frac{1}{2} \left(\sum_{k=0}^n ((-1)^k \binom{n}{k} p^k (1-p)^{n-k}) + 1 \right) \\ &= \frac{1}{2} \left(\sum_{k=0}^n \left(\binom{n}{k} (-p)^k (1-p)^{n-k} \right) + 1 \right) = \frac{1}{2}(1 - 2p)^n + \frac{1}{2}. \end{aligned}$$

1573

□

Claim 4.23. Assume $0 < p < \frac{1}{2}$, the mutual information score between X_i and X_j that are $k = |i - j|$ steps apart can directly be computed by

$$\begin{aligned} I(X_i; X_j) &= \left(\frac{1}{2} + \frac{1}{2}(1 - 2p)^k \right) \log_2 \left(1 + (1 - 2p)^k \right) \\ &\quad + \left(\frac{1}{2} - \frac{1}{2}(1 - 2p)^k \right) \log_2 \left(1 - (1 - 2p)^k \right). \end{aligned} \quad (28)$$

1574 *Proof.* We shortly write $p_{00}(k) = \mathbf{P}[X_i = 0, X_j = 0]$ and $p_0 = \mathbf{P}[X_i = 0]$ (resp $p_{11}(k)$,
 1575 $p_{10}(k)$ and $p_{01}(k)$, p_1). Furthermore, denote $p_{\text{even}}(k) = \mathbf{P}_{Z \sim \text{Bin}(k, p)}[Z \text{ is even}]$ and
 1576 $p_{\text{odd}}(k)$ as proven above. Note that we can compute these probabilities:

$$\begin{aligned} p_{00}(k) &= \mathbf{P}[X_i = 0 | X_j = 0] \cdot \mathbf{P}[X_j = 0] = p_{\text{even}}(k) \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{4} \cdot (1 - 2p)^k, \\ p_{10}(k) &= \mathbf{P}[X_i = 1 | X_j = 0] \cdot \mathbf{P}[X_j = 0] = p_{\text{odd}}(k) \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4} (1 - 2p)^k. \end{aligned}$$

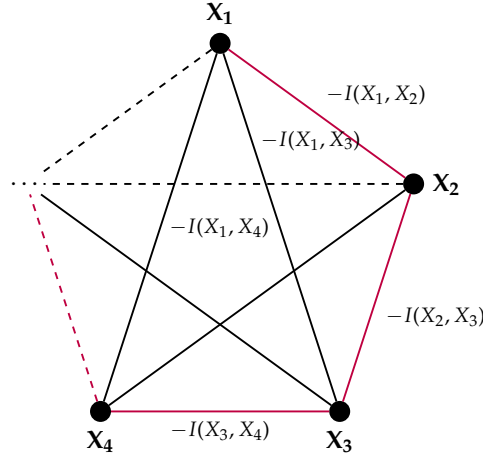


Figure 4.3: An extract of the complete graph encoding the mutual information between the random variables $X_1, \dots, X_n$ described in Definition 4.5 and used in Section 4.5. The weights encode the negated mutual information corresponding to the described process. The **MST** is formed by the vertices on the path $P(X_1, X_2, \dots)$. In our experiment with $n = 1000$ vertices and the flip probability $p = 0.05$, we have $-I(X_1, X_2) = I(X_2, X_3) = \dots \approx -0.7136$, $-I(X_1, X_3) = I(X_2, X_4) = \dots \approx -0.5471$ and $-I(X_1, X_4) \dots \approx -0.4277$.

1577 We trivially get $p_{11}(k) = p_{00}(k)$ and $p_{01}(k) = p_{10}(k)$ by symmetry. Then, we can
 1578 compute the mutual information directly.

$$\begin{aligned}
 I(X_i; X_j) &= p_{00}(k) \log_2 \frac{p_{00}(k)}{p_0 p_0} + p_{01}(k) \log_2 \frac{p_{01}(k)}{p_0 p_1} \\
 &\quad + p_{10}(k) \log_2 \frac{p_{10}(k)}{p_1 p_0} + p_{11}(k) \log_2 \frac{p_{11}(k)}{p_1 p_1} \\
 &= p_{00}(k) \log_2(4p_{00}(k)) + p_{01}(k) \log_2(4p_{01}(k)) \\
 &\quad + p_{10}(k) \log_2(4p_{10}(k)) + p_{11}(k) \log_2(4p_{11}(k)) \\
 &= 2p_{00}(k) \log_2(4p_{00}(k)) + 2p_{01}(k) \log_2(4p_{01}(k)) \\
 &= \left(\frac{1}{2} + \frac{1}{2}(1-2p)^k \right) \log_2 \left(1 + (1-2p)^k \right) \\
 &\quad + \left(\frac{1}{2} - \frac{1}{2}(1-2p)^k \right) \log_2 \left(1 - (1-2p)^k \right).
 \end{aligned}$$

1579

□

This section studies fundamental graph optimization problems under differential privacy (DP) and shows new, reconstruction-based lower bounds. We consider a graph $G = (V, E, \mathbf{w})$ where the vertex set V and edges E are public and the weights $\mathbf{w} : E \rightarrow \mathbb{R}$ must be kept differentially private under an ℓ_1 neighboring relation. For the problems of releasing a minimum-weight spanning tree and a minimum-weight perfect matching, we show new, tight error bounds of $\Omega(n \cdot \log(m/n)/\epsilon)$ on worst-case graphs with n vertices and $m > 2n$ edges. The upper bounds are known pure DP algorithms while the new lower bound holds even under approximate (ϵ, δ) -DP as long as $\delta \leq (n/m)^{\Omega(1)}$. Our lower bounds improve the $\Omega(n/\epsilon)$ lower bounds of Sealfon (PODS '16).

The fact that approximate DP does not reduce error for Minimum-Weight Spanning Tree (MST) under the ℓ_1 neighboring relation contrasts with the recent upper bound of Pagh et al. (PODS '25, Chapter 4) which shows that approximate DP allows much better error under the ℓ_∞ neighboring relation. Going beyond worst-case graphs, we give lower bounds for large families of sparse graphs with expansion properties. We show a lower bound of $\Omega(n/\epsilon)$ for the minimum spanning tree for any graph where the minimum cut is at least $\Omega(\log(n))$. Finally, we consider the problem of private hierarchical clustering under Dasgupta's cost function (STOC '16) and show the first approximate DP lower bound parameterized by the minimum weight of a balanced cut. This extends lower bounds of Deng et al. (ICLR '25) to general graphs and to approximate DP.

Graph-structured data is a fundamental abstraction in modern data management. Weighted graphs naturally arise, for example, from relational data describing transportation networks, communication infrastructure, recommendation systems, similarity relations, and interactions between entities. Many important analysis tasks on such data reduce to classical graph optimization problems, including computing sparse connectivity structures such as minimum spanning trees, assignment structures such as minimum-weight perfect matchings, and multiscale summaries such as hierarchical clusterings. These structures serve as compact summaries that support downstream tasks such as routing, indexing, similarity search, visualization, and unsupervised learning.

In many applications, however, the edge weights encode sensitive aggregate information derived from individuals. For example, edge weights may represent traffic intensities collected from users, communication frequencies between populations, similarity scores derived from behavioral data, or compatibility measures computed from medical records. In such settings, a single individual may affect multiple edge weights simultaneously, but only by a small amount in each coordinate. Compared to edge-level privacy models that treat edges as discrete records, the ℓ_1 model naturally captures continuous-valued and aggregate data contributions, making it particularly well suited for weighted graph optimization

problems. This motivated Sealfon [Sea16] to study differential privacy under an ℓ_1 neighboring relation on *edge weights*, where neighboring weighted graphs differ by bounded total change in their edge-weight vectors. Other researchers have applied this appealing privacy model to other graph problems, including hierarchical clustering [Den+25] and all pairs shortest paths distances [Che+23].

Examples. In a transportation network, an **MST** can represent a sparse backbone connecting cities or distribution centers using low-cost routes inferred from aggregate traffic data. Publishing such a structure enables routing, infrastructure planning, and resilience analysis while dramatically compressing the underlying weighted network. Similarly, in a recommendation or similarity graph, a hierarchical clustering, also known as a dendrogram, provides a navigable taxonomy of users, products, or documents derived from sensitive behavioral correlations. These structures often involve sensitive data, such as road traffic data or user browsing activity, and thus are good candidates for private data release.

PROBLEM SETTING In this section, we continue studying the problems of **MINIMUM SPANNING TREE (MST)**, **MINIMUM-WEIGHT PERFECT MATCHING (MWPM)**, and **HIERARCHICAL CLUSTERING (HC)** in the edge-weight model of differential privacy (DP). The privacy constraint requires that the output distribution of $\mathcal{A}$ is similar for neighboring weight $\mathbf{w}, \mathbf{w}'$ vectors that are close in ℓ_1 distance. Again, we measure our utility as the expected additive error obtained by difference between the cost of a released solution and the optimum.

OUR CONTRIBUTIONS

We derive *two* types of **reconstruction-based** lower bounds for **approximate DP** and the ℓ_1 neighboring relationship. We first consider lower bounds on explicit **worst-case** topologies. For MST and MWPM, we provide lower bounds showing that the expected additive error must grow with a factor of $\Omega(n \cdot \log(\frac{m}{n}))$. These lower bounds work via a new reconstruction-attack setup using a dataset from a large domain where more information can be reconstructed compared to a binary dataset.

This lifts the prior lower bound of $\Omega(n)$ by a logarithmic factor and tightly matches the upper bound obtained by the simple input perturbation algorithm that adds noise to each weight independently [Aam26; Sea16]. The fact that input perturbation is an optimal approach for these problems is additional evidence that input perturbation is often a good approach to graph optimization problems, as also seen in Chapter 4. Furthermore, the fact that this upper bound under pure differential privacy ($\delta = 0$) can be matched by lower bounds shown under approximate differential privacy (with polynomially small δ) is somewhat surprising given the polynomial separation by a factor of $\mathcal{O}(\sqrt{n})$ between pure and approximate differential privacy for **MINIMUM SPANNING TREE** under the ℓ_∞ neighboring relation [Pag+25].

The lower bounds in the previous paragraph, and indeed many other lower bounds in edge-weight DP, use worst-case topologies that are not always realistic in a real-world setting. For example, the lower bounds for MST (with $\delta > 0$) and HC use a doubly-connected star graph or the complete graph as their worst-case topology, respectively (see Figure 5.1), leaving it open whether better performance is possible on sparser or more realistic topologies.

Our second type of lower bounds show that it is not possible to achieve much better performance for large classes of sparse topologies. For MST, we show that *any*

Problem	Error Bound	Reference	Graph type	Parameters
MST	$\mathcal{O}(n \cdot \log n)$	[Sea16]	Any	$\delta = 0$
	$\mathcal{O}(n \cdot \log(m/n))$	[Aam26]	Any	$\delta = 0$
	$\Omega(n)$	[Sea16]	Multi-edge star graph	$\delta < c_\epsilon$
	$\Omega(D)$	[HT24]	Diameter D	$\delta = 0$
	$\Omega(n \cdot \log(m/n))$	Theorem 5.3	Worst-case	$\delta < (n/m)^{\Omega(1)}$
	$\Omega(n)$	Corollary 5.8	Min-cut $\geq 5 \log(n)$	$\delta < \frac{c_\epsilon}{m}$
MWPM	$\mathcal{O}(n \cdot \log n)$	[Sea16]	Any	$\delta = 0$
	$\mathcal{O}(n \cdot \log(m/n))$	[Aam26]	Any	$\delta = 0$
	$\Omega(n)$	[Sea16]	Collection of 4-cycles	$\delta < c_\epsilon$
	$\Omega(n \cdot \log(m/n))$	Theorem 5.10	Worst-case	$\delta < (n/m)^{\Omega(1)}$
HC	$\mathcal{O}(n^2 \log n)$	[Imo+23]	Any	$\delta = 0$
	$\Omega(n^2)$	[Den+25]	Complete graph	$\delta = 0$
	$\Omega(n^2)$	Theorem 5.18	Sparse expander	$\delta < \frac{c_\epsilon}{m}$

Figure 5.1: Upper and lower bounds for the error of differentially private graph optimization problems: MST, MWPM, and HC under the ℓ_1 neighboring relationship parameterized by the number of vertices n and edges m . Upper bounds hold for pure differential privacy ($\delta = 0$), while lower bounds are for (ϵ, δ) -DP. All bounds omit a multiplicative factor $1/\epsilon$ for readability; $c_\epsilon > 0$ is a constant that depends only on ϵ .

topology with *minimum cut* at least $\Omega(\log(n))$ must incur error $\Omega(n)$, differing from the lower bound of the worst-case topology by a $\log(\frac{m}{n})$ factor. This complements a lower bound of Hladík and Tetek [HT24] which establishes that the error must grow linearly with the diameter of the topology but holds for pure DP only. For HC under Dasgupta’s cost function [Das16], we obtain a fully-parameterized lower bound that grows with the expansion (in the sense of expander graphs), and decreases with the maximum degree of the graph. For constant-degree regular expander graphs, our lower bound is $\Omega(n^2)$, which matches the state-of-the-art lower bound of Deng, Gao, Upadhyay, Wang, and Zhou [Den+25] and Imola, Epasto, Mahdian, Cohen-Addad, and Mirrokni [Imo+23] but generalizes it to approximate DP and beyond the complete graph.

A summary of all our new results together with known bounds appears in Figure 5.1. Note that the error bounds depend only on graph parameters, so the error is independent of the weight (or cost) of a solution. Thus these problems can be solved privately with good error bounds whenever the weight (or cost) of an optimal solution is sufficiently large compared to the error bound.

TECHNIQUES Our lower bounds work via *reconstruction attacks*, where a private dataset is encoded into the weights of the graph topology, and based on the solution to the graph problem, a decoding process attempts to reconstruct the dataset. If the error of the graph algorithm is low enough, we show that the reconstruction will be successful, contradicting DP. We introduce several new techniques in designing these reconstruction attacks.

For our lower bounds on worst-case topologies, our sharper bounds come from encoding a dataset in $[m/n]^n$ rather than a binary dataset, which allows for more

information to be reconstructed. To do this, we need to use a new encoding scheme on a dense topology. For MINIMUM SPANNING TREE, we use a complete bipartite graph whose left vertices represent coordinates and whose right vertices represent possible values, assigning low weight to the edge connecting each coordinate i to its value x_i . Thus, a sufficiently accurate spanning tree reveals the value of many coordinates, contradicting the reconstruction guarantees imposed by differential privacy. Using the same construction again for MINIMUM-WEIGHT PERFECT MATCHING, we note that a *perfect* matching can not encode the same value multiple times. Therefore, we draw a connection to the classic balls-into-bins problem showing that a large fraction of the dataset is collision-free with high probability. It turns out to be sufficient to run the reconstruction attack on this non-colliding subset of the random dataset.

As recently observed by Aamand [Aam26] the known lower bounds for MST and MWPM are matched by Sealfon’s input perturbation algorithms that simply add Laplace noise to each edge weight and run the non-private optimization algorithm as post-processing. For completeness, we include a simple proof of the improved bound of Aamand [Aam26]. Interestingly, our worst-case lower bound asymptotically matches this upper bound, even when parameterized by the number of vertices *and* edges.

For our second type of lower bound, we focus on reconstructing binary datasets and encode them *randomly* into the edge weights of a given topology. This requires analyzing the behavior of the random weights using ideas from random graph theory. At a high level, we need to establish that a low-cost optimal solution exists in the random weights, and that the decoding procedure will reveal many more 0-edges than 1-edges. For MINIMUM SPANNING TREE, it is enough to assume that the topology has min-cut at least $\log(n)$; a union bound over all cuts in the graph will then ensure that a MST of weight 0 exists. Then, a simple decoding procedure where the algorithm guesses all data points in the tree have weight 0 is an effective reconstruction attack.

For HIERARCHICAL CLUSTERING, we have to further subsample the edges at a rate of $\frac{1}{d}$ from the topology (assuming the topology is d -regular) before encoding the dataset; this ensures that a low-cost clustering exists. Given a returned HC, the decoder guesses that all data points in edges crossing the top-most balanced cut (Lemma 5.16) of the HC are 0. This opens up an interesting challenge where it is possible that returned low-cost HC adversarially has few sampled 0-weight edges crossing its balanced cut, which would reveal too little information in the dataset. To circumvent this, we leverage the fact that the HC algorithm cannot distinguish between sampled 0-weight edges and non-sampled edges, meaning that any tree it returns cannot adversarially avoid sampled edges, and it will reveal sufficiently many 0s in the dataset.

Our lower bounds also use generalized reductions from differential privacy to reconstruction rates (Section 5.1), which may find more application in the edge weight setting. It remains an interesting open question to show tight bounds for larger values of δ .

RELATED WORK

GRAPH OPTIMIZATION PROBLEMS. MINIMUM SPANNING TREE and Minimum-Weight (Perfect) Matching are classic graph optimization problems, with applications including tree-structured graphical models, synthetic data generation, and market or network matching. HIERARCHICAL CLUSTERING is an unsupervised learn-

ing technique that - unlike *flat* clustering methods such as k -means - organizes data into clusters at multiple levels of granularity. The problem dates back more than half a century [War63] and has applications across many different domains, including network analysis [LRU14], text analysis [SKK00], and biology [JS68; Sne05; Die+15; Eis+98; Sot+03].

RECONSTRUCTION ATTACKS. Our lower-bound proofs are part of a long line of reconstruction attacks in differential privacy. The seminal work of Dinur and Nissim [DN03] showed that answering too many queries too accurately can allow reconstruction of most of a private database. Subsequent work developed this connection into general lower-bound methods for private data release, including iterative constructions [GRU12], lower bounds based on reconstruction and fingerprinting ideas [De12], and robust traceability arguments [Dwo+15]. In the graph setting, Sealfon [Sea16] and Eden, Liu, Raskhodnikova, and Smith [Ede+25] achieve lower bounds via reconstruction attacks; we improve on the results of Sealfon, while the latter result applies to a different privacy model.

PRIVATE GRAPH ALGORITHMS. Early graph-DP work studied edge-level statistics such as degree distributions [Hay+09], while node-DP was formalized and studied by Kasiviswanathan, Nissim, Raskhodnikova, and Smith [Kas+13]. Under edge-level DP, Eliás, Kapralov, Kulkarni, and Lee [Eli+20] gave near-optimal algorithms for approximating cuts via synthetic graphs, and Liu, Upadhyay, and Zou [LUZ24] obtained optimal bounds for several other cut-related private graph approximation tasks. In a *local* version of edge-DP where each node releases its own perturbed view of the graph, Eden, Liu, Raskhodnikova, and Smith [Ede+25] studied triangle counting and used reconstruction-style lower-bound techniques. Private matching in a different privacy model and objective, has been studied in discrete allocation and graph models [Hsu+14; Din+25]. These privacy models are distinct from the edge-weight model, and the techniques do not easily carry over.

EDGE-WEIGHT PRIVATE GRAPH OPTIMIZATION. The edge-weight model was introduced by Sealfon [Sea16], modeling the situation where graph structure is public but weights are sensitive. He studied differentially private release of approximate shortest paths and distances, and also gave algorithms and lower bounds for releasing MINIMUM SPANNING TREE, SINGLE-SOURCE SHORTEST-PATH TREES, and Minimum-Weight (Perfect) Matching under the ℓ_1 neighboring relation on weights. A recent line of work has studied private release of *all-pairs shortest-path distances* under the ℓ_1 neighboring relation, with lower bounds shown using discrepancy-based methods [FLL22; Che+23; Bod+24]. MST was studied under the ℓ_∞ neighboring relation in Hladík and Tetek [HT24] and Chapter 4, showing tight error bounds in pure and approximate DP settings, respectively. To our knowledge, [hladik_2024](#) provide the only other topology-dependent lower bounds for edge-weight DP, showing the error of MST for the ℓ_1 neighboring relation must grow with $\Omega(D)$, where D is the diameter of the topology. Their argument uses the packing technique, which restricts it to pure DP ($\delta = 0$).

PRIVATE HIERARCHICAL CLUSTERING. The non-private objective framework for hierarchical clustering was introduced by Dasgupta [Das16] and recently won a STOC Test of Time Award [ACM26]. It was further developed by Cohen-Addad, Kanade, Mallmann-Trenn, and Mathieu [Coh+19] and the differentially private version was first studied by Imola, Epasto, Mahdian, Cohen-Addad, and Mirrokni

[Imo+23], who gave algorithms and proved a packing-based $\Omega(n^2/\epsilon)$ additive-error lower bound for pure edge DP. Deng, Gao, Upadhyay, Wang, and Zhou [Den+25] subsequently studied hierarchical clustering in the edge-weight model and showed that hard instances can be embedded into the weights of the complete graph, implying an $\Omega(n^2/\epsilon)$ lower bound under pure DP. Our hierarchical-clustering lower bound follows the edge-weight line of Deng, Gao, Upadhyay, Wang, and Zhou [Den+25], but strengthens the picture by applying to approximate DP and to beyond the complete topology. They also derived an algorithm achieving $\tilde{O}(\frac{1}{\epsilon})$ multiplicative error for graphs in which every edge weight is at least 1; we do not compare with this algorithm as this assumption is too specialized for our purposes.

5.1 PRELIMINARIES

For two vectors $\mathbf{w}, \mathbf{w}' \in \mathbb{R}^m$, we define $d_H(\mathbf{w}, \mathbf{w}') := \sum_{i=1}^m \mathbb{1}[w_i \neq w'_i]$ to be the *Hamming distance*, i.e., the number of coordinates in which they differ. For sets S, S' we extend the definition by the symmetric difference $d_H(S, S') := |S \setminus S'| + |S' \setminus S|$. We write $\mathbf{x} \sim_H \mathbf{x}'$ if $d_H(\mathbf{x}, \mathbf{x}') \leq 1$.

GRAPH TERMINOLOGY We consider a finite, simple and undirected graph $G = (V, E, \mathbf{w})$, where the set of n vertices V and the set of m edges E are public and the weight vector $\mathbf{w} = (w_1, \dots, w_m) \in \mathbb{R}^m$ is private. We use $\mathbf{w} \in \mathbb{R}^m$ and $w : E \rightarrow \mathbb{R}$ interchangeably to denote the edge weights. We refer to $F = (V, E)$ as the *topology* of G . Throughout the paper, we assume $m > n$ and that F is connected. We denote by $\mathcal{G}$ (respectively $\mathcal{G}_\omega$) the family of all unweighted (weighted) graphs.

WEIGHTS OF SETS AND CUTS For a subset of edges $S \subseteq E$ of edges, we denote the *weight* as $w(S) = \sum_{e \in S} w_e$. We refer to a partition A, B of V as a *cut* of G . The *weight* of the cut is given by $w(A, B) := \sum_{uv \in E \cap (A \times B)} w_{uv}$. Similarly, for any edge set $E' \subseteq E$, we denote $E'(A, B) = |\{(u, v) \in E' : u \in A, v \in B\}|$ to be the size of the cut, or the number of edges crossing between A, B . We refer to any cut A, B satisfying $\frac{1}{3}n \leq |A|, |B| \leq \frac{2}{3}n$ as a *balanced cut*.

SPANNING TREES AND PERFECT MATCHINGS A *spanning tree* is an acyclic subset $T \subseteq E$ of size $n - 1$ making the graph connected. A *matching* $M \subseteq E$ is a set of pairwise vertex-disjoint edges, i.e., $e \cap e' = \emptyset$ for any two distinct $e, e' \in M$. For a matching M and a subset $S \subseteq V$, we let $S(M)$ denote those vertices in S that are incident to an edge of M . A *perfect matching* is a matching that covers every vertex of V . We denote the set of all spanning trees and perfect matchings of G as $\mathcal{S}(G)$ and $\mathcal{P}(G)$, respectively. A *minimum spanning tree* (MST) is a spanning tree T^* that minimizes $w(T)$ among all spanning trees $T \in \mathcal{S}(G)$; analogously, a *minimum weight perfect matching* (MWPM) is a perfect matching $M^* \in \mathcal{P}(G)$ that minimizes $w(M)$ among all $M \in \mathcal{P}(G)$.

HIERARCHICAL CLUSTERING A *hierarchical clustering* (HC) for a graph G is represented by a rooted tree T whose leaves correspond to V , and where each internal node corresponds to the union of the leaves in its subtree. Traversing the tree from the root to the leaves corresponds to successively refining the clustering. In seminal work [Das16], Dasgupta introduced a cost function $\text{cost}_w(T)$ evaluating the quality of a hierarchical clustering, setting the stage for optimization. Intuitively,

1839 it charges the total weight separated at each merge times the size of the subtree at
1840 that merge, encouraging less weight to cross higher merges. Formally,

Definition 5.1 ([Das16] Dasgupta’s cost objective). *Given an input graph $G = (V, E)$ together with weights $w : E \rightarrow \mathbb{R}_{\geq 0}$ and a hierarchical clustering tree T , the Dasgupta cost function cost_w is given by*

$$\text{cost}_w(T) := \sum_{\{u,v\} \in E} |T[u \vee v]| \cdot w(\{u, v\}), \quad (1)$$

1841 where $|T[u \vee v]|$ denotes the number of leaves in the subtree induced by the
1842 lowest common ancestor of u and v in T . As noted by Dasgupta, we may assume
1843 without loss of generality that T is a rooted *binary* tree, because every non-binary
1844 hierarchical clustering tree can be transformed into a binary one without increasing
1845 $\text{cost}_w(T)$. We illustrate this cost function on a small example in Appendix 5.7.

1846 **RECONSTRUCTION ATTACKS.** Our DP lower bounds will be shown via re-
1847 construction attacks, which are algorithms that try to reconstruct the private
1848 dataset $\mathbf{x}$ based on the algorithm output. If the attack reconstructs coordinates of $\mathbf{x}$,
1849 they demonstrate an impossibility for DP, and any reasonable notion of privacy
1850 (see [Vad17] for an overview of reconstruction attacks).

1851 We will use two types of reconstruction attacks in our lower bounds. In the first
1852 type, we consider an algorithm that attempts to reconstruct a fixed coordinate i of
1853 $\mathbf{x}$, and show that any DP algorithm cannot be too successful at doing so. This is a
1854 slight generalization of Sealfon [Sea16, Lemma 5.3] to general data domains $\mathcal{X}$.

1855 **Lemma 5.1.** *Let $\mathcal{B} : \mathcal{X}^d \rightarrow \mathcal{X}^d$ be any mechanism that is (ϵ, δ) -differentially private
1856 under the Hamming neighborhood relationship. Then for a random dataset $\mathbf{x} \leftarrow \mathcal{X}^d$ drawn
1857 uniformly at random, we have for each $i \in [d]$, we have $\mathbf{P}[\mathcal{B}(\mathbf{x})_i = x_i] \leq \frac{\exp(\epsilon)}{|\mathcal{X}|} + \delta$.*

1858 As we will see, using a data domain of size $|\mathcal{X}| = \frac{m}{n}$ will be critical to obtaining
1859 tight lower bounds. We also consider reconstructing an entire set $I \subseteq [d]$ of indices
1860 of a binary dataset $\mathbf{x}$. An attack is then a mechanism $\mathcal{B} : \{0, 1\}^d \rightarrow 2^{[d]} \times \{0, 1\}^d$,
1861 and its success is measured by

$$\text{Succ}(\mathcal{B}) = \frac{1}{2} - \frac{\mathbb{E}[d_H(\mathbf{x}|_I, \mathbf{y}|_I)]}{\mathbb{E}[|I|]}, \quad (2)$$

1862 where $\mathbf{x} \sim \{0, 1\}^d$ and $(I, \mathbf{y}) := \mathcal{B}(\mathbf{x})$, and $\mathbf{x}|_I$ indicate the subvector of $\mathbf{x}$ on indices
1863 I . Intuitively, success is the normalized error that randomly guessing I would
1864 yield (namely $\frac{1}{2}$) minus the normalized error made by $\mathcal{B}$ in the set I . DP limits the
1865 possible success as follows.

1866 **Theorem 5.2.** *Let $\mathcal{B} : \{0, 1\}^d \rightarrow 2^{[d]} \times \{0, 1\}^d$ be any (ϵ, δ) -DP algorithm and let
1867 $\mathbf{x} \sim \{0, 1\}^d$ be drawn uniformly at random. Then, $\text{Succ}(\mathcal{B}) \leq \frac{e^\epsilon - 1}{2(e^\epsilon + 1)} + \frac{d\delta}{e^\epsilon + 1}$.*

1868 The proof appears in Section 5.7.1. It differs from prior reconstruction attacks
1869 because I is itself a differentially private output. When $\delta = o(\frac{1}{d})$, the second term
1870 becomes negligible, and when $\epsilon < 1$, we have that $\text{Succ}(\mathcal{B}) = O(\epsilon)$.

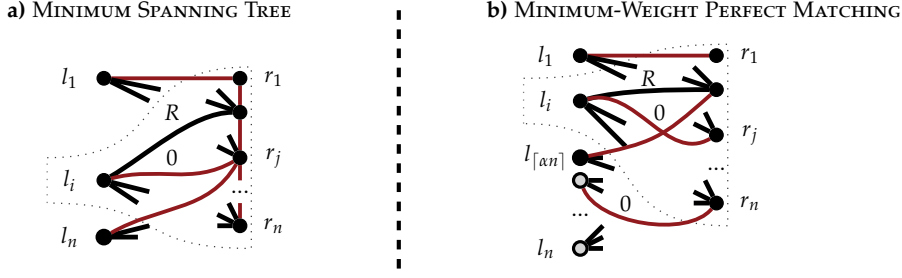


Figure 5.2: **a)** Encoding the vector $\mathbf{x} = (x_1, \dots, x_n)$ into the minimum-weight spanning tree. **b)** The encoding for min-weight perfect matching for a vector of length $\lceil \alpha n \rceil$. We pad the left side with the gray vertices. In both cases, each vertex r_j represents the integer j , and we set the weight of the edge from l_i to r_j to 0, if $x_i = j$ and otherwise to $R = \mathcal{O}(\frac{1}{\epsilon} \ln n)$.

5.2 PRIVATE MINIMUM-WEIGHT SPANNING TREES

In Section 5.2.1, we show that there is a worst-case topology F on which the additive error of **MST** under DP is tight in $\Theta((n/\epsilon) \cdot \log(m/n))$ (for small enough δ) achieved by the Laplace mechanism. Going beyond worst-case graphs, in Section 5.2.2, we show that the lower bound of $\Omega(n/\epsilon)$ due to [Sea16] holds on all graph topologies with minimum cut $\Omega(\log(n))$.

5.2.1 Tight Lower Bounds on Worst-Case Topologies

Assuming $\delta \leq (n/m)^{-\Omega(1)}$, we now close the gaps between the known upper and the lower bounds. The idea is to encode a random vector $\mathbf{x} \in [n]^d$ into the **MST** of a dense graph G so that releasing an overly accurate **MST** by a differentially private mechanism would contradict the reconstruction rate in Lemma 5.1. As a by-product, we also get a lower bound for ϵ -DP, obtained by taking the limit of δ towards 0 recovering the packing-based lower bound obtained by Hladík and Tetek [HT24].

Theorem 5.3 (Worst-Case Lower Bound MST). *Let $\epsilon > 0$ and $\delta \leq (n/m)^{\Omega(1)}$ where $m > 2n$. There exists a graph topology $F = (V, E)$ and a distribution $\mathcal{D}_\omega$ of weights such that for any (ϵ, δ) -DP algorithm $\mathcal{A}$ that outputs an approximate **MST** T under the ℓ_1 neighboring relationship, if the weights $\mathbf{w} \sim \mathcal{D}_\omega$ then the expected additive error satisfies*

$$w(T) - w(T^*) \geq \Omega((n/\epsilon) \cdot \min(\ln(1/\delta), \ln(m/n))),$$

where T^* is the optimal MST.

We will start with the case $m = n^2$, and focus on the general case later. The full encoding scheme appears in Figure 5.2. Throughout the construction, we assume the topology $F \in \mathcal{G}$ to be public.

ENCODING Let $\text{Enc}_R : [n]^n \rightarrow \mathcal{G}_\omega$ be the encoding function that takes some dataset $\mathbf{x} \in [n]^n$ and returns a weighted graph that encodes the dataset into the MST. The parameter $R > 1$ is used to control the edge weights. We construct a new connected graph $G = (V_1 \cup V_2, E, \mathbf{w})$ with $2n$ vertices and $n^2 + (n - 1)$ edges in the following way: Let $V_1 = \{\ell_1, \dots, \ell_n\}$ and $V_2 = \{r_1, \dots, r_n\}$ be two vertex sets

each of size n . Each vertex $\ell_i \in V_1$ represents one coordinate of $\mathbf{x}$ and has edges to each vertex in V_2 . Set $w(\{\ell_i, r_{x_i}\}) := 0$ and $w(\{\ell_i, r_j\}) := R$ for $j \neq x_i$. We ensure connectivity by adding a path through the vertices in V_2 with zero weighted edges (contributing the $n - 1$ extra edges).

DECODING Let $\text{Dec}_G : \mathcal{S}(G) \rightarrow [n]^n$ be a decoder and assume that the topology G was produced by the encoder (to simplify notation we assume F is public as soon as the instance has been created).

Given a spanning tree $T \in \mathcal{S}(G)$, we reconstruct the vector $\mathbf{y} \in [n]^n$ as follows: for each $i \in [n]$, choose any $r_j \in V_2$ with $\{\ell_i, r_j\} \in T$ and set $y_i := j$. Note that any vertex in V_1 must have at least one edge in T by construction, and therefore, the reconstruction for each coordinate is well-defined. In case there are multiple, we break ties arbitrarily.

Now define the attack $\text{Rec} : [n]^n \rightarrow [n]^n$ as $\text{Rec}_{\mathcal{A}_{\text{MST}}, R}(\mathbf{x}) := \text{Dec}_G(\mathcal{A}_{\text{MST}}(\text{Enc}_R(\mathbf{x})))$ where $\mathcal{A}_{\text{MST}}$ is any (possibly randomized) **MST** algorithm. By the construction above, it is clear that an optimal **MST** algorithm $\mathcal{A}_{\text{OPT}}$ recovers all coordinates perfectly, i.e. $\text{Rec}_{\mathcal{A}_{\text{OPT}}, R}(\mathbf{x}) = \mathbf{x}$. This leads to the following two observations.

Observation 5.4. *Let $R > 0$ and $G = (F, \mathbf{w})$ be obtained from $\text{Enc}_R(\mathbf{x})$. If $T \in \mathcal{S}(F)$ is any and T^* the optimal spanning tree of G , then*

$$d_H(\text{Dec}_F(T), \mathbf{x}) \leq \frac{w(T) - w(T^*)}{R} = \frac{w(T)}{R},$$

Indeed, $w(T^*) = 0$, and whenever $\text{Dec}_F(T)_i \neq x_i$, the tree T contains an edge $\{\ell_i, r_j\}$ with $(j \neq x_i)$, which has weight R . These bad edges are distinct for different i , so the number of decoding errors is at most $w(T)/R$.

Observation 5.5 (Induced ℓ_1). *For every $\mathbf{x} \sim_H \mathbf{x}'$ and $R > 0$, setting $k := 2\lceil R \rceil$, there is a chain of ℓ_1 -neighboring graphs, i.e. $\text{Enc}_R(\mathbf{x}) := G_0 \sim_1 G_1 \sim_1 \dots \sim_1 G_k := \text{Enc}_R(\mathbf{x}')$.*

Each consecutive G_i is obtained by successively changing a single weight by one. By group privacy (see e.g. [Vad17, Lemma 7.2.2]), this means there is a $(2\lceil R \rceil \epsilon, 2\lceil R \rceil e^{2\lceil R \rceil \epsilon} \delta)$ -DP guarantee on $\text{Enc}_R(\mathbf{x})$ under the *Hamming* neighboring relationship. We are now ready to prove Theorem 5.3.

Proof of Theorem 5.3. Suppose for contradiction, that there exists an (ϵ, δ) -DP **MST** algorithm $\mathcal{A}_{\text{MST}}$ with expected additive error less than $\frac{n}{100} \cdot R$. Assume a random dataset $\mathbf{x} \sim \text{Uni}([n]^n)$. Note that the decoding step is simply post-processing, and hence $\text{Rec}_{\mathcal{A}_{\text{MST}}, R}$ parameterized by $\mathcal{A}_{\text{MST}}$ and R is $(2\lceil R \rceil \epsilon, 2\lceil R \rceil e^{2\lceil R \rceil \epsilon} \delta)$ -DP. Now set $R := \min\left(\frac{\ln n}{2c\epsilon}, \frac{\ln(1/\delta)}{2c\epsilon}\right)$ for some $c > 1$. Combining with Observation 5.4 and by the utility guarantee of $\mathcal{A}_{\text{MST}}$, we would leak at least $\frac{99}{100} \cdot n$ of the input coordinates in expectation. Let $\mathbf{y} := \text{Rec}_{\mathcal{A}_{\text{MST}}, R}(\mathbf{x})$ and by Lemma 5.1,

$$\begin{aligned} \mathbf{P}[\mathbf{y}_i = \mathbf{x}_i] &\leq \frac{e^{2\lceil R \rceil \epsilon}}{n} + 2\lceil R \rceil e^{2\lceil R \rceil \epsilon} \delta \\ &\leq \frac{1}{n} \exp\left(\ln(n^{1/c})\right) + \frac{\ln(n)}{\epsilon c} \exp\left(\ln(n^{1/c})\right) \delta \\ &= n^{(1-c)/c} + \frac{\ln(n)}{\epsilon c} n^{1/c} \delta. \end{aligned}$$

By our assumption on $\delta = n^{-\Omega(1)}$, the second term vanishes. Linearity of expectation gives

$$\mathbb{E} \left[\sum_{i \in [n]} (\mathbb{1}[y_i = x_i]) \right] \leq n \cdot n^{(1-c)/c} = n^{1/c}.$$

In particular, for a sufficiently large constant c , we contradict the utility guarantee of $\mathcal{A}_{\text{MST}}$. Note also that we require $R \leq \frac{\ln n}{2c\epsilon}$ to get any meaningful probability. Thus, we have shown a lower bound of $\Omega((n/\epsilon) \ln n)$ in the dense case ($m = \Theta(n^2)$), for sufficiently small δ .

It remains to show that this technique can be parameterized by the number of edges m to yield a bound of $\Omega((n/\epsilon) \cdot \ln(m/n))$ for any $2n \leq m \leq n^2$. Suppose, we want a lower bound for graphs with m edges and $2n$ vertices. Without loss of generality, we treat m/n as an integer. We can build n^2/m many subgraphs on $2(m/n)$ vertices (constructed as described above for the dense case with n replaced by m/n). To make the full graph connected, we add a path through one representative vertex from each subgraph, adding only $\mathcal{O}(n^2/m)$ low-cost edges that would only affect the bound by a constant. Ignoring this lower-order term, observe that the number of edges is exactly $(m/n)^2 \cdot (n^2/m) = m$ (we ignore edges on the path of length $\leq n$).

For $i \in [n^2/m]$, let each subgraph encode an i.i.d. random vector $\mathbf{x}^{(i)} \in [m/n]^{m/n}$ of length m/n . Since an (ϵ, δ) -DP algorithm satisfies the same privacy guarantees on the weights of each individual subgraph, the previous lower bound applied to a single subgraph states that we cannot reconstruct a single $\mathbf{x}^{(k)}$ with expected error better than $\Omega(\frac{m}{n\epsilon} \cdot \ln \frac{m}{n})$. By linearity of expectation, we get a lower bound on the expected total error of $\Omega(\frac{m}{n\epsilon} \cdot \ln(\frac{m}{n}) \cdot \frac{n^2}{m}) = \Omega(\frac{n}{\epsilon} \cdot \ln(\frac{m}{n}))$. $\square$

5.2.2 Lower Bounds for General Topology Classes

We will design a subset reconstruction attack that uses any given approximate DP **MST** algorithm $\mathcal{A}_{\text{MST}}$. Assuming the additive cost of $\mathcal{A}_{\text{MST}}$ is sufficiently low, the reconstruction success will be high enough to form a contradiction with Theorem 5.2 and establish an error lower bound for any DP algorithm.

Our reconstruction attack works as follows: the dataset is drawn from $\mathbf{x} \sim \text{Uni}(\{0, 1\}^m)$, and then arbitrarily encoded into E with a labeling function $\ell : E \rightarrow [m]$. The weights are set as $w(e) = R x_{\ell(e)}$, where R is a factor used to amplify the weight that we will choose later. Then, $\mathcal{A}_{\text{MST}}$ is run on $(V, E, \mathbf{w})$ to produce a tree $T \subseteq E$. The returned tree T contains edges whose weight is more likely to be 0, and thus the reconstructed set is set to be $I = \{\ell(e) : e \in T\}$, and all guesses $\{y_i : i \in I\}$ are $y_i = 0$. This procedure appears in Algorithm 7.

We will show that if $\mathcal{A}_{\text{MST}}$ has sufficiently low additive error, then this attack has a success rate much higher than 0. Specifically, we will show $\mathbb{E}[\sum_{i \in I} x_i] \leq 0.1 \mathbb{E}[|I|]$. By construction, $w(T) = R \sum_{i \in I} x_i$. Using the additive error guarantee of $\mathcal{A}_{\text{MST}}$, we have $\mathbb{E}[\sum_{i \in I} x_i] = \frac{1}{R} \mathbb{E}[w(T)] \leq \frac{1}{R} w(T^*) + \frac{1}{10} n$. Thus, we need to show that the *optimal MST* in $(V, E, \mathbf{w})$ is likely to have low weight. This is where we assume that the *minimum cut* of F is not too small. Recall that the minimum cut is given by $\min_{A \sqcup B = V} E(A, B)$. We use a concentration inequality, and a union bound over all cuts, to argue that if the mincut of F is at least $5 \log n$, then with high probability all cuts have an edge labeled 0. This immediately implies that an **MST** of weight 0 exists. In order for the union bound to work, we use a result of Karger [Kar93] bounding the number of approximate minimum cuts, as these cuts have higher failure probability. Formally,

Algorithm 7 Subset Reconstruction Attack Rec^{MST} for MST

Require: Public graph topology $F = (V, E)$, dataset $\mathbf{x} \in \{0, 1\}^m$,
parameter $R > 0$, algorithm $\mathcal{A}_{\text{MST}}$.

- 1: Initialize weights $\mathbf{w} \in \{0, R\}^m$.
- 2: Let $\ell : E \rightarrow [m]$ be any bijection from E to $[m]$ labels.
- 3: Set $w(e) := Rx_{\ell(e)}$ for all $e \in E$. $\triangleright$ Encode the dataset into the weights
- 4: Let $G = (F, \mathbf{w})$.
- 5: Compute minimum spanning tree $T := \mathcal{A}_{\text{MST}}(G)$.
- 6: Initialize reconstruction set $I = \{\ell(e) : e \in T\} \subseteq [m]$.
- 7: **for each** $i \in I$, guess $y_i = 0$.
- 8: **return** Reconstruction set I , reconstructed vector $\mathbf{y} = \{y_i : i \in I\}$.

Lemma 5.6. Suppose the graph topology $F = (V, E)$ has minimum cut $\lambda \geq 5 \log(n)$, and the random dataset $\mathbf{x} \sim \text{Uni}(\{0, 1\}^m)$ is embedded into $\mathbf{w}$ via an arbitrary bijection. Then, with probability at least $1 - \frac{3}{n^2}$, there exists a zero-cost spanning tree $T \subseteq E$, i.e. $w(T) = 0$.

The proof appears in Section 5.7.2. Note that $\mathbb{E}[|I|] = |I| = n - 1$ always. Having established bounds on both $\mathbb{E}[\sum_{i \in I} x_i]$ and $\mathbb{E}[|I|]$, we can show a reconstruction attack lower bound:

Theorem 5.7. Let (V, E) be a public graph topology with minimum cut at least $5 \log(n)$. Suppose that $\mathcal{A}_{\text{MST}}$ attains expected additive error $\leq \frac{R}{10}n$ for some $R > 0$. Then $\text{Succ}(\text{Rec}^{\text{MST}}) \geq 0.4$.

Proof. Let $\mathcal{E}$ denote the event that the optimal MST has weight that is not 0. We have that

$$\begin{aligned} \mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] &= \mathbb{E} \left[\sum_{i \in I} x_i \right] \leq \mathbb{E} \left[\sum_{i \in I} x_i \mid \neg \mathcal{E} \right] + \mathbf{P}[\mathcal{E}] \mathbb{E} \left[\sum_{i \in I} x_i \mid \mathcal{E} \right] \\ &\leq \mathbb{E} \left[\sum_{i \in I} x_i \mid \neg \mathcal{E} \right] + \mathbf{P}[\mathcal{E}] \cdot n \cdot R. \end{aligned}$$

By Lemma 5.6, we know $\mathbf{P}[\mathcal{E}] \leq \frac{3}{n^2}$. Furthermore, by the construction of I , we know $R \sum_{i \in I} x_i = w(T)$. Given that $\neg \mathcal{E}$ occurs, we have $\mathbb{E}[w(T)] \leq \frac{R}{10}n$ by the error guarantee of $\mathcal{A}_{\text{MST}}$. Thus, the above bound is $\frac{n}{10} + \mathcal{O}(\frac{R}{n})$. $\square$

Theorem 5.7 and Lemma 5.2 immediately imply the following corollary:

Corollary 5.8. For any topology $F = (V, E)$ with minimum cut at least $5 \log(n)$, there is no (ϵ, δ) -DP algorithm $\mathcal{A}_{\text{MST}}$ attaining $\frac{1}{10\epsilon}n$ additive error for any $\epsilon \leq 1, \delta \leq \frac{\epsilon \cdot 0.01}{m}$.

Proof. Suppose the contrary, and instantiate Rec^{MST} with $\mathcal{A}_{\text{MST}}$ and $R = \frac{1}{\epsilon}$. By group privacy, Rec^{MST} satisfies $(\epsilon \lceil R \rceil, \delta \lceil R \rceil e^{\epsilon \lceil R \rceil})$ -DP, which means it satisfies $(2, \frac{10}{\epsilon} \delta)$ -DP as $\epsilon \lceil R \rceil < 2$. By Lemma 5.2, this means it can only be a $\frac{1}{2} - \frac{1}{1+e^2} + m\delta < 0.39$ -successful reconstruction attack, yet this contradicts Theorem 5.7. $\square$

While looser than Theorem 5.3 by a $\log(\frac{m}{n})$ factor, this result establishes that the error of MST must still grow linearly under (ϵ, δ) -DP for many topologies. We

note that this proof would still work if F had *any* linear-sized subgraph with large enough min-cut.

5.3 MINIMUM-WEIGHT PERFECT MATCHINGS

We will now show multiple lower bounds for the problem of Minimum-Weight (Perfect) Matching (MWPM).

5.3.1 Reconstruction-Based Lower Bound

As any perfect matching has size $n/2$, the upper bound in Theorem 5.25 immediately carries over to Minimum-Weight (Perfect) Matching as well. We now show that the resulting additive error of $\Theta((n/\epsilon) \cdot \ln(\frac{m}{n}))$ is tight for this problem as well (for sufficiently small δ). The encoding will be slightly different for MST because all coordinates with the same value would be encoded by the same vertex, and a matching can only recover one of those. To fix this, we prove a standard balls and bins result stating that if x has length $n' = \alpha n$ for some small $0 < \alpha \leq 1$, only a α^2 fraction of a random $\mathbf{x} \sim \text{Uni}([n]^{\lceil \alpha n \rceil})$ has collisions with high probability. The proof to the following statement uses a standard concentration inequality.

Lemma 5.9 (Collision-free). *Let $(X_1, \dots, X_d) \sim \text{Uni}([n]^d)$ with $d = \lceil \alpha n \rceil$ for a constant $0 < \alpha \leq 1$. Then, with probability at least $1 - \exp(-\Omega(\alpha^3 n))$, the number of indices i for which there exists some $j \neq i$ with $X_i = X_j$ is $\mathcal{O}(\alpha^2 n)$.*

Proof. Define $\tilde{X}_i = \mathbb{1}[\exists j \neq i : X_j = X_i]$ and let $Y = \sum_{i=1}^d \tilde{X}_i$. By a union bound, $\mathbb{E}[\tilde{X}_i] \leq \sum_{j \neq i} \mathbb{P}[X_j = X_i] = (d-1)/n$, so that $\mathbb{E}[Y] \leq d(d-1)/n = \alpha^2 n - \alpha \leq \alpha^2 n$. Since changing a single X_i affects at most two of the $\tilde{X}_i$'s, Y is 2-Lipschitz. By McDiarmid's inequality [McD89], for any $t > 0$, $\mathbb{P}[Y \geq \mathbb{E}[Y] + t] \leq \exp(-2t^2/(4d))$. Setting $t = \mathbb{E}[Y]$ gives $\mathbb{P}[Y \geq 2\mathbb{E}[Y]] \leq \exp(-\mathbb{E}[Y]^2/(2d)) = \exp(-\Omega(\alpha^3 n))$. Thus, with probability at least $1 - \exp(-\Omega(\alpha^3 n))$, we have $Y \leq 2\mathbb{E}[Y] = \mathcal{O}(\alpha^2 n)$. $\square$

Theorem 5.10 (Lower bound MWPM). *Let $\epsilon > 0$, $\delta \leq (n/m)^{\Omega(1)}$, and $m > 2n$. Then there exists a graph topology $F = (V, E)$ and a distribution of weights $\mathcal{D}_\omega$ on weights of E such that for any (ϵ, δ) -DP protocol $\mathcal{B}$ that outputs an approximate minimum-weight perfect matching M under the ℓ_1 neighboring relationship, if the weights $\mathbf{w} \sim \mathcal{D}_\omega$, then the expected additive error satisfies*

$$w(M) - w(M^*) \geq \Omega((n/\epsilon) \cdot \min(\ln(1/\delta), \ln(m/n))),$$

where M^* is an optimal minimum-weight perfect matching.

Fix some constant $0 < \alpha \leq 1$ and let R again be a parameter that controls the weights. Now let $\text{Enc}_R : [n]^d \rightarrow \mathcal{G}_\omega$ be constructed as follows. Let $\mathbf{x} \in [n]^d$ be some vector of length $d = \alpha n$ and $0 < \alpha \leq 1$. Then the encoder creates a complete bipartite graph $G = (V_1 \cup V_2, E, \mathbf{w})$ with bipartitions $V_1 = \{\ell_1, \dots, \ell_n\}$ and $V_2 = \{r_1, \dots, r_n\}$. Each of the first d vertices $\ell_i \in V_1$ represents a single coordinate of $\mathbf{x}$. For each $i \in [d]$, we set $w(\{\ell_i, r_{x_i}\}) := 0$ and $w(\{\ell_i, r_j\}) := R$ for $j \neq x_i$. We treat the remaining $(1 - \alpha)n$ vertices in V_1 as dummy nodes that simply get zero-weight edges to all vertices in V_2 . Furthermore, let $\text{Dec}_G : \mathcal{P}(G) \rightarrow [n]^d$ and $\text{Rec}_G : [n]^d \rightarrow [n]^d$ be analogously defined as in Section 5.2. Find this construction also in Figure 5.2.

Using this encoding, the optimal solution is determined by the number of collisions C in $\mathbf{x}$ and is not necessarily 0 as before. By Lemma 5.9, we can make C

arbitrarily small with probability exponentially small in n . The key observation is that there are still $\Theta(n)$ many coordinates that do not collide. Every non-colliding coordinate can be recovered from a correct edge in some perfect matching M . Hence, we get this observation.

Observation 5.11. Let $G = (F, \mathbf{w})$ be obtained from $\text{Enc}_R(\mathbf{x})$ and let

$$C = |\{i \in [d] \mid \exists j \neq i, x_i = x_j\}|$$

be the number of colliding coordinates. If $M \in \mathcal{P}(F)$ is any perfect matching, then

$$d_H(\text{Dec}_F(M), \mathbf{x}) \leq \frac{w(M) - w(M^*)}{R} + C,$$

where M^* is an optimal Minimum-Weight (Perfect) Matching of G .

We now have all the ingredients ready to prove the theorem. The proof is similar to Theorem 5.3. First, draw $\mathbf{x} \sim \text{Uni}([n]^{\alpha n})$ and fix a small constant $0 < \alpha \leq 1$. WLOG, assume that αn is an integer. Then $C \leq \alpha^2 n$ with probability $1 - \exp(-\Omega(\alpha^3 n))$ by Lemma 5.9.

Suppose for contradiction that there exists an (ϵ, δ) -DPMMinimum-Weight (Perfect) Matching algorithm $\mathcal{B}$ with expected additive error $\mathbb{E}[w(M) - w(M^*)] < 0.01 \alpha n R$. Then by Observation 5.11, $\mathbb{E}[d_H(\text{Dec}_G(M), \mathbf{x})] \leq 0.01 \alpha n + \alpha^2 n$. So $\mathcal{B}$ correctly recovers at least $\alpha n - 0.01 \alpha n - \alpha^2 n = \alpha n(0.99 - \alpha)$ coordinates in expectation.

Observe that for some fixed $R > 1$, Observation 5.5 holds again, and the induced ℓ_1 -distance of two encoded graphs differs by at most $2\lceil R \rceil$ if the encoded vectors are Hamming neighbors. Hence, by Lemma 5.1,

$$\begin{aligned} \mathbb{P}[\text{Rec}_{\mathcal{A}, R}(\mathbf{x})_i = x_i] &\leq \frac{e^{2\lceil R \rceil \epsilon}}{n} + 2\lceil R \rceil e^{2\lceil R \rceil \epsilon} \delta \\ &\leq \frac{1}{n} \exp\left(\ln(n^{1/c})\right) + \frac{\ln(n)}{\epsilon c} \exp\left(\ln n^{1/c}\right) \delta \\ &= n^{(1-c)/c} + \frac{\ln(n)}{\epsilon c} n^{1/c} \delta. \end{aligned}$$

Assuming $\delta = n^{-\Omega(1)}$, the second term is $o(1/n)$ and vanishes. Using linearity of expectation over all $i \in [\alpha n]$,

$$\mathbb{E}\left[\sum_{i \in [\alpha n]} \mathbb{1}[\text{Rec}_{\mathcal{A}, R}(\mathbf{x})_i = x_i]\right] \leq \left(n^{(1-c)/c} \cdot \alpha n\right) = \alpha \cdot n^{1/c},$$

For sufficiently large $c > 1$, this is sublinear in n , contradicting the assumed utility guarantee of $\Theta(\alpha n)$. Finally, to obtain a bound that also depends on the number of edges, we use the same decomposition technique as in the proof of Theorem 5.3 with simplification that we don't need to connect the subgraphs.

5.3.2 Packing-Based Lower Bound

We now show tight error bounds for MWPM under pure-DP for both the ℓ_1 and ℓ_∞ neighboring relationships using a similar argument to the one proposed by Hladík and Tetek [HT24]. The authors explicitly asked whether their technique can be used for other problems as well. The presented lower bound of $\Omega(\frac{n^2}{\epsilon} \ln n)$ matches the one the authors gave for private MST. The proof also works for ℓ_1 and recovers the reconstruction-based lower bound we gave in Theorem 5.10.

For the following, let $\mathcal{P}(G)$ be the set of all perfect matchings in some graph G . Furthermore, define the Hamming distance of two perfect matchings M, M' to be $d_H(M, M') := |M \setminus M'|$. The key observation is that similar to their setting, we can again greedily construct an exponential-sized set of dissimilar perfect matchings where each pair has a large Hamming distance of $d = \Theta(n)$. While [HT24] used a complete graph for constructing their family of hard instances, we will use the complete bipartite graph because it makes it easier to argue about perfect matchings. Define M^* to be the optimal minimum-weight perfect matching on $(G, \mathbf{w})$ (the graph will be clear from the context), and $w(M) := \sum_{e \in M} \mathbf{w}_e$ the weight of a matching, so that the additive error of M is $w(M) - w(M^*)$.

We denote by $\mathcal{L}_{(G, \mathbf{w})}(\alpha)$ the set of all perfect matchings with additive error less than some α on some graph $(G, \mathbf{w})$.

Lemma 5.12 ([Vad17; HT24] Packing Argument for Matchings). *Given a collection of weights $\mathcal{W} \subseteq \mathbb{R}^m$ each at induced distance at most r from some fixed weight vector $\mathbf{w}_0 \in \mathbb{R}^m$. Fix some α for which all $\mathcal{L}_{(G, \mathbf{w})}(\alpha)$ are disjoint. Then for any ε -DP MWPM mechanism $\mathcal{M}$, there exist weights $\mathbf{w} \in \mathcal{W}$ such that, drawing $M \leftarrow \mathcal{M}(G, \mathbf{w})$,*

$$\mathbf{P}[M \in \mathcal{L}_{(G, \mathbf{w})}(\alpha)] = \mathbf{P}[w(M) \leq w(M^*) + \alpha] \leq \frac{\exp(r \cdot \varepsilon)}{|\mathcal{W}|},$$

where M^* is the optimal MWPM on $(G, \mathbf{w})$.

In our cases, we will use the ℓ_1 and ℓ_∞ neighboring relationships on the weight vectors. We are now ready to prove our lower bound.

Theorem 5.13 (Lower Bound ε -DP). *Let G be the complete bipartite graph $K_{n,n}$ on $2n$ vertices. Let $S \subseteq \mathcal{P}(G)$ be a set of perfect matchings with $d_H(M, M') \geq n/4$ for all distinct $M, M' \in S$ and size at least $|S| \geq e^{(n/4) \cdot \ln n}$. Then for any ε -DP algorithm $\mathcal{M}$ that privately releases a MWPM $M \leftarrow \mathcal{M}(G, \mathbf{w})$, there exist weights $\mathbf{w}$ such that*

$$\frac{1}{\sqrt{|S|}} \geq \begin{cases} \mathbf{P}\left[w(M) \leq w(M^*) + \frac{1}{128} \cdot \left(\frac{n}{\varepsilon} \cdot \ln(n) - 8\right)\right] & (\text{wrt. } \ell_1) \\ \mathbf{P}\left[w(M) \leq w(M^*) + \frac{1}{64} \cdot \left(\frac{n^2}{\varepsilon} \cdot \ln(n) - 8n\right)\right] & (\text{wrt. } \ell_\infty), \end{cases}$$

where M^* is the optimal MWPM on $(G, \mathbf{w})$.

Proof. Let $G = (V, E)$ be the complete bipartite graph on $2n$ vertices and let $S \subseteq \mathcal{P}(G)$ be such that each pair of perfect matchings is at least d apart in Hamming distance. For each $S_i \in S$, we create a corresponding weight function $\mathbf{w}_i$ where we set the weight of an edge to some value 0, if it is part of this matching and $R \in \mathbb{R}_+$ otherwise.

Now for each $S_i \in S$ we create a corresponding weight function $\mathbf{w}_i$ where we set a weight to 0, if the corresponding edge was part of the matching and to some $R > 0$ otherwise (with value to be chosen later). We first prove the claim for the ℓ_∞ neighboring relationship. Fix some arbitrary $\mathbf{w}_0$ and note that all other weight functions have $\|\mathbf{w} - \mathbf{w}_0\|_\infty \leq \lceil R \rceil \leq R + 1$. We can show that the sets $\mathcal{L}_{(G, \mathbf{w}_i)}(\frac{1}{2}Rd)$ of matchings with error at most $\frac{1}{2}dR$ are pairwise disjoint.

To see this, fix two distinct hard instances with weights $\mathbf{w}$ and $\mathbf{w}'$ where the optimal perfect matchings $d_H(M_1, M_2) > d$. Now assume for the sake of contra-

2089 diction that there exists some matching $B \in \mathcal{L}_{(G, \mathbf{w})}(\frac{1}{2}Rd) \cap \mathcal{L}_{(G, \mathbf{w}')}(\frac{1}{2}Rd)$. Then for
 2090 $\alpha = \frac{1}{2} \cdot Rd$ we have both $d_H(M_1, B) < \frac{d}{2}$ and $d_H(M_2, B) < \frac{d}{2}$ by the way the weights
 2091 are chosen. This would imply $d_H(M_1, M_2) < d$, immediately contradicting the
 2092 assumed Hamming distance between M_1 and M_2 . We can now apply Lemma 5.12.
 2093 Let $\mathcal{M}$ be any MWPM algorithm and draw $T \leftarrow \mathcal{M}(G, \mathbf{w})$. Then,

$$\mathbf{P} \left[w(T) \leq w(T^*) + \frac{d \cdot R}{2} \right] \leq \exp((R+1) \cdot \varepsilon) \cdot \frac{1}{|S|} := \alpha, \quad (3)$$

2094 Now set $R := \frac{1}{2\varepsilon} \cdot \ln|S| - 1$ so that $\exp((R+1)\varepsilon) = \sqrt{|S|}$ and hence $\alpha = 1/\sqrt{|S|}$.
 2095 Using $d = n/4$, Lemma 5.14 implies the existence of a set of size at least $e^{(n/4) \ln n}$
 2096 with pairwise Hamming distance at least $n/4$. Hence, we can match the constants
 2097 given in the theorem statement:

$$\begin{aligned} \frac{dR}{2} &= \frac{n}{8} \cdot \left(\frac{n \cdot \ln n}{8\varepsilon} - 1 \right) \\ &= \frac{1}{64} \cdot \left(\frac{n^2}{\varepsilon} \ln n - 8n \right) = \Theta\left(\frac{n^2}{\varepsilon} \ln n\right). \end{aligned}$$

2098 For the ℓ_1 neighboring relationship, first observe that all weights have $\|\mathbf{w} - \mathbf{w}_0\|_1 \leq$
 2099 $\lceil 2 \cdot R \cdot n \rceil$. Then setting $\alpha = \frac{\exp((2Rn+1)\varepsilon)}{|S|}$ allows us to prove the claimed lower
 bound we already showed in Chapter 5. It can be easily verified that setting
 $R := \frac{1}{4\varepsilon n} \ln|S| - \frac{1}{2n}$ gives the desired result. $\square$

Chapter 5 gives a
 (ε, δ) -DP bound that
 recovers this pure-DP
 LB by taking the
 limit of $\delta \rightarrow 0$.

We will now show that such a set S exists and can even be constructed.

Lemma 5.14 (Volume of a d -ball around perfect matching). *For the complete bipartite graph G , any fixed perfect matching M and any integer $d \leq n$, we have*

$$|\{M' \mid d_H(M, M') \leq d\}| \leq n^{d+1}.$$

2103 *Proof.* We can give a direct combinatorial proof by observing that we first pick
 2104 some subset of one of the bipartitions of size at most d that we want to rewire.
 2105 Then there are at most $d!$ many ways to remap those selected vertices. Hence,

$$\begin{aligned} |\{M' \in \mathcal{P}(G) \mid d_H(M, M') \leq d\}| &\leq \sum_{i=1}^{\lceil d \rceil} \binom{n}{i} i! \\ &\leq d \cdot \frac{n!}{(n-d)!} \leq n^{d+1}. \end{aligned} \quad \square$$

Lemma 5.15 (Large set of dissimilar matchings). *Let G be the complete bipartite graph G . Then we can construct a set $S \subseteq \mathcal{P}(G)$ of size $e^{(n/4) \cdot \ln n}$, such that $d_H(M, M') \geq n/4$ for all distinct $M, M' \in S$.*

2106 *Proof.* We provide a greedy argument similar to [HT24]. We can generate a set
 2107 $S \subseteq \mathcal{P}(G)$ of size at least $n! / n^{d+1}$ with pairwise Hamming distance at least d and
 2108 $|S| \geq \frac{n!}{n^{d+1}}$ for some integer $d \leq n$.

2109 Therefore, first initialize a set $S := \emptyset$ and let U_i be the set of matchings that
 2110 might still be added to S . Set $U_0 := \mathcal{P}(G)$ and repeat the following process as long
 2111 as U_{i-1} is non-empty: In each step i , select any $M \in U_{i-1}$ and add it to S . Then, set

U_i to be U_{i-1} without those matchings that are closer than d in Hamming distance to M . When this process stops, we necessarily have $|S| \geq \frac{n!}{n^{d+1}}$, because in each step, we removed at most n^{d+1} elements by Lemma 5.14 and the number of perfect matchings on a complete bipartite graph with $2n$ vertices is exactly n factorial. Now, we can set $d = \frac{1}{4}n$ (or any other small enough constant) and we get

$$\begin{aligned} |S| &\geq \frac{n!}{n^{d+1}} \geq \frac{(n/e)^n}{n^{n/4+1}} = e^{(3n/4-1)\ln n - n} \\ &\geq e^{(n/4) \cdot \ln n} = e^{\Omega(n \cdot \ln n)}, \end{aligned}$$

where we used the trivial lower bound of $n! \geq (n/e)^n$ by a Stirling approximation and we assume sufficiently large n for convenience where the linear n lower-order term is dominated. $\square$

5.4 HIERARCHICAL CLUSTERING UNDER DASGUPTA'S COST FUNCTION

In this section, we design the first (subset) reconstruction-attack based lower bound for hierarchical clustering. Observe that our approach for MST, where the dataset is encoded into every edge of the topology, does not work here. This is because, by a simple concentration argument, every cut in the graph will have weight approximately half the total size of the cut, preventing any cheap hierarchical clustering from existing. Instead, we use a sparse dataset encoding which requires new technical ideas to work.

Before describing the attack, we introduce new graph notation. For a topology $F = (V, E)$, let $d_{\max}(F)$ denote the maximum degree vertex in F . Also, let $\phi(F)$ denote the minimum-size balanced cut in F ; i.e. $\phi(F) = \min_{A, B \text{ balanced cut}} |E(A, B)|$ where $|E(A, B)|$ denotes the number of edges crossing the cut. For any n and $d \geq 3$, there exist *spectral expander graphs* where every node has degree d and $\phi(F) \geq \Omega(nd)$ [Vad12]. These are the graphs where our bounds will be tightest.

To ensure the existence of a low-cost hierarchical clustering, we will encode a random dataset into a random *sparse* subset, where each edge is sampled with probability $\leq 1/d_{\max}(F)$. After privately computing the hierarchical clustering, we use the *induced balanced-cut* (Lemma 5.16) to try to reconstruct edges carrying encoded data that cross this cut; these edges will form the set I . The error guarantee is sufficient to establish that $\mathbb{E}[\sum_{i \in I} x_i]$ is low. However, because the encoding is now sparse, it becomes harder to argue that $\mathbb{E}[|I|]$ is sufficiently large. To do this, we make the observation that the clustering algorithm cannot distinguish encoded vs. non-encoded edges among the edges of weight 0, and thus the number of encoded edges in the cut is not much lower than its expectation by another concentration argument. This enables us to prove a lower bound of $\Omega(\frac{\phi(F)}{\varepsilon d_{\max}(F)} n)$ that grows with the *minimum* balanced cut size. Somewhat counterintuitively, the lower bound shrinks, rather than grows, with the maximum degree, which is a byproduct of our sparse encoding. Nonetheless, this bound applies to many topologies, and is tight up to a logarithmic factor for sparse expanders (see discussion in Section 5.4.2).

5.4.1 Reconstruction Attack Outline

Our reconstruction attack encodes the dataset into a random subsample of edges $E' \subseteq E$, where each edge $e \in E$ is selected i.i.d. with probability $p = \frac{0.5}{d_{\max}(F)}$. The dataset is encoded by taking an arbitrary bijection $\ell : E' \rightarrow [k]$ (a label function) where $k = |E'|$, and setting $w(e) := R \cdot x_{\ell(e)}$ for all $e \in E'$, and $w(e) := 0$ for all

Algorithm 8 Reconstruction Attack Rec^{HC} for Hierarchical Clustering

Require: Public graph topology $F = (V, E)$, dataset $\mathbf{x} \in \{0, 1\}^m$, parameter R ,
 HC algorithm $\mathcal{A}_{\text{HC}}$

- 1: Initialize weight function $w : E \rightarrow \{0, R\}$.
- 2: Subsample $E' \subseteq E$ by selecting each edge from E with $p = 0.5 \frac{1}{d_{\max}}$. Let $k = |E'|$.
- 3: Let $\ell : E' \rightarrow [k]$ be any bijection from E' to $[k]$ labels.
- 4: Set $w(e) := R \cdot x_{\ell(e)}$ for all $e \in E'$ and $w(e) := 0$ for all $e \in E \setminus E'$. $\triangleright$ Encode $\mathbf{x}$ into the weights
- 5: Compute hierarchical clustering $T := \mathcal{A}_{\text{HC}}((F, w))$.
- 6: Compute induced balanced cut $(A, B) := \text{HCToBC}(T)$. $\triangleright$ See Lemma 5.16
- 7: Initialize reconstruction set $I = \{\ell(e) : e \in E'(A, B)\} \subseteq [n]$.
- 8: **for each** $i \in I$, guess $y_i = 0$.
- 9: **return** Reconstruction set I , reconstructed vector $\mathbf{y} = \{y_i : i \in I\}$.

2154 $e \in E \setminus E'$. This sparse encoding ensures that there is a low-cost hierarchical
 2155 clustering. Our reconstruction attack will not use the entire returned hierarchical
 2156 clustering T to reconstruct data. Instead, it will use only the edges in the induced
 2157 balanced cut of T , which is defined in the following lemma from [Das16]. Their
 2158 proof is constructive and forms the balanced cut with a procedure similar to a
 2159 heavy-light decomposition. We refer to the balanced cut guaranteed by Lemma 5.16
 2160 as the *induced* balanced cut of T .

Lemma 5.16 ([Das16, Lemma 11]). *For any hierarchical clustering T , there exists a balanced cut A, B with $|A|, |B| \in (\frac{n}{3}, \frac{2n}{3})$ such that $\frac{w(A, B)}{|A| \cdot |B|} \leq \frac{27}{4n^3} \cdot \text{cost}_w(T)$. Moreover, there exists a procedure $\text{HCToBC}(T)$ which outputs the cut in linear time.*

2161 The attack proceeds by running a hierarchical clustering algorithm $\mathcal{A}_{\text{HC}}$ on the
 2162 graph (V, E, w) , obtaining a low-cost hierarchical clustering T , and computing the
 2163 balanced cut A, B of T (which, by Lemma 5.16, is also guaranteed to have low
 2164 cost). Then, it uses the reconstruction set $I = \{\ell(e) : e \in E'(A, B)\}$ (i.e. those data
 2165 elements which are encoded to edges that cross A, B), and guesses $y_i = 0$ for every
 2166 $i \in I$. This reconstruction procedure, Rec^{HC} , appears in Algorithm 8.

2167 An important observation for our analysis is the following: The edges $e \in E'$
 2168 such that $x_{\ell(e)} = 0$ appear the same as the edges $e \in E \setminus E'$ in the weight function
 2169 (both appear as 0), and thus the returned tree T gives us no further information to
 2170 which of these two sets an edge of weight 0 belongs. Formally, define $E'_0 = \{e \in E' : x_{\ell(e)} = 0\}$ and $E'_1 = \{e \in E' : x_{\ell(e)} = 1\}$. The observation can be stated as:

2172 **Observation 5.17.** *In Algorithm 8, the tree T , and subsequent post-processing of it, are*
 2173 *conditionally independent of E'_0 given E'_1 .*

2174 We will use this observation to show that there are many edges in $E'_0(A, B)$, just
 2175 due to the randomness of E' , and thus I will be sufficiently large.

2176 5.4.2 Reconstruction Attack Lower Bound

2177 We will now prove the following reconstruction attack.

Theorem 5.18 (Lower Bounds HC). *Let $F = (V, E)$ be a public graph topology such that $d_{\max}(F) \geq 8$, and $\phi(F) \geq 10000d_{\max}(F) \log n$. Suppose HC is a hierarchical clustering algorithm with additive error $\leq R \cdot \frac{n\phi(F)}{400d_{\max}(F)}$ for some $R > 0$. Then, $\text{Succ}(\text{Rec}^{\text{HC}}) \geq 0.4$. Furthermore, the expected size of the reconstructed set is at least $\frac{\phi(F)}{8d_{\max}(F)}$.*

This yields the following immediate corollary whose proof is identical to Corollary 5.8:

Corollary 5.19. *Let $F = (V, E)$ be a public graph topology such that $d_{\max}(F) \geq 8$ and $\phi(F) \geq 8n$, and $\frac{\phi(F)}{d_{\max}(F)} \geq 10000 \log(n)$. Then, for any $\varepsilon \leq 1, \delta \leq \frac{0.01\varepsilon}{m}$, there is no (ε, δ) -DP algorithm for hierarchical clustering with error less than $\frac{n\phi(F)}{400\varepsilon d_{\max}(F)}$.*

Corollary 5.19 is tightest when F is a d -regular spectral expander graph where $\phi(F) \geq \Omega(nd)$. In this case, the lower bound of $\Omega(\frac{n^2}{\varepsilon})$ is the same as that for the complete graph and tight with the upper bound in [Imo+23] up to $\log(n)$ factors. Observe that our lower bound argument can be applied to any subgraph of F by just ignoring edges and vertices, and thus it is possible to remove a small number of high-degree nodes and derive a stronger lower bound on a subgraph.

We now prove Theorem 5.18 and therefore need some supporting lemmas. First, we will show the optimal cost of clustering the encoded graph is just $\mathcal{O}(n \cdot \log(n))$ with high probability. This comes from the fact that the components in E' w.h.p. will have size $\mathcal{O}(\log(n))$.

First, we will show that the connected components in the sampled edges E' have size $\mathcal{O}(\log n)$.

Lemma 5.20 (Largest components). *Suppose $F = (V, E)$ is a graph and let H be the subgraph obtained by keeping each edge independently with probability $p \leq \frac{1-\alpha}{d_{\max}(F)}$ for some $\alpha \in (0, 1)$. Then the maximal size of a connected component in H is $\frac{2-\alpha}{\alpha^2} 4 \log n$ with probability at least $1 - n^{-3}$.*

The proof uses standard techniques and is deferred to Section 5.7.3.

Lemma 5.21 (Existence of a good Dasgupta cluster). *Suppose $F = (V, E)$ is a graph and let $\alpha \in (0, 1)$ be a constant. Now, assign weights $w(e) := 1$ with probability $\frac{1-\alpha}{d_{\max}(F)}$ and $w(e) := 0$ otherwise for each edge $e \in E$ independently. Then with probability at least $1 - 2n^{-3}$, there exists a clustering tree T , such that $\text{cost}_w(T) \leq \frac{2-\alpha}{\alpha^2} \cdot 4n \log(n)$.*

Proof. Let $H = (V, E_H)$ be the random subgraph consisting of the sampled edges, i.e. $E_H = \{e \in E \mid w(e) = 1\}$ and note that only edges in E_H contribute to the Dasgupta cost in G . This is equivalent to analyzing the connected components $C_1, \dots, C_k \subseteq V$ of H separately because the overall Dasgupta cost decomposes to a sum of the cost within each C_i .

Now, observe that any C_i can only contribute an additive term of at most $|C_i| \cdot m_i$ where m_i is the number of edges in component C_i . The worst case can be achieved if each of the m_i edges has to cross the balanced cut, thus adding cost of at most $|C_i|$ each.

Therefore, we have for the optimal Dasgupta clustering tree $\mathcal{T}$,

$$\text{cost}_w(\mathcal{T}) \leq \sum_{i \in [k]} |C_i| \cdot m_i \leq \max_{i \in [k]} |C_i| \cdot \sum_{i \in [k]} m_i = \max_{i \in [k]} |C_i| \cdot |E_H|.$$

By Lemma 5.20, with probability at least $1 - n^{-3}$, each $|C_i| \leq \frac{2-\alpha}{\alpha^2} \cdot 4 \log n$. Furthermore, $|E_H| \sim \text{Bin}(\frac{nd}{2}, \frac{1-\alpha}{d})$ and by a Chernoff bound with probability at least $1 - \exp(-\Omega(n))$, we have $|E_H| \leq n$. A union bound over both high probability events yields the desired result. $\square$

The number of mistakes made in the set I is given by $\sum_{i \in I} x_i = |E'_1(A, B)|$. We use the previous bound on the optimal clustering cost to convert the additive error of HC into an absolute bound on $|E'_1(A, B)|$.

Lemma 5.22. *Suppose Algorithm 8 is instantiated with an algorithm $\mathcal{A}_{\text{HC}}$ with expected additive error at most $C_F R n$ for a constant C_F (that may depend on n, F). Then, the balanced cut (A, B) returned by $\text{HCToBC}(\mathcal{T})$ has expected size $\mathbb{E}[|E'_1(A, B)|] \leq 2C_F + 50 \log(n)$.*

Proof. By the additive error guarantee of HC, we have for fixed weights w ,

$$\mathbb{E}[\text{cost}_w(T)|w] \leq \text{cost}_w(T^*) + C_F R n.$$

Recall that $R > 0$ is the weight, that we put on the subsampled edges that encode an one of the dataset. Note that in Algorithm 8, we use $\alpha = 0.5$ and then $\frac{2-\alpha}{\alpha^2} = 6$. By Lemma 5.21, with probability at least $1 - 2n^{-3}$, we have that w has an optimal clustering cost of $(6 \cdot 4) \cdot R n \log n$. In the worst case, the cost of clustering a complete graph is $n^3 R$. Thus, we can use conditional expectation to show $\mathbb{E}[\text{cost}_w(T)] \leq 24n \log(n) + 2n^{-3}n^3 R + C_F R n \leq 25R n \log(n) + C_F R n$. Furthermore, by Lemma 5.16, the balanced cut A, B satisfies

$$w(A, B) \leq \frac{27}{4n^3} |A| |B| \text{cost}_w(T) \leq \frac{27}{16n} \text{cost}_w(T),$$

because $|A| |B|$ maximizes in the interval $(\frac{n}{3}, \frac{2n}{3})$ exactly if both have size exactly $\frac{n}{2}$. The last step is to observe that $w(A, B) = R \cdot |E'_1(A, B)|$ due to the way the data is embedded. Hence, we get the following in expectation,

$$\begin{aligned} R \cdot \mathbb{E}[|E'_1(A, B)|] &= \mathbb{E}[w(A, B)] \leq \frac{27}{16n} \mathbb{E}[\text{cost}_w(T)] \\ &\leq \frac{27}{16n} (C_F R n + 25R n \log n) \\ &\leq R(2C_F + 50 \log n). \end{aligned}$$

Dividing by R yields the result. $\square$

Next, we need to show that $|I|$ is sufficiently large, as this will show that I is a successful reconstruction set. As $|I| = |E'_0(A, B)| + |E'_1(A, B)|$, and we have already upper bounded $|E'_1(A, B)|$, we will lower bound $|E'_0(A, B)|$. Precisely, we lower bound $\mathbb{E}[|E'_0(A, B)| | E'_1]$ since conditioning on E'_1 allows us to consider A, B as a constant per Observation 5.17. Under this conditioning, the edges in E'_0 are distributed multinomially, allowing us to do a straightforward Chernoff bound.

Lemma 5.23. *Suppose $\mathbf{x}$ and E' are sampled as they are in Algorithm 8. Then, for any balanced cut A_0, B_0 , we have that $\mathbb{E}[|E'_0(A_0, B_0)| | E'_1] \geq \frac{p}{2} |\tilde{E}(A_0, B_0)|$ where $\tilde{E} = E \setminus E'_1$*

2241 *Proof.* (Of Lemma 5.23) By construction, each edge $e \in E$ is independently assigned
 2242 to $E \setminus E'$, E'_0 and E'_1 with probabilities $(1 - p, \frac{p}{2}, \frac{p}{2})$. Thus, conditioned on E'_1 , each
 2243 remaining edge in $e \in E \setminus E'_1$ is independently assigned to $E \setminus E'$ or E'_0 with
 2244 probability $(\frac{1-p}{1-p/2}, \frac{p/2}{1-p/2})$. Thus,

$$\begin{aligned} \mathbb{E}[E'_0(A_0, B_0) \mid E'_1] &= \frac{p/2}{1-p/2} |(E \setminus E'_1)(A_0, B_0)| \\ &\geq \frac{p}{2} |(E \setminus E'_1)(A_0, B_0)|. \end{aligned} \quad \square$$

2245 The prior bound of $|\tilde{E}(A_0, B_0)|$ is not completely independent of E'_1 ; the following
 2246 result shows it is almost always bigger than $\frac{2}{3}|E(A_0, B_0)|$, giving us a simple bound
 2247 to work with.

Lemma 5.24. Suppose $d_{\max} \geq 8$ and $\phi(F) \geq 8n$. With probability $1 - 2^{-0.2n}$, for all balanced cuts A_0, B_0 , we have $|E'_1(A_0, B_0)| \leq \frac{1}{3}|E(A_0, B_0)|$.

2248 *Proof.* (Of Lemma 5.24) $E'_1(A_0, B_0)$ is a binomial distribution on $E(A_0, B_0)$ with
 2249 probability $\frac{p}{2}$. Thus, it suffices to compute a tail bound $\mathbf{P}[X \geq \frac{m}{3}]$, where $m =$
 2250 $E(A_0, B_0)$ and $X \sim \text{Bin}(m, \frac{p}{2})$. By Chernoff's bound, we know $\mathbf{P}[X \geq \frac{mp}{2} + t] \leq$
 2251 $\exp(-\frac{t^2}{mp+2t/3})$ for all $t \geq 0$. By assumption we know $p = 0.5 \frac{1}{d_{\max}} \leq \frac{1}{8}$. Taking
 2252 $t = \frac{m}{4}$ the above bound is $\mathbf{P}[X \geq \frac{m}{16} + \frac{m}{4}] \leq \exp(-\frac{m^2/16}{m/4+m/6}) < \exp(-\frac{m}{8})$. Because
 2253 $m \geq \phi(F) \geq 8n$, we have $\exp(-\frac{m}{8}) \leq \exp(-n)$, the failure probability over all 2^n
 2254 balanced cuts is at most $2^n e^{-n} \leq 2^{-0.2n}$. $\square$

2255 We are now ready to put it all together and prove our main reconstruction lower
 2256 bound for HC.

2257 **Theorem 5.18** (Lower Bounds HC). Let $F = (V, E)$ be a public graph topology such that
 2258 $d_{\max}(F) \geq 8$, and $\phi(F) \geq 10000d_{\max}(F) \log n$. Suppose HC is a hierarchical clustering
 2259 algorithm with additive error $\leq R \cdot \frac{n\phi(F)}{400d_{\max}(F)}$ for some $R > 0$. Then, $\text{Succ}(\text{Rec}^{\text{HC}}) \geq 0.4$.

2260 Furthermore, the expected size of the reconstructed set is at least $\frac{\phi(F)}{8d_{\max}(F)}$.

2261 *Proof.* Observe that, based on the construction of the set I and y , we have,

$$\sum_{i \in I} \mathbb{1}[x_i \neq y_i] = E'_1(A, B), \text{ and } |I| = E'_1(A, B) + E'_0(A, B).$$

2262 We will prove that $9 \mathbb{E}[E'_1(A, B)] \leq \mathbb{E}[E'_0(A, B)]$, as this implies

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] \leq 0.1 \mathbb{E}[|I|],$$

2263 meaning the attack is 0.4 successful. Because HC has additive error $\frac{\phi(F)R}{400d_{\max}(F)}n$,
 2264 by Lemma 5.22, we have $\mathbb{E}[|E'_1(A, B)|] \leq \frac{\phi(F)}{200d_{\max}(F)} + 50 \log(n)$. By assumption, we
 2265 know that

$$50 \log(n) \leq \frac{\phi(F)}{200d_{\max}(F)}.$$

Therefore, $\mathbb{E}[|E'_1(A, B)|] \leq \frac{\phi(F)}{100d_{\max}(F)}$. By the law of total expectation, we have
 $\mathbb{E}[|E'_0(A, B)|] = \mathbb{E}[\mathbb{E}[|E'_0(A, B)| | E'_1]]$. We can bound the inner quantity,

$$\begin{aligned} \mathbb{E}[|E'_0(A, B)| | E'_1] &= \mathbb{E}_{A,B} [\mathbb{E}[|E'_0(A, B)| | A, B, E'_1]] \\ &\geq \min_{A_0, B_0 \text{ balanced cut}} \mathbb{E}[|E'_0(A_0, B_0)| | A = A_0, B = B_0, E'_1] \\ &= \min_{A_0, B_0 \text{ balanced cut}} \mathbb{E}[|E'_0(A_0, B_0)| | E'_1] \quad (\text{by Observation 5.17}) \\ &\geq \frac{p}{2} \min_{A_0, B_0 \text{ balanced cut}} |(E \setminus E'_1)(A_0, B_0)|. \quad (\text{by Lemma 5.23}). \end{aligned}$$

Now, we will expand the whole expectation. In the following, let $\mathcal{E}$ be the event in Lemma 5.24; i.e. that for all balanced cuts A_0, B_0 , we have

$$|E'_1(A_0, B_0)| \leq \frac{1}{3} |E(A_0, B_0)|.$$

Hence,

$$\begin{aligned} \mathbb{E}[|E'_0(A, B)|] &\geq \mathbb{E} \left[\frac{p}{2} \min_{A_0, B_0 \text{ balanced cut}} |(E \setminus E'_1)(A_0, B_0)| \right] \\ &\geq \mathbf{P}[\mathcal{E}] \mathbb{E} \left[\frac{p}{2} \min_{A_0, B_0 \text{ balanced cut}} |(E \setminus E'_1)(A_0, B_0)| \middle| \mathcal{E} \right] \\ &\geq (1 - 2^{-0.2n}) \frac{p}{2} \mathbb{E} \left[\min_{A_0, B_0 \text{ balanced cut}} \frac{2}{3} |E(A_0, B_0)| \middle| \mathcal{E} \right] \quad (\text{by Lemma 5.24}) \\ &> \frac{p}{4} \phi(G) = \frac{\phi(F)}{8d_{\max}}. \end{aligned}$$

Recalling that $\mathbb{E}[|E'_1(A, B)|] \leq \frac{\phi(F)}{100d_{\max}(F)}$ and $\mathbb{E}[|E'_0(A, B)|] > \frac{\phi(F)}{8d_{\max}(F)}$. Putting this together, we can obtain

$$9 \mathbb{E}[|E'_1(A, B)|] \leq \frac{9\phi(F)}{100d_{\max}(F)} < \frac{\phi(F)}{8d_{\max}(F)} < \mathbb{E}[|E'_0(A, B)|].$$

□

5.5 PRIVATE ALGORITHMS WITH TIGHT ERROR

We now detail algorithms with superior error on **MST** and **Minimum-Weight (Perfect) Matching** that appear in ongoing work [Aam26] demonstrating that the lower bounds of Sections 5.2.1 and 5.3 are tight. The algorithm is a simple input privatization, adding Laplace noise to each of the edge weights independently. The novelty lies in the analysis: instead of bounding the maximum noise added to individual edge weights as in prior work [Sea16; HT24; Pag+25], it uses a concentration bound on the total noise of any single spanning tree and then applies a union bound. We can phrase it even more general: any graph problem that outputs k edges can achieve additive error $\mathcal{O}(\frac{k}{\epsilon} \cdot \ln(\frac{m}{k}))$, because one can bound $\binom{m}{k} \leq (\frac{em}{k})^k$. In the cases of MSTs and MWPMs where $k = \Theta(n)$, this gives an upper bound with a logarithmic dependence of $\ln(\frac{m}{n})$. The proof appears in Section 5.7.4.

Theorem 5.25 ([Aam26]). *Let $G = (V, E, \mathbf{w})$ be a graph and let $\mathcal{S}(G)$ be a family of solutions such that every $S \in \mathcal{S}(G)$ satisfies $S \subseteq E$ and $|S| = k$. Consider the problem of minimizing $w(S)$ among all $S \in \mathcal{S}(G)$. Then there exists an ϵ -DP algorithm $\mathcal{A}$ that outputs a feasible solution $S \in \mathcal{S}(G)$, such that with probability at least $1 - n^{-\Omega(1)}$,*

$$w(S) - w(S^*) \leq \mathcal{O}\left(\frac{k}{\epsilon} \cdot \ln\left(\frac{m}{k}\right)\right),$$

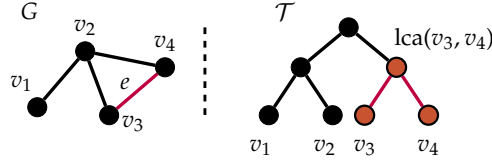


Figure 5.3: An example of a hierarchical clustering T with $\text{cost}_w(T) = 12$ for a graph $G = (V, E)$ assuming unit weights. The edge e has cost 2 as the lowest common ancestor $\text{lca}(v_3, v_4)$ has exactly two leaves in its subtree of T . Furthermore, cutting the tree at $\text{lca}(v_3, v_4)$ would give a balanced cut.

where $S^* = \arg \min_{\tilde{S} \in \mathcal{S}(G)} w(\tilde{S})$ is any optimal solution.

5.6 CONCLUSION AND FUTURE WORK

We first showed that for **MST** and Minimum-Weight (Perfect) Matching there exist worst-case topologies where the error is tight with the upper bounds. We also proved lower bounds for **MST** and HC on large classes of topologies which are close to the upper bounds, off by just a $\log(n)$ factor.

For MWPM, it remains open whether lower bounds can be derived for a large class of topologies, and what specific property that class should satisfy. For HC, closing the gap between the best lower bound of $\Omega(\frac{1}{\epsilon} \cdot n^2)$ and upper bound of $\mathcal{O}(\frac{1}{\epsilon} \cdot n^2 \cdot \log(n))$ is an interesting question, as well as deriving a polynomial-time algorithm that achieves this additive error (rather than using the exponential mechanism). Finally, we believe that both our lower bound techniques for obtaining tight bounds and bounds on general topologies are likely to carry over to other graph problems as well.

5.7 APPENDIX: RECONSTRUCTION-LOWER BOUNDS

In Figure 5.3, we illustrate the Dasgupta cost function, $\text{cost}_w(T)$, for a given hierarchical clustering tree T . We are now ready to prove the proofs we removed from the main part.

Lemma 5.1. *Let $\mathcal{B} : \mathcal{X}^d \rightarrow \mathcal{X}^d$ be any mechanism that is (ϵ, δ) -differentially private under the Hamming neighborhood relationship. Then for a random dataset $\mathbf{x} \leftarrow \mathcal{X}^d$ drawn uniformly at random, we have for each $i \in [d]$, we have $\mathbf{P}[\mathcal{B}(\mathbf{x})_i = x_i] \leq \frac{\exp(\epsilon)}{|\mathcal{X}|} + \delta$.*

Proof. Let $n \in \mathbb{N}$. We index the elements in $\mathcal{X}$ by $\{1, \dots, n\}$. For some arbitrarily chosen $i \in [d]$, assume a random vector $\mathbf{x} = \mathbf{x}_{<i} \oplus X_i \oplus \mathbf{x}_{>i} \sim \text{Uni}([n]^d)$ where $\oplus$ denotes vector concatenation, and we use the subscript $\mathbf{x}_{<i}$ and $\mathbf{x}_{>i}$ to split the vector at index i . Now we can bound the probability that the i 'th coordinate of the output of $\mathcal{B}(\mathbf{x})$ leaks its corresponding input x_i :

$$\begin{aligned}
\mathbf{P}[\mathcal{B}(\mathbf{x})_i = x_i] &= \frac{1}{|\mathcal{X}|^{d-1}} \sum_{\mathbf{x}_{<i} \in [n]^{i-1}} \sum_{\mathbf{x}_{>i} \in [n]^{d-i}} \left(\mathbf{P}_{X_i \sim \text{Uni}([n])} [\mathcal{B}(\mathbf{x}_{<i} \oplus x_i \oplus \mathbf{x}_{>i})_i = x_i] \right) \\
&\leq |\mathcal{X}|^{1-d} \sum_{\mathbf{x}_{<i} \in [n]^{i-1}} \sum_{\mathbf{x}_{>i} \in [n]^{d-i}} \left(e^\epsilon \mathbf{P}_{x_i \sim \text{Uni}([n])} [\mathcal{B}(\mathbf{x}_{<i} \oplus (1) \oplus \mathbf{x}_{>i})_i = x_i] + \delta \right) \quad (4) \\
&\leq |\mathcal{X}|^{1-d} \sum_{\mathbf{x}_{<i} \in [n]^{i-1}} \sum_{\mathbf{x}_{>i} \in [n]^{d-i}} \left(e^\epsilon \cdot \frac{1}{n} + \delta \right) \quad (5) \\
&\leq \frac{e^\epsilon}{|\mathcal{X}|} + \delta.
\end{aligned}$$

2316 In line 4, we use the fact that each X_i is i.i.d. and in step 5, we use the privacy
 2317 guarantees of the mechanism B and flip to a neighboring dataset where we
 2318 explicitly set $X_i = 1$ under the hamming adjacency relation. $\square$

2319 5.7.1 Proof of Lemma 5.2

2320 First, we start with a supporting lemma.

Lemma 5.26. Suppose P, Q are distributions on a discrete set $\mathcal{X}$ satisfying

$$\mathbf{P}_{X \sim P}[X \in S] \leq e^\epsilon \mathbf{P}_{Y \sim Q}[Y \in S] + \delta \text{ for all } S \subseteq \mathcal{X}.$$

Then, there exists a function $\delta(x) : \mathcal{X} \rightarrow [0, \infty)$ satisfying

1. $\mathbf{P}_{X \sim P}[X = x] \leq e^\epsilon \mathbf{P}_{X \sim Q}[X = x] + \delta(x)$, and
2. $\sum_{x \in \mathcal{X}} \delta(x) \leq \delta$.

2321 *Proof.* We denote $P(x) = \mathbf{P}_{X \sim P}[X = x]$ (and similar for Q).

2322 We take the function $\delta(x) = \max\{P(x) - e^\epsilon Q(x), 0\}$. Condition 1 is satisfied
 2323 because $P(x) = e^\epsilon Q(x) + (P(x) - e^\epsilon Q(x)) \leq e^\epsilon Q(x) + \delta(x)$.

2324 For condition 2, let $N \subseteq \mathcal{X}$ denote the set where $P(x) - e^\epsilon Q(x) \geq 0$. On $x \in N$,
 2325 we have $\delta(x) = P(x) - e^\epsilon Q(x)$, and on $x \in \mathcal{X} \setminus N$, we have $\delta(x) = 0$. Thus,

$$\sum_{x \in \mathcal{X}} \delta(x) = \sum_{x \in N} \delta(x) = \sum_{x \in N} P(x) - e^\epsilon Q(x) = \mathbf{P}_{X \sim P}[X \in N] - e^\epsilon \mathbf{P}_{Y \sim Q}[Y \in N] \leq \delta.$$

2326 $\square$

2327 **Theorem 5.2.** Let $\mathcal{B} : \{0, 1\}^d \rightarrow 2^{[d]} \times \{0, 1\}^d$ be any (ϵ, δ) -DP algorithm and let
 2328 $\mathbf{x} \sim \{0, 1\}^d$ be drawn uniformly at random. Then, $\text{Succ}(\mathcal{B}) \leq \frac{e^\epsilon - 1}{2(e^\epsilon + 1)} + \frac{d\delta}{e^\epsilon + 1}$.

2329 *Proof.* It suffices to show that

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] \geq \frac{1}{1 + e^\epsilon} \mathbb{E}[|I|] - \frac{1}{1 + e^\epsilon} d\delta.$$

2330 We use conditional expectation to write

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] = \mathbb{E} \left[\mathbb{E} \left[\sum_{i \in I} |y_i - x_i| \middle| I \right] \right] = \mathbb{E} \left[\sum_{i \in I} \mathbb{E}[|y_i - x_i| \mid I] \right].$$

2331 We will form a bound $\mathbb{E}[|y_i - x_i| | I]$ for each i . To do this, we expand it as follows
 2332 by conditional probability,

$$\begin{aligned} \mathbb{E}[|y_i - x_i| | I] &= \mathbf{P}[x_i = 1, y_i = 0 | I] + \mathbf{P}[x_i = 0, y_i = 1 | I] \\ &= \frac{\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]}{\mathbf{P}[I]}. \end{aligned}$$

2333 In the following, a sum over the variable x implicitly ranges over $\{0, 1\}^n$, and a
 2334 sum over y implicitly ranges over $\{0, 1\}^I$. We can write

$$\mathbf{P}[x_i = a, y_i = b, I] = \sum_{\substack{x: x_i = a \\ y: y_i = b}} 2^{-n} \mathbf{P}[\mathcal{A}(x) = (I, y)] = 2^{-n} \sum_{x: x_i = a} \mathbf{P}[\mathcal{A}(x) \in (I, \mathcal{Y}_{i,b})],$$

2335 where $\mathcal{Y}_{i,b} = \{y \in \{0, 1\}^I : y_i = b\}$. For a vector $x \in \{0, 1\}^n$, let $x^{(i \rightarrow 0)}$ and $x^{(i \rightarrow 1)}$
 2336 denote x with its i th bit set to 0 (resp. 1). Because each $\mathcal{A}(x^{(i \rightarrow 0)})$ and $\mathcal{A}(x^{(i \rightarrow 1)})$
 2337 satisfy (ε, δ) -DP, by Lemma 5.26, there exists a non-negative function $\delta_{x \uparrow i}(I, b)$ such
 2338 that

$$\mathbf{P}[\mathcal{A}(x^{(i \rightarrow 0)}) \in (I, \mathcal{Y}_{i,b})] \leq e^\varepsilon \mathbf{P}[\mathcal{A}(x^{(i \rightarrow 1)}) \in (I, \mathcal{Y}_{i,b})] + \delta_{x \uparrow i}(I, b),$$

2339 and $\sum_{I \subseteq [n]} \sum_{b \in \{0,1\}} \delta_{x \uparrow i}(I, b) \leq \delta$. Similarly, there exists a non-negative function
 2340 $\delta_{x \downarrow i}(I, b)$ such that

$$\mathbf{P}[\mathcal{A}(x^{(i \rightarrow 1)}) \in (I, \mathcal{Y}_{i,b})] \leq e^\varepsilon \mathbf{P}[\mathcal{A}(x^{(i \rightarrow 0)}) \in (I, \mathcal{Y}_{i,b})] + \delta_{x \downarrow i}(I, b)$$

2341 and $\sum_{I \subseteq [n]} \sum_{b \in \{0,1\}} \delta_{x \downarrow i}(I, b) \leq \delta$. By summing the first inequality over $x \in \{0, 1\}^n$
 2342 with $x_i = 0$ (and the second over x with $x_i = 1$), we obtain

$$\begin{aligned} \mathbf{P}[x_i = 0, y_i = 0, I] &= 2^{-n} \sum_{x: x_i = 0} \mathbf{P}[\mathcal{A}(x) \in (I, \mathcal{Y}_{i,0})] \\ &\leq 2^{-n} \sum_{x: x_i = 1} (e^\varepsilon \mathbf{P}[\mathcal{A}(x) \in (I, \mathcal{Y}_{i,0})] + \delta_{x \uparrow i}(I, 0)) \\ &= e^\varepsilon \mathbf{P}[x_i = 1, y_i = 0, I] + \delta_{i,1}^\uparrow(I, 0), \end{aligned}$$

2343 where $\delta_{i,a}^\uparrow(I, b)$ is defined to be $2^{-n} \sum_{x: x_i = a} \delta_{x \uparrow i}(I, b)$. Rearranging, we have

$$\mathbf{P}[x_i = 1, y_i = 0, I] \geq e^{-\varepsilon} \mathbf{P}[x_i = 0, y_i = 0, I] - e^{-\varepsilon} \delta_{i,1}^\uparrow(I, 0).$$

2344 Similarly, we can write

$$\mathbf{P}[x_i = 0, y_i = 1, I] \geq e^{-\varepsilon} \mathbf{P}[x_i = 1, y_i = 1, I] - e^{-\varepsilon} \delta_{i,0}^\downarrow(I, 1),$$

2345 where $\delta_{i,a}^\downarrow(I, b) = 2^{-n} \sum_{x: x_i=a} \delta_{x \downarrow i}(I, b)$. This means that

$$\begin{aligned}
& \mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I] \\
&= \frac{1}{1+e^\varepsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\quad + \frac{e^\varepsilon}{1+e^\varepsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\geq \frac{1}{1+e^\varepsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\quad + \frac{e^\varepsilon}{1+e^\varepsilon} (e^{-\varepsilon} \mathbf{P}[x_i = 0, y_i = 0, I] - e^{-\varepsilon} \delta_{i,1}^\uparrow(I, 0) \\
&\quad + e^{-\varepsilon} \mathbf{P}[x_i = 1, y_i = 1, I] - e^{-\varepsilon} \delta_{i,0}^\downarrow(I, 1)) \\
&= \frac{1}{1+e^\varepsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\quad + \frac{1}{1+e^\varepsilon} (\mathbf{P}[x_i = 0, y_i = 0, I] - \delta_{i,1}^\uparrow(I, 0) + \mathbf{P}[x_i = 1, y_i = 1, I] - \delta_{i,0}^\downarrow(I, 1)) \\
&= \frac{1}{1+e^\varepsilon} \mathbf{P}[I] - \frac{1}{1+e^\varepsilon} (\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)),
\end{aligned}$$

2346 where the last step follows from the law of total probability. Thus,

$$\begin{aligned}
\mathbb{E}[|y_i - x_i| | I] &= \frac{\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]}{\mathbf{P}[I]} \\
&\geq \frac{1}{1+e^\varepsilon} - \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{(1+e^\varepsilon) \mathbf{P}[I]}.
\end{aligned}$$

2347 Finally, we bound the entire expectation as

$$\begin{aligned}
\mathbb{E} \left[\sum_{i \in I} \mathbb{E}[|y_i - x_i| | I] \right] &\geq \mathbb{E} \left[\sum_{i \in I} \left(\frac{1}{1+e^\varepsilon} - \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{(1+e^\varepsilon) \mathbf{P}[I]} \right) \right] \\
&= \sum_{I \subseteq [n]} \mathbf{P}[I] \sum_{i \in I} \left(\frac{1}{1+e^\varepsilon} - \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{(1+e^\varepsilon) \mathbf{P}[I]} \right) \\
&= \frac{1}{1+e^\varepsilon} \sum_{I \subseteq [n]} \mathbf{P}[I] \cdot |I| - \frac{1}{1+e^\varepsilon} \sum_{I \subseteq [n]} \mathbf{P}[I] \sum_{i \in I} \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{\mathbf{P}[I]} \\
&= \frac{1}{1+e^\varepsilon} \mathbb{E}[|I|] - \frac{1}{1+e^\varepsilon} \sum_{I \subseteq [n]} \sum_{i \in I} \delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1). \tag{6}
\end{aligned}$$

The second term simplifies to

$$\begin{aligned}
& \sum_{I \subseteq [n]} \sum_{i \in I} \delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1) \\
&= \sum_{I \subseteq [n]} \sum_{i \in I} 2^{-n} \left(\sum_{x: x_i=1} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \delta_{x \downarrow i}(I, 1) \right) \\
&= \sum_{i=1}^n \sum_{I \subseteq [n]} \mathbb{1}[i \in I] \cdot 2^{-n} \left(\sum_{x: x_i=1} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \delta_{x \downarrow i}(I, 1) \right) \\
&\leq \sum_{i=1}^n \sum_{I \subseteq [n]} 2^{-n} \left(\sum_{x: x_i=1} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \delta_{x \downarrow i}(I, 1) \right) \\
&= \sum_{i=1}^n 2^{-n} \left(\sum_{x: x_i=1} \sum_{I \subseteq [n]} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \sum_{I \subseteq [n]} \delta_{x \downarrow i}(I, 1) \right) \\
&\leq \sum_{i=1}^n 2^{-n} \left(\sum_{x: x_i=1} \delta + \sum_{x: x_i=0} \delta \right) = \sum_{i=1}^n \delta = n\delta.
\end{aligned}$$

Plugging back into (6), we obtain

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{E} [|y_i - x_i| | I] \right] \geq \frac{\mathbb{E} [|I|]}{1 + e^\epsilon} - \frac{1}{1 + e^\epsilon} n\delta,$$

giving the answer. $\square$

5.7.2 Proof of Lemma 5.6

Let $E_0 \subseteq E$ denote the embedded edges where $x_{\ell(e)} = 0$. By the embedding procedure, we know E_0 is a Bernoulli subsample of E with probability $\frac{1}{2}$. It suffices to show that all cuts in the subgraph induced by E_0 are non-empty, as this will mean it is possible to construct a spanning tree entirely in E_0 . In other words, we show that each cut in the graph has at least one zero-edge sampled with high probability, which, by a standard argument, implies that there must exist a zero-cost spanning tree. Now let $\mathcal{E}$ denote the *complement* event; i.e., that there exists a cut A, B such that $E_0(A, B) = 0$. We will upper bound $\mathbf{P}[\mathcal{E}]$ by a union bound over all cuts. Let λ denote the minimum cut size in (V, E) , and let N_i denote the number of cuts A, B such that $E(A, B) = i$. The probability that a cut with value i has no edges sampled is given by 2^{-i} . By the union bound, we have

$$\mathbf{P}[\mathcal{E}] \leq \sum_{i=\lambda}^{|E|} 2^{-i} N_i.$$

To bound this, we use [Kar93, Theorem 6.2] which shows that $N_i \leq n^{\lceil 2i/\lambda \rceil} \leq n \cdot n^{2i/\lambda}$. Plugging this in, we have by a geometric sum

$$\begin{aligned}
\sum_{i=\lambda}^{|E|} 2^{-i} N_i &\leq \sum_{i=\lambda}^{\infty} 2^{-i} n \cdot n^{2i/\lambda} \leq n \sum_{i=\lambda}^{\infty} \left(\frac{n^{2/\lambda}}{2} \right)^i \\
&= n \frac{\left(\frac{n^{2/\lambda}}{2} \right)^\lambda}{1 - \left(\frac{n^{2/\lambda}}{2} \right)} = \frac{n^3}{\left(1 - \frac{1}{2} (n^{2/\lambda}) \right) \cdot 2^\lambda}.
\end{aligned}$$

Using $\lambda \geq 5 \log(n)$, it holds that $n^3 2^{-\lambda} < \frac{1}{n^2}$, and $\frac{1}{2} \cdot n^{2/\lambda} \leq \frac{2^{2/5}}{2} < \frac{2}{3}$, so the denominator $(1 - \frac{1}{2} n^{2/\lambda}) \geq \frac{1}{3}$. Thus, the probability of failure is at most $\frac{3}{n^2}$. $\square$

2367 5.7.3 Proof of Lemma 5.20

2368 We are using some standard results in random graph theory to prove the following
2369 result. Additionally, we need the following definitions.

Definition 5.2 (Stochastic domination). *A random variable X is stochastically dominated by a random variable Y , if $\mathbf{P}[X \geq a] \leq \mathbf{P}[Y \geq a]$ for all a . A coupling of X and Y is a random vector $(\tilde{X}, \tilde{Y})$ such that the marginal distributions coincide with X and Y respectively.*

2370 The following theorem due to Strassen [Str65] shows that stochastic dominance
2371 is equal to the existence of a monotone coupling.

Lemma 5.27 ([Str65]). *A random variable X stochastically dominates a random variable Y , if and only if there exists a (monotone) coupling $(\tilde{X}, \tilde{Y})$ of X and Y such that $\mathbf{P}[\tilde{X} \geq \tilde{Y}] = 1$.*

2372 Now, we are ready to prove Lemma 5.20.

2373 **Lemma 5.20** (Largest components). *Suppose $F = (V, E)$ is a graph and let H be the
2374 subgraph obtained by keeping each edge independently with probability $p \leq \frac{1-\alpha}{d_{\max}(F)}$ for
2375 some $\alpha \in (0, 1)$. Then the maximal size of a connected component in H is $\frac{2-\alpha}{\alpha^2} 4 \log n$ with
2376 probability at least $1 - n^{-3}$.*

2377 *Proof.* We prove this result assuming that the graph is d -regular, so $d_{\max}(F) = d$.
2378 The proof for a general graph is similar. Let $\alpha \in (0, 1)$, set $p := \frac{1-\alpha}{d}$ and let C_v
2379 denote the connected component of a fixed vertex v in H and WLOG assume that
2380 our edge subsampling is exploring C_v by BFS starting in v . If $0 \leq m_t \leq d$ is the
2381 number of neighbors of the currently explored vertex at step $t \geq 1$ that have not
2382 already been explored, then observe that the size of the frontier conditioned on the
2383 past $\mathcal{F}$, can be described as

$$X_0 = 1, \quad X_t = X_{t-1} - 1 + Z_t \quad \text{where } Z_t \mid \mathcal{F}_t \sim \text{Bin}(m_t, p),$$

2384 because we keep each neighbor that is not already included with probability p
2385 independently.

2386 Intuitively, cycles can only reduce the number of newly discovered vertices,
2387 so the BFS exploration is dominated by the process that simply ignores them.
2388 Therefore, define the process

$$Y_0 = 1, \quad Y_t = Y_{t-1} - 1 + \tilde{Z}_t \quad \text{where } \tilde{Z}_t \sim \text{Bin}(d, p), \quad (7)$$

2389 We claim that for each $t \geq 1$, the random variable Y_t conditional on the past,
2390 stochastically dominates X_t . Therefore, we will explicitly give a coupling $(\tilde{X}, \tilde{Y})$
2391 of X and Y and apply Lemma 5.27. Set $\tilde{X} = X$ and draw $U_t \mid \mathcal{F}_t \sim \text{Bin}(d - m_t, p)$
2392 independently of Z_t . Now, let $\tilde{Y}_t = \tilde{Y}_{t-1} - 1 + Z_t + U_t$ and consider the coupling
2393 $(\tilde{X}, \tilde{Y})$. Clearly, the marginal distributions match because $Z_t + U_t \sim \text{Bin}(d, p)$ by the
2394 convolution of binomials. We show by induction that for each $t \geq 0$, $\tilde{X}_t \leq \tilde{Y}_t$. The
2395 base cases $X_0, Y_0 = 1$ are trivial. Now, observe that for $Z_t \sim \text{Bin}(m_t, p)$, we have

$$\begin{aligned} \tilde{X}_t &= \tilde{X}_{t-1} - 1 + Z_t \leq \tilde{Y}_{t-1} - 1 + Z_t \\ &\leq \tilde{Y}_{t-1} - 1 + Z_t + U_t = \tilde{Y}_t, \end{aligned}$$

where the last line follows from the non-negativity of the binomial distribution.

Hence, Y_t gives a valid upper bound on how large the component grows, i.e.

$$\mathbf{P}[|C_v| \geq t] = \mathbf{P}[X_t \geq 1] \leq \mathbf{P}[Y_t \geq 1].$$

Because Y_t is a sum of t binomials, by a standard convolution result, we get

$$Y_t = \sum_{k=1}^t \tilde{Z}_k - t,$$

which is distributed as $Y_t \sim \text{Bin}(td, p) - t$. Now, let $\mu = t(1 - \alpha)$. By a standard

Chernoff bound, $\mathbf{P}[\text{Bin}(td, p) \geq (1 + \gamma)\mu] \leq \exp(-\frac{\gamma^2 \mu}{2 + \gamma})$ for all $\gamma > 0$. Rearranging the term, yields

$$\mathbf{P}[\text{Bin}(td, p) \geq t] \leq \exp\left(-\frac{(t - \mu)^2}{t + \mu}\right) = \exp\left(-t \cdot \frac{\alpha^2}{2 - \alpha}\right). \quad (8)$$

To make this decrease with $\frac{1}{n^3}$, we choose $t = \frac{2 - \alpha}{\alpha^2} \cdot 4 \ln(n)$. Then we get by a union bound

$$\begin{aligned} \mathbf{P}[\exists v : |C_v| \geq t] &\leq n \exp(-4 \ln(n)) \\ &= n^{-3}. \end{aligned}$$

Hence, with probability at least $1 - n^{-3}$, no component has size larger than $\frac{2 - \alpha}{\alpha^2} \cdot 4 \log n$. $\square$

5.7.4 Proof of Theorem 5.25

Theorem 5.25 ([Aam26]). Let $G = (V, E, \mathbf{w})$ be a graph and let $\mathcal{S}(G)$ be a family of solutions such that every $S \in \mathcal{S}(G)$ satisfies $S \subseteq E$ and $|S| = k$. Consider the problem of minimizing $w(S)$ among all $S \in \mathcal{S}(G)$. Then there exists an ε -DP algorithm $\mathcal{A}$ that outputs a feasible solution $S \in \mathcal{S}(G)$, such that with probability at least $1 - n^{-\Omega(1)}$,

$$w(S) - w(S^*) \leq \mathcal{O}\left(\frac{k}{\varepsilon} \cdot \ln\left(\frac{m}{k}\right)\right),$$

where $S^* = \arg \min_{\tilde{S} \in \mathcal{S}(G)} w(\tilde{S})$ is any optimal solution.

The following proof is due to [Aam26]. Consider the algorithm that adds noise $Z_e \sim \text{Lap}(1/\varepsilon)$ to each edge independently. Denote these weights with $\mathbf{w}'$. This mechanism is ε -DP under the ℓ_1 neighboring relationship by a standard result [Dwo+o6b]. Now let $T \subseteq \mathcal{S}(G)$ be an optimal solution over the noisy weights $\mathbf{w}'$ and T^* be an optimal over the original weights $\mathbf{w}$. Let $Z_T = \sum_{t \in T} Z_t$ be the total noise of any T and let $M = \max_{T \in \mathcal{S}(G)} |Z_T|$. Then,

$$w(T') \leq w'(T') + M \leq w'(T^*) + M \leq w(T^*) + 2 \cdot M,$$

by the same inequalities as used in [Sea16]. Note that the sub-exponential norm of each of the i.i.d. Laplace random variables Z_e is $\|Z_e\|_{\psi_1} \leq \frac{2}{\varepsilon}$. By a standard Bernstein inequality [Ver18], we get for each T for any $t > 0$ and an absolute constant $c > 0$,

$$\mathbf{P}[|Z_T| \geq t] \leq 2 \exp\left(-c \min\left(\frac{\varepsilon^2 t^2}{n}, \varepsilon t\right)\right),$$

2422 Now setting $t = c \cdot (k/\varepsilon) \ln(m/k)$ for a sufficiently large constant c , the tail above
 2423 becomes dominated by the linear term $\exp(-\Omega(n \ln(m/k)))$. Clearly, $|\mathcal{S}(G)| \leq$
 2424 $\binom{n}{k} \leq \left(\frac{em}{k}\right)^k$. A union bound over $\mathcal{S}(G)$ yields,

$$\mathbf{P} \left[\max_{T \in \mathcal{S}(G)} |Z_T| \geq C \frac{k}{\varepsilon} \ln(m/k) \right] \leq n^{-\Omega(1)},$$

2425 Combined with the chain of inequalities above, this gives

$$w(T') - w(T^*) \leq \mathcal{O}((k/\varepsilon) \ln(m/k))$$

2426 with probability at least $1 - n^{-\Omega(1)}$.

We may optimally describe a message with a particular amount of distortion and later decide that the message needs to be specified more accurately. Then, when an addendum to the original message is sent we hope that this refinement is as efficient as if the more strict requirements had been known at the start.

Successive Refinement of Information — [EC91]

LOSSLESS PRIVATE MULTIPLE RELEASE

Published at International Conference on Machine Learning (ICML) 2025

Joint work with Joel Daniel Andersson, Rasmus Pagh and Boel Nelson

Koufogiannis et al. (2016) showed a *gradual release* result for Laplace noise-based differentially private mechanisms: given an ϵ -DP release, a new release with privacy parameter $\epsilon' > \epsilon$ can be computed such that the combined privacy loss of both releases is at most ϵ' and the distribution of the latter is the same as a single release with parameter ϵ' . They also showed gradual release techniques for Gaussian noise, later also explored by Whitehouse et al. (2022). In this chapter, we consider a more general *multiple release* setting in which analysts hold private releases with different privacy parameters corresponding to different access/trust levels. These releases are determined one by one, with privacy parameters in arbitrary order. A multiple release is *lossless* if having access to a subset S of the releases has the same privacy guarantee as the least private release in S , and each release has the same distribution as a single release with the same privacy parameter. Our main result is that lossless multiple release is possible for a large class of additive noise mechanisms. For the Gaussian mechanism we give a simple method for lossless multiple release with a short, self-contained analysis that does not require knowledge of the mathematics of Brownian motion. We also present lossless multiple release for the Laplace and Poisson mechanisms. Finally, we consider how to efficiently do gradual release of sparse histograms, and present a mechanism with running time independent of the number of dimensions.

In deployments of differential privacy, it may be hard to determine the appropriate privacy budget to grant an analyst since it depends on trust and accuracy assumptions that may change over time. A system may also have different security clearance levels—a data analyst might have a higher clearance level than a developer, but both might require access to *some* statistics. Similarly, a company that wants to release, for example, user statistics could make an accurate release for their own data analysts, a less accurate release for external consultants, and include an even less accurate release in a report for shareholders or other external actors. A different example setting is users that want to sell their data on data markets: users could let the accuracy of the release depend on how much they are paid, and use different budgets for different releases.

Another usage scenario, relevant in distributed or federated settings, is *local differential privacy* where the data owner could be sharing a differentially private function of their data with multiple servers, and the set of servers might change over time. Here the privacy budget might depend on the server: for example, a

2473 patient may trust their local hospital more than their national hospitals, but might
 2474 not trust the hospitals not to collude by sharing data among themselves.

2475 **MULTIPLE RELEASES** These scenarios motivate creating *multiple releases* with
 2476 different privacy budgets aimed at different analysts. However, these releases
 2477 should be *coordinated* such that a group of analysts who combine their information
 2478 do not gain more knowledge about the input than the most knowledgeable member
 2479 of the group. This kind of collusion resilience was first studied by Xiao, Tao,
 2480 and Chen [XTC09] with a non-DP privacy objective—we refer to their work for
 2481 additional motivation.

2482 A related aspect is that we may want to provide an analyst with a less private,
 2483 more accurate release after the trust we place in them increases. In this case, we
 2484 want the accuracy of the latest release to match the accuracy that can be obtained,
 2485 given the combined privacy budget of both releases. That is, no additional cost
 2486 should be incurred for making two releases rather than one. For example, an
 2487 external consultant that later gets employed directly by the company should be
 2488 able to get access to the more accurate release without constituting a privacy
 2489 violation or requiring an increased privacy budget to reach the same accuracy.

Figuratively speaking,
 imagine turning a
 “knob” that controls
 the privacy level.

2490 **BASELINE** It is possible to create multiple releases for *any* differentially private
 2491 mechanism if we are willing to increase the privacy budget by a constant factor. In
 2492 particular, we can create a sequence of independent releases with geometrically
 2493 increasing privacy parameters, referred to as “ ϵ -doubling” by Ligett, Neel, Roth,
 2494 Waggoner, and Wu [Lig+17], and provide each analyst with the most accurate
 2495 release they are entitled to. Using composition results, the combined privacy
 2496 budget for the information given to a set of analysts is a constant factor from the
 2497 highest budget of a single analyst in the group. In this chapter, we study how to
 2498 make multiple releases in a *lossless* way without accuracy or privacy penalty.

2499 6.0.1 Related Work

2500 Koufogiannis, Han, and Pappas [KHP16] introduced the concept of (lossless)
 2501 *gradual release*, which makes it possible to increase the privacy budget with *no loss*
 2502 in accuracy. They also considered privacy tightening for the Laplace mechanism,
 2503 where successive releases are increasingly private. In the journal version of the
 2504 paper, they also introduce gradual release for the Gaussian mechanism under
 2505 approximate DP which is based on the machinery of Brownian motion.

2506 This technique was later applied in a noise reduction framework by Ligett, Neel,
 2507 Roth, Waggoner, and Wu [Lig+17] with the goal of producing mechanisms with
 2508 ex-post privacy, i.e., where the privacy budget is set based on what is required to
 2509 achieve a desired accuracy level. Follow-up work by Whitehouse, Ramdas, Wu,
 2510 and Rogers [Whi+22] gave results for noise reduction under approximate DP
 2511 using Brownian motion. These works were motivated by work on privacy filters
 2512 and odometers [Rog+16], keeping track of privacy budgets over time, rather than
 2513 multiple release settings. More recently, Pan [Pan24] demonstrated lossless gradual
 2514 release for *Randomized Response*.

2515 Giris, Data, Chaudhuri, Fragouli, and Diggavi [Gir+20] studied privacy-utility-
 2516 randomness trade-offs for a multi-analyst setting under local differential privacy.
 2517 Here each user has a sample $X_i \sim D$ and makes a single public release of X_i . Each
 2518 analyst then utilizes the public release (together with shared randomness among
 2519 the users) to estimate D at a more accurate level than what the public release

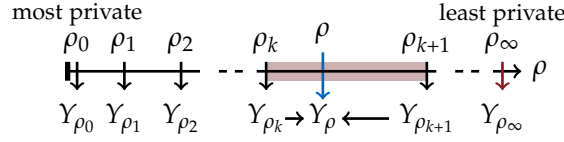


Figure 6.1: The idea behind lossless multiple release. For concreteness we consider additive noise mechanism and zero-concentrated differential privacy. Each Y_{ρ_i} denotes a noisy estimate, and to release a new estimate with $\rho > 0$, we can combine the adjacent estimates for ρ_k and ρ_{k+1} together with some fresh noise to obtain a new release Y_ρ that is exactly ρ -zero concentrated DP (ρ -zCDP). Note that these estimates do not need to be strictly increasing (or decreasing) but can be released in any order. Furthermore, releasing any subset of these estimates is exactly $\max(S)$ - ρ -zCDP, where S is the set of privacy parameters.

alone would allow. Later Zhang and He [ZH23] also used lossless gradual release techniques for the Gaussian mechanism in designing a database system supporting analysts with different privilege levels.

Similar research questions have been investigated outside of the DP literature. Xiao, Tao, and Chen [XTC09] studied releasing a sensitive dataset where each element is kept with probability p , and otherwise sampled uniformly from the universe. Li, Chen, Li, and Zhang [Li+12] considered privatizing data by additive Gaussian noise of scale σ . Both works dealt with arbitrary sequences of parameters (probabilities p or noise scalings σ), and demonstrated how to correlate releases to guarantee that (1) each release matches the single-release case and, (2) limiting the sensitive information derived from combining releases. Our method for adaptively producing Gaussian releases deviates from [Li+12], in that ours does not need to maintain any covariance matrix for past releases. This makes our approach more time- and space-efficient.

Finally, we mention the classic work of Equitz and Cover [EC91] who studied *successive refinement of information* where a source is encoded once and then refined in later stages without decreasing the rate compared to a one-shot optimal code. They showed that this is possible precisely when the sequence of reconstructions can be written as a Markov chain. While their work did not focus on privacy, successive refinement is conceptually related to lossless gradual release. Indeed, parts of the mathematical machinery was independently re-discovered by subsequent work on lossless gradual release.

BASIC TECHNIQUE We illustrate our framework in the setting of additive Gaussian noise and ρ -zero-concentrated differential privacy (ρ -zCDP). We want to release a private estimate of a real-valued query $f : \mathcal{X} \rightarrow \mathbb{R}$ with ℓ_2 -sensitivity $\Delta_2 = 1$ using the Gaussian mechanism. Now consider the simple case where one wants to privately release an estimate with two privacy levels $\rho < \rho'$. Then releasing $Y_\rho = f(\mathbf{x}) + \mathcal{N}(0, \frac{1}{2\rho})$ together with $Y_{\rho'-\rho} = f(\mathbf{x}) + \mathcal{N}(0, \frac{1}{2(\rho'-\rho)})$ is ρ' -zCDP by composition. That is, instead of using a privacy budget of $\rho + \rho'$ for independently releasing both estimates, the overall budget spent is just the maximum of both, which is ρ' . Now observe that these two samples can be used to obtain two samples that *exactly* match the desired utility as well, if we combine them in specific way.

By standard properties of the Gaussian distribution, it is easy to see that the linear combination $Y = \frac{\rho}{\rho'} \cdot Y_\rho + \frac{\rho'-\rho}{\rho'} \cdot Y_{\rho'-\rho} \sim \mathcal{N}\left(f(\mathbf{x}), \frac{1}{2\rho'}\right)$ which is exactly what we need for a release with budget ρ' .

To do multiple lossless releases for more general additive noise mechanisms, it turns out that one can always combine existing releases with fresh noise as

see definitions in Section 5.1, and recall that releasing $\mathcal{N}(0, 1/(2\rho))$ satisfies ρ -zCDP.

illustrated in Figure 6.1. In fact, to make any new release with privacy parameter ρ , it suffices to have saved the two “adjacent” releases whose privacy parameters are closest to ρ . Our approach applies more generally to a class of (independent) additive noise mechanisms that support gradual release, but the releases can be made in any order, and we do not require the set of releases to be known in advance.

OUR CONTRIBUTIONS We introduce a framework for *lossless multiple release*, generalizing past work on gradual release. In addition to allowing for making multiple private releases and having the privacy loss only scale with the least private release, we impose no specific ordering on the privacy parameters used. Furthermore, we formalize a general theorem (Section 6.3.2) showing that lossless multiple release is possible for a large class of mechanisms based on adding i.i.d. noise to each coordinate whose distribution satisfies a *convolution preorder*, as defined next.

Definition 6.1 (Convolution preorder). *A family of real-valued distributions $\mathcal{D}(\rho)$ parameterized by $\rho \in \mathbb{R}_+$ is said to satisfy a convolution preorder, if given $\mathcal{D}(\rho_1)$ and $\mathcal{D}(\rho_2)$ for $\rho_1 < \rho_2$, there exists a distribution $\mathcal{C}(\rho_2, \rho_1)$ such that $\mathcal{D}(\rho_2) * \mathcal{C}(\rho_2, \rho_1) = \mathcal{D}(\rho_1)$.*

This relation is a natural way of stating that the distributions become more noisy as the parameter ρ gets smaller. We are unaware of any existing term in the literature but the definition is closely related to *convolution order* [Sto]. Examples of noise distributions satisfying Definition 6.1 include the Laplace, Poisson, and the Gaussian mechanisms, parameterized by a decreasing function of their variance.

Theorem 6.1 (Meta theorem, informal). *Let $\mathcal{A}_{f,\rho} : \mathcal{X} \rightarrow \mathbb{R}^d$ be a mechanism that adds independent, identically distributed noise to coordinates of a function $f : \mathcal{X} \rightarrow \mathbb{R}^d$. If the noise distribution of $\mathcal{A}_{f,\rho}$ satisfies convolution preorder, then there exists an algorithm enabling lossless multiple release. Also, any invertible post-processing of $\mathcal{A}_{f,\rho}$ preserves this property.*

For the Gaussian mechanism in particular, we show how to get lossless multiple release by only using basic properties of Gaussians, and we show similar results for the Laplace and Poisson mechanisms from first principles. We give concrete instantiations of *Gaussian sparse histograms* (Section 6.4.2) and *factorization mechanisms* (Section 6.4.1).

6.1 THREAT MODEL AND GOALS

Our setting is a multi-user system with a set of analysts/users S , all able to query the same dataset using a differentially private mechanism. User s has their own security clearance level with corresponding privacy budget ρ_s . This setting allows for multi-level security where each user belongs to a security clearance level, like in the classic Bell-LaPadula model [BLP76], where users with high clearance levels have higher values of ρ . A common example of such clearance levels is using increasing levels with labels such as *public*, *restricted*, *confidential*, and *top secret*. The user with the highest clearance in the system has a privacy budget of $\max_{s \in S}(\rho_s)$, which we denote $\rho_{\max}$.

We consider an adversary who gains partial knowledge, that is, an adversary that sees *some* of the releases. Our goal is to design a differentially private mechanism such that an adversary with access to releases from some users $S' \subseteq S$, can learn *at most* what they could have learned from a release with privacy parameter $\max_{s' \in S'}(\rho_{s'})$. This means that even by compromising or colluding with more users, the adversary's knowledge may not increase. In case an adversary observes all releases (e.g., by compromising all users), the privacy loss would be bounded by $\rho_{\max}$.

However, our goal is to design a mechanism where the releases are *lossless* in the sense that the noise distribution from multiple releases with a combined budget $\rho_{\max}$ would be indistinguishable from one single release with the privacy budget $\rho_{\max}$. In other words, the privacy loss should be determined by $\rho_{\max}$, while independent releases would usually have privacy parameter $\sum_{s \in S} \rho_s$ due to composition.

6.2 GAUSSIAN LOSSLESS MULTIPLE RELEASE

Extending the work in [KHP16; Li+12], we demonstrate next that in the case of the Gaussian mechanism, providing lossless multiple release is clean and follows immediately from simple properties of Gaussians. Throughout the paper, we use the symbol Y to be a random variable that depends on the private dataset and Z to be one that does not.

We first consider the gradual release setting where an increasingly accurate estimate is released, and the overall privacy loss is determined solely by the latest, least private release. In Lemma 6.4, we drop this restriction, allowing releases in any order while still guaranteeing that the overall privacy guarantee is the maximum ρ value provided. For simplicity, we consider the one-dimensional case, where we want to release some query $f : \mathcal{X} \rightarrow \mathbb{R}$. The d -dimensional setting is handled by sampling each coordinate independently.

GETTING STARTED Foreshadowing the usage of ρ -zCDP, we initially use $1/(2\rho)$ for denoting the variance of a Gaussian. Our inquiry starts with a basic observation about inverse-variance weighting of Gaussians.

Lemma 6.2. Let $Y_\rho \sim \mathcal{N}(\beta, \frac{1}{2\rho})$ and $Y_{\rho'} \sim \mathcal{N}(\beta, \frac{1}{2\rho'})$ where $\beta \in \mathbb{R}$ and $\rho, \rho' > 0$. Then,

$$Y = \frac{\rho}{\rho + \rho'} Y_\rho + \frac{\rho'}{\rho + \rho'} Y_{\rho'} \sim \mathcal{N}\left(\beta, \frac{1}{2(\rho + \rho')}\right).$$

Proof. The mean of Y is immediate by the fact that Y is a weighted-average of Y_ρ and $Y_{\rho'}$, both of mean β . For the variance, direct computation yields:

$$\text{Var}[Y] = \frac{\rho^2/(2\rho) + \rho'^2/(2\rho')}{(\rho + \rho')^2} = \frac{1}{2(\rho + \rho')}.$$

As the sum of two Gaussians is itself Gaussian, we are done. $\square$

The essence of what is being claimed is that a Gaussian of variance $\frac{1}{2\rho}$ and another Gaussian of variance $\frac{1}{2\rho'}$ with the same mean can be combined into a new Gaussian with variance $\frac{1}{2(\rho + \rho')}$ and the same mean. Inspired by this, we can repeatedly invoke the lemma for the following result.

Lemma 6.3. Given $0 < \rho_1 < \dots < \rho_m$ and $\beta \in \mathbb{R}$ define $Y_{\rho_1}, \dots, Y_{\rho_m} \in \mathbb{R}$ where $Y_{\rho_1} \sim \mathcal{N}(\beta, \frac{1}{2\rho_1})$, and for $k > 1$:

$$Y_{\rho_{k+1}} = \frac{\rho_k}{\rho_{k+1}} Y_{\rho_k} + \frac{\rho_{k+1} - \rho_k}{\rho_{k+1}} \cdot \mathcal{N}\left(\beta, \frac{1}{2(\rho_{k+1} - \rho_k)}\right).$$

Then for any $i \in [m]$, we have $Y_{\rho_i} \sim \mathcal{N}(\beta, \frac{1}{2\rho_i})$ and for any $j \in [m]$, we have $\text{Cov}(Y_{\rho_i}, Y_{\rho_j}) = \frac{1}{2 \cdot \max(\rho_i, \rho_j)}$.

Proof. We first show the distribution of Y_{ρ_k} by induction on k . Assume that $Y_{\rho_k} \sim \mathcal{N}(\beta, \frac{1}{2\rho_k})$, which is true for the base case of $k = 1$. Note that the inductive step follows immediately from invoking Lemma 6.2. For the covariance, note that we can expand the expressions inside the covariance and “throw away” the independent noise added in each recurrence. Assuming $i \leq j$ we get $\text{Cov}(Y_{\rho_i}, Y_{\rho_j}) = \text{Cov}(Y_{\rho_i}, \frac{\rho_i}{\rho_j} Y_{\rho_i}) = \frac{\rho_i}{\rho_j} \text{Var}[Y_{\rho_i}] = \frac{1}{2\rho_j}$, implying the stated covariance. $\square$

Note that jointly Gaussian random variables are uniquely determined by their mean vector and covariance matrix. We will use this property in the next section to show that a different mechanisms that allows to create these releases in *arbitrary order* matches the distribution one would obtain by simply sequentially creating these releases as described in Lemma 6.3.

RELEASES IN ARBITRARY ORDER The Gaussian sequence in Lemma 6.3 will be the basis for our lossless multiple release version of the Gaussian mechanism. What the lemma does not address is generating the sequence in arbitrary order: Statically, given the full sequence $(\rho_k)_{k \in [m]}$, we can generate the Gaussians, but what if we receive them one-by-one and in arbitrary order? We will address this next.

Lemma 6.4. For $\rho_\infty \in \mathbb{R}_+ \cup \{\infty\}$ and $\beta \in \mathbb{R}$, let $M = \{(0, \infty), (\rho_\infty, Y_{\rho_\infty})\}$ where $Y_{\rho_\infty} \in \mathcal{N}(\beta, \frac{1}{2\rho_\infty})$. Consider a finite subset $S \subset (0, \rho_\infty)$ and the following process that runs for $|S|$ iterations:

1. Pick an arbitrary $\rho \in S$ and delete it from S ;
2. Let $(\rho_l, Y_{\rho_l}), (\rho_r, Y_{\rho_r}) \in M$ where $\rho \in (\rho_l, \rho_r)$ and $\rho_r - \rho_l$ is minimal;
3. Sample $Z \sim \mathcal{N}\left(0, \frac{(1 - \rho_l/\rho)(1/\rho - 1/\rho_r)}{2(1 - \rho_l/\rho_r)}\right)$, let

$$Y_\rho = \frac{1 - \rho_l/\rho}{1 - \rho_l/\rho_r} Y_{\rho_r} + \frac{\rho_l/\rho - \rho_l/\rho_r}{1 - \rho_l/\rho_r} Y_{\rho_l} + Z,$$

and add (ρ, Y_ρ) to M .

Then the sequence of random variables generated by the process has the same distribution as described in Lemma 6.3.

Proof sketch. The argument is inductive. Under the hypothesis that all values generated up to the given point have the distribution described by Lemma 6.3, we argue

that the newly generated value does too. The argument considers the four different cases for ρ_l, ρ_r , e.g., $\rho_l = 0, \rho_r \neq \rho_\infty$. As the proof involves tedious computation we refer the reader to Section 6.6.3 for the formal proof. $\square$

FORMALIZING LOSSLESS MULTIPLE RELEASE Lemma 6.4 will constitute the basis for our implementation of lossless multiple release, but we have yet to formally define this notion. We do so next.

Definition 6.2 (Lossless multiple release). Let $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$ be a family of mechanisms on a domain $\mathcal{X}$, indexed by a privacy parameter $\rho \in \mathbb{R}_+$. We say that $M : \mathcal{X} \times \mathbb{R}_+ \rightarrow \mathcal{Y}$ implements $\mathcal{M}_\rho$ with lossless multiple release if for every $x \in \mathcal{X}$ it satisfies:

1. $\forall \rho: M(x, \rho)$ and $\mathcal{M}_\rho(x)$ are identically distributed.
2. For every finite subset $S \subset \mathbb{R}_+$, processed in arbitrary order by M , and $y \in \mathcal{Y}$, conditioned on $M(x, \max(S)) = y$ the joint distribution of $(M(x, \rho))_{\rho \in S}$ is uniquely determined by y and S .

Functionally, a mechanism meets the definition if its outputs can be correlated such that for any set of outputs, their joint distribution can be viewed as (randomized) post-processing of the least private release. An implementation necessarily has to store information about releases that have been made, i.e., the sequence of inputs to M and the corresponding outputs, to fulfill the requirement that releases for different privacy parameters are correlated. When M only supports outputting releases for a sequence of *increasing* privacy parameters, we call it *lossless gradual release*.

Having stated the definition for lossless multiple release, consider Algorithm 9. It implements Lemma 6.4, and the idea is visualized in Figure 6.1.

Algorithm 9 GaussianMultipleRelease

Parameters: ℓ_2 sensitivity Δ_2

Require: Set of releases M , privacy parameter ρ

- 1: Find $(\rho_k, Y_{\rho_k}), (\rho_{k+1}, Y_{\rho_{k+1}}) \in M$ such that
 - 2: $\rho \in [\rho_k, \rho_{k+1})$ and $\forall (\rho', \cdot) \in M : \rho' \notin (\rho_k, \rho_{k+1})$
 - 3: Sample $Z_\rho \sim \mathcal{N}(0, \Delta_2^2 \cdot \frac{(1-\rho_k/\rho) \cdot (1/\rho - 1/\rho_{k+1})}{2(1-\rho_k/\rho_{k+1})})$
 - 4: Let $Y_\rho \doteq Z_\rho + \frac{(1-\rho_k/\rho)Y_{\rho_{k+1}} + (\rho_k/\rho - \rho_k/\rho_{k+1})Y_{\rho_k}}{1-\rho_k/\rho_{k+1}}$
 - 5: Add $M = M \cup \{(\rho, Y_\rho)\}$
 - 6: **Return** Y_ρ
-

Corollary 6.5. With M initialized as $M = \{(0, \infty), (\infty, f(\mathbf{x}))\}$, Algorithm 9 implements the Gaussian mechanism with lossless multiple release.

Proof. Let $\{Y_\rho\}_{\rho \in S}$ be the set of outputs produced by Algorithm 9 on receiving the set S of privacy parameters in arbitrary order. Observe that the algorithm is implementing the (adaptive) sampling in Lemma 6.4, and so it produces outputs with the same distribution as Lemma 6.3. Property 1 of Definition 6.2 follows immediately from observing that $Y_\rho \sim \mathcal{N}(f(x), \frac{1}{2\rho})$. For property 2, note that every release in Lemma 6.3 can be viewed as randomized post-processing of the least private release. One way to see this is to note that for any two releases Y_ρ and $Y_{\rho'}$

where $\rho < \rho'$, we have that $\text{Cov}(Y_\rho, Y_{\rho'}) = \text{Var}[Y_{\rho'}]$, implying that $Y_\rho = Y_{\rho'} + Z$ for aptly scaled zero-mean Gaussian noise Z . $\square$

6.3 EXTENDING TO INDEPENDENT ADDITIVE NOISE

We will next show that lossless multiple release holds for a larger class of mechanisms. To proceed we recall the notion of an Independent Additive Noise Mechanism (**IANM**).

Definition 6.3. We define an **IANM** $\mathcal{A}_{f,\rho} : \mathcal{X} \rightarrow \mathbb{R}^d$ as a mechanism of the form

$$\mathcal{A}_{f,\rho}(\mathbf{x}) = f(\mathbf{x}) + Z, \quad Z \sim \mathcal{D}(\rho),$$

where $\mathcal{D}(\rho)$ is a probability distribution parameterized in ρ , that draws a d -dimensional vector with i.i.d. samples.

It turns out that any **IANM** mechanism $\mathcal{A}_{f,\rho}$ satisfying Definition 6.1 supports lossless multiple release.

Lemma 6.6. Any **IANM** mechanism $\mathcal{A}_{f,\rho}$ with noise distribution $\mathcal{D}(\rho)$ satisfying convolution preorder can be implemented with lossless multiple release.

Proof. The proof is for the one-dimensional case, but the same argument can be invoked for the multidimensional case as each coordinate has independent noise. We begin by proving that given $S = \{\rho_k : k \in [m]\} \subset \mathbb{R}^+$, where $\rho_1 < \dots < \rho_m$, it is possible to construct a set of releases $\{Y_{\rho_k} : k \in [m]\}$ that satisfy Definition 6.2. By Definition 6.1, it is possible to express each Y_{ρ_k} as

$$Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_m} + \sum_{j=k}^{m-1} W_{\rho_{j+1}, \rho_j}, \quad (1)$$

where $Z_{\rho_m} \sim \mathcal{D}(\rho_m)$ and $W_{\rho_{j+1}, \rho_j} \sim \mathcal{C}(\rho_{j+1}, \rho_j)$ are sampled independently. To prove that $Y_{\rho_k} \stackrel{d}{=} \mathcal{A}_{f,\rho_k}(\mathbf{x})$, observe that Y_{ρ_k} is distributed as $f(\mathbf{x})$ plus a random variable drawn from $\mathcal{D}(\rho_m) * \mathcal{C}(\rho_m, \rho_{m-1}) * \dots * \mathcal{C}(\rho_{k+1}, \rho_k) = \mathcal{D}(\rho_k)$, as needed. For the second property in Definition 6.2, observe that conditioning on $Y_{\rho_m} = y$ we can express every other Y_{ρ_k} for $k \in [m-1]$ as

$$Y_{\rho_k} = y + \sum_{j=k}^{m-1} W_{\rho_{j+1}, \rho_j}.$$

As a result, the joint distribution on $(Y_{\rho_1}, \dots, Y_{\rho_m})_{|Y_{\rho_m}=y}$ is uniquely determined by y , proving the second property.

We will now argue (inductively) that we can produce these releases adaptively. Let $S_1 \subset S_2 \dots \subset S_m = S$ be a sequence of subsets of S where $|S_i| = i$. For the base case of S_1 we can make a single release using $\mathcal{A}_{f,\rho}$. For our inductive hypothesis, assume we have produced releases corresponding to all the privacy parameters in S_n . Now, for S_{n+1} , we re-label the privacy parameters such that $S_{n+1} = \{\rho_k\}_{k \in [n+1]}$ where $\rho_1 < \dots < \rho_{n+1}$. For the unique $\rho_i \in S_{n+1} \setminus S_n$, we can conditionally sample it from the joint distribution over $(Y_{\rho_k})_{k \in [n+1]}$, conditioned on the value of each release made in the previous round. By induction, it follows that we can *adaptively* release S . $\square$

6.3.1 Sampling in Concrete Settings

Lemma 6.6 proves the existence of a sampling procedure for lossless multiple release. In Section 6.2 we showed a sampling procedure in the particular case of Gaussian noise. In Section 6.6.1 we show that the structure of Algorithm 9 holds for general IANM mechanisms. Namely, if the privacy parameters we support come from a bounded range $(\rho_0, \rho_\infty) \subset \mathbb{R}_+$, then the corresponding algorithm has a similar structure (see Algorithm 13).

More precisely, consider two neighboring releases Y_{ρ_k} and $Y_{\rho_{k+1}}$ with noise parameters ρ_k and ρ_{k+1} , respectively, and a new lossless release Y_ρ with parameter $\rho \in [\rho_k, \rho_{k+1}]$. We can use (1) on this set of $m+1$ releases and condition on the values of the m previous releases $Y_{\rho_1}, \dots, Y_{\rho_m}$. Next, write $W_{\rho_{k+1}, \rho_k} = W_1 + W_2$ where $W_1 = Y_\rho - Y_{\rho_{k+1}}$ and $W_2 = Y_{\rho_k} - Y_\rho$. In Section 6.6.1, we show that sampling Y_ρ can be reduced to the following task:

$$\text{Sample } W_1 \text{ conditioned on } W_1 + W_2 = Y_{\rho_k} - Y_{\rho_{k+1}}. \quad (2)$$

EXAMPLE: POISSON MECHANISM Consider the IANM mechanism using the Poisson distribution, $\text{Poi}(\lambda)$. This mechanism has the property that noise is always a non-negative integer, making it a natural noise distribution for integer vectors in settings where negative noise is undesirable. Appendix 6.6.5 states some basic privacy properties of the Poisson mechanism.

The family of Poisson distributions parameterized by $\rho = 1/\lambda$ satisfies convolution preorder since for any $\lambda_1 > \lambda_2$, $\text{Poi}(\lambda_1) = \text{Poi}(\lambda_1 - \lambda_2) * \text{Poi}(\lambda_2)$. To adaptively perform private lossless multiple release of a value $f(\mathbf{x})$ using the Poisson mechanism and parameters $\lambda_1 > \dots > \lambda_m$ we notice that the “bridging” distribution in the k^{th} term of the sum in (1) has distribution $\text{Poi}(\lambda_k - \lambda_{k+1})$. Note that without conditioning on $Y_{\rho_1}, \dots, Y_{\rho_m}$, $W_1 \sim \text{Poi}(\lambda - \lambda_{k+1})$ and $W_2 \sim \text{Poi}(\lambda_k - \lambda)$. By Lemma 6.22 in the appendix we have that the sampling in (2) reduces to $W_1 \sim \text{Bin}(N, p)$ for $N = Y_{\rho_k} - Y_{\rho_{k+1}}$ and $p = (\lambda - \lambda_{k+1})/(\lambda_k - \lambda_{k+1})$.

EXAMPLE: LAPLACE MECHANISM Koufogiannis, Han, and Pappas [KHP16] have already shown that the Laplace distribution parameterized by $\rho = 1/b$ satisfies convolution preorder for any scale parameters $b_1 > b_2$. Let $\text{LapBridge}(b_2, b_1)$ be the probability distribution that draws 0 with probability b_2^2/b_1^2 and from $\text{Lap}(0, b_1)$ with the remaining probability. Then $\text{Lap}(0, b_2) * \text{LapBridge}(b_2, b_1) = \text{Lap}(0, b_1)$ exactly. To implement lossless release via the sampling in (2), it turns out that the conditional distribution is a mixture $W_1 \sim \text{LapBridge}(b, b_2)$ of three distributions: With some probability W_1 is equal to either 0 or $Y_{\rho_k} - Y_{\rho_{k+1}}$, and otherwise it is sampled from the convolution of two Laplace distributions. We also show that the related exponential distribution satisfies convolution preorder, see Section 6.6.6 for the full details.

6.3.2 Lossless Multiple Release as a Blackbox

Inspired by Corollary 15 in Koufogiannis, Han, and Pappas [KHP16], we provide a meta theorem for lossless multiple release based on IANM mechanisms. The central component is the following lemma, showing that the class of lossless multiple release mechanisms is closed under invertible post-processing.

Lemma 6.7. *Let $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$ satisfy lossless multiple release, and let $H : \mathcal{Y} \rightarrow \mathcal{Y}'$ be an invertible function. Then $\mathcal{M}'_\rho = H \circ \mathcal{M}_\rho$ also satisfies lossless multiple release.*

Proof. Let $M : \mathcal{X} \times \mathbb{R}_+ \rightarrow \mathcal{Y}$ be an implementation of $\mathcal{M}_\rho$ satisfying lossless multiple release, and let $M' = H \circ M$, which we will argue implements lossless multiple release. The only property that does not trivially hold for M' is the second property of Definition 6.2. For a set $S \subset \mathbb{R}_+$, note that to condition on $M'(\mathbf{x}, \max(S)) = y$ is equivalent to conditioning on $H^{-1}(M'(\mathbf{x}, \max(S))) = M(\mathbf{x}, \max(S)) = H^{-1}(y) \in \mathcal{Y}$. We thus have that conditioning $M'(\mathbf{x}, \max(S)) = y$ implies that the joint distribution over $(M(\mathbf{x}, \rho))_{\rho \in S}$ is fully determined, and consequently,

$$(H(M(\mathbf{x}, \rho)))_{\rho \in S} = (M'(\mathbf{x}, \rho))_{\rho \in S}.$$

This finishes the proof. $\square$

Theorem 6.8 (Meta theorem). *Let $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$ be a family of mechanisms on a domain $\mathcal{X}$, indexed by a privacy parameter $\rho \in \mathbb{R}_+$. Furthermore, assume $\mathcal{M}_\rho$ can be decomposed as $\mathcal{M}_\rho = H \circ \mathcal{A}_{f,\rho}$ where*

- $\mathcal{A}_{f,\rho} : \mathcal{X} \rightarrow \mathbb{R}^d$ is an independent add. noise mechanism for releasing $f : \mathcal{X} \rightarrow \mathbb{R}^d$ with privacy parameter ρ and with noise distribution satisfying a convolution preorder (Definition 6.1);
- $H : \mathbb{R}^d \rightarrow \mathcal{Y}$ is an invertible post-processing step;

Then $\mathcal{M}_\rho$ can be implemented with lossless multiple release.

Proof. The theorem follows from invoking Lemma 6.6 together with Lemma 6.7. $\square$

SUPPORTING NON-INVERTIBLE POST-PROCESSING To support non-invertible post-processing, we also introduce a weaker notion: *weakly lossless multiple release*.

Definition 6.4 (Weakly Lossless Multiple Release). *A mechanism $\mathcal{M}_\rho$ supports weakly lossless multiple release if it can be written as $\mathcal{M}_\rho = H \circ \mathcal{M}'_\rho$ where $\mathcal{M}'_\rho$ supports lossless multiple release and H is an arbitrary function (possibly chosen from some distribution).*

We define *weakly lossless gradual release* analogously. It follows that these classes of mechanisms are closed under all post-processing. While this notion is indeed weaker, any algorithm $M(\mathbf{x}, \rho)$ implementing weakly lossless multiple release for a ρ -private mechanism $\mathcal{M}_\rho(\mathbf{x})$, will have the property that the set of releases $(M(\mathbf{x}, \rho))_{\rho \in S}$ are $\max(S)$ -private, if ρ -privacy is closed under post-processing. Examples of such privacy notions are ϵ -DP and ρ -zCDP. We get the following immediate corollary to Theorem 6.8.

Corollary 6.9. *If the function H in Theorem 6.8 is not invertible, then $\mathcal{M}_\rho = H \circ \mathcal{A}_{f,\rho}$ supports weakly lossless multiple release.*

Weakly lossless multiple release will play a role in the next section, where we consider non-invertible post-processing such as truncation.

6.4 APPLICATIONS

In this section, we describe two applications supported by our framework for lossless multiple release:

- Factorization mechanisms [Li+15], where we want to privately release a linear query $A\mathbf{x}$, and
- Sparse Gaussian histograms, also known as stability histograms [Wil+24; GAT20].

We will make use of both *lossless* multiple release (Definition 6.2), and its weaker variant (Definition 6.4). This will be necessary since, e.g., the truncation used for sparse Gaussian histograms is not an invertible function, and so not covered by Theorem 6.8.

Because the noise generation for histograms on large domains is expensive, we give a dimension-independent algorithm that works in the gradual release setting. Throughout, we state our results as post-processings of the Gaussian mechanism, but analogous results hold for any other IANM mechanism meeting Definition 6.1.

6.4.1 Lossless Multiple Release Factorization Mechanism

Let A be a query matrix and fix a public factorization $A = LR$. Denote the *factorization mechanism* [Li+15] as $\mathcal{F}_\rho(\mathbf{x}) = A\mathbf{x} + L\mathbf{z} = L(R\mathbf{x} + \mathbf{z})$ on some private dataset $\mathbf{x}$ where $\mathbf{z}$ is a vector drawn from a distribution $\mathcal{D}(\rho)$ satisfying Definition 6.1 parameterized by ρ , typically Gaussian or Laplace noise.

Lemma 6.10. $\mathcal{F}_\rho$ can be implemented with weakly lossless multiple release, and lossless multiple release if L has a left-inverse.

Proof. Observe that $\mathcal{F}_\rho(\mathbf{x}) = L \circ \mathcal{A}_{R,\rho}(\mathbf{x})$ for an IAN mechanism $\mathcal{A}_{R,\rho}(\mathbf{x}) = R\mathbf{x} + \mathbf{z}$ where $\mathbf{z} \sim \mathcal{D}(\rho)$ for a $\mathcal{D}(\rho)$ satisfying Definition 6.1. The result follows from Theorem 6.8 and Corollary 6.9, and noting that the linear map L is invertible exactly when L has a left-inverse. $\square$

Besides proving existence in Lemma 6.10, we also give an explicit instantiation in Algorithm 10 using the Gaussian mechanism.

Algorithm 10 FactorizationMultipleRelease

Parameters: Factorization $A = LR$, ℓ_2 sensitivity Δ_2

Require: Set of releases M , privacy parameter ρ

- 1: Find $(\rho_k, Y_{\rho_k}), (\rho_{k+1}, Y_{\rho_{k+1}}) \in M$ such that
- 2: $\rho \in [\rho_k, \rho_{k+1})$ and $\forall (\rho', \cdot) \in M : \rho' \notin (\rho_k, \rho_{k+1})$.
- 3: $Z_\rho \sim \mathcal{N}\left(0, \Delta_2^2 \cdot \frac{(1-\rho_k/\rho) \cdot (1/\rho - 1/\rho_{k+1})}{2(1-\rho_k/\rho_{k+1})}\right)^d$.
- 4: Let $Y_\rho \doteq L \cdot Z_\rho + \frac{(1-\rho_k/\rho)Y_{\rho_{k+1}} + (\rho_k/\rho - \rho_k/\rho_{k+1})Y_{\rho_k}}{1-\rho_k/\rho_{k+1}}$.
- 5: Set $M = M \cup \{(\rho, Y_\rho)\}$.
- 6: **Return** Z .

Lemma 6.11. *With initialization $M = \{(0, \infty), (\infty, A\mathbf{x})\}$, Algorithm 10 implements the Gaussian noise factorization mechanism $\mathcal{F}_\rho$ with (weakly) lossless multiple release.*

Proof sketch. Note that the algorithm is practically a copy of Algorithm 9, but for the specific sensitive function $R\mathbf{x}$. The only structural difference is that the linearity of the post-processing L allows for storing correlated Gaussian noise directly in M . $\square$

6.4.2 Weakly Lossless Multiple Release of Histograms

We will now describe an example where the post-processing is neither linear nor invertible: releasing sparse (Gaussian) histograms [Kor+09; GAT20; BW18]. Let $\mathbf{x} = (X_i)^n$ be a dataset of n users, and we want to privately release the histogram $H(\mathbf{x}) = \sum_{i=1}^n X_i$ where $X_i \in \{0, 1\}^d$ and a user can contribute to l distinct counts and the Gaussian mechanism which scales proportional to $\sqrt{l}$ is preferable. In many natural settings, the domain size d can be very large, and therefore, the resulting histogram is usually very sparse, $k = \|H(\mathbf{x})\|_0 \ll d$. Releasing a noisy histogram where noise has been added to *each* coordinate would destroy the sparsity. One can cope by introducing a threshold $\tau > 0$ where only noisy counts that exceed τ are released. τ is usually set high enough such that the noise to a zero count will (after thresholding with high probability) still be zero. We denote the *support* of the histogram as $U(H(\mathbf{x})) = \{i \in [d] : H(\mathbf{x})_i \neq 0\}$ as the index of the non-zero coordinates. Define the thresholding function $T_\tau : \mathbb{R}^d \rightarrow \mathbb{R}^d$ as:

$$T_\tau(\mathbf{x}) = (y_i)_{i=1}^d, \quad \text{where} \quad y_i = \begin{cases} x_i, & \text{if } x_i \geq \tau \\ 0, & \text{otherwise} \end{cases}.$$

Denote the sparse histogram mechanism as $\mathcal{H}_\rho(\mathbf{x}) = T_\tau(H(\mathbf{x}) + Z)$ where $Z \sim \mathcal{D}(\rho)$ is a distribution satisfying a convolution preorder (Definition 6.1).

Lemma 6.12. *$\mathcal{H}_\rho$ can be implemented with weakly lossless multiple release.*

Proof. Observe that $\mathcal{H}_\rho$ can be expressed as a composite function $T_\tau \circ \mathcal{A}_{\rho,f}(\mathbf{x})$ for an IANM mechanism $\mathcal{A}_{\rho,f}(\mathbf{x})$ meeting Definition 6.1. The statement thus follows from Corollary 6.9. $\square$

It is straightforward to implement weakly lossless gradual release for stability histograms using our framework with time and space complexity linear in dimension d ; see Algorithm 11. The following lemma is proved in Section 6.6.4.

Lemma 6.13. *With initialization $M = \{(0, 0)\}$, Algorithm 11 implements $\mathcal{H}_\rho$ with weakly lossless gradual release.*

Algorithm 11 HistogramGradualRelease

Parameters: ℓ_2 sensitivity Δ_2

Require: histogram $H(\mathbf{x})$, set of releases M ,
privacy parameter ρ , threshold τ

- 1: Extract the single element $(\rho', Z') \in M$.
 - 2: Sample $\tilde{Z} \sim \mathcal{N}\left(0, \frac{1}{2}\Delta_2^2(\rho - \rho')\right)^d$.
 - 3: Let $Z = \frac{\rho'}{\rho}Z' + \frac{1}{\rho}\tilde{Z}$.
 - 4: **for** each $i \in [d]$ **do**
 - 5: **if** $H(\mathbf{x})_i + Z_i > \tau$ **then** $Y_i = H(\mathbf{x})_i + Z_i$.
 - 6: **else** $Y_i = 0$.
 - 7: Set $M = \{(\rho, Z)\}$
 - 8: **Return** Y
-

IMPROVING EFFICIENCY FOR WEAKLY LOSSLESS GRADUAL RELEASE Koro-
lova, Kenthapadi, Mishra, and Ntoulas [Kor+09] showed that under approximate
differential privacy, one can skip the noise generation for zero counts if thresh-
olding is enabled, adding only a small probability for infinite privacy loss if this
coordinate is non-zero in a neighboring dataset. Unfortunately, this approach
does not fit our meta theorem, so instead, we apply a technique due to Cormode,
Procopiu, Srivastava, and Tran [Cor+12] that efficiently simulates the noise dis-
tribution of those zero counts that would exceed the threshold and thus gives an
identical distribution to the truncated noisy histogram. The procedure is described
in the following lemma.

Lemma 6.14. *For a fixed noise distribution $\mathcal{D}$, the output distribution of releasing $\mathcal{T}_\tau(H(\mathbf{x}) + Z)$ with $Z \sim \mathcal{D}$ are independent noise samples is equal to the following process: (a) add noise to every non-zero coordinate in $H(\mathbf{x})_i$ for $i \in U(H(\mathbf{x}))$ and then (b) draw $q \sim \text{Binomial}(d - k, p)$ where $p = \mathbf{P}_{Z \sim \mathcal{D}}[Z > \tau]$. (c) Sample a subset $Q \subseteq [d] \setminus U(H(\mathbf{x}))$ of size $|Q| = q$ uniformly at random and (d) set the noise for $H(\mathbf{x})$ to $Z \sim \mathcal{D}$ conditioned on being above the threshold τ .*

Proof sketch. A formal argument can be found in Cormode, Procopiu, Srivastava, and Tran [Cor+12], here we just provide a sketch. It is clear that every entry in the support gets the correct output distribution. Furthermore, the number of zero counts q whose noise exceeds τ follows a binomial distribution, and we can add conditional noise to a uniformly drawn subset of size q . $\square$

Algorithm 14 in Section 6.6.4 implements the efficient routine described by Lemma 6.14 under weakly lossless gradual release. The key challenge is that the probability for a histogram count to exceed the threshold in a given round now depends on whether it exceeded the threshold in any prior round. Intuitively, all zero counts that have never exceeded the threshold have an equal probability to exceed the threshold in any given round, and so the simulation technique

in Lemma 6.14 can be used for these counts (with different probabilities and conditional distributions). However, once any zero count exceeds the threshold, it will have to be treated the same as non-zero counts in all future rounds.

The utility and privacy of Algorithm 14 is proved by arguing that its output distribution matches that of Algorithm 11. A proof sketch for the following lemma is given in Section 6.6.4.

Lemma 6.15. *Let $H(\mathbf{x})$ be a histogram over $[d]$, $\rho_1, \dots, \rho_m$ be a sequence increasing of privacy budgets, $\tau_1, \dots, \tau_m$ be a sequence of thresholds and $\Delta_2 > 0$ the ℓ_2 -sensitivity of the histogram. Also let the sequences of outputs $(Y^{(1)}, \dots, Y^{(m)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(m)})$ be derived from running Algorithm 11 and Algorithm 14 respectively with the preceding parameters as input. Then the sequences $(Y^{(1)}, \dots, Y^{(m)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(m)})$ are identically distributed.*

The benefit of the efficient sampling technique is that the number of sampled Gaussians will *never* exceed that of the naïve approach and can potentially be in the order of the sparsity depending on the parameter regime.

Lemma 6.16. *Let c be the number of zero-counts in the histogram output by Algorithm 14 least once over its execution. Then Algorithm 14 will sample $(k + c)m$ (truncated or otherwise) Gaussians.*

We again refer the reader to Section 6.6.4 for a proof.

6.5 CONCLUSION AND OPEN QUESTIONS

We have initiated a systematic study of differential privacy with multiple releases, motivated by settings in which many levels of privacy or trust may co-exist. The main message is that it is possible to generalize a large class of lossless gradual release techniques to this setting, even when releases are determined online and in no particular order. For mechanisms based on Gaussian, Laplace, or Poisson additive noise we give simple and efficient sampling procedures for creating new lossless releases. In particular, we are able to do lossless multiple release for any factorization mechanism using invertible matrices. Finally, we consider algorithmic challenges related to lossless gradual release and show that private sparse histograms may be computed much more efficiently than what a direct application of our general results would imply.

There are still many open questions concerning mechanisms that do not inherit their privacy guarantee from an IANMM mechanism. In particular, it would be interesting to determine under which conditions the exponential mechanism [MT07] supports lossless multiple release. Koufogiannis, Han, and Pappas [KHP16] conjecture that this is always possible. Other central private algorithms, such as report noisy max [DR14] also do not have known lossless multiple release mechanisms, even in the gradual release setting. A final challenge we would like to mention is implementing our mechanisms on a finite computer, e.g., creating a multiple release version of the discrete Gaussian mechanism [CKS20]. An appealing approach would be to base this on Poisson noise, which meets the technical conditions for our framework and approaches the Gaussian distribution in the limit.

Algorithm 12 GenericMultipleRelease

Parameters: Noise distribution family $\mathcal{D}(\rho)$ satisfying a convolution preorder, bridging noise distribution $\mathcal{C}(\rho, \rho')$ where $\mathcal{D}(\rho') * \mathcal{C}(\rho', \rho) = \mathcal{D}(\rho)$ for $0 < \rho < \rho'$.

Require: Sensitive value $f(\mathbf{x})$, new privacy parameter ρ_k , set of past releases $M = \{(\rho_i, Y_{\rho_i}) : i \in [m] \setminus \{k\}\}$

- 1: **if** $M = \emptyset$ **then**
- 2: Sample Z_{ρ_k} from $\mathcal{D}(\rho_k)$.
- 3: Set $Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_k}$.
- 4: **else if** $k = 1$ **then**
- 5: Sample W_{ρ_{k+1}, ρ_k} from $\mathcal{C}(\rho_{k+1}, \rho_k)$.
- 6: Set $Y_{\rho_k} = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$.
- 7: **else if** $k \in (1, m)$ **then**
- 8: Draw $W_{\rho_{k+1}, \rho_k} \sim \mathcal{C}(\rho_{k+1}, \rho_k)$ conditioned on $W_{\rho_{k+1}, \rho_k} + W_{\rho_k, \rho_{k-1}} = Y_{\rho_{k-1}} - Y_{\rho_{k+1}}$.
- 9: Set $Y_{\rho_k} = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$.
- 10: **else if** $k = m$ **then**
- 11: Sample Z_{ρ_k} from $\mathcal{D}(\rho_k)$ conditioned on $Z_{\rho_k} + W_{\rho_k, \rho_{k-1}} = Y_{\rho_{k-1}} - f(\mathbf{x})$.
- 12: Set $Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_k}$.
- 13: Set $M = M \cup \{(\rho_k, Y_{\rho_k})\}$.
- 14: **Return** Y_{ρ_k} .

6.6 APPENDIX

6.6.1 On Lossless Multiple Release Sampling

Lemma 6.6 makes no claim about how easy it is to sample noise from the conditional noise distribution, but guarantees its existence. We will use this section for discussing this sampling in greater detail. Consider the joint distribution of the releases defined by Equation (1) from the proof of Lemma 6.6, re-stated below for convenience:

$$Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_m} + \sum_{j=k}^{m-1} W_{\rho_{j+1}, \rho_j}, \quad k = 1, \dots, m, \quad (1)$$

where Y_{ρ_k} is the ρ_k -private release, $Z_{\rho_m} \sim \mathcal{D}(\rho_m)$ and $W_{\rho_{j+1}, \rho_j} \sim \mathcal{C}(\rho_{j+1}, \rho_j)$. Recall that $\mathcal{C}(\rho', \rho)$ is the unique distribution where for any $0 < \rho < \rho'$ we have $\mathcal{D}(\rho) = \mathcal{D}(\rho') * \mathcal{C}(\rho', \rho)$. As argued for in the proof of Lemma 6.6, to make a new ρ -private release, we consider the joint distribution, and sample the new releases conditioned on all past releases.

Formally, for a set of privacy parameters $\rho_1 < \dots < \rho_m$, we are interested in releasing ρ_k for a single $k \in [m]$, conditioned on the set of past releases $M = \{(\rho_i, Y_{\rho_i}) : i \in [m] \setminus \{k\}\}$. We give pseudocode for this in Algorithm 12, and a lemma for its correctness.

Lemma 6.17. *Initializing $M = \emptyset$ and then running Algorithm 12 for a set of privacy parameters $\{\rho_i : i \in [m]\}$ (processed in arbitrary order), will produce a set of outputs $\{Y_{\rho_i} : i \in [m]\}$ with distribution given by Equation (1).*

Proof. For $M = \emptyset$, we have that $Y_{\rho_k} = f(\mathbf{x}) + \mathcal{D}(\rho_k) = \mathcal{A}_{f, \rho_k}(\mathbf{x})$ on lines 2-3, as expected, where $\mathcal{A}$ is our IANM noise mechanism releasing f with ρ -privacy. For

Algorithm 13 SimplifiedGenericMultipleRelease

Parameters: Noise distribution family $\mathcal{D}(\rho)$ satisfying a convolution preorder, bridging noise distribution $\mathcal{C}(\rho, \rho')$ where $\mathcal{D}(\rho') * \mathcal{C}(\rho', \rho) = \mathcal{D}(\rho)$ for $0 < \rho < \rho'$.

Require: Privacy parameter ρ , set of releases M .

- 1: Find $(\rho_k, Y_{\rho_k}), (\rho_{k+1}, Y_{\rho_{k+1}}) \in M$ such that $\rho \in [\rho_k, \rho_{k+1})$ and $\forall (\rho', \cdot) \in M : \rho' \notin (\rho_k, \rho_{k+1})$.
- 2: Sample $W_{\rho_{k+1}, \rho}$ from $\mathcal{C}(\rho_{k+1}, \rho)$ conditioned on $W_{\rho_{k+1}, \rho} + W_{\rho, \rho_k} = Y_{\rho_k} - Y_{\rho_{k+1}}$.
- 3: Set $Y_\rho = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho}$.
- 4: Set $M = M \cup \{(\rho, Y_\rho)\}$.
- 5: **Return** Y_ρ

the remaining cases, assume M contains $m - 1$ releases with the correct joint distribution, and we are generating a new release at privacy level ρ_k . We will argue that Algorithm 12 produces a Y_{ρ_k} whose joint distribution with the releases in M matches (1). For $k = 1$, observe that $Y_{\rho_1} = Y_{\rho_2} + W_{\rho_2, \rho_1}$, and so lines 5-6 are correct. For $1 < k < m$, note that $Y_{\rho_k} = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$ and $Y_{\rho_{k-1}} = Y_{\rho_k} + W_{\rho_k, \rho_{k-1}}$, which combined allows us to identify the correct conditional distribution. Y_{ρ_k} is given by $Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$, conditioned on $W_{\rho_{k+1}, \rho_k} + W_{\rho_k, \rho_{k-1}} = Y_{\rho_{k-1}} - Y_{\rho_{k+1}}$, matching lines 8-9. For the final case of $k = m$, then $Y_{\rho_m} = f(\mathbf{x}) + Z_{\rho_m}$ and we have that $Y_{\rho_{m-1}} = Y_{\rho_m} + W_{\rho_m, \rho_{m-1}}$. It follows that the correct noise distribution is $Y_{\rho_m} = f(\mathbf{x}) + Z_{\rho_m}$, conditioned on $Z_{\rho_m} + W_{\rho_m, \rho_{m-1}} = Y_{\rho_{m-1}} - f(\mathbf{x})$, matching lines 11-12. An inductive argument identical to that given in the proof of Lemma 6.6 completes the proof. $\square$

6.6.2 Simplification and Removing Dependency on the Dataset

Note that $f(\mathbf{x})$ is only used in Algorithm 12 when a more accurate release is generated (lines 1 and 10). In the remaining cases we are only adding noise to past releases. This already allows us to simplify the algorithm, and argue for not having to store $f(\mathbf{x})$ in memory indefinitely. The algorithm reduces to the case (see line 7 of Algorithm 12) where only the two closest releases are combined into a new one. Such an algorithm is given in Algorithm 13, together with Lemma 6.18.

Lemma 6.18. *Let $M = \{(\rho_0, Y_{\rho_0}), (\rho_\infty, Y_{\rho_\infty})\}$ for $Y_{\rho_\infty} = \mathcal{A}_{\rho_\infty, f}(\mathbf{x})$, and $Y_{\rho_0} = Y_{\rho_\infty} + W_{\rho_\infty, \rho_0}$. Then running Algorithm 13 with input M and a set of privacy parameters $\{\rho_i : i \in [m]\} \subset (\rho_0, \rho_\infty)$ (processed in arbitrary order), will produce a set of outputs $\{Y_{\rho_i} : i \in [m]\}$ with distribution given by Equation (1). Moreover, the set M will at all times satisfy ρ_∞ -privacy.*

Proof. The statement follows from carefully comparing to Algorithm 12. Let $S = (\rho_1, \dots, \rho_m)$ be a sequence of noise values from the lemma statement, and let $O = (Y_{\rho_1}, \dots, Y_{\rho_m})$ be the outputs. Consider a second sequence $S' = (\rho_\infty, \rho_0, \rho_1, \dots, \rho_m)$, and consider the corresponding sequence of outputs $O' = (Y'_{\rho_\infty}, Y'_{\rho_0}, Y'_{\rho_1}, \dots, Y'_{\rho_m})$ from inputting S' and $M' = \emptyset$ to Algorithm 12. We can directly check that $Y_{\rho_0} \stackrel{d}{=} Y'_{\rho_0}$ and $Y_{\rho_\infty} \stackrel{d}{=} Y'_{\rho_\infty}$. For the remaining outputs, note that just before each algorithm is called, their internal states M and M' are also identically distributed. Now, the fact that each $\rho_i \in (\rho_0, \rho_\infty)$ implies that the remaining outputs $Y'_{\rho_1}, \dots, Y'_{\rho_m}$ are generated from lines 10-12 in Algorithm 12. Checking carefully, lines 1-3 in Algorithm 13 are implementing the same routine, and since $M \stackrel{d}{=} M'$, it follows

that $(Y_{\rho_1}, \dots, Y_{\rho_m}) \stackrel{d}{=} (Y'_{\rho_1}, \dots, Y'_{\rho_m})$, and so the statement follows from Lemma 6.17. The last statement on the ρ_∞ -privacy of M follows from Algorithm 13 implements Equation (1), and so each release in M can at any time during execution be viewed as randomized post-processing of Y_∞ . $\square$

Essentially, if we commit to supporting a bounded range of privacy parameters then we get a simpler algorithm. Lemma 6.18 also says something more: If we commit to supporting a lowest level of privacy ρ_∞ , then Algorithm 13 can be implemented in such a way that its internal state is ρ_∞ -private. After initializing M using the sensitive function $f(\mathbf{x})$, we can erase $f(\mathbf{x})$ from memory and Y_∞ will contain enough private information to generate all future releases. This could prove useful in settings where the time between releases is large, and we want to limit the private information leaked if the state of the algorithm were to be compromised. This can be compared with the gradual release setting, where natural implementations would require consistent access to $f(\mathbf{x})$.

6.6.3 Proof for Gaussian Lossless Multiple Release

Proof of Lemma 6.4. To argue that the values generated by the process match the distribution of the privacy tightening approach as specified in Lemma 6.3, we prove it inductively for an increasing subsets of releases. A d -dimensional Gaussian is uniquely defined by its variance and covariance structure. Therefore, it is sufficient to show that they exactly match the ones in Lemma 6.3

Let $S_n = \{\rho_k : k \in [n-1]\}$ where $0 < \rho_1 < \dots < \rho_n < \rho_\infty$ be the set of values in S for which we have generated Gaussians at the start of the n^{th} round of the process. Note that we re-label the ρ 's between the rounds such that $\rho_k \in S_n$ always is the k^{th} smallest value in S_n . Our argument will proceed by induction: assume that all the Gaussians $\{Y_{\rho_k} : k \in [n]\}$ generated after n rounds have the distribution given by Lemma 6.3. Then we will show that $\{Y_{\rho_k} : k \in [n-1]\} \cup \{Y_\rho\}$ has the same distribution as predicted by $S_n \cup \{\rho\}$ from invoking Lemma 6.3.

We begin with our base case. For $n = 1$, we have that $\rho_l = 0$ and $\rho_r = \rho_\infty$, and so $Y_\rho = Y_{\rho_\infty} + \mathcal{N}(0, \frac{1}{2\rho} - \frac{1}{2\rho_\infty})$. Since $Y_{\rho_\infty} \sim \mathcal{N}(\beta, \frac{1}{2\rho_\infty})$, we have that $Y_\rho \sim \mathcal{N}(\beta, \frac{1}{2\rho})$, as expected, and so the base case passes.

For $n \geq 2$, we first consider the following cases.

Case 1: $\rho_l = 0, \rho_r = \rho_1 \in S_n$. In this case, $Y_\rho = Y_{\rho_1} + \mathcal{N}(0, \frac{1}{2\rho} - \frac{1}{2\rho_1})$, and so $Y_\rho \sim \mathcal{N}(\beta, \frac{1}{2\rho})$. Since $\rho < \rho_1$, we have for all $i \in [n-1]$,

$$\text{Cov}(Y_\rho, Y_{\rho_i}) = \text{Cov}(Y_{\rho_1}, Y_{\rho_i}) = \frac{1}{2\rho_i}$$

as expected for the release with the smallest value in $\{\rho\} \cup S_n$.

Case 2: $\rho_l = \rho_{n-1}, \rho_r = \rho_\infty = \infty$. We have to deal with the case where we have set $\rho_\infty = \infty$ separately, as it is a bit of a trick. Note that we get

$$Y_\rho = \frac{(1 - \rho_{n-1})\beta + \rho_{n-1}Y_{\rho_{n-1}}}{\rho} + \mathcal{N}\left(0, \frac{1 - \rho_{n-1}/\rho}{2\rho}\right),$$

since we have $Y_{\rho_\infty} = \beta$ in this case. The final result is once more a sum of Gaussians, with mean β , and for the variance we can explicitly compute that $\text{Var}[Y_\rho] =$

$$\frac{\rho_{n-1}^2}{2\rho_{n-1}\rho^2} - \frac{\rho - \rho_{n-1}}{2\rho^2} = \frac{1}{2\rho}, \text{ as expected.}$$

For the covariance, for $i \in [n-1]$: $\text{Cov}(Y_\rho, Y_{\rho_i}) = \frac{\rho_{n-1}}{\rho} \text{Cov}(Y_{\rho_{n-1}}, Y_{\rho_i}) = \frac{1}{2\rho}$ as expected for the largest value in $\{\rho\} \cup S_n$, and so we are done. Now we consider

the most general case, where a new element is squeezed in the chain of existing releases.

Case 3: $\rho_l \in S_{n-1}$ and $\rho_r \neq \infty$. We begin with computing the variance of Y using the hypothesis $\text{Var}[Y_{\rho_r}] = \frac{1}{2\rho_r}$ and $\text{Var}[Y_{\rho_l}] = \frac{1}{2\rho_l}$ and $\text{Cov}[Y_{\rho_r}, Y_{\rho_l}] = \frac{1}{2\rho_r}$. We first define for brevity the coefficients

$$a = \frac{1 - \rho_l/\rho}{1 - \rho_l/\rho_r}, \quad \text{and} \quad b = \frac{\rho_l/\rho - \rho_l/\rho_r}{1 - \rho_l/\rho_r} = \frac{\rho_l(\rho_r - \rho)}{\rho(\rho_r - \rho_l)}. \quad (3)$$

Observe that the noise term $Z \sim \mathcal{N}(0, \sigma^2)$ with $\sigma^2 = \frac{(\rho - \rho_l)(\rho_r - \rho)}{2\rho^2(\rho_r - \rho_l)}$ is independent of Y_{ρ_l} and Y_{ρ_r} while these RV's are correlated. Therefore, by standard identities and the hypothesis, we get

$$\begin{aligned} \text{Var}[Y_\rho] &= a^2 \cdot \frac{1}{2\rho_r} + b^2 \cdot \frac{1}{2\rho_l} + 2ab \cdot \text{Cov}[Y_{\rho_r}, Y_{\rho_l}] + \sigma^2 \\ &= \frac{a^2 + 2ab}{2\rho_r} + \frac{b^2}{2\rho_l} + \sigma^2 \\ &= \frac{(a+b)^2 - b^2}{2\rho_r} + \frac{b^2}{2\rho_l} + \sigma^2. \end{aligned}$$

Now, observe that the coefficients form a convex combination, i.e. $a + b = 1$. Hence, we get as desired,

$$\begin{aligned} \text{Var}[Y_\rho] &= \frac{1}{2\rho_r} + b^2 \cdot \left(\frac{1}{2\rho_l} - \frac{1}{2\rho_r} \right) + \sigma^2 \\ &= \frac{1}{2\rho_r} + \left(\frac{\rho_l(\rho_r - \rho)}{\rho(\rho_r - \rho_l)} \right)^2 \cdot \left(\frac{1}{2\rho_l} - \frac{1}{2\rho_r} \right) + \sigma^2 \\ &= \frac{1}{2\rho_r} + \frac{\rho_l(\rho_r - \rho)^2}{2\rho^2(\rho_r - \rho_l)\rho_r} + \frac{(\rho - \rho_l)(\rho_r - \rho)}{2\rho^2(\rho_r - \rho_l)} \\ &= \frac{1}{2\rho_r} + \left(\frac{(\rho_r - \rho)(\rho_l(\rho_r - \rho) + \rho_r(\rho - \rho_l))}{2\rho^2(\rho_r - \rho_l)\rho_r} \right) \\ &= \frac{1}{2\rho_r} + \frac{(\rho_r - \rho)\rho(\rho_r - \rho_l)}{\rho^2(\rho_r - \rho_l)2\rho_r} \\ &= \frac{\rho + \rho_r - \rho}{2\rho_r\rho} = \frac{1}{2\rho}. \end{aligned}$$

What remains to show is that the covariances match up, which we do next. We start with the case where $\rho_r \neq \rho_\infty$, and so there exists $\rho_k, \rho_{k+1} \in S_n$ such that $\rho \in (\rho_k, \rho_{k+1})$. For $j \in [n-1]$, we therefore have that

$$\begin{aligned} \text{Cov}(Y_\rho, Y_{\rho_j}) &= \text{Cov}\left(\frac{(\rho_l^{-1} - \rho^{-1})Y_{\rho_r} + (\rho^{-1} - \rho_r^{-1})Y_{\rho_l}}{2(\rho_l^{-1} - \rho_r^{-1})}, Y_{\rho_j} \right) \\ &= \frac{1}{2(\rho_l^{-1} - \rho_r^{-1})} \left[(\rho_l^{-1} - \rho^{-1})\text{Cov}(Y_{\rho_r}, Y_{\rho_j}) + (\rho^{-1} - \rho_r^{-1})\text{Cov}(Y_{\rho_l}, Y_{\rho_j}) \right] \\ &= \frac{1}{2(\rho_l^{-1} - \rho_r^{-1})} \left[(\rho_l^{-1} - \rho^{-1})\min(\rho_r^{-1}, \rho_j^{-1}) + (\rho^{-1} - \rho_r^{-1})\min(\rho_l^{-1}, \rho_j^{-1}) \right]. \end{aligned}$$

We consider the cases of $j \leq k$ and $j > k$ separately. For $j \leq k$, we have that

$$\text{Cov}(Y_\rho, Y_{\rho_j}) = \frac{1}{2(\rho_l^{-1} - \rho_r^{-1})} \left[(\rho_l^{-1} - \rho^{-1})\rho_r^{-1} + (\rho^{-1} - \rho_r^{-1})\rho_l^{-1} \right] = \frac{1}{2\rho},$$

and similarly for $j \geq k+1$ we get

$$\text{Cov}(Y_\rho, Y_{\rho_j}) = \frac{1}{2(\rho_k^{-1} - \rho_{k+1}^{-1})} \left[(\rho_k^{-1} - \rho^{-1})\rho_j^{-1} + (\rho^{-1} - \rho_{k+1}^{-1})\rho_j^{-1} \right] = \frac{1}{2\rho_j}.$$

It follows that $\text{Cov}(Y_\rho, Y_{\rho_j}) = \frac{\Delta_2^2}{2\max(\rho, \rho_j)}$ for $j \in [m]$, and so we are done with this part of the covariances.

For the last step, we consider what happens when $\rho_r = \rho_\infty < \infty$. In this case, $\rho_l = \rho_{n-1}$ and so for $j \in [n-1]$:

$$\begin{aligned} \text{Cov}(Y_\rho, Y_{\rho_j}) &= \text{Cov} \left(\frac{(1 - \rho_{n-1}/\rho)Y_{\rho_\infty} + (\rho_{n-1}/\rho - \rho_{n-1}/\rho_\infty)Y_{\rho_{n-1}}}{2(1 - \rho_{n-1}/\rho_\infty)}, Y_{\rho_j} \right) \\ &= \frac{1 - \rho_{n-1}/\rho}{2(1 - \rho_{n-1}/\rho_\infty)} \cdot \text{Cov}(Y_{\rho_\infty}, Y_{\rho_j}) + \frac{\rho_{n-1}/\rho - \rho_{n-1}/\rho_\infty}{2(1 - \rho_{n-1}/\rho_\infty)} \cdot \text{Cov}(Y_{\rho_{n-1}}, Y_{\rho_j}) \\ &= \frac{1 - \rho_{n-1}/\rho}{2(\rho_\infty - \rho_{n-1})} + \frac{1/\rho - 1/\rho_\infty}{2(1 - \rho_{n-1}/\rho_\infty)} = \frac{1 - \rho_{n-1}/\rho + \rho_\infty/\rho - 1}{2(\rho_\infty - \rho_{n-1})} = \frac{1}{2\rho}, \end{aligned}$$

and now we are done. $\square$

6.6.4 Details on (Efficient) Weakly Lossless Gradual Release of Sparse Gaussian Histograms

Algorithm 11 implements *weakly lossless gradual release* of private histograms. We prove this next.

Proof of Lemma 6.13. Observe that for any increasing sequence $(\rho_k)_{k \in m}$, the corresponding sequence of variables Z on line 3 produced by the algorithm, have the same distribution as in Lemma 6.2 for $\beta = 0$. The Y that is ultimately returned, however, is a post-processing of $H(\mathbf{x}) + Z$, which has the same distribution as Lemma 6.2 for $\beta = H(\mathbf{x})$. Therefore Algorithm 11 is implementing lossless gradual release for the Gaussian mechanism applied to $H(\mathbf{x})$, combined with a non-invertible post-processing. The lemma statement follows from Corollary 6.9. $\square$

Note that one *only* has to store the noisy terms from the preceding round to implement Algorithm 11. Nevertheless, it might be infeasible, say, when the domain is really large, to sample the noise for the zero coordinates. Algorithm 14 uses the computational trick in Lemma 6.14 to speed up this computation. The algorithm is static, where the privacy budgets are fixed upfront but can easily be converted to

an online algorithm. Recall that $U(H(\mathbf{x})) = \{i \in [d] : H(\mathbf{x})_i \neq 0\}$ is the support of the histogram.

Algorithm 14 EfficientHistogramGradualRelease

Parameters: ℓ_2 -sensitivity Δ_2

Require: Private histogram $H(\mathbf{x})$, privacy budgets $\rho_1 < \dots < \rho_m$, thresholds $\tau_1, \dots, \tau_m$

```

1: Let  $S^{(0)} = U(H(\mathbf{x}))$  // Tracking counts that have already been released
2: Let  $Z^{(0)} = \{0\}^d$  and  $\rho_0 = 0$ .
3:  $\forall r \in [m]$  : let distribution  $\mathcal{D}^{(r)} = \mathcal{N}\left(0, \frac{1}{2}\Delta_2^2(\rho_r - \rho_{r-1})\right)$ .
4: for each round  $r \in [m]$  do
5:   Initialize  $Y^{(r)} = \{0\}^d$ .
6:   for each tracked count in preceding round  $j \in S^{(r-1)}$  do
7:     Draw fresh noise  $\tilde{Z}_j^{(r)} \sim \mathcal{D}^{(r)}$ .
8:     Update aggregate noise  $Z_j^{(r)} = \frac{\rho_{r-1}}{\rho_r} Z_j^{(r-1)} + \frac{1}{\rho_r} \tilde{Z}_j^{(r)}$ .
9:     if  $H(\mathbf{x})_j + Z_j^{(r)} > \tau_r$  then  $Y_j^{(r)} = H(\mathbf{x})_j + Z_j^{(r)}$ .
10:  Let  $\tilde{Z}^{(k)} \sim \mathcal{D}^{(k)}$  for  $k \in [r]$  be random variables.
11:  Compute  $p^{(r)} = \mathbf{P}\left[\sum_{k=1}^r \tilde{Z}^{(k)} > \tau_r \rho_r \mid \{\forall \ell \in [r-1] : \sum_{k=1}^\ell \tilde{Z}^{(k)} \leq \rho_\ell \tau_\ell\}\right]$ .
12:  Draw  $q \sim \text{Bin}\left(d - |S^{(r-1)}|, p^{(r)}\right)$ .
13:  Select a subset  $Q \subseteq [d] \setminus S^{(r-1)}$  uniformly at random of size  $q$ .
14:  for each index  $j \in Q$  do
15:    Initialize  $Z_j^{(r)} = 0$ .
16:    for  $k \in [r-1]$  do
17:      Draw fresh noise  $\tilde{Z}_j^{(k)} \sim \mathcal{D}^{(k)}$  conditioned on  $\tilde{Z}_j^{(k)} \leq \rho_k \tau_k - Z_j^{(r)}$ .
18:      Update aggregate noise  $Z_j^{(r)} = Z_j^{(r)} + \tilde{Z}_j^{(k)}$ .
19:      Draw fresh noise  $\tilde{Z}_j^{(k)} \sim \mathcal{D}^{(k)}$  conditioned on  $\tilde{Z}_j^{(k)} > \rho_k \tau_k - Z_j^{(r)}$ .
20:      Update aggregate noise  $Z_j^{(r)} = Z_j^{(r)} + \tilde{Z}_j^{(k)}$ .
21:     $Y_j^{(r)} = H(\mathbf{x})_j + Z_j^{(r)}$  // Guaranteed to be above the threshold.
22:  Set  $S^{(r)} = S^{(r-1)} \cup Q$ .
23:  output private histogram  $Y^{(r)}$ .
```

We proceed to give proofs for Lemma 6.15 and 6.16.

Proof sketch for Lemma 6.15. The claim directly follows an inductive argument. For the base case, note that the first iteration of Algorithm 14 is running the same routine as described by Lemma 6.14, and so $\hat{Y}^{(1)}$ must be identically distributed to $Y^{(1)}$. For our inductive hypothesis, assume that the subsequences $(Y^{(1)}, \dots, Y^{(k)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(k)})$ are identically distributed. Note that for the $(k+1)^{\text{st}}$ release, Algorithm 14 handles all the true nonzero counts and every zero count that has ever exceeded the threshold in a past round in the same way. These counts in $\hat{Y}^{(k+1)}$ and $Y^{(k+1)}$ clearly have the same distribution.

Now, note that each of the remaining zero counts have up to and including the k^{th} round been reported as zero in each round, and so their probability of exceeding the threshold, $p^{(k+1)}$, should be equal across all of them.

The sampling performed by Algorithm 14 is structured in the same manner as Lemma 6.14, but with sampling probability $p^{(k+1)}$, and the noise terms added for any zero count exceeding the threshold, are different. For the distributions to match, the probability of exceeding the threshold and the noise distribution for an element chosen to exceed the threshold are more complex. The probability of exceeding the threshold is conditioned on prefixes of Gaussians, which correctly simulates the probability of not exceeding the threshold at any prior round until first doing so in the $(k+1)^{\text{st}}$ one. The noise added once selected to exceed the threshold is a sum of truncated Gaussians, which simulate the same event. As the principle is built on the same logic as Lemma 6.14, we have that $(Y^{(1)}, \dots, Y^{(k+1)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(k+1)})$ are identically distributed, and so the claim is true by induction. $\square$

Proof of Lemma 6.16. Note that each of the k non-zero true counts will, in each round, have fresh noise added on line 7. If a non-zero count is selected by the binomial sampling in the k^{th} round, the for-loop starting on line 17 will result in sampling k noise terms, after which for future rounds it will get treated as a non-zero count. It follows that non-zero entries contribute km samples, and zero counts exceeding the threshold contribute cm samples. $\square$

6.6.5 The Poisson Mechanism

We are not aware of any explicit statements in the literature on the privacy guarantees obtained by adding Poisson distributed noise to a d -dimensional vector of integers. However, since the Poisson distribution is the limiting distribution of binomial distributions with the same mean $\lambda = Np$, where N is the number of trials, such bounds can be derived from existing bounds on the binomial mechanism. For the sake of completeness, we include such statements based on the following theorem from [Aga+18]:

Theorem 6.19 (Agarwal, Suresh, Yu, Kumar, and McMahan [Aga+18]). For any δ , parameters N, p and sensitivity bounds $\Delta_1, \Delta_2, \Delta_\infty$ such that

$$Np(1-p) \geq \max \left(23 \log \left(\frac{10d}{\delta} \right), 2\Delta_\infty \right),$$

the d -dimensional Binomial mechanism is (ϵ, δ) -differentially private for

$$\epsilon = \frac{\Delta_2 \sqrt{2 \log \frac{1.25}{\delta}}}{\sqrt{Np(1-p)}} + \frac{\Delta_2 c_p \sqrt{\log \frac{10}{\delta}} + \Delta_1 b_p}{Np(1-p)(1-\delta/10)} + \frac{\frac{2}{3}\Delta_\infty \log \frac{1.25}{\delta} + \Delta_\infty d_p \log \frac{20d}{\delta} \log \frac{10}{\delta}}{Np(1-p)},$$

where

$$d_p \triangleq \frac{4}{3} \cdot (p^2 + (1-p)^2), \quad b_p \triangleq \frac{2(p^2 + (1-p)^2)}{3} + (1-2p), \\ c_p \triangleq \sqrt{2} \left(3p^3 + 3(1-p)^3 + 2p^2 + 2(1-p)^2 \right).$$

3034 Setting $p = \lambda/N$ and considering the limiting bound when $N \rightarrow \infty$, we get:

Lemma 6.20 (Privacy guarantees of the Poisson Mechanism). The d -dimensional Poisson mechanism with parameter $\lambda > \max(23 \log(10d/\delta), 2\Delta_\infty)$, is (ϵ, δ) -DP with

$$\epsilon = \frac{\Delta_2 \sqrt{2 \log \frac{1.25}{\delta}}}{\sqrt{\lambda}} + \frac{5\sqrt{2} \Delta_2 \sqrt{\log \frac{10}{\delta}} + \frac{5}{3}\Delta_1}{\lambda(1-\delta/10)} \\ + \frac{\frac{2}{3}\Delta_\infty \log \frac{1.25}{\delta} + \frac{4}{3}\Delta_\infty \log \frac{20d}{\delta} \log \frac{10}{\delta}}{\lambda}.$$

3035 A simpler expression can be derived for unit sensitivities by relaxing the con-
3036 stants and assuming that δ is not too large:

3037 **Corollary 6.21** (Simplified upper bound with unit sensitivities). Assume that $\Delta_1 =$
3038 $\Delta_2 = \Delta_\infty = 1$. Then for $\delta < 1/100$, the d -dimensional Poisson mechanism with parameter
3039 $\lambda > 23 \log(10d/\delta)$ is (ϵ, δ) -differentially private for $\epsilon = \frac{\sqrt{2 \log \frac{1.25}{\delta}}}{\sqrt{\lambda}} + \frac{2 \log \frac{20d}{\delta} \log \frac{10}{\delta}}{\lambda}$.

3040 We will need the following lemma that determines the sampling step of private
3041 lossless multiple release for the Poisson mechanism:

Lemma 6.22. Let $X_1 \sim \text{Poi}(\lambda_1)$ and $X_2 \sim \text{Poi}(\lambda_2)$ be independent Poisson random variables. Then, for any nonnegative integer k , the conditional distribution of X_1 given $X_1 + X_2 = k$ is

$$X_1 \mid (X_1 + X_2 = k) \sim \text{Binomial} \left(k, \frac{\lambda_1}{\lambda_1 + \lambda_2} \right).$$

3042 *Proof.* Since X_1 and X_2 are independent, their joint probability mass function is for
3043 all $x = 0, 1, \dots, k$

$$PX_1 = x, X_2 = k - x = e^{-(\lambda_1 + \lambda_2)} \frac{\lambda_1^x}{x!} \frac{\lambda_2^{k-x}}{(k-x)!}.$$

Moreover, the sum $X_1 + X_2$ is Poisson with parameter $\lambda_1 + \lambda_2$, so that

$$\mathbf{P}[X_1 + X_2 = k] = e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}.$$

Thus, by the definition of conditional probability,

$$\begin{aligned} \mathbf{P}[X_1 = x \mid X_1 + X_2 = k] &= \frac{\mathbf{P}[X_1 = x, X_2 = k - x]}{\mathbf{P}[X_1 + X_2 = k]} = \frac{e^{-(\lambda_1 + \lambda_2)} \frac{\lambda_1^x}{x!} \frac{\lambda_2^{k-x}}{(k-x)!}}{e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}} \\ &= \binom{k}{x} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{k-x}. \end{aligned}$$

This is precisely the probability mass function of a Binomial random variable with parameters k and $\frac{\lambda_1}{\lambda_1 + \lambda_2}$. $\square$

6.6.6 The Laplace Mechanism

Koufogiannis, Han, and Pappas [KHP16] showed how to do lossless *gradual* releases for the Laplace mechanism and supports either tightening the privacy guarantees or loosening them. We will now strengthen this result by exactly showing how to do them in arbitrary order, similar to what was done in Section 6.3 for the Gaussian mechanism. We will first show that the Laplace distribution satisfies Definition 6.1. This already implies the existence of an algorithm supporting gradual lossless release via Lemma 6.6. After, we will derive how to sample a new release with scale parameter b given two distinct releases with scaling $b_2 < b$ and $b < b_1$.

Lemma 6.23 ([KKP01] Characteristic function of Laplace). *Let X be Laplace random variable with probability density function as in Definition A.2, then for all $t \in \mathbb{R}$, the characteristic function of X is*

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} \frac{1}{2b} e^{-|x|/b} dx = \frac{1}{1 + b^2 t^2}.$$

Proof. By a simple integration:

$$\begin{aligned} \mathbb{E}[e^{itX}] &= \frac{1}{2b} \int_{-\infty}^{\infty} e^{itx - |x|/b} dx = \frac{1}{2b} \left(\int_{-\infty}^0 e^{(1/b + ti)x} dx + \int_0^{\infty} e^{(-1/b + ti)x} dx \right) \\ &= \frac{1}{2b} \left(\frac{1}{1/b + ti} + \frac{1}{1/b - ti} \right) = \frac{1}{1 + b^2 t^2}. \end{aligned} \quad \square$$

We will now show that the zero-centered Laplace distribution with scale parameter b satisfies convolution preorder (Definition 6.1). Note that a larger value of b corresponds to a more private release by definition. Building on this, we will condition on the two closest releases to create a new one in the middle, as done in Lemma 6.4 for the Gaussian. The following fact was already shown in Koufogiannis, Han, and Pappas [KHP16], but we give a proof here for completeness using a technique based on characteristic functions as done by Equitz and Cover [EC91].

Claim 6.24 (Convolution preorder: Laplace). Fix $b_2 < b_1 \in \mathbb{R}^+$ let $X \sim \text{Lap}(0, b_2)$ and draw

$$W = \begin{cases} 0 & \text{with probability } b_2^2/b_1^2, \\ \text{Lap}(0, b_1) & \text{otherwise} \end{cases}, \quad \text{then } X + W \sim \text{Lap}(0, b_1).$$

Proof. We know that the characteristic function of X is $\varphi_X(t) = \frac{1}{1+b_2^2 t^2}$ and for the convolution $\varphi_{X+W} = \frac{1}{1+b_1^2 t^2}$. Because of independence, the convolution is defined for all t as:

$$\begin{aligned} \varphi_X(t)\varphi_W(t) &= \varphi_{X+W}(t) \\ \Leftrightarrow \varphi_W(t) &= \frac{1+b_2^2 t^2}{1+b_1^2 t^2} = \frac{b_1^2 + b_1^2 b_2^2 t^2}{b_1^2(1+b_1^2 t^2)} = \frac{b_2^2(1+b_1^2 t^2) + b_1^2 - b_2^2}{b_1^2(1+b_1^2 t^2)} \\ &= \frac{b_2^2}{b_1^2} + \left(1 - \frac{b_2^2}{b_1^2}\right) \cdot \frac{1}{1+b_1^2 t^2}. \end{aligned}$$

The last expression encodes the convex combination of the claimed mixture distribution because $\frac{1}{1+b_1^2 t^2}$ is again the characteristic function of a zero-centered Laplace distribution with scaling parameter b_1 . $\square$

We next prove a simple result about the convolution of two Laplace distributions (compare also eq., 2.3.23 of Kotz, Kozubowski, and Podgórski [KKPo1]).

Lemma 6.25 (Convolution). For two fixed scaling parameters $b_1 \neq b_2 \in \mathbb{R}^+$, let $X_1 \sim \text{Lap}(0, b_1)$ and $X_2 \sim \text{Lap}(0, b_2)$. Then the density of $X_1 + X_2$ is given by

$$(f_{b_1} * f_{b_2})(t) = \frac{1}{2(b_1^2 - b_2^2)} \left(b_1 e^{-|t|/b_1} - b_2 e^{-|t|/b_2} \right).$$

Proof. Assume $t \geq 0$ and note that the other case follows by symmetry. We compute the density of the convolution by a straightforward integration:

$$\begin{aligned} (f_{b_1} * f_{b_2})(t) &= \int_{-\infty}^{\infty} f_{b_1}(x) f_{b_2}(t-x) dx = \frac{1}{4b_1 b_2} \int_{-\infty}^{\infty} \exp\left(-\frac{|x|}{b_1} - \frac{|t-x|}{b_2}\right) dx \\ &= \frac{1}{4b_1 b_2} \cdot \left(\int_{-\infty}^0 e^{\frac{x}{b_1} - \frac{t-x}{b_2}} dx + \int_0^t e^{-\frac{x}{b_1} - \frac{t-x}{b_2}} dx + \int_t^{\infty} e^{-\frac{x}{b_1} - \frac{x-t}{b_2}} dx \right) \\ &= \frac{1}{4b_1 b_2} \cdot \left(\left[\frac{\exp(x/b_1 - (t-x)/b_2)}{b_1^{-1} + b_2^{-1}} \right]_{-\infty}^0 \right. \\ &\quad \left. + \left[\frac{\exp(-x/b_1 - (t-x)/b_2)}{b_2^{-1} - b_1^{-1}} \right]_0^t + \left[\frac{\exp(-x/b_1 - (x-t)/b_2)}{-b_2^{-1} - b_1^{-1}} \right]_t^{\infty} \right) \\ &= \frac{1}{4} \cdot \left(\frac{\exp(-t/b_2)}{b_1 + b_2} + \frac{\exp(-t/b_1) - \exp(-t/b_2)}{b_1 - b_2} + \frac{\exp(-t/b_1)}{b_1 + b_2} \right) \\ &= \frac{1}{2(b_1^2 - b_2^2)} \left(b_1 e^{-t/b_1} - b_2 e^{-t/b_2} \right). \quad \square \end{aligned}$$

Now, we are ready to show that the Laplace mechanism can be implemented with multiple releases.

Lemma 6.26 (Multiple release Laplace). For fixed $0 < b_2 < b < b_1$, let $\mu_1 = b^2/b_1^2$ and $\mu_2 = b^2/b_2^2$ and $D_1 \sim \text{Ber}(\mu_1)$ and $D_2 \sim \text{Ber}(\mu_2)$. Furthermore, let

$$X_1 = \begin{cases} 0 & \text{if } D_1 = 1 \\ \text{Lap}(0, b) & \text{otherwise} \end{cases} \quad \text{and} \quad X_2 = \begin{cases} 0 & \text{if } D_2 = 1 \\ \text{Lap}(0, b_2) & \text{otherwise} \end{cases}.$$

Then we have that $\mathbf{P}[X_1 = 0 \mid X_1 + X_2 = 0] = 1$, and for every real number $k \neq 0$ the distribution of X_1 conditioned on $X_1 + X_2 = k$ is:

$$X_1 \mid (X_1 + X_2 = k) \sim \begin{cases} 0 & w.p. \mu_1 \cdot (1 - \mu_2) \cdot \frac{\exp(-|k|/b_2)}{2b_2 \cdot f_{X_1+X_2}(k)} \\ k & w.p. (1 - \mu_1) \cdot \mu_2 \cdot \frac{\exp(-|k|/b)}{2b \cdot f_{X_1+X_2}(k)} \\ H(b, b_2, k) & w.p. (1 - \mu_1) \cdot (1 - \mu_2) \cdot \frac{(f_b * f_{b_2})(k)}{f_{X_1+X_2}(k)} \end{cases} \quad (4)$$

where 0 and k denote the constant distributions, $H(b, b_2, k)$ is the probability distribution with probability density function $h(x) = \frac{f_b(x)f_{b_2}(k-x)}{(f_b * f_{b_2})(k)}$,

$$\begin{aligned} f_{X_1+X_2}(k) &= \frac{(1 - \mu_1)\mu_2}{2b} e^{-|k|/b} \frac{(1 - \mu_1)(1 - \mu_2)}{2(b^2 - b_2^2)} (be^{-|k|/b} - b_2e^{-|k|/b_2}) \\ &\quad + \mu_1\mu_2\delta_0(x) + \frac{\mu_1(1 - \mu_2)}{2b_2} e^{-|k|/b_2}, \text{ and} \\ (f_b * f_{b_2})(k) &= \frac{1}{2(b_1^2 - b_2^2)} (b_1e^{-|k|/b_1} - b_2e^{-|k|/b_2}), \end{aligned}$$

where δ_0 is the Dirac delta function.

3077 *Proof.* First we consider $\mathbf{P}[X_1 = 0 \mid X_1 + X_2 = 0]$, arguing that

$$\mathbf{P}[X_1 = 0 \mid X_1 + X_2 = 0] = \frac{\mathbf{P}[X_1 = 0] \mathbf{P}[X_2 = 0]}{\mathbf{P}[X_1 + X_2 = 0]} = \frac{\mu_1\mu_2}{\mu_1\mu_2 + 0} = 1.$$

3078 To see why the first equality holds, split $\mathbf{P}[X_1 + X_2 = 0]$ into each of the four
3079 combinations of discrete/continuous for X_1, X_2 . The only non-zero contribution
3080 to the probability mass comes from the discrete/discrete case, as the remaining
3081 cases contribute mass proportional to the probability that a continuous distribution
3082 assumes an exact value, which is zero.

3083 Next we turn to the distribution of X_1 conditioned on $X_1 + X_2 = k$ where $k \neq 0$.
3084 Denote by f_b the probability density function of $\text{Lap}(0, b)$. Note that the mixture
3085 densities become

$$\begin{aligned} f_{X_1}(x) &= \mu_1\delta_0(x) + (1 - \mu_1)f_b(x), \\ f_{X_2}(x) &= \mu_2\delta_0(x) + (1 - \mu_2)f_{b_2}(x). \end{aligned}$$

3086 Before analyzing the different cases separately, we compute the pdf of the convolu-
3087 tion $X_1 + X_2$.

3088 **Convolution** $f_{X_1+X_2}$.

3089 Note that we have to take care of a subtle technicality: the case that both X_1
3090 and X_2 are zero from the discrete part can only happen when we condition on

3091 $X_1 + X_2 = 0$. We can now give the density of the convolution $X_1 + X_2$ at any real
3092 point x :

$$\begin{aligned}
 f_{X_1+X_2}(x) &= \sum_{(d_1, d_2) \in \{0,1\}^2} f_{X_1+X_2|D_1=d_1, D_2=d_2}(x) \cdot \mathbf{P}[D_1 = d_1 \wedge D_2 = d_2] \\
 &= \mu_1(1 - \mu_2) \cdot f_{b_2}(x) + (1 - \mu_1)\mu_2 \cdot f_b(x) + (1 - \mu_1)(1 - \mu_2) \cdot (f_b * f_{b_2})(x) \\
 &\quad + \mu_1\mu_2\delta_0(x) \\
 &= \frac{\mu_1(1 - \mu_2)}{2b_2} e^{-|x|/b_2} + \frac{(1 - \mu_1)\mu_2}{2b} e^{-|x|/b} \\
 &\quad + (1 - \mu_1)(1 - \mu_2) \frac{1}{2(b^2 - b_2^2)} \left(b e^{-|x|/b} - b_2 e^{-|x|/b_2} \right) + \mu_1\mu_2\delta_0(x),
 \end{aligned}$$

3093 where the third term follows from Lemma 6.25. Note that the last term only
3094 contributes when $x = 0$.

3095 We can now show how the sampling procedure in the claim is justified. We first
3096 assume $k \neq 0$ and analyze three possible cases how $X_1 + X_2$ is built up: Either both
3097 of them are drawn from the (continuous) Laplace distribution or exactly one. (The
3098 case where both are from their respective discrete parts can only happen if $k = 0$,
3099 analyzed above.)

3100 **Case 1:** $X_1 = 0$ and $X_2 = k$.

3101 $X_2 = k$ is necessarily from its continuous part. By the definition of conditional
3102 probability, we have for $k \neq 0$:

$$\begin{aligned}
 \mathbf{P}[X_1 = 0 | X_1 + X_2 = k, k \neq 0] &= \frac{\mathbf{P}[X_1 = 0 \wedge X_1 + X_2 = k]}{\mathbf{P}[X_1 + X_2 = k]} = \frac{\mathbf{P}[X_1 = 0] \mathbf{P}[X_2 = k]}{\mathbf{P}[X_1 + X_2 = k]} \\
 &= \mu_1 \frac{f_{X_2}(k)}{f_{X_1+X_2}(k)} = \frac{\mu_1(1 - \mu_2)}{2b_2 f_{X_1+X_2}(k)} \cdot e^{-|k|/b_2} := p_1.
 \end{aligned}$$

3103 For the third equality, we simply used definition of a probability density function
3104 via its limit:

$$\lim_{\Delta \rightarrow 0^+} \frac{\mathbf{P}[X_2 \in [k - \Delta, k + \Delta]]}{\mathbf{P}[X_1 + X_2 \in [k - \Delta, k + \Delta]]} = \frac{f_{X_2}(k)}{f_{X_1+X_2}(k)}.$$

3105 **Case 2:** $X_1 = k$ and $X_2 = 0$.

3106 Now assume the flipped case. By a similar argument:

$$\begin{aligned}
 p_2 := \mathbf{P}[X_1 = k | X_1 + X_2 = k] &= \frac{\mathbf{P}[X_1 = k, X_1 + X_2 = k]}{\mathbf{P}[X_1 + X_2 = k]} \\
 &= \frac{\mathbf{P}[X_1 = k] \mathbf{P}[X_2 = 0]}{\mathbf{P}[X_1 + X_2 = k]} = \frac{(1 - \mu_1)\mu_2}{2b f_{X_1+X_2}(k)} \cdot e^{-|k|/b}.
 \end{aligned}$$

3107 **Case 3:** $X_1 \neq 0, X_2 \neq 0, X_1 + X_2 = k$.

3108 In the remaining case, both X_1 and X_2 are independently sampled from continuous
3109 Laplace distributions with probability density functions $f_b(x)$ and $f_{b_2}(x)$. Therefore,
3110 with the remaining probability $1 - (p_1 + p_2)$, we know that X_1 is sampled according
3111 to the following conditional probability density function:

$$f_{X_1}(x) |_{X_1+X_2=k} = \frac{f_{X_1, X_1+X_2}(x, k)}{(f_b * f_{b_2})(k)} = \frac{f_b(x) f_{b_2}(k - x)}{(f_b * f_{b_2})(k)},$$

where the last line follows from the trivial identity $X_1 + X_2 = k \Leftrightarrow X_2 = k - X_1$.
 This is a valid probability density function because f is trivially non-negative due
 to its parts being non-negative and furthermore

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{f_b(x)f_{b_2}(x-k)}{(f_b * f_{b_2})(k)} dx &= \frac{1}{(f_b * f_{b_2})(k)} \int_{-\infty}^{\infty} f_b(x)f_{b_2}(k-x) dx \\ &= \frac{(f_b * f_{b_2})(k)}{(f_b * f_{b_2})(k)} = 1. \end{aligned}$$

What is left is to verify that the probabilities in Equation (4) indeed add up to
 one for $k \neq 0$:

$$\begin{aligned} &\frac{(1-\mu_1)(1-\mu_2)(f_b * f_{b_2})(k)}{f_{X_1+X_2}(k)} + \frac{(1-\mu_1)\mu_2 \exp(-|k|/b)}{2bf_{X_1+X_2}(k)} + \frac{\mu_1(1-\mu_2) \exp(-|k|/b_2)}{2b_2f_{X_1+X_2}(k)} \\ &= \frac{1}{f_{X_1+X_2}(k)} \cdot \\ &\quad \underbrace{\left((1-\mu_1)(1-\mu_2)(f_b * f_{b_2})(k) + \frac{(1-\mu_1)\mu_2 \exp(-|k|/b)}{2b} + \frac{\mu_1(1-\mu_2) \exp(-|k|/b_2)}{2b_2} \right)}_{f_{X_1+X_2}(k) \text{ for } k \neq 0} \\ &= 1 \end{aligned} \quad \square$$

6.6.7 Showing Convolution Preorder for Exponential Noise

The exponential distribution is closely related to the Laplace distribution but gives
 poor differential privacy guarantees. Nevertheless, it still serves as a building block
 for some private mechanisms, e.g., Report-Noisy-Max [Din+21]. We show next that
 it also satisfies a convolution preorder.

Definition 6.5 (Exponential distribution characteristic function). *The characteristic function of the Exponential Distribution is given by $\varphi_X(t) = \mathbb{E}[e^{itX}] = \frac{\lambda}{\lambda - it}$*

Claim 6.27 (Convolution preorder: Exponential distribution). *Fix $\lambda_1 < \lambda_2 \in \mathbb{R}_+$, let $X \sim \text{Exp}(\lambda_2)$ and draw*

$$W = \begin{cases} 0 & \text{with probability } \lambda_1/\lambda_2, \\ \text{Exp}(\lambda_1) & \text{otherwise} \end{cases}, \quad \text{then } X + W \sim \text{Exp}(\lambda_1).$$

Proof. Using the same trick as in the proof of Lemma 6.24, we have that

$$\begin{aligned} \varphi_W(t) &= \frac{\varphi_{X+W}(t)}{\varphi_X(t)} = \frac{\lambda_1}{\lambda_2} \cdot \frac{\lambda_2 - it}{\lambda_1 - it} \\ &= \frac{\lambda_1}{\lambda_2} \left(1 + \frac{\lambda_2 - \lambda_1}{\lambda_1 - it} \right) = \frac{\lambda_1}{\lambda_2} + \left(1 - \frac{\lambda_1}{\lambda_2} \right) \varphi_{X+W}(t), \end{aligned}$$

where the final expression is the characteristic function of the claimed mixture
 distribution. $\square$

6.6.8 Convolution Preorder for Discrete Laplace

Alongside the Discrete Gaussian, the Discrete Laplace is fundamental with huge practical impact because it enables implementation on a finite machine. The existence of a convolution preorder for the Laplace distribution already suggests that something similar might be possible for the Discrete Laplace as well. Indeed, the following proof uses the same base idea of studying the characteristic function.

The following result is not included in the published version of the paper. The discrete Laplace distribution $\text{DLap}(\alpha)$ is a two-sided geometric distribution with probability mass function $p(x) = \frac{1-\alpha}{1+\alpha} \cdot \alpha^{|x|}$ for $\alpha \in (0, 1)$ and $x \in \mathbb{Z}$. It is well known that the discrete counterpart to the continuous Laplace mechanism satisfies pure DP if one releases $f(\mathbf{x}) + \text{DLap}(\alpha)$ for $\alpha = \exp(-\varepsilon/\Delta)$ where Δ is the ℓ_1 sensitivity of the query f [GRS09]. In the following, we will need the characteristic function of the Discrete Laplace which we quickly reprove for convenience.

Claim 6.28 (Characteristic function of the Discrete Laplace). *For $\alpha \in (0, 1)$, the characteristic function of $X \sim \text{DLap}(\alpha)$ is for all t :*

$$\phi_X(t) = \frac{(1-\alpha)^2}{1-2\alpha \cos(t) + \alpha^2}.$$

Proof.

$$\begin{aligned} \phi_X(t) &= \mathbb{E}[\exp(itX)] = \frac{1-\alpha}{1+\alpha} \cdot \left(1 + \sum_{k \geq 1} e^{itk} \alpha^k + \sum_{k \geq 1} e^{-itk} \alpha^k \right) \\ &= \frac{1-\alpha}{1+\alpha} \cdot \left(1 + \sum_{k \geq 1} (e^{it} \alpha)^k + \sum_{k \geq 1} (e^{-it} \alpha)^k \right) \\ &= \frac{1-\alpha}{1+\alpha} \cdot \left(1 + \frac{e^{it} \alpha}{1 - e^{it} \alpha} + \frac{e^{-it} \alpha}{1 - e^{-it} \alpha} \right) \\ &= \frac{1-\alpha}{1+\alpha} \cdot \left(\frac{(1 - e^{it} \alpha)(1 - e^{-it} \alpha) + (e^{it} \alpha)(1 - e^{-it} \alpha) + e^{-it} \alpha(1 - e^{it} \alpha)}{(1 - e^{it} \alpha)(1 - e^{-it} \alpha)} \right) \\ &= \frac{(1-\alpha)^2}{1-2\alpha \cos(t) + \alpha^2}, \end{aligned}$$

where we used the identity $e^{it} + e^{-it} = 2 \cos(t)$ by Euler's formula, and the convergence of the geometric sum is justified because the convergence radius is $|\alpha e^{it}| = \alpha \sqrt{\cos(t)^2 + \sin(t)^2} = \alpha < 1$ by assumption on α (and similarly for $|\alpha e^{-it}|$). $\square$

Claim 6.29 (Deconvolution Laplace). *Fix two scaling parameters $0 < \alpha_x \leq \alpha_z < 1$. Let $X \sim \text{DLap}(\alpha_x)$ and $Z \sim \text{DLap}(\alpha_z)$. Then $\phi_X(t)\phi_Y(t) = \phi_Z(t)$ if we set,*

$$\phi_Y(t) = \psi(t)\psi(-t) \quad \text{with } \psi(t) = \frac{1-\alpha_z}{1-\alpha_x} \cdot \frac{1-\alpha_x \exp(it)}{1-\alpha_z \exp(it)}. \quad (5)$$

Proof. Plugging in Lemma 6.28 gives,

$$\phi_Y(t) = \frac{\phi_Z(t)}{\phi_X(t)} = \frac{(\alpha_z - 1)^2}{(\alpha_x - 1)^2} \cdot \frac{1 - 2\alpha_x \cos(t) + \alpha_x^2}{1 - 2\alpha_z \cos(t) + \alpha_z^2}.$$

3143 Observe that the characteristic function of this deconvolution inherently is a
 3144 convolution of some other distribution. Therefore, using the identity $2 \cos(t) =$
 3145 $\exp(it) + \exp(-it)$ gives the factorization in the claim:

$$\begin{aligned}\phi_Y(t) &= \left(\frac{1 - \alpha_z}{1 - \alpha_x} \right)^2 \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)} \cdot \frac{1 - \alpha_x \exp(-it)}{1 - \alpha_z \exp(-it)} \\ &= \left(\frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)} \right) \left(\frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(-it)}{1 - \alpha_z \exp(-it)} \right) \\ &= \psi(t)\psi(-t) \quad \text{for } \psi(t) = \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)}.\end{aligned}$$

3146

□

Theorem 6.30 (Sampling from Deconvolution). *Fix $0 < \alpha_x \leq \alpha_z < 1$ scaling parameters as before. Let $X \sim \text{DLap}(\alpha_x)$ and Y_1, Y_2 be iid, independent of X as*

$$\mathbf{P}[Y_i = 0] = \frac{1 - \alpha_z}{1 - \alpha_x}, \quad \mathbf{P}[Y_i = k] = \left(1 - \frac{1 - \alpha_z}{1 - \alpha_x}\right) (1 - \alpha_z) \alpha_z^{k-1}, \quad k \geq 1.$$

Equivalently,

$$Y_1, Y_2 \sim \begin{cases} 0 & \text{with probability } \frac{1 - \alpha_z}{1 - \alpha_x}, \\ \text{Geom}(1 - \alpha_z) & \text{otherwise} \end{cases}.$$

Then $X + (Y_1 - Y_2) \sim \text{DLap}(\alpha_z)$.

3147 *Proof.* We will now analyze the distribution with the characteristic function $\psi(t)$
 3148 more carefully.

$$\begin{aligned}\psi(t) &= \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)} \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \left(1 + \frac{(\alpha_z - \alpha_x) \exp(it)}{1 - \alpha_z \exp(it)} \right) \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} \exp(0 \cdot it) \cdot \left(1 + (\alpha_z - \alpha_x) \sum_{k=1}^{\infty} \left(\alpha_z^{k-1} \exp(kit) \right) \right) \quad (\text{Geom series}) \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \sum_{k=1}^{\infty} \left(\alpha_z^{k-1} \exp(kit) \right) \\ &= \underbrace{\frac{1 - \alpha_z}{1 - \alpha_x}}_{a_1} + \underbrace{\frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \exp(it)}_{a_2} \\ &\quad + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \sum_{k=2}^{\infty} \left(\alpha_z^{k-1} \exp(kit) \right).\end{aligned}\tag{6}$$

Observe that these coefficients a_i (without the complex exponential part) are by definition non-negative and sum up to

$$\begin{aligned} \sum_{i=1}^{\infty} a_i &= \frac{1 - \alpha_z}{1 - \alpha_x} + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \sum_{k=1}^{\infty} \alpha_z^{k-1} \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \frac{1}{1 - \alpha_z} \\ &= \frac{1 - \alpha_z + \alpha_z - \alpha_x}{1 - \alpha_x} = 1. \end{aligned}$$

Recall that $\exp(ita)$ is the characteristic function of the Dirac delta δ_a , the distribution that outputs a with probability one. Therefore, we can interpret $\psi(t)$ in Equation (6) as the distribution that draws 0 with probability $\frac{1 - \alpha_z}{1 - \alpha_x}$ and otherwise some $k \in \mathbb{N}_{>0}$ with probability

$$\begin{aligned} \tilde{p} &= \alpha_z^{k-1} \left(\frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \right) \cdot \left(1 - \frac{1 - \alpha_z}{1 - \alpha_x} \right)^{-1} \\ &= \alpha_z^{k-1} \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \cdot \frac{1 - \alpha_x}{\alpha_z - \alpha_x} \\ &= \alpha_z^{k-1} (1 - \alpha_z). \end{aligned}$$

Observe that the second case (conditioned on not zero) is actually a draw from a geometric distribution $\text{Geom}(1 - \alpha_z)$! Observe by independence,

$$\phi_{Y_1 - Y_2}(t) = \phi_{Y_1}(t) \phi_{Y_2}(-t) = \psi(t) \psi(-t).$$

Hence, it follows that $X + (Y_1 - Y_2) \sim \text{DLap}(\alpha_z)$. □

6.7 EXTENSION: NON-ADDITIVE NOISE MECHANISMS

The following results are ongoing work that had not yet been published when this thesis was submitted. Nevertheless, we believe that they complement the previous results as they describe an alternative way of viewing Private Lossless Multiple Release (PLMR) in terms of a Markov chain. The main intuition is that the *memoryless property* of a Markov chain exactly corresponds to the post-processing characterization we had before.

6.7.1 Markov Chain Formulation

Recall that lossless multiple release requires that, for any subset of the privacy levels and in any order, the entire collection of releases can be reproduced by post-processing the least private one alone. This is precisely the memoryless property of a Markov chain, which lets us replace reasoning about the joint distribution of all releases with a single transition kernel between adjacent privacy levels.

Lemma 6.31 (Lossless Multiple Release $\Leftrightarrow$ (Discrete-Time) Markov process). *Consider a private mechanism $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$. Then $\mathcal{M}_\rho$ can be implemented with lossless multiple release, if and only if, for every $x \in \mathcal{X}$ and every finite $S \subset \mathbb{R}_+$, there exists a Markov process $(Y_\rho)_{\rho \in S}$ where $Y_\rho \stackrel{d}{=} \mathcal{M}_\rho(x)$.*

Proof. $PLMR \Rightarrow Markov \text{ process}$. Let M denote an implementation of PLMR for $\mathcal{M}_\rho$. Consider the sequence $(Y_\rho)_{\rho \in S} \equiv (M(x, \rho))_{\rho \in S}$. By Property 1, $M(x, \rho) \stackrel{d}{=} \mathcal{M}_\rho(x)$. To check the Markov property, consider any $\rho_1 < \rho_2 < \rho_3$ in S . Using Property 2 for every $S' \subseteq \{\rho_1, \rho_2, \rho_3\} \subseteq S$ the joint distribution over $(Y_\rho)_{\rho \in S'}$ is determined by Y_{ρ_3} , hence. Thus, considering every input order, the following objects are determined:

$$Y_{\rho_1} | Y_{\rho_2}, \quad Y_{\rho_1} | Y_{\rho_3}, \quad Y_{\rho_2} | Y_{\rho_3},$$

and so, from first principles, $Y_{\rho_1} | Y_{\rho_2}$ is independent of Y_{ρ_3} , proving that the Markov property holds. Hence $(Y_\rho)_{\rho \in S}$ is a Markov process.

$Markov \text{ process} \Rightarrow PLMR$. Define M as the mechanism that, for a fixed $x \in \mathcal{X}$ and on receiving a privacy parameter ρ outputs $M(x, \rho) \equiv Y_\rho$. Property 1 of Definition 6.2 immediately holds as $Y_\rho \stackrel{d}{=} \mathcal{M}_\rho(x)$. Property 2 is implied by the Markov property of $(Y_\rho)_{\rho \in S}$ together with the chain rule. $\square$

We can use this result to give a very elegant characterization of k -ary *Randomized Response* under arbitrary releases. This immediately extends the (arguably more complicated) analysis for gradual release [Pan24]. Recall the following definition of randomized response.

Definition 6.6 ([War65] k -ary Randomized Response). *Let $x \in [k]$ be a single private value and fix some privacy parameter $\epsilon > 0$. We define k -ary randomized response $kRandResp(\epsilon, x)$ to be the mechanism that releases*

$$Y_\epsilon = \begin{cases} x & \text{with probability } \frac{\exp(\epsilon)}{\exp(\epsilon) + k - 1} \\ z \in [k] \setminus \{x\} \text{ uniformly at random} & \text{otherwise.} \end{cases}$$

We will now show how an output $Y_1 \sim RandResp(\epsilon_2, X)$ can be (data-independently) post-processed, into a sample of $RandResp(\epsilon_1, X)$ using a stochastic kernel. In the following, we denote by $\tanh$ the hyperbolic tangent.

Lemma 6.32 (Stochastic Kernel k -ary RR). *Let $x \in [k]$ and fix two privacy parameters $0 \leq \epsilon' \leq \epsilon$. Now draw $Y_\epsilon \sim kRandResp(\epsilon, x)$ and sample conditionally*

$$Y_{\epsilon'} | (Y_\epsilon = y_\epsilon) \sim \begin{cases} y_\epsilon & \text{with probability } 1 - p_{\text{flip}}(\epsilon, \epsilon') \\ z \in [k] \setminus \{y_\epsilon\} \text{ uni. at random} & \text{with total probability } p_{\text{flip}}(\epsilon, \epsilon') \end{cases},$$

where

$$p_{\text{flip}}(\epsilon, \epsilon') = \frac{k-1}{k} \cdot \left(1 - \frac{(\exp(\epsilon') - 1) \cdot (\exp(\epsilon) + k - 1)}{(\exp(\epsilon) - 1) \cdot (\exp(\epsilon') + k - 1)} \right).$$

Then we have $Y_{\epsilon'} \sim RandResp(\epsilon', x)$.

Proof. Fix some private dataset $x \in [k]$. In the following, we denote with

$$r_\epsilon = \frac{\exp(\epsilon)}{\exp(\epsilon) + k - 1} \quad \text{and} \quad r_{\epsilon'} = \frac{\exp(\epsilon')}{\exp(\epsilon') + k - 1}$$

the probability that no flip is taking place. Rewrite the mechanism in terms of a stochastic kernel with some unknown flip probabilities $p_{\text{flip}}(\epsilon, \epsilon')$ that only depend on the two privacy parameters ϵ and ϵ' :

$$K^{\epsilon \rightarrow \epsilon'} := \begin{pmatrix} 1 - p_{\text{flip}}(\epsilon, \epsilon') & \frac{p_{\text{flip}}(\epsilon, \epsilon')}{k-1} & \frac{p_{\text{flip}}(\epsilon, \epsilon')}{k-1} \\ \frac{p_{\text{flip}}(\epsilon, \epsilon')}{k-1} & 1 - p_{\text{flip}}(\epsilon, \epsilon') & \dots \\ \vdots & \vdots & \vdots \\ \frac{p_{\text{flip}}(\epsilon, \epsilon')}{k-1} & \dots & 1 - p_{\text{flip}}(\epsilon, \epsilon') \end{pmatrix}.$$

Observe that the law of the mechanism is exactly

$$M_\epsilon(x) = \begin{pmatrix} (\exp(\epsilon) + k - 1)^{-1} \\ \dots \\ \frac{\exp(\epsilon)}{\exp(\epsilon) + k - 1} \\ \dots \\ (\exp(\epsilon) + k - 1)^{-1} \end{pmatrix} = \begin{pmatrix} q_\epsilon \\ \dots \\ r_\epsilon \\ \dots \\ q_\epsilon \end{pmatrix},$$

where r_ϵ appears at the coordinate $x \in [k]$. Now observe that for $x \in [k]$, we have

$$\begin{aligned} (K^{\epsilon \rightarrow \epsilon'} \cdot M_\epsilon(x))_x &= \left(1 - p_{\text{flip}}(\epsilon, \epsilon')\right) r_\epsilon + p_{\text{flip}}(\epsilon, \epsilon') \cdot q_\epsilon \\ &= r_\epsilon - p_{\text{flip}}(\epsilon, \epsilon') \cdot (r_\epsilon - q_\epsilon). \end{aligned}$$

Now we need to solve $(K^{\epsilon \rightarrow \epsilon'} \cdot M_\epsilon(x))_x = r_{\epsilon'}$ to get the correct perturbation probabilities for $M_{\epsilon'}$. Hence, substituting back yields

$$p_{\text{flip}}(\epsilon, \epsilon') = \frac{r_\epsilon - r_{\epsilon'}}{r_\epsilon - q_\epsilon} = \frac{k-1}{k} \cdot \left(1 - \frac{(\exp(\epsilon') - 1) \cdot (\exp(\epsilon) + k - 1)}{(\exp(\epsilon) - 1) \cdot (\exp(\epsilon') + k - 1)}\right).$$

By symmetry, all other coordinates are equal and one can easily verify that $K^{\epsilon \rightarrow \epsilon'} \cdot M_\epsilon(x) = M_{\epsilon'}(x)$ for all $x \in [k]$, verifying the probabilities in the statement. $\square$

Note that K satisfies the condition of the *Chapan-Kolmogorov Equation* [MS17]. Therefore, it is a consistent Markov transition family, i.e. $K^{\epsilon \rightarrow \epsilon'} = K^{\epsilon \rightarrow \tilde{\epsilon}} \cdot K^{\tilde{\epsilon} \rightarrow \epsilon'}$ for all $0 \leq \epsilon' \leq \tilde{\epsilon} \leq \epsilon$. Note that this makes the whole chain consistent in a sense that allows releases in *arbitrary* order. We will make use of this property in Lemma 6.32. Note that privacy tightening might still be possible.

Now we get a simple closed form for the binary case.

Lemma 6.33 (Privacy Tightening Binary RR). *Let $x \in \{0, 1\}$ and fix two privacy parameters $0 \leq \epsilon' \leq \epsilon$. Now draw $Y_\epsilon \sim \text{RandResp}(\epsilon, x)$ and sample conditionally*

$$Y_{\epsilon'} = \begin{cases} 1 - Y_\epsilon & \text{with probability } p_{\text{flip}}(\epsilon, \epsilon') \\ Y_\epsilon & \text{otherwise} \end{cases}, \text{ where } p_{\text{flip}}(\epsilon, \epsilon') := \frac{1}{2} \left(1 - \frac{\tanh(\epsilon'/2)}{\tanh(\epsilon/2)}\right).$$

Then we have $Y_{\epsilon'} \sim \text{RandResp}(\epsilon', x)$.

Proof. Note that, by definition of the hyperbolic tangent $\tanh(a/2) := \frac{\exp(a) - 1}{\exp(a) + 1}$.

Hence for $k = 2$, we immediately get by basic algebra

$$p_{\text{flip}}(\varepsilon, \varepsilon') = \frac{1}{2} \left(1 - \frac{\tanh(\varepsilon'/2)}{\tanh(\varepsilon/2)} \right).$$

□

RELEASE IN ARBITRARY ORDER We will now show that we can also produce them in any order. Consider the case where two releases $Y_1 \rightarrow Y_2$ with corresponding privacy levels $0 < \varepsilon_1 < \varepsilon_2$ have been published by the Markov kernel, $K^{\varepsilon_2 \rightarrow \varepsilon_1}$ and both of them are distributed as single releases from $Y_1 \sim \text{kRandResp}(\varepsilon_1, x)$ and $Y_2 \sim \text{kRandResp}(\varepsilon_2, x)$ respectively. We will now show how to create a new release whose privacy level $\varepsilon \in (\varepsilon_1, \varepsilon_2)$ lies in between but preserves the Markov chain $Y_2 \rightarrow Y \rightarrow Y_1$ while also $Y \sim \text{kRandResp}(\varepsilon, x)$ as desired.

Lemma 6.34 (Arbitrary Release k -ary RR). *Let $x \in [k]$ and fix privacy parameters $0 < \varepsilon_1 \leq \varepsilon \leq \varepsilon_2$. As already defined in Lemma 6.32,*

$$K^{\eta \rightarrow \eta'}(i, j) = \begin{cases} 1 - p_{\text{flip}}(\eta, \eta') & i = j \\ p_{\text{flip}}(\eta, \eta') / (k - 1) & \text{otherwise} \end{cases}, \text{ with}$$

$$p_{\text{flip}}(\varepsilon, \varepsilon') = \frac{q_\varepsilon - r'_\varepsilon}{q_\varepsilon - r_\varepsilon} = \frac{k - 1}{k} \cdot \left(1 - \frac{(\exp(\varepsilon') - 1) \cdot (\exp(\varepsilon) + k - 1)}{(\exp(\varepsilon) - 1) \cdot (\exp(\varepsilon') + k - 1)} \right).$$

Assume a Markov process $Y_2 \rightarrow Y_1$ has produced

$$Y_2 \sim \text{kRandResp}(\varepsilon_2, x), \text{ and}$$

$$Y_1 \mid Y_2 = y_2 \sim K^{\varepsilon_2 \rightarrow \varepsilon_1}(y_2, \cdot)$$

If we conditionally on sample

$$\mathbf{P}[Y = y \mid Y_1 = a, Y_2 = b] = \frac{K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a)}{K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a)},$$

then

1. $Y_2 \rightarrow Y \rightarrow Y_1$ forms a (discrete-time) Markov process, and furthermore,
2. the marginal distribution $Y \sim \text{kRandResp}(\varepsilon, x)$ is correct.

Proof. Fix the private dataset $x \in [k]$ and let Y_1, Y, Y_2 sampled as described. Furthermore, denote with $M_\varepsilon(x)$ the probability vector of the mechanism given x . Note that by Lemma 6.33, we know that Y_1 and Y_2 are distributed as independent releases of $\text{kRandResp}(\varepsilon_1, x)$ and $\text{kRandResp}(\varepsilon_2, x)$. Furthermore, by construction, $Y_1 \rightarrow Y_2$ forms a Markov chain, and hence the joint distribution of Y_1, Y_2 factorizes, i.e. $\mathbf{P}[Y_1 = a, Y_2 = b] = \mathbf{P}[Y_1 = a \mid Y_2 = b] \mathbf{P}[Y_2 = b]$. We first show that $Y_1 \rightarrow Y \rightarrow Y_2$ forms a Markov chain with corresponding kernels $K^{\varepsilon_2 \rightarrow \varepsilon}, K^{\varepsilon \rightarrow \varepsilon_1}$.

To do so, we show that the joint distribution satisfies the Markov property, i.e., it factorizes out. Therefore, fix $y, a, b \in [k]$ and observe by Bayes's Theorem,

$$\mathbf{P}[Y_1 = a, Y = y, Y_2 = b] = \mathbf{P}[Y = y \mid Y_1 = a, Y_2 = b] \mathbf{P}[Y_1 = a, Y_2 = b] \quad (7)$$

$$= \frac{K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a)}{K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a)} \cdot K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a) \cdot \mathbf{P}[Y_2 = b] \quad (8)$$

$$= K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a) \mathbf{P}[Y_2 = b], \quad (9)$$

where the last line shows the desired factorization.

It is left to show that $Y \sim \text{kRandResp}(\varepsilon, x)$ also has the correct marginal distribution. By marginalization, we have for all y ,

$$\begin{aligned} \mathbf{P}[Y = y] &= \sum_{ab} \frac{K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a)}{K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a)} \cdot K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a) \cdot \mathbf{P}[Y_2 = b] \\ &= \sum_{ab} K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a) = \sum_a K^{\varepsilon \rightarrow \varepsilon_1}(y, a) \cdot \mathbf{P}[Y = y] \\ &= \mathbf{P}[Y = y]. \end{aligned}$$

□

Note that in the first step, we have used the fact that the kernel decomposes by fulfilling the *Chapman-Kolmogorov Equation*. Find a short empirical evaluation on Figure 6.2. In this example, we adaptively call the k -ary *Randomized Response* with three different privacy parameters, where the privacy level of the last one lies in between the first two. We then compare the empirical histogram with the direct releases.

6.7.2 Impossibility Results

One open question from the published version was whether there are mechanisms where PMLR is impossible. Here, we answer this question in the affirmative while again using the Markov chain formulation.

Lemma 6.35. *There exists a mechanism for which Private Multiple Lossless Releases is not possible.*

Proof. Consider an output space of three elements and let $M : \mathcal{X} \rightarrow \mathcal{Y}$ be the exponential mechanism that on dataset D_i , releases $r \in [3]$ with probability $\propto \exp(\frac{\varepsilon}{2} u_{D_i}(r))$. Furthermore, let D_1, D_2, D_3 be three neighboring datasets together with utilities, $u_1 = (0, 0, 0)$ and $u_2 = (0, 0, 1)$ and $u_3 = (1, 1, 0)$. We will show that multiple releases are not possible by trying to construct a Markov chain that supports PMLR. Assume $\varepsilon_1 = \ln 8$ and $\varepsilon_2 = \ln 9$. Then the marginal probabilities are

$$p_{D_1}^{\varepsilon_1} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \quad p_{D_2}^{\varepsilon_1} = \left(\frac{1}{5}, \frac{1}{5}, \frac{3}{5}\right) \quad p_{D_3}^{\varepsilon_1} = \left(\frac{3}{7}, \frac{3}{7}, \frac{1}{7}\right) \quad (10)$$

$$p_{D_2}^{\varepsilon_2} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \quad p_{D_2}^{\varepsilon_2} = \left(\frac{1}{4}, \frac{1}{4}, \frac{2}{4}\right) \quad p_{D_3}^{\varepsilon_2} = \left(\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right) \quad (11)$$

Now consider the following subset of the constraints of the probability of releasing element 3 under ε_2 . Assume for contradiction there exists a (data-independent) stochastic kernel $K = (k_{ij})_{ij} \in \mathbb{R}^{3 \times 3}$ such that

$$p_{D_i}^{\varepsilon_1} = K \cdot p_{D_i}^{\varepsilon_2} \quad \forall i \in [3].$$

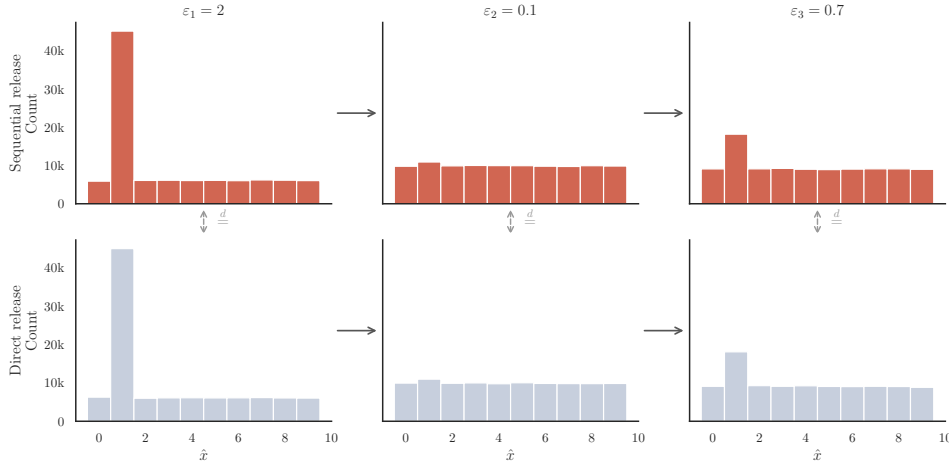


Figure 6.2: Empirical evaluation of sampling from the explicit kernel for 10-ary Randomized Response on 10^6 independent runs on the dataset $\mathbf{x} = 1$. **(top)** The correlated sampling procedure from the kernel. The overall privacy budget for releasing all samples is purely determined by the least private release, i.e. ϵ_3 . **(bottom)** independent releases using overall privacy budget of $\epsilon_1 + \epsilon_2 + \epsilon_3 = 2.8$.

One can easily verify that this system of linear equations is infeasible. $\square$

We leave a further exploration of this direction to future work. It may also be worth investigating possible connections to the lower bounds in [EC91] in the context of *Successive Refinement of Information*.

FUTURE WORK Furthermore, one could relax PLMR by dropping the strict requirement that every subset of releases must be as private as the least private release within that subset. This requirement was primarily motivated by the on-line setting, where privacy levels arrive one at a time and the set of releases is dynamically changing. One could also imagine a setting, where all of those levels are already specified (offline setting), and one only requires the whole output sequence to be as private as the least private release, without imposing any stricter constraints on the subsets of this sequence. this relaxed problem can be formulated as a linear program over the full joint distribution of (a finite output space of) releases. The constraints enforce (a) the prescribed marginal distribution of each individual release and (b) the desired privacy guarantee for the joint output. One can then minimize the smallest privacy parameter ϵ' for which the constraints are feasible. The resulting linear program has a size exponential in the number of requested privacy levels, but it may still be useful for small sets of releases.

“Yeah, yeah,” I say. “Invisible.
When you call me, I immediately know that it is you
because you’re the only one I know who hides his number.”

— The Kangaroo Chronicles, Marc-Uwe Kling

THE CORRELATED SPARSE GAUSSIAN HISTOGRAM MECHANISM

Published at FORC 2025

Joint work with Christian Lebeda

We consider the problem of releasing a sparse histogram under (ϵ, δ) -differential privacy. The stability histogram independently adds noise from a Laplace or Gaussian distribution to the non-zero entries and removes those noisy counts below a threshold. Thereby, the introduction of new non-zero values between neighboring histograms is only revealed with probability at most δ , and typically, the value of the threshold dominates the error of the mechanism. We consider the variant of the stability histogram with Gaussian noise.

Recent works ([Joseph and Yu, COLT ’24] and [Lebeda, SOSA ’25]) reduced the error for private histograms using correlated Gaussian noise. However, these techniques can not be directly applied in the very sparse setting. Instead, we adopt Lebeda’s technique and show that adding correlated noise to the non-zero counts only allows us to reduce the magnitude of noise when we have a sparsity bound. This, in turn, allows us to use a lower threshold by up to a factor of $1/2$ compared to the non-correlated noise mechanism. We then extend our mechanism to a setting without a known bound on sparsity. Additionally, we show that correlated noise can give a similar improvement for the more practical discrete Gaussian mechanism.

Releasing approximate counts is a common task in differential private data analysis. For example, the task of releasing a search engine log, where one wants to differentially private release the number of users who have searched each possible search term. The full domain of search terms is too large to work with directly, but the vector of all search counts is extremely sparse as most phrases are never searched. The standard approach to exploit sparsity while ensuring differential privacy is the *stability histogram* [BV19; Kor+09; BNS16; LAP22; GAT20; Göt+12; Wil+20]. The idea is to preserve sparsity by filtering out zero counts and then adding noise from a suitable probability distribution to the remaining ones. Unfortunately, some filtered-out counts could be non-zero in a neighboring dataset, and thus, revealing any of them would violate privacy. To limit this privacy violation, the stability histogram introduces a threshold τ and releases only those noisy counts that exceed $1 + \tau$. This way, we might still reveal the true dataset among a pair of neighboring datasets if one of these counters exceeds the threshold, but with an appropriate choice of τ , this is extremely unlikely to happen.

In our setting, one single user may contribute to many counts; hence, using the *Gaussian Sparse Histogram Mechanism* (GSHM) [GAT20; Wil+24] might be preferable. The GSHM is similar to the stability histogram, but it replaces noise from the Laplace distribution with Gaussian noise. To analyze the (ϵ, δ) guarantees of the

GSHM, one has to find an upper bound for δ , which is influenced by the standard Gaussian mechanism and the small probability of infinite privacy loss that can happen when the noise exceeds τ . We denote these two quantities as δ_{gauss} and δ_{inf} . Recently, Wilkins, Kifer, Zhang, and Karrer [Wil+24] gave exact privacy guarantees of the GSHM, which improves over the analysis by GAT [GAT20] where δ_{gauss} and δ_{inf} were simply summed. Their main contribution is a more intricate case distinction of a single user's impact on the values of δ_{gauss} and δ_{inf} . This allows them to use a lower threshold with the same privacy parameters. Their improvement lies in the constants, but it can be significant in practical applications because the tighter analysis essentially gives better utility at no privacy cost.

Our goal is to reduce the error further. Because the previous analysis is exact, we must exploit some additional structure of the problem. Consider a d -dimensional histogram $H(\mathbf{x}) = \sum_{i=1}^n X_i$ for $\mathbf{x} = (X_1, \dots, X_n)$ for users with data $X_i \in \{0, 1\}^d$. Notice that *adding* a single user to $\mathbf{x}$ can only increase counts in $H(\mathbf{x})$, whereas *removing* one can only decrease counts. Lebeda [Leb25] recently showed how to exploit this monotonicity by adding a small amount of correlated noise. This reduces the total magnitude of noise almost by a factor of 2 compared to the standard Gaussian Mechanism. So, it is natural to ask if we can adapt this technique to the setting when the histogram is sparse. This motivates our main research question:

Question: When can we take advantage of monotonicity and use correlated noise to improve the *Gaussian Sparse Histogram Mechanism*?

OUR CONTRIBUTIONS We answer this question by introducing the *Correlated Stability Histogram* (CSH). Building on the work by Lebeda [Leb25], we extend their framework to the setting of releasing a histogram under a sparsity constraint, that is $\|H(\mathbf{x})\|_0 \leq k$ for some known k . This is a natural setting that occurs e.g. for Misra-Gries sketches, where k is the size of the sketch [MG82; LT23]. Furthermore, enforcing this sparsity constraint is necessary to achieve the structure required to benefit from correlated noise.

Correlated GSHM. We introduce the *Correlated Stability Histogram* (CSH), a variant of the GSHM using correlated noise. Our algorithm achieves a better utility-privacy trade-off than GSHM for k -sparse histograms. Similar to the result of Lebeda [Leb25], we show that correlated noise can reduce the error by almost a factor of 2 at no additional privacy cost. In particular, our main result lies in the following (informally stated) theorem:

Theorem 7.1 (The Correlated Stability Histogram Mechanism (Informal)). *Let $H(\mathbf{x}) = \sum_i^n X_i$ denote a histogram with bounded sparsity, where $\mathbf{x} = (X_1, \dots, X_n)$ and $X_i \in \{0, 1\}^d$. If the GSHM privately releases $H(\mathbf{x})$ under (ϵ, δ) -DP with noise magnitude σ and removes noisy counters below a threshold τ , then the Correlated Stability Histogram Mechanism (CSH) releases $H(\mathbf{x})$ also under (ϵ, δ) -DP with noise of magnitude $\sigma/2 + o(1)$ and removes noisy counters below a threshold $\tau/2 + o(1)$.*

As a baseline, we first analyze our mechanism using the *add-the-deltas* [GAT20] approach and show that even this approach outperforms the exact analysis in Wilson, Zhang, Lam, Desfontaines, Simmons-Marengo, and Gipson [Wil+20] in many settings. We then turn our attention to a tighter analysis, which uses an intricate

case distinction to upper bound the parameter δ . Furthermore, we complement our approach with the following results:

Generalization & Extensions. We extend our mechanism to multiple other settings, including extensions considered in Wilkins, Kifer, Zhang, and Karrer [Wil+24]. First, we generalize our approach to allow an additional threshold that filters out infrequent data in a pre-processing step. Second, we discuss how other aggregate database queries can be included in our mechanism. Last, we generalize to top- k counting queries when we have no bound for $\|H(\mathbf{x})\|_0$.

Discrete Gaussian Noise. Our mechanism achieves the same improvement over GSHM when noise is sampled from the discrete Gaussian rather than the continuous distribution. We present a simple modification to make our mechanism compatible with discrete noise. This has practical relevance even for the dense setting considered by Lebeda [Leb25].

Organization. The rest of the section is organized as follows. Section 7.1 introduces the problem formally and reviews the required background. Section 7.2 introduces the *Correlated Stability Histogram*, our main algorithmic contribution for the sparse case. Furthermore, in Section 7.3 we generalize our approach to the non-sparse setting. The extensions of our technique to match the setting of Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] are discussed in Section 7.5. In Section 7.6, we show how to adapt our mechanism for discrete Gaussian noise. Finally, we conclude the paper in Section 6.5 and discuss open problems for future work.

Additionally, numerical evaluations in Section 7.4 confirm our claim that we improve the error over previous techniques.

7.1 PRELIMINARIES AND BACKGROUND

Given a dataset $\mathbf{x} \in \mathcal{U}^*$, where $\mathcal{U}^* := \bigcup_{m=0}^{\infty} \mathcal{U}^m$ is the set of datasets with any size, we want to perform an aggregate query $\mathcal{A}$ under differential privacy. We focus on settings where each user has a set of elements, and we want to estimate the count of all elements over the dataset. Therefore, we consider a dataset $\mathbf{x} = (X_1, \dots, X_n)$ of n data points where each $X_i \in \{0, 1\}^d$. Our goal is to output a private estimate of the histogram $H(\mathbf{x}) \in \mathbb{N}^d$, where $H(\mathbf{x}) = \sum_{i=1}^n X_i^1$. For any vector $\mathbf{h} \in \mathbb{R}^d$, we define the *support* as $U(\mathbf{h}) = \{i \in [d] : H_i \neq 0\}$ and denote the ℓ_0 norm as $\|\mathbf{h}\|_0 := |U(\mathbf{h})|$, being the number of non-zeroes in $\mathbf{h}$. In this work, we focus on settings where the dimension d is very large or infinite.

Proposed by Dwork, McSherry, Nissim, and Smith [Dwo+06b], differential privacy (DP) is a property of a randomized mechanism. The intuition behind differential privacy is that privacy is preserved by ensuring that the output distribution does not depend much on any individual's data. Note that by this definition $|\mathbf{x}| = n$ and $|\mathbf{x}'| = n \pm 1$:

Definition 7.1 (Neighboring datasets). *A pair of datasets $\mathbf{x} \sim \mathbf{x}'$ is neighboring if there exists an i such that either*

$$\mathbf{x} = (X'_1, \dots, X'_{i-1}, X'_{i+1}, \dots, X'_{n+1}) \quad \text{or} \quad \mathbf{x}' = (X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n).$$

¹ Note that in the differential privacy literature the term *histogram* is often used for the special case where users each have a single element. We use it for consistency with related work (e.g. [Wil+24]). The setting we consider is closely related to the problem of releasing one-way marginals.

In this chapter, we consider both the ℓ_2 sensitivity and, more generally, the exact sensitivity space of the queries.

Definition 7.2 (Sensitivity space and ℓ_2 sensitivity). *The sensitivity space of a deterministic function $f : \mathcal{U}^* \rightarrow \mathbb{R}^d$ is the set $\Delta f = \{f(\mathbf{x}) - f(\mathbf{x}') \in \mathbb{R}^d \mid \mathbf{x} \sim \mathbf{x}'\}$ and denote $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$ as the ℓ_2 norm of any $\mathbf{x} \in \mathbb{R}^d$. Then the ℓ_2 sensitivity of f is defined as*

$$\Delta_2 f = \max_{\mathbf{x} \sim \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_2 = \max_{\mathbf{x} \in \Delta f} \|\mathbf{x}\|_2.$$

The standard Gaussian mechanism adds continuous noise from a Gaussian distribution to $f(\mathbf{x})$ with magnitude scaled according to Lemma 7.2.

Lemma 7.2 ([CKS20, Theorem 8] The Analytical Gaussian Mechanism). *Let $f : \mathcal{U}^* \rightarrow \mathbb{R}^d$ denote a function with ℓ_2 sensitivity at most Δ_2 . Then the mechanism that outputs $f(X) + (Z_1, \dots, Z_d)$, where each $Z_i \sim \mathcal{N}(0, \sigma^2)$ are independent and identically distributed, satisfies (ϵ, δ) -differential privacy if the following inequality holds*

$$\delta \geq \Phi\left(\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right) - e^\epsilon \Phi\left(-\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right),$$

where $\Phi(X)$ denotes the cdf of the Gaussian distribution.

7.1.1 The Gaussian Sparse Histogram Mechanism

We consider the problem of privately releasing the histogram $H(\mathbf{x})$ of a dataset $\mathbf{x} = (X_1, \dots, X_n)$ where each $X_i \in \{0, 1\}^d$ under differential privacy. The standard techniques for releasing a private histogram are the well-known Laplace mechanism [Dwo+06b] and Gaussian mechanism [Dwo+06a; CKS20], which achieve ϵ - and (ϵ, δ) -DP, respectively. Although this works well for dense data, it is unsuited for very sparse data where $\|H(\mathbf{x})\|_0 \ll d$ because adding noise to *all entries* increases both the maximum error as well as the space and time requirements. Furthermore, the mechanisms are undefined for infinite domains ($d = \infty$).



Figure 7.1: Examples of different kinds of neighboring datasets for the Gaussian Sparse Histogram Mechanism where a single user can contribute to at most four counters, thus $\|X_i\|_0 \leq 4$. These counters are depicted in green. **a)** For the example on the left, the mechanism behaves exactly as running the Gaussian mechanism on a restricted domain. **b)** In the case in the middle, we only have to bound the probability that one of the green elements together with the additive noise term exceeds the threshold $1 + \tau$. **c)** The case on the right is the most difficult case for the privacy analysis because the overall δ value depends on both kinds of changes.

Algorithm 15 The Gaussian Sparse Histogram Mechanism (GSHM)**Require:** Parameters σ and τ , histogram $H(\mathbf{x}) \in \mathbb{N}^d$.

- 1: Let $\tilde{\mathbf{h}} = \{0\}^d$.
- 2: **for** each $i \in [d]$ where $H(\mathbf{x})_i \neq 0$ **do**
- 3: Sample $Z_i \sim \mathcal{N}(0, \sigma^2)$.
- 4: **if** $H(\mathbf{x})_i + Z_i > 1 + \tau$ **then**
- 5: Set $\tilde{H}_i = H(\mathbf{x})_i + Z_i$.
- 6: **Release** $\tilde{\mathbf{h}}$.

In the classic sparse histogram setting where $\|X_i\|_0 = 1$, the preferred technique under (ϵ, δ) -DP is the *stability histogram* which combines Laplace noise with a thresholding technique (see [BV19; Kor+09; BNS16; LAP22; GAT20; Göt+12; Wil+20]).

In this chapter, we consider the *Gaussian Sparse Histogram Mechanism* (see [GAT20; Wil+24]), which replaces Laplace noise in the stability histogram with Gaussian noise. This is often preferred when users can contribute multiple items as the magnitude of noise is scaled by the ℓ_2 sensitivity instead of the ℓ_1 sensitivity. The GSHM adds Gaussian noise to each non-zero counter of $H(\mathbf{x})$ and removes all counters below a threshold $1 + \tau$. Find the pseudocode in Algorithm 15. We discuss the impact of the parameter τ below.

To get (ϵ, δ) differential privacy guarantees, we observe that there are two sources of privacy loss that have to be accounted for by the value of δ : δ_{gauss} from the Gaussian noise itself and the probability of infinite privacy loss δ_{inf} , when a zero count is deterministically ignored in one dataset, but possibly released in a neighboring one. These events have infinite privacy loss because they can only occur for one of the datasets. Therefore, δ_{inf} bounds the possibility of outputting a counter that is not present in the neighboring dataset. Figure 7.1 gives some intuition about the role of δ_{gauss} and δ_{inf} . Throughout this section, we assume that $\|X_i\|_0 \leq k$ for some known integer $k \ll d$.

7.1.2 The add-the-deltas Approach

The following approach appeared in a technical report in the *Google differential privacy library* [GAT20]. Similar techniques have been used elsewhere in the literature (e.g. [Wil+20]). We refer to this technique as *add-the-deltas*. We have to account for the privacy loss due to the magnitude of the Gaussian noise. Therefore, the value of δ_{gauss} is typically found by considering the worst-case effect of a user that only changes non-zero counters in both histograms. The value of δ_{gauss} follows from applying Lemma 7.2 (compare also Figure 7.1a).

The event of infinite privacy loss is captured by δ_{inf} . This is bounded by considering the worst-case scenario of changing k zero-counters to 1 and the fact that infinite privacy loss occurs exactly when any of these k counters exceed the threshold in the non-zero dataset (compare Figure 7.1b).

The observation for the *add-the-deltas* approach is that $\delta_{\text{gauss}} + \delta_{\text{inf}}$ is a valid upper bound on the overall δ value, and hence the condition of Lemma 7.3 is sufficient to achieve differential privacy.

Lemma 7.3 ([GAT20] *add-the-deltas*). *If pairs of neighboring histograms differs in*

at most k counters then the Gaussian Sparse Histogram Mechanism with parameters σ and τ satisfies $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -differential privacy where

$$\delta_{\text{gauss}} = \Phi\left(\frac{\sqrt{k}}{2\sigma} - \frac{\epsilon\sigma}{\sqrt{k}}\right) - e^\epsilon \Phi\left(-\frac{\sqrt{k}}{2\sigma} - \frac{\epsilon\sigma}{\sqrt{k}}\right), \quad \text{and} \quad \delta_{\text{inf}} = 1 - \Phi\left(\frac{\tau}{\sigma}\right)^k.$$

7.1.3 Exact Analysis by Taking the Max over the Sensitivity Space

While the *add-the-deltas* approach is sufficient to bound δ it does not give the tightest possible parameters. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] were able to derive an exact value for δ . Compared to the *add-the-deltas* approach above, the key insight is that the two worst-case scenarios cannot occur simultaneously for any pair of neighboring histograms. This means that each counter either contributes to δ_{gauss} or δ_{inf} , but never to both. In the extreme cases, either all k counters flip from zero to one, from one to zero, or none of them do between neighboring datasets. Thus, we only need to consider a single source of privacy loss. In the other (mixed) cases, we have to consider both sources of privacy loss, but the change is smaller than for the worst-case pair of datasets. A small example is depicted in Figure 7.1. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] used this fact to reduce the threshold required to satisfy the given privacy parameters using a tighter analysis. We now restate their main result: the exact privacy analysis of the GSHM.

Lemma 7.4 ([Wil+24] Exact Privacy Analysis of the GSHM). *If pairs of neighboring histograms differs in at most k counters then the Gaussian Sparse Histogram Mechanism with parameters σ and τ satisfies (ϵ, δ) -differential privacy where $\gamma(j) = (k - j) \log \Phi\left(\frac{\tau}{\sigma}\right)$ and*

$$\begin{aligned} \delta \geq & \max \left[1 - \Phi\left(\frac{\tau}{\sigma}\right)^k, \right. \\ & \max_{j \in [k]} 1 - \Phi\left(\frac{\tau}{\sigma}\right)^{k-j} + \Phi\left(\frac{\tau}{\sigma}\right)^{k-j} \\ & \quad \cdot \left[\Phi\left(\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon - \gamma(j))\sigma}{\sqrt{j}}\right) - e^{\epsilon - \gamma(j)} \Phi\left(-\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon - \gamma(j))\sigma}{\sqrt{j}}\right) \right], \\ & \left. \max_{j \in [k]} \Phi\left(\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon + \gamma(j))\sigma}{\sqrt{j}}\right) - e^{\epsilon + \gamma(j)} \Phi\left(-\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon + \gamma(j))\sigma}{\sqrt{j}}\right) \right]. \end{aligned}$$

Note that we use a slightly different convention from Wilkins, Kifer, Zhang, and Karrer [Wil+24]. We use a threshold of $1 + \tau$ rather than τ . Furthermore, they also consider a more general mechanism that allows for optional aggregation queries and an additional threshold parameter. Although our work can easily be adapted to this setting as well, these extensions are not the focus of our main work to simplify presentation. A discussion of the extensions can be found in Section 7.5.

7.1.4 The Correlated Gaussian Mechanism

Our work builds on a recent result by Lebeda Lebeda [Leb25] about using correlated noise to improve the utility of the Gaussian mechanism under the add-remove

neighboring relationship. They show that when answering d counting queries, adding a small amount of correlated noise to all queries can reduce the magnitude of Gaussian noise by almost half. We restate their main result for (ϵ, δ) -differential privacy with a short proof as they use another privacy definition.

Lemma 7.5 ([Leb25] The Correlated Gaussian Mechanism). *Let $H(\mathbf{x}) := \sum_{i=1}^n X_i$ where $X_i \in \{0, 1\}^d$. Then the mechanism that outputs $H(\mathbf{x}) + Z_{\text{cor}} \mathbf{1}^d + (Z_1, \dots, Z_d)$ satisfies (ϵ, δ) -DP. Here $\mathbf{1}^d$ is the d -dimensional vector of all ones, $Z_{\text{cor}} \sim \mathcal{N}(0, \sigma^2/\gamma)$, and each $Z_i \sim \mathcal{N}(0, \sigma^2)$ where*

$$\delta \geq \Phi\left(\frac{\sqrt{d+\gamma}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{d+\gamma}}\right) - e^\epsilon \Phi\left(-\frac{\sqrt{d+\gamma}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{d+\gamma}}\right).$$

Proof. This follows from combining the inequality for the standard Gaussian mechanism (see Lemma 7.2) with [Leb25, Lemma 3.5]. Furthermore, if we set $\gamma = \sqrt{d}$ as in [Leb25, Theorem 3.1] we minimize the total magnitude of noise. Notice that the value of σ scales with $\sqrt{d}$ for the standard Gaussian mechanism and it scales with $\frac{1}{2}\sqrt{d+\gamma} = \frac{1}{2}\sqrt{d+\sqrt{d}}$ here. We add two noise samples to each query, and the total error for each data point scales with

$$\frac{1}{2}\sqrt{d+\gamma+d/\gamma+1} = (\sqrt{d}+1)/2.$$

□

Concurrently with the result discussed above, Joseph and Yu [JY24] considered the setting where user contributions are sparse such that $\|X_i\|_0 \leq k$ for some $k \leq d/2$. They give an algorithm that adds the optimal amount of correlated Gaussian noise for any d and $k \leq d/2$. It is natural to ask whether this algorithm can improve error for our setting since they focus on a sparse setting. However, their improvement factor over the standard Gaussian mechanism depends on the sparsity. Naively applying their technique for the setting of the GSHM where $k \ll d$ yields no practical improvements. We instead focus on adapting the technique from [Leb25]. We discuss a more restrictive setting where the technique of Joseph and Yu [JY24] applies in Section 6.5.

7.2 ALGORITHMIC FRAMEWORK

We are now ready to introduce our main contribution, a variant of the Gaussian Sparse Histogram Mechanism using correlated noise. We first introduce our definition of k -sparse monotonic histograms. Throughout this section, we assume that all histograms are both k -sparse and monotonic. Intuitively, we define monotonicity on a histogram in a way that captures the setting where the counts are either all increasing or all decreasing. Observe that due to the monotonicity constraint, we have that the supports U and U' for two neighboring histograms satisfy either $U \subseteq U'$ or $U' \subseteq U$. We use this property in the privacy proofs later. The histograms are also monotonic in [Wil+24], but they do not require k -sparsity. We provide a mechanism for a setting where the histograms are not k -sparse in Section 7.3. There we enforce the sparsity constraint using a simple pre-processing step. Both sparsity and monotonicity are required in order to achieve the structure between neighboring histograms required to benefit from correlated noise.

Algorithm 16 Correlated Stability Histogram (CSH)**Require:** Parameters k, σ and τ , histogram $H(\mathbf{x}) \in \mathbb{N}^d$ where $\|H(\mathbf{x})\|_0 \leq k$.

- 1: Let $\tilde{\mathbf{h}} = \{0\}^d$.
- 2: Sample $Z_{\text{corr}} \sim \mathcal{N}(0, \sigma^2 / \sqrt{k})$.
- 3: **for** each $i \in [d]$ where $H(\mathbf{x})_i \neq 0$ **do**
- 4: Sample $Z_i \sim \mathcal{N}(0, \sigma^2)$.
- 5: **if** $H(\mathbf{x})_i + Z_i + Z_{\text{corr}} > 1 + \tau$ **then**
- 6: Set $\tilde{H}_i = H(\mathbf{x})_i + Z_i + Z_{\text{corr}}$.
- 7: **Release** $\tilde{H}$.

Definition 7.3 (k -sparse monotonic histogram). *We assume that the input histogram is k -sparse. That is, for any dataset $\mathbf{x}$ we have $\|H(\mathbf{x})\|_0 \leq k$. Furthermore, the sensitivity space of H is $\{0, 1\}^d \cup \{0, -1\}^d$. That is, the difference between counters of neighboring histograms are either non-decreasing or non-increasing.*

One example of k -sparse monotonic histograms is Misra-Gries sketches. Merging Misra-Gries sketches is common in practical applications. The sensitivity space of merged Misra-Gries sketches of size k exactly matches Definition 7.3 (See Lebeda and Tetek [LT23]). Our algorithm is more general than Misra-Gries sketches and satisfies differential privacy as long as the structure between neighboring histograms holds for all pairs of neighboring datasets.

Notice that Definition 7.3 implies that neighboring histograms differ in at most k counters. As such, we can release the histogram using the standard Gaussian Sparse Histogram Mechanism. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] already uses the fact that counters are either non-decreasing or non-increasing in their analysis. We intend to further take advantage of the monotonicity by adding a small amount of correlated noise to all non-zero counters. This allows us to reduce the total magnitude of noise similar to [Leb25]. The reduced magnitude of noise in turn allows us to reduce the threshold required for privacy. The pseudocode for our mechanism is in Algorithm 16.

We first give privacy guarantees for Algorithm 16 in a (relatively) simple closed form similar to the *add-the-deltas* approach. Later, we give tighter bounds using a more complicated analysis similar to Lemma 7.4. Unfortunately, the proofs from previous work in either case rely on the fact that all noisy counters are independent. That is clearly not the case for our mechanism because the value of Z_{corr} is added to all entries. Instead, we use different techniques to give similar results, starting with the *add-the-deltas* approach. The following lemma gives a general bound for the event that one of j correlated noisy terms exceeds a threshold τ . We use this in the proof for both approaches later.

Lemma 7.6 (Upper bound for Correlated Noise). *Let $Z_{\text{corr}} \sim \mathcal{N}(0, \sigma^2 / \sqrt{k})$ be a single sample for a real $k > 0$ together with j additional samples $Z_1, \dots, Z_j \sim \mathcal{N}(0, \sigma^2)$. Then*

for any $\tau > 0$, we have $\mathbf{P}[\exists i \in [j] : Z_{\text{corr}} + Z_i > \tau] \leq 1 - \Phi\left(\frac{\tau}{\sigma(k^{-1/4} + 1)}\right)^{j+1}$.

Proof. We first give a bound for Z_{corr} and the Z_i 's independently, which is sufficient for us to bound the probability of this event. First, observe that the j uncorrelated

terms $Z_i \sim \mathcal{N}(0, \sigma^2)$ are independent, and we are interested in bounding the maximum of them and hence:

$$\begin{aligned} \mathbf{P} \left[\max_{Z_1, \dots, Z_j} Z_i \leq \frac{\tau}{1 + k^{-1/4}} \right] &= \mathbf{P} \left[\mathcal{N}(0, \sigma^2) \leq \frac{\tau}{1 + k^{-1/4}} \right]^j \\ &= \Phi \left(\frac{\tau}{\sigma (1 + k^{-1/4})} \right)^j. \end{aligned}$$

Similarly, we have for $Z_{\text{corr}} \sim N(0, \sigma^2/\sqrt{k})$ that

$$\mathbf{P} \left[Z_{\text{corr}} \leq \frac{\tau k^{-1/4}}{1 + k^{-1/4}} \right] = \Phi \left(\frac{\tau k^{-1/4}}{\sigma (1 + k^{-1/4}) \cdot k^{-1/4}} \right) = \Phi \left(\frac{\tau}{\sigma (1 + k^{-1/4})} \right).$$

Notice that if both $Z_i \leq \tau/(1 + k^{-1/4})$ and $Z_{\text{corr}} \leq \tau k^{-1/4}/(1 + k^{-1/4})$ hold then $Z_{\text{corr}} + Z_i \leq \tau$. As such, we can prove that the lemma holds since,

$$\mathbf{P} \left[Z_{\text{corr}} + \max_{Z_1, \dots, Z_j} Z_i > \tau \right] \leq \mathbf{P} \left[Z_{\text{corr}} > \frac{\tau k^{-1/4}}{1 + k^{-1/4}} \vee \max_{Z_1, \dots, Z_j} Z_i > \frac{\tau}{1 + k^{-1/4}} \right] \quad (1)$$

$$= 1 - \mathbf{P} \left[Z_{\text{corr}} \leq \frac{\tau k^{-1/4}}{1 + k^{-1/4}} \right] \cdot \mathbf{P} \left[\max_{Z_1, \dots, Z_j} Z_i \leq \frac{\tau}{1 + k^{-1/4}} \right] \quad (2)$$

$$= 1 - \Phi \left(\frac{\tau}{\sigma (1 + k^{-1/4})} \right)^{j+1},$$

where step (1) holds by a union bound and step (2) holds because the random variables Z_{corr} and $\tilde{Z} = \max_{Z_1, \dots, Z_j} Z_i$ are independent. $\square$

7.2.1 The add-the-deltas Analysis

The following Theorem 7.7 proves privacy guarantees of our mechanism using a similar technique than the one proposed by the GAT [GAT20] and Wilson, Zhang, Lam, Desfontaines, Simmons-Marengo, and Gipson [Wil+20] which is known as *add-the-deltas*. The total value for δ is split between δ_{gauss} and δ_{inf} which accounts for the two types of privacy loss that are relevant to the mechanism. Like with the GSHM, these values are found by considering worst-case pairs of neighboring histograms for each case. However, a pair of neighboring histograms cannot be worst-case for both values as seen in Figure 7.1 which motivates the tighter analysis later in the section.

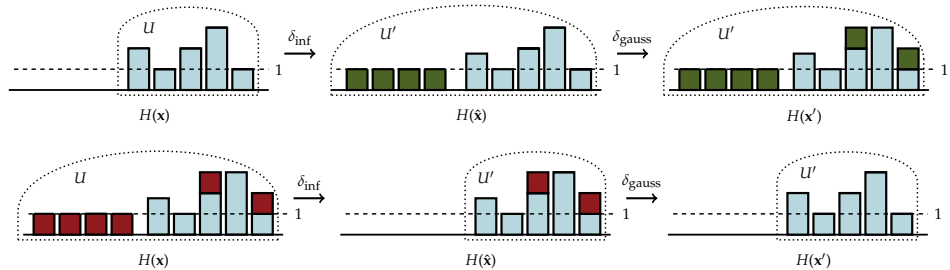


Figure 7.2: The separation technique of the δ_{gauss} and δ_{inf} used in Theorems 7.7 and 7.9. The idea is to construct an intermediate histogram $H(\hat{\mathbf{x}})$ with the same support U' as $H(\mathbf{x}')$ but only reflect the changes that can cause infinite privacy loss between $H(\mathbf{x})$ and $H(\mathbf{x}')$.

Theorem 7.7 (*add-the-deltas technique*). *Alg. 16 satisfies $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -dp for k -sparse monotonic histograms where*

$$\delta_{\text{gauss}} = \Phi\left(\frac{\sqrt{k+\sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k+\sqrt{k}}}\right) - e^\epsilon \Phi\left(-\frac{\sqrt{k+\sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k+\sqrt{k}}}\right)$$

$$\delta_{\text{inf}} = 1 - \Phi\left(\frac{\tau}{\sigma(1+k^{-1/4})}\right)^{k+1}.$$

Proof. By Definition 2.4, the lemma holds if for any pair of neighboring k -sparse monotonic histograms $H(\mathbf{x})$ and $H(\mathbf{x}')$ and all sets of outputs Y we have

$$\mathbf{P}[CSH(H(\mathbf{x})) \in Y] \leq e^\epsilon \mathbf{P}[CSH(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}} + \delta_{\text{inf}}.$$

We prove that the inequality above holds by introducing a third histogram. This new histogram is constructed such that it is between $H(\mathbf{x})$ and $H(\mathbf{x}')$. We first state the desired properties of this histogram and then show that such a histogram must exist for all neighboring histograms. Assume for now that there exists a histogram $H(\hat{\mathbf{x}}) \in \mathbb{N}^d$ where the following two inequalities hold for any set of outputs Y :

$$\mathbf{P}[CSH(H(\hat{\mathbf{x}})) \in Y] \leq e^\epsilon \mathbf{P}[CSH(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}}, \quad (3)$$

$$\mathbf{P}[CSH(H(\mathbf{x})) \in Y] \leq \mathbf{P}[CSH(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}}. \quad (4)$$

Then the inequality we need for the lemma follows immediately since

$$\begin{aligned} \mathbf{P}[CSH(H(\mathbf{x})) \in Y] &\leq \mathbf{P}[CSH(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}} \\ &\leq e^\epsilon \mathbf{P}[CSH(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}} + \delta_{\text{inf}}. \end{aligned}$$

Next, we show how to construct $H(\hat{\mathbf{x}})$ from $H(\mathbf{x})$ and $H(\mathbf{x}')$, and finally we show that each of two inequalities holds using this construction. Let $U = \{i \in [d] : H(\mathbf{x})_i \neq 0\}$ denote the support of $H(\mathbf{x})$ and define similarly U' and $\hat{U}$. We construct $H(\hat{\mathbf{x}})$ such that it has the same support as $H(\mathbf{x}')$, that is $U' = \hat{U}$. For all $i \in (U \cap U')$ we set $H(\hat{\mathbf{x}})_i = H(\mathbf{x})_i$ and for all $i \notin (U \cap U')$ we set $H(\hat{\mathbf{x}})_i = H(\mathbf{x}')_i$. In other words, we construct $H(\hat{\mathbf{x}})$ such that (1) $H(\mathbf{x}')$ and $H(\hat{\mathbf{x}})$ only differ in entries that are in both U and U' (2) $H(\mathbf{x})$ and $H(\hat{\mathbf{x}})$ only differ in entries that are in only one of U and U' . This allows us to analyze each case separately to derive our values for δ_{gauss} and δ_{inf} . An example of this construction is shown in Figure 7.2.

We start with the case of δ_{gauss} using $H(\mathbf{x}')$ and $H(\hat{\mathbf{x}})$. Let CSH' denote a new mechanism equivalent to Algorithm 16 except that the condition on line 5 is removed. Notice that with $H(\mathbf{x}')$ or $H(\hat{\mathbf{x}})$ as inputs CSH' is equivalent to the Correlated Gaussian Mechanism restricted to U' . Since $|U'| \leq k$ we have by Lemma 7.5 that

$$\mathbf{P}[CSH'(H(\hat{\mathbf{x}})) \in Y] \leq e^\epsilon \mathbf{P}[CSH'(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}}.$$

Notice that if we post-process the output of CSH' by removing entries below $1 + \tau$, then the output distribution is equivalent to CSH . Equation (3) therefore holds because post-processing does not affect differential privacy guarantees (see Definition 2.4).

The histograms $H(\mathbf{x})$ and $H(\hat{\mathbf{x}})$ only differ in entries that all have a count of 1 in one of the histograms while they all have a count of 0 in the other. δ_{inf} accounts for the event where any such counter exceeds the threshold because the

distributions are identical for the shared support. The probability of this event is increasing in the number of different elements between $H(\mathbf{x})$ and $H(\hat{\mathbf{x}})$ and therefore, the worst case happens for neighboring datasets such that $H(\mathbf{x}) = \mathbf{1}^k$ and $H(\hat{\mathbf{x}}) = \mathbf{0}^k$. Note that this bound also hold when zero counters in $H(\mathbf{x})$ are non-zero in $H(\hat{\mathbf{x}})$. We focus on one direction because the proof is almost identical for the symmetric case. We bound the probability of outputting any entries from $H(\mathbf{x})$, that is $\tilde{H}_i \doteq 1 + Z_{\text{corr}} + Z_i > 1 + \tau$ for at least one $i \in (U \setminus U')$. The bound for δ_{inf} follows from setting $j = k$ in Lemma 7.6. $\square$

In the Lemma 7.8 below, we show how to directly compute the required threshold based on the result in Theorem 7.7.

Lemma 7.8 (Computing the threshold τ). *For a fixed privacy budget ϵ, δ and parameters k and σ , the add-the-deltas technique requires*

$$\tau \geq \Phi^{-1} \left(\sqrt[k+1]{1 - \delta - \delta_{\text{gauss}}} \right) \cdot \left(1 + k^{-1/4} \right) \sigma,$$

where Φ^{-1} is the inverse CDF of the normal distribution and δ_{gauss} is defined as in Theorem 7.7.

Proof. Observe $\delta - \delta_{\text{gauss}} \geq \delta_{\text{inf}} = 1 - \Phi\left(\frac{\tau}{\sigma(1+k^{-1/4})}\right)^{k+1}$. Taking Φ^{-1} on both sides yields $\frac{\tau}{\sigma(1+k^{-1/4})} \geq \Phi^{-1} \left(\sqrt[k+1]{1 - \delta - \delta_{\text{gauss}}} \right)$. Multiplying by $\sigma \left(1 + k^{-1/4} \right)$ proves the claim. $\square$

7.2.2 Tighter Analysis

Next, we carry out a more careful analysis that considers all elements from the sensitivity space similar to Wilkins, Kifer, Zhang, and Karrer [Wil+24]. As discussed above, we cannot directly translate the analysis by Wilkins et al. because they rely on independence between each entry. We refer the full proof to the full version of the paper [LR24], and we only give a short intuition behind the theorem here.

Theorem 7.9 (Tighter Analysis). *For fixed k, σ and τ , we define*

$$\gamma(j) = \min \left(\sqrt{j}, \frac{1}{2} \sqrt{j + \sqrt{k}} \right), \quad \psi(m) = \Phi \left(\frac{\tau}{(1 + k^{-1/4}) \cdot \sigma} \right)^{m+1}, \text{ and}$$

$$\hat{\epsilon}(j) = \epsilon + \ln(\psi(k - j)).$$

Then Algorithm 16 satisfies (ϵ, δ) -DP for k -sparse monotonic histograms, where

$$\max \left[1 - \psi(k), \Phi \left(\frac{\sqrt{k + \sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k + \sqrt{k}}} \right) - e^\epsilon \Phi \left(-\frac{\sqrt{k + \sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k + \sqrt{k}}} \right), \right. \\ \max_{j \in [k-1]} 1 - \psi(k - j) + \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\epsilon\sigma}{\gamma(j)} \right) - e^\epsilon \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\epsilon\sigma}{\gamma(j)} \right), \\ \left. \max_{j \in [k-1]} \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\hat{\epsilon}(j)\sigma}{\gamma(j)} \right) - e^{\hat{\epsilon}(j)} \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\hat{\epsilon}(j)\sigma}{\gamma(j)} \right) \right] \leq \delta.$$

The result relies on a case by case analysis where each term in the maximum corresponds to a specific difference between neighboring histograms. The first two terms covers cases when we only have to consider either the infinite privacy loss

Algorithm 17 Top-k Mechanism using Correlated Gaussian Noise

Require: Parameters k , σ , and τ , dataset $\mathbf{x} = (X_1, \dots, X_n)$ where $X_i \in \{0, 1\}^d$.

- 1: Let $H(\mathbf{x}) = \sum_{i=1}^n X_i$.
- 2: Let $\tilde{H}(\mathbf{x}) = \{0\}^d$.
- 3: **for** each $i \in [d]$ **do**
- 4: Set $\tilde{H}(\mathbf{x})_i = \max(0, H(\mathbf{x})_i - H(\mathbf{x})^{(k+1)})$.
- 5: **Release** $\text{CSH}(k, \sigma, \tau, \tilde{H}(\mathbf{x}))$.

event or the Gaussian noise, respectively. The remaining terms cover the differences when we have to account for both Gaussian noise and the threshold similar to case c) in Figure 7.1. The first of the internal maximum covers the cases where $U \supset U'$ and $|U'| \leq j$, while the second covers the cases where $U \subset U'$ and $|U| \leq j$. For each case in our analysis, we split up the impact from Gaussian noise and the threshold. Together, these cases cover all elements in the sensitivity space for k -sparse monotonic histograms.

7.3 TOP-K COUNTING QUERIES

The privacy guarantees of Algorithm 16 are conditioned on the histogram being k -sparse for all datasets. Here we present a technique when we do not have this guarantee. Specifically, we consider the setting when the input is a dataset $\mathbf{x} = (X_1, \dots, X_n)$ where $X_i \in \{0, 1\}^d$. We want to release a private estimate of $H(\mathbf{x}) = \sum_{i=1}^n X_i$, and X_i can have any number of non-zero entries. We use superscript to denote the elements of the histogram in descending order with ties broken arbitrarily such that $H(\mathbf{x})^{(1)} \geq H(\mathbf{x})^{(2)} \geq \dots \geq H(\mathbf{x})^{(d-1)} \geq H(\mathbf{x})^{(d)}$. This setting is studied in a line of work for *Private top-k selection*: [DR19; Ros97; QSZ21; MT07; BU17; WZ24].

Our algorithm in this setting relies on a simple pre-processing step. We first find the value of the $(k+1)$ 'th largest entry in the histogram. We then subtract that value from all entries in the histogram and remove negative counts. This gives us a new histogram which we use as input for Algorithm 16. We show that this histogram is both k -sparse and monotonic which implies that the mechanism has the same privacy guarantees as Algorithm 16.

Lemma 7.10. *The function $\tilde{H} : \{0, 1\}^{n \times d} \rightarrow \mathbb{N}^d$ in Algorithm 17 produces k -sparse monotonic histograms.*

Proof. By Definition 7.3 we have to show two properties of $\tilde{H}$. It must hold for any $\mathbf{x}$ that $\|\tilde{H}(\mathbf{x})\|_0 \leq k$ and for any neighboring pair $\mathbf{x} \sim \mathbf{x}'$ we have $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, 1\}^d$ or $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, -1\}^d$.

The sparsity claim is easy to see. Any counter that is not strictly larger than $H(\mathbf{x})^{(k+1)}$ is removed in line 4. By definition of $H(\mathbf{x}')^{(k+1)}$, there are at most k such entries.

To prove the monotonicity property, we must show that counters in $\tilde{H}$ either all increase or all decrease by at most one. We only give the proof for the case where $\mathbf{x}$ is constructed by adding one data point to $\mathbf{x}'$. The proof is symmetric for the case where $\mathbf{x}$ is created by removing one data point from $\mathbf{x}'$.

We first partition $H(\mathbf{x}')$ into three sets: Let $U = \{i \in [d] \mid H(\mathbf{x}')_i > H(\mathbf{x}')^{(k+1)}\}$, $M = \{i \in [d] \mid H(\mathbf{x}')_i = H(\mathbf{x}')^{(k+1)}\}$ and $L = [d] \setminus \{U, M\}$. Because we have $H(\mathbf{x})_i - H(\mathbf{x}')_i \in \{0, 1\}$ for all $i \in [d]$, also $H(\mathbf{x})^{(k+1)} - H(\mathbf{x}')^{(k+1)} \in \{0, 1\}$. Note that

the $(k + 1)$ 'th largest entry might not represent the same element, but we only care about the value.

Consider the case where $H(\mathbf{x})^{(k+1)} = H(\mathbf{x}')^{(k+1)}$. We see that for all $l \in L$: $\tilde{H}(\mathbf{x})_l = \tilde{H}(\mathbf{x}')_l = 0$, and for all $u \in U$ we have $\tilde{H}(\mathbf{x}')_u - \tilde{H}(\mathbf{x})_u = H(\mathbf{x}')_u - H(\mathbf{x})^{(k+1)} - H(\mathbf{x})_u + H(\mathbf{x}')^{(k+1)} \in \{0, 1\}$ because we subtract the same value and increment $H(\mathbf{x}')_u$ by at most one. Some elements $m \in M$ might now be exactly one larger than $H(\mathbf{x}')^{(k+1)}$ in $H(\mathbf{x})$, but still $\tilde{H}(\mathbf{x})_m \in \{0, 1\}$ by definition. Thus we have that $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, 1\}^d$.

Now, consider the case where $H(\mathbf{x})^{(k+1)} = H(\mathbf{x}')^{(k+1)} + 1$. No element $l \in L$ can become larger or equal to the new $(k + 1)$ 'th largest element, so still $\tilde{H}(\mathbf{x}')_l = \tilde{H}(\mathbf{x})_l = 0$. Elements in $u \in U$ get reduced by at most one for $H(\mathbf{x})$, because they either are increased together with the $H(\mathbf{x}')^{(k+1)}$ and then $\tilde{H}(\mathbf{x})_u - \tilde{H}(\mathbf{x}')_u = 0$, or they stay the same in which case $\tilde{H}(\mathbf{x})_u = \tilde{H}(\mathbf{x}')_u - 1$. One can also see that for $m \in M$, $H(\mathbf{x})_m - H(\mathbf{x})^{(k+1)} \in \{0, -1\}$, depending on whether or not they increase together with $H(\mathbf{x})^{(k+1)}$. As such we have $\tilde{H}(\mathbf{x})_m = 0$ for all $m \in M$. Therefore, $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, -1\}^d$. $\square$

Since Algorithm 17 simply returns the output of running Algorithm 16 with a k -sparse monotonic histogram it has the same privacy guarantees.

Corollary 7.11. *The privacy guarantees specified by Theorems 7.7 and 7.9 hold for Algorithm 17.*

Algorithm 17 introduces bias when subtracting $H(\mathbf{x})^{(k+1)}$ from each counter as a pre-processing step. If we have access to a private estimate of $H(\mathbf{x})^{(k+1)}$ we can add it to all counters of the output as post-processing. Since this estimate would be used for multiple counters, similar to Z_{corr} , we might want to use additional privacy budget to get a more accurate estimate. This can be included directly in the privacy analysis. If we e.g. release $\tilde{H}(\mathbf{x})^{(k+1)} = H(\mathbf{x})^{(k+1)} + \mathcal{N}(0, \sigma^2/\sqrt{k})$ the privacy guarantees from Theorem 7.7 hold if we change $\sqrt{k} + \sqrt{k}$ to $\sqrt{k} + 5\sqrt{k}$ in δ_{gauss} .

7.4 NUMERICAL EVALUATIONS

To back up our theoretical claims, we compare the error of the *Correlated Stability Histogram* against both the *add-the-deltas* approach [GAT20] and the *exact* analysis [Wil+24] of the uncorrelated GSHM. We consider the error in both privacy analysis approaches for our mechanisms as well (Theorems 7.7 and 7.9).

We compare the parameters of the mechanisms when releasing k -sparse monotonic histograms. We ran the experiments shown in Figure 7.3 with the same privacy parameters as in Wilkins, Kifer, Zhang, and Karrer [Wil+24]², which is $\epsilon = 0.35$ and $\delta = 10^{-5}$. Following their approach, we plot the minimum τ such that each mechanism satisfies (ϵ, δ) -DP for a given magnitude of noise. Note that our setting of $\|H(\mathbf{x})\|_{0 \leq k}$ differs from [Wil+24], so the experiments are not directly comparable.

RESULTS. The first plot shown in Figure 7.3a resembles the same $k = 51914$ used in [Wil+24, Figure 1 (A)]. In this setting, one can lower the threshold by approximately 43%. Because our technique favors large k (we have to scale the noise with $\frac{1}{2}\sqrt{k} + \sqrt{k}$ instead of $\sqrt{k}$), we also include the case where $k = 10$ is small in Figure 7.3b. The plots show that we make some small improvements

² in turn based on the privacy guarantees of the *Facebook URL dataset* [Mes+20].

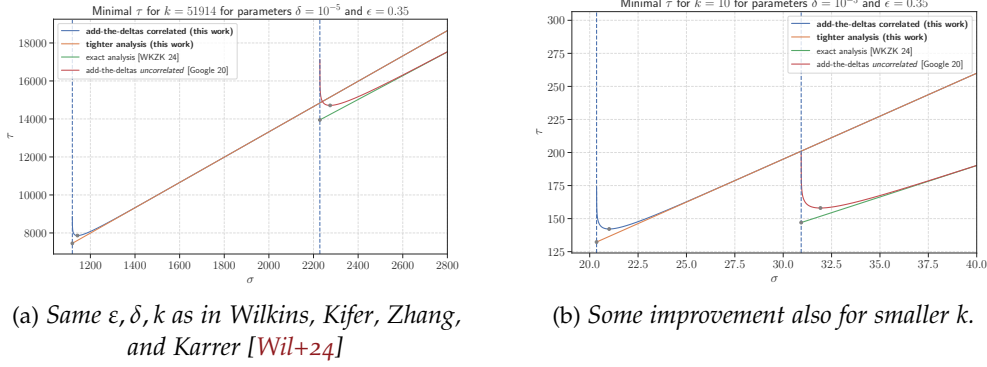


Figure 7.3: The results of our experiments. Using the same parameters as in [Wil+24], the graphs show the minimum τ required to get $(0.35, 10^{-5})$ -DP guarantees for a noise level σ . The green line denotes the tight analysis of Wilkins, Kifer, Zhang, and Karrer [Wil+24], the red shows the add-the-deltas [GAT20] approach and the blue and orange lines are our results. The marked points denote the minimum τ for each technique. **a)** Uses the same parameters as in Wilkins, Kifer, Zhang, and Karrer [Wil+24]. As high values of k are preferable for our mechanism, we bring down the threshold from ≈ 13950 to ≈ 7860 , lowering it by $\approx 43\%$. **b)** We get some small improvement even for small k values. Note that since our mechanism adds two noise samples the plot shows the total magnitude of noise.

even in that case. Even in that case, our looser *add-the-deltas* analysis for the CSH beats Wilkins, Kifer, Zhang, and Karrer [Wil+24]. Note that the values of σ in Algorithm 15 and Algorithm 16 are not directly comparable, because we add noise twice in Algorithm 16. For a fair comparison, we instead use the total magnitude of noise in the plots. For the *Correlated Sparse Histogram Mechanism* we plot the value of $(\sigma^2 + \sigma^2/\sqrt{k})^{1/2} = (1 + 1/\sqrt{k})^{1/2}\sigma$. The dotted lines indicated the minimum magnitude of total noise for which GSHM and *Correlated Sparse Histogram Mechanism* satisfy (ϵ, δ) -DP.

7.5 EXTENSIONS

The setting in Wilkins, Kifer, Zhang, and Karrer [Wil+24] is slightly different from ours. Here we discuss how to adapt our technique to their setting and consider two extensions of the GSHM we did not include in the pseudocode of Algorithm 15.

ADDITIONAL SPARSITY THRESHOLD. The mechanism of Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] employs a second threshold $\tau' < \tau$ that allows to filter out infrequent data in a pre-processing step: All counters below τ' are removed before adding noise. The high threshold τ is then used to remove noisy counters similar to both Algorithms 15 and 16. This generalized setup allows them to account for pre-processing steps that the privacy expert has no control over. Our setting corresponds to the case where $\tau' = 1$, and it is straightforward to incorporate this constraint into our mechanism: For a given τ' , we simply have to replace τ with the difference between the two thresholds in all theorems. In fact, this situation is very similar to our result in Section 7.3 where we remove all values below a lower threshold $\tau' = 1 + H(\mathbf{x})^{(k+1)}$ before adding noise.

Note that the lower threshold of Wilkins, Kifer, Zhang, and Karrer [Wil+24] is assumed to be data independent. If it is data-dependent, one must take care not to

3709 violate privacy. The privacy guarantees hold if we can give guarantees about the
 3710 structure of pre-processed neighboring datasets similar to Lemma 7.10.

3711 AGGREGATOR FUNCTIONS. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer
 3712 [Wil+24], consider a setting where we are given an aggregator function $\mathcal{A}$ which
 3713 returns a vector when applied to any dataset $\mathbf{x}$.

3714 If any $j \in [d]$ in the privatized released histogram $\tilde{H}(\mathbf{x})_j$, exceeds the threshold,
 3715 the aggregator function is applied to all data points where $\mathbf{x}_{i,j} = 1$. The aggregated
 3716 values are then privatized by adding Gaussian noise which shape and magnitude
 3717 can be set independently of the magnitude of noise added to the count. The
 3718 privatized aggregate is then released along with $\tilde{H}(\mathbf{x})_j$. This setup is motivated by
 3719 *group-by* database operations where we first want to privately estimate the count
 3720 of each group together with some aggregates modeled by $\mathcal{A}$.

3721 We did not consider aggregate functions in our analysis and instead focused on
 3722 the classical setting, where we only wanted to privately release a sparse histogram
 3723 while ignoring zero counters. Wilkins, Kifer, Zhang, and Karrer account for the
 3724 impact of the aggregator functions in the equivalent of δ_{gauss} in [Wil+24, Theorem
 3725 5.4, Corollary 5.4.1].

3726 In short, the effect of aggregators can be accounted for as an increase in the ℓ_2
 3727 sensitivity using a transformation of the aggregate function. The same technique
 3728 could be applied to our mechanism. This would give us a similar improvement
 3729 as Lemma 7.8 over the standard GSHM. It's not as clear how the mechanisms
 3730 compare under the tighter analyses.

3731 The core idea behind the correlated noise can also be used to improve some
 3732 aggregate queries. This can be used even in settings where we do not want to add
 3733 correlated noise to the counts. The mechanism of Lebeda uses an estimate of the
 3734 dataset size $\tilde{n}$ to reduce the independent noise added to each query. They spent
 3735 part of the privacy budget to estimate $\tilde{n}$. However, in our setting, we already have
 3736 a private estimate for the dataset size for the aggregate query in $\tilde{H}(\mathbf{x})_j$.

3737 If each query is a *sum* query where users have a value between 0 and some
 3738 value $C \in \mathbb{R}$, we can reduce the magnitude of noise by a factor of 2 by adding
 3739 $(\tilde{n} - n)C/2$ to each sum (It follows from [Leb25, Lemma 3.6]). We can use a similar
 3740 trick if $\mathcal{A}$ computes a sum of values in some range $[L, U]$. The standard approach
 3741 would be to add noise with magnitude scaled to $\max(|L|, |U|)$ (see e.g. [Wil+20,
 3742 Table 1]). Instead, we can recenter values around zero. As an example, $\mathcal{A}$ contains
 3743 a sum query with values in $[100, 200]$. We subtract $(L + U)/2$ from each point so
 3744 that we instead sum values in $[-50, 50]$ and reduce the sensitivity from 200 to 50.
 3745 As post-processing, we add $150 \cdot \tilde{H}(\mathbf{x})_j$ to the estimate of the new sum. The error
 3746 then depends on $\tilde{H}(\mathbf{x})_j - H(\mathbf{x})_j$. As with the Correlated Gaussian Mechanism, we
 3747 gain an advantage over estimating the sum directly if $\mathcal{A}$ contains multiple such
 3748 sums because we can reuse the estimate $\tilde{H}(\mathbf{x})_j$ for each sum.

3749 7.6 FROM THEORY TOWARDS PRACTICE: DISCRETE GAUSSIAN NOISE

3750 All mechanisms discussed in this chapter achieve privacy guarantees using contin-
 3751 uous Gaussian noise which is a standard tool in the differential privacy literature.
 3752 However, real numbers cannot be represented exactly on computers which makes
 3753 the implementation challenging. We can instead use the Discrete Gaussian Mecha-
 3754 nism [CKS20]. In this section, we modify the technique by Lebeda Lebeda [Leb25]
 3755 to make the mechanism compatible with discrete Gaussian noise $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ and
 3756 then provide privacy guarantees for the discrete analogue to Algorithm 16 using

Algorithm 18 The Discrete Correlated Gaussian Mechanism**Require:** Parameters k and σ , monotonic histogram $H(\mathbf{x}) \in \mathbb{N}^k$.

- 1: Let $\tilde{\mathbf{h}} = \{0\}^k$.
- 2: Sample $Z_{\text{corr}} \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k})$.
- 3: **for** each $i \in [k]$ **do**
- 4: Sample $Z_i \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2)$.
- 5: Set $\tilde{H}_i = H(\mathbf{x})_i + (Z_i + Z_{\text{corr}})/2$.
- 6: **Release** $\tilde{H}$.

ρ -zero concentrated DP (ρ -zCDP). We first list required privacy guarantees of the *Multivariate Discrete Gaussian* as defined in Definition A.6.

Lemma 7.12 ([CKS20, Theorem 2.13] Multivariate Discrete Gaussian Mechanism). *Let $\sigma_1, \dots, \sigma_d > 0$ and $\rho > 0$ and let $q : \mathcal{U}^* \rightarrow \mathbb{Z}^d$ satisfy $\sum_{i \in [d]} (q(\mathbf{x})_i - q(\mathbf{x}')_i)^2 / \sigma_i^2 \leq 2\rho$ for all neighboring datasets $\mathbf{x} \sim \mathbf{x}'$. Define a randomized algorithm $M : \mathcal{U}^* \rightarrow \mathbb{Z}^d$ by $M(\mathbf{x}) = q(\mathbf{x}) + Z$ where $Z_i \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_i^2)$ independently for each $i \in [d]$. Then M satisfies ρ -zCDP.*

Next, we present a discrete variant of the Correlated Gaussian Mechanism and prove that it satisfies ρ -zCDP. We can then convert the privacy guarantees to approximate differential privacy. We combine this with the *add-the-deltas* technique for a discrete variant of Algorithm 16. This approach does not give the tightest privacy parameters, but it is significantly simpler than a direct analysis of approximate differential privacy for multivariate discrete Gaussian noise (see [CKS20, Theorem 2.14]). The primary contribution in this section is a simple change that makes the correlated Gaussian mechanism compatible with the discrete Gaussian. We leave providing tighter analysis of the privacy guarantees open for future work.

Lemma 7.13. *Algorithm 18 satisfies ρ -zCDP where $\rho = (k + \sqrt{k})/(8\sigma^2)$.*

Proof. Fix any pair of neighboring histograms $H(\mathbf{x})$ and $H(\mathbf{x}')$. We prove the lemma for the case where $H(\mathbf{x}) - H(\mathbf{x}') \in \{0, 1\}^k$. The proof is symmetric when $H(\mathbf{x}) - H(\mathbf{x}') \in \{0, -1\}^k$. Construct a new pair of histograms $\hat{H}(\mathbf{x}), \hat{H}(\mathbf{x}') \in \mathbb{N}^{k+1}$ such that $\hat{H}(\mathbf{x})_i = 2 \cdot H(\mathbf{x})_i$ for all $i \in [k]$ and $\hat{H}(\mathbf{x})_{k+1} = 0$. We set $\hat{H}(\mathbf{x}')_i = 2 \cdot H(\mathbf{x}')_i - 1$ for all $i \in [k]$ and finally $\hat{H}(\mathbf{x}')_{k+1} = 1$.

We clearly have that $\hat{H}(\mathbf{x}) \in \mathbb{Z}^{k+1}$ and $\hat{H}(\mathbf{x}') \in \mathbb{Z}^{k+1}$ for all possible $H(\mathbf{x})$ as required by Lemma 7.12. Now, we set $\sigma_i^2 = 4\sigma^2$ for $i \in [k]$ and $\sigma_{k+1}^2 = 4\sigma^2/\sqrt{k}$. We constructed $\hat{H}(\mathbf{x})$ and $\hat{H}(\mathbf{x}')$ such that they differ by at most 1 in all entries for any pair of neighboring histograms. We therefore have that

$$\sum_{i \in [k+1]} (\hat{H}(\mathbf{x})_i - \hat{H}(\mathbf{x}')_i)^2 / \sigma_i^2 \leq \sum_{i \in [k+1]} 1 / \sigma_i^2 = k / (4\sigma^2) + \sqrt{k} / (4\sigma^2) = 2\rho.$$

By Lemma 7.12 we have that $D_\alpha(M(\mathbf{x}) || M(\mathbf{x}')) \leq \rho\alpha$ for all $\alpha > 1$, where $M(\mathbf{x}) = \hat{H}(\mathbf{x}) + Z$ and $Z_i \sim \mathcal{N}(0, \sigma_i^2)$. The output of $M(\mathbf{x})$ can be post-processed such that $\tilde{H}_i = (M(\mathbf{x})_i + M(\mathbf{x})_{k+1})/2$. Notice that such a post-processing gives us the same output distribution as Algorithm 18 for both $H(\mathbf{x})$ and $H(\mathbf{x}')$. The algorithm therefore satisfies ρ -zCDP because the α -Rényi divergence cannot increase from post-processing by Definition 2.9. $\square$

Note that the scaling step is crucial for the privacy proof. It would not be sufficient to simply replace $Z_i \sim \mathcal{N}(0, \sigma^2)$ with $Z_i \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$. We need to sample noise in discrete steps of length $1/2$ instead of length 1. Otherwise, the trick of centering the differences between the histograms using the $k+1$ entry as an offset does not work. If we prefer a mechanism that always outputs integers, we can simply post-process the output. Next, we give the privacy guarantees of the Correlated Stability Histogram when using discrete noise.

Lemma 7.14. *Let $Z_{\text{corr}} \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k})$ and $Z_1, \dots, Z_k \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2)$. Then for any $\tau > 0$ we have that:*

$$1 - \mathbf{P} \left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k}) \leq \frac{2\tau k^{-1/4}}{k^{-1/4} + 1} \right] \cdot \mathbf{P} \left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2) \leq \frac{2\tau}{k^{-1/4} + 1} \right]^k.$$

Proof. Similar to the proof of Lemma 7.6: if $Z_{\text{corr}} \leq 2\tau k^{-1/4}/(k^{-1/4} + 1)$ and $\max_{Z_1, \dots, Z_j} Z_i = \tilde{Z} \leq 2\tau/(k^{-1/4} + 1)$ then the sums cannot be above 2τ . The only difference is that we have to split up the probabilities. In the discrete case the probabilities slightly change when we rescale. $\square$

Finally, we can replace the Gaussian noise in Algorithm 16 with discrete Gaussian noise. Using the *add-the-deltas* approach and the lemmas above we get that.

Theorem 7.15. *For parameters k, σ , and τ , consider the mechanism $\mathcal{M}: \mathbb{N}^d \rightarrow \mathbb{Q}^d$ that given a histogram $H(\mathbf{x})$: (1) runs Algorithm 18 with parameters k and σ restricted to the support of $H(\mathbf{x})$ (2) removes all noisy counts less than or equal to $1 + \tau$. Then $\mathcal{M}$ satisfies $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -DP for k -sparse monotonic histograms, where δ_{gauss} is such that $(k + \sqrt{k})/(8\sigma^2)$ - ρ -zCDP implies $(\epsilon, \delta_{\text{gauss}})$ -DP and*

$$\delta_{\text{inf}} = 1 - \mathbf{P} \left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k}) \leq \frac{2\tau k^{-1/4}}{k^{-1/4} + 1} \right] \cdot \mathbf{P} \left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2) \leq \frac{2\tau}{k^{-1/4} + 1} \right]^k.$$

Proof. The proof relies on the construction of an intermediate histogram from the proof of Theorem 7.7. We have that

$$\begin{aligned} \mathbf{P}[\mathcal{M}(H(\mathbf{x})) \in Y] &\leq \mathbf{P}[\mathcal{M}(H(\tilde{\mathbf{x}})) \in Y] + \delta_{\text{inf}} \\ &\leq e^\epsilon \mathbf{P}[\mathcal{M}(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}} + \delta_{\text{inf}}, \end{aligned}$$

where the first inequality follows from Lemma 7.14 and the second inequality follows from Lemma 7.13. Both inequalities rely on the fact that the histograms are k -sparse. $\square$

7.7 CONCLUSION AND OPEN PROBLEMS

We introduced the *Correlated Stability Histogram* for the setting of k -sparse monotonic histograms and provided privacy guarantees using the *add-the-deltas* approach and a more fine-grained case-by-case analysis. We show that our mechanism outperforms the state-of-the-art – the *Gaussian Sparse Histogram Mechanism* – and improves the utility by up to a factor of 2. In addition to various extensions, we

enriched our theoretical contributions with a step towards practice by including a version that works with discrete Gaussian noise.

Unlike the previous work [Wil+24], our bound in Theorem 7.9 is not tight. It would be interesting to derive exact bounds for our mechanism as well. Finally, we point out that the uncorrelated GSHM is still preferred in some settings. The *Correlated Sparse Histogram Mechanism* requires an upper bound on $\|H(\mathbf{x})\|_0$, whereas GSHM requires a bound on $\|X_i\|_0$. If we have access to a bound $\|X_i\|_0 \leq k$ but no sparsity bound, we can apply our technique from Section 7.3. This approach works well when the dataset is close to k -sparse, but if the dataset is large with many high counters the pre-processing step can introduce high error. Furthermore, if histograms are k -sparse, but we additionally know that $\|X_i\|_0 \leq m$, for some $m < k$, our improvement factor changes. If $m \leq k/4$, the GSHM has a lower error than the CSH. However, in that setting, it might still be possible to reduce the error using the technique of Joseph and Yu [JY24]. We leave exploring this regime for future work.

7.8 APPENDIX: PROOF OF THEOREM 7.9 (TIGHTER ANALYSIS)

Before proving the main theorem, we need a useful lemma that can be used to bound the effect of the magnitude for the Gaussian noise on the value of δ .

Lemma 7.16. *Let $H(\mathbf{x})$ and $H(\mathbf{x}')$ be a pair of j -sparse monotonic histograms with the same support. Then for Algorithm 16 with parameters $k \geq j$, σ and τ , we have*

$$\mathbf{P}[\text{CSH}(H(\mathbf{x})) \in Y] \leq e^\varepsilon \cdot \mathbf{P}[\text{CSH}(H(\mathbf{x}')) \in Y] + \delta$$

for any $\varepsilon \geq 0$ where $\gamma = \min\left(\frac{1}{2}\sqrt{j + \sqrt{k}}, \sqrt{j}\right)$ and

$$\delta \geq \Phi\left(\frac{\gamma}{2\sigma} - \frac{\varepsilon\sigma}{\gamma}\right) - e^\varepsilon \Phi\left(-\frac{\gamma}{2\sigma} - \frac{\varepsilon\sigma}{\gamma}\right).$$

Proof. We show that the statement holds for each case of the minimum for γ separately. In each case, we show that the output distribution is equivalent to post-processing a Gaussian noise mechanism. We start with the case of $\gamma = \frac{1}{2}\sqrt{j + \sqrt{k}}$. Notice that by Lemma 7.5, the inequality holds if we apply the correlated Gaussian mechanism to histograms with at most j entries and set the magnitude of Z_{cor} to $\sigma/\sqrt{k}$. Notice that the output distributions of CSH for both $H(\mathbf{x})$ or $H(\mathbf{x}')$ as input is equivalent to running the Correlated Gaussian Mechanism restricted to their support followed by a post-processing step of removing values below $1 + \tau$.

Similarly, for the case of $\gamma = \sqrt{j}$ notice that $\|H(\mathbf{x}) - H(\mathbf{x}')\|_2 \leq \gamma$. Therefore, we can output an (ε, δ) -DP estimate of the histogram using the Standard Gaussian Mechanism with the parameters chosen according to the lemma. Now, notice that if we apply the Standard Gaussian Mechanism to the histograms, we can recover the output distribution of CSH by post-processing. We first sample noise $Z_{\text{cor}} \sim \mathcal{N}(0, \sigma/\sqrt{k})$ and add the value to all entries, and then removing all noise counts below $1 + \tau$ and any count not in the support of the histograms.

The lemma thus holds since DP is preserved under post-processing. $\square$

We are now ready to proof the theorem which we restate for convenience.

3842 **Theorem 7.9 (Tighter Analysis).** For fixed k, σ and τ , we define

$$\gamma(j) = \min \left(\sqrt{j}, \frac{1}{2} \sqrt{j + \sqrt{k}} \right), \quad \psi(m) = \Phi \left(\frac{\tau}{(1 + k^{-1/4}) \cdot \sigma} \right)^{m+1}, \text{ and}$$

$$\hat{\varepsilon}(j) = \varepsilon + \ln(\psi(k - j)).$$

3843 Then Algorithm 16 satisfies (ε, δ) -DP for k -sparse monotonic histograms, where

$$\begin{aligned} \max & \left[1 - \psi(k), \Phi \left(\frac{\sqrt{k + \sqrt{k}}}{4\sigma} - \frac{2\varepsilon\sigma}{\sqrt{k + \sqrt{k}}} \right) - e^\varepsilon \Phi \left(-\frac{\sqrt{k + \sqrt{k}}}{4\sigma} - \frac{2\varepsilon\sigma}{\sqrt{k + \sqrt{k}}} \right), \right. \\ & \max_{j \in [k-1]} 1 - \psi(k - j) + \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\varepsilon\sigma}{\gamma(j)} \right) - e^\varepsilon \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\varepsilon\sigma}{\gamma(j)} \right), \\ & \left. \max_{j \in [k-1]} \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\hat{\varepsilon}(j)\sigma}{\gamma(j)} \right) - e^{\hat{\varepsilon}(j)} \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\hat{\varepsilon}(j)\sigma}{\gamma(j)} \right) \right] \leq \delta. \end{aligned}$$

3844 *Proof.* We prove the theorem by considering all values of the sensitivity space of
3845 k -sparse monotonic histograms. We start with the simplest cases. If either $U = \emptyset$
3846 or $U' = \emptyset$ we only have to consider the infinite privacy loss event. This gives us
3847 the first term of the max by Lemma 7.6. Similarly, when $U = U'$, we get the second
3848 term of the max using the proof for δ_{gauss} in Theorem 7.7.

3849 For the remaining cases, we have to consider both the impact of Gaussian
3850 noise and the potential for the infinite privacy loss event similar to Theorem 7.7.
3851 We analyze the two cases of $U \subset U'$ and $U' \subset U$ separately. Without loss of
3852 generality, we assume that $\max(|U|, |U'|) = k$, since this is the worst-case for
3853 the privacy parameters. Furthermore, we have to analyze the mechanism when
3854 $\min(|U|, |U'|) = j$ for each possible $j \in [k - 1]$.

3855 First, we consider the case where $U' \subset U$. We construct an intermediate his-
3856 togram $H(\hat{\mathbf{x}})$ as in the proof of Theorem 7.7. The histogram $H(\hat{\mathbf{x}})$ has support
3857 U' , and $H(\hat{\mathbf{x}})_i = H(\mathbf{x})_i$ for all $i \in U'$. Since $H(\hat{\mathbf{x}})$ and $H(\mathbf{x}')$ are a pair of j -sparse
3858 monotonic histograms we have by Lemma 7.16 that

$$\mathbf{P}[\text{CSH}(H(\hat{\mathbf{x}})) \in Y] \leq e^\varepsilon \mathbf{P}[\text{CSH}(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}}(j),$$

3859 where

$$\delta_{\text{gauss}}(j) = \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\varepsilon\sigma}{\gamma(j)} \right) - e^\varepsilon \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\varepsilon\sigma}{\gamma(j)} \right).$$

3860 The histograms $H(\hat{\mathbf{x}})$ and $H(\mathbf{x})$ only differs the entries $U \setminus U'$. Since $|U \setminus U'| = (k - j)$
3861 we have that

$$\mathbf{P}[\text{CSH}(H(\mathbf{x})) \in Y] \leq \mathbf{P}[\text{CSH}(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}}(j),$$

3862 where $\delta_{\text{inf}}(j) = 1 - \psi(k - j)$ follows from Lemma 7.6.

3863 We recover the third term in the max using the values for $\delta_{\text{inf}}(j)$ and $\delta_{\text{gauss}}(j)$
3864 and the fact that

$$\begin{aligned} \mathbf{P}[\text{CSH}(H(\mathbf{x})) \in Y] & \leq \mathbf{P}[\text{CSH}(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}}(j) \\ & \leq e^\varepsilon \mathbf{P}[\text{CSH}(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}}(j) + \delta_{\text{inf}}(j). \end{aligned}$$

3865 Finally, we consider the case where $U \subset U'$. This is the most technically involved
3866 case in the proof. Here, the intermediate histogram $H(\hat{\mathbf{x}})$ does not give us the
3867 bound we want because it is not k -sparse. Instead, we use an alternative repre-
3868 sentation of each histogram similar to the technique used in [Leb25]. We consider

a mechanism for this representation and show that the bound holds when the alternative representations are used as input for this mechanism. We then show that we can recover the distribution of Algorithm 16 using post-processing. We first give the proof when $\gamma = \frac{1}{2}\sqrt{j + \sqrt{k}}$ and then show that it also holds for $\gamma = \sqrt{j}$.

Assume without loss of generality that $U = [j]$ and $U' = [k]$ (we can always reorder the elements of the new representation, but this assumption makes the presentation simpler). We construct a vector $\mathbf{g} \in \mathbb{R}^{k+1}$ from $H(\mathbf{x})$ such that $\mathbf{g}_i = H(\mathbf{x})_i$ for each $i \in [j]$ and $\mathbf{g}_i = 0$ for any $i > j$. For $\mathbf{g}' \in \mathbb{R}^{k+1}$ we set $\mathbf{g}'_i = H(\mathbf{x}')_i - 1/2$ for all $i \in [k]$ and we set $\mathbf{g}'_{k+1} = \frac{1}{2}k^{1/4}$.

Now, we consider two mechanisms with input $\mathbb{R}^{k+1}$. The first mechanism $\mathcal{G}$ ignore entries from index $j+1$ to k and adds noise independently from $\mathcal{N}(0, \sigma^2)$ to all entries with index $i \in [j]$ as well as the $k+1$ 'th entry. We want to show that $\mathbf{P}[\mathcal{G}(\mathbf{g}) \in Y] \leq e^{\hat{\epsilon}(j)} \cdot \mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y] + \hat{\delta}(j)$ for all Y where

$$\hat{\delta}(j) = \Phi\left(\frac{\frac{1}{2}\sqrt{j + \sqrt{k}}}{2\sigma} - \frac{\hat{\epsilon}(j)\sigma}{\frac{1}{2}\sqrt{j + \sqrt{k}}}\right) - e^{\hat{\epsilon}(j)}\Phi\left(-\frac{\frac{1}{2}\sqrt{j + \sqrt{k}}}{2\sigma} - \frac{\hat{\epsilon}(j)\sigma}{\frac{1}{2}\sqrt{j + \sqrt{k}}}\right).$$

Notice that $\mathcal{G}$ is equivalent to the standard Gaussian mechanism restricted to the domain $[j] \cup \{k+1\}$. At the same time, the distribution of $\mathcal{G}(\mathbf{g}')$ is equivalent to running $\mathcal{G}(\hat{\mathbf{g}})$ for a third vector $\hat{\mathbf{g}}$ where $\hat{\mathbf{g}}_i = \mathbf{g}_i$ for $i \in ([j] \cup \{k+1\})$ and $\hat{\mathbf{g}}_i = 0$ for $j < i \leq k$. The inequality above therefore follows from Lemma 7.2 if the ℓ_2 distance between $\mathbf{g}$ and $\hat{\mathbf{g}}$ is at most $\frac{1}{2}\sqrt{j + \sqrt{k}}$ for any neighboring histograms. The absolute difference between $\mathbf{g}_i$ and $\hat{\mathbf{g}}_i$ is $1/2$ for each $i \in [j]$, while $\mathbf{g}_{k+1}$ and $\hat{\mathbf{g}}_{k+1}$ differ by $k^{1/4}/2$. The ℓ_2 distance between the vectors is therefore always $\|\mathbf{g} - \hat{\mathbf{g}}\|_2 = (\sum_{i \in [k+1]} (\mathbf{g}_i - \hat{\mathbf{g}}_i)^2)^{1/2} = ((k^{1/4}/2)^2 + \sum_{i \in [j]} (1/2)^2)^{1/2} = \frac{1}{2}\sqrt{j + \sqrt{k}}$.

Next, consider a mechanism $\mathcal{F}$ that adds noise independently from $\mathcal{N}(0, \sigma^2)$ to all non-zero entries of a vector. Let $\tilde{\mathbf{g}}$ be the noisy vector after adding noise to a vector $\mathbf{g}$. If $\{j+1, \dots, k\}$ is in support of $\mathbf{g}$ we check if $\tilde{\mathbf{g}}_{k+1}/k^{1/4} + \tilde{\mathbf{g}}_i > \tau$ for any $i \in \{j+1, \dots, k\}$. In that case, we output $\perp$; otherwise, we set all such entries to 0 and output the noisy vector $\tilde{\mathbf{g}}$. Here, we use the output $\perp$ to group all events corresponding to the infinite privacy loss in the original representation.

We want to show that $\mathbf{P}[\mathcal{F}(\mathbf{g}) \in Y] \leq e^{\epsilon} \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y] + \hat{\delta}(j)$ for all Y . It is sufficient to show that the inequality holds for the set $Y_\epsilon = \{y \mid e^\epsilon \leq \mathcal{F}(\mathbf{g})(y) / \mathcal{F}(\mathbf{g}')(y)\}$, where $\mathcal{F}(\mathbf{g})(y)$ denotes the density at y (see [CKS20, Theorem 5]). Now, if $\mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\epsilon] \leq \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\epsilon] / \psi(k-j)$ the inequality holds because $\mathcal{G}(\mathbf{g})$ and $\mathcal{F}(\mathbf{g})$ has the same distribution and

$$\begin{aligned} \mathbf{P}[\mathcal{F}(\mathbf{g}) \in Y_\epsilon] &= \mathbf{P}[\mathcal{G}(\mathbf{g}) \in Y_\epsilon] \leq e^{\hat{\epsilon}} \mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\epsilon] + \hat{\delta}(j) \\ &\leq e^{\epsilon + \ln(\psi(k-j))} (\mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\epsilon] / \psi(k-j)) + \hat{\delta}(j) \\ &= e^{\epsilon} \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\epsilon] + \hat{\delta}(j), \end{aligned}$$

where the last equality follows from $1/\psi(k-j) = e^{-\ln(\psi(k-j))}$.

The missing piece above is to show that $\mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\epsilon] \leq \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\epsilon] / \psi(k-j)$. The equivalent inequality is easy to see in the non-correlated variant of the Gaussian Sparse Histogram Mechanism because Y_ϵ does not contain $\perp$ so $\mathbf{P}[\mathcal{F}'(\mathbf{g}') \in Y_\epsilon] = \mathbf{P}[\mathcal{G}'(\mathbf{g}') \in Y_\epsilon] \mathbf{P}[\top]$, where $\top$ is the complement of $\perp$, and $\mathcal{F}'$ and $\mathcal{G}'$ are the equivalents to $\mathcal{F}$ and $\mathcal{G}$ without the $(k+1)$ 'th entry. The value of $\psi(k-j)$ is a lower bound on the probability of $\top$ occurring. Here we need to be a bit more careful here, but the inequality still holds because $\top$ occurring increases the probability

that $\tilde{\mathbf{g}}'_{k+1}$ is small, which leads to a higher privacy loss event. We sketch the argument below

$$\begin{aligned} \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\varepsilon] / \psi(k-j) &= \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\varepsilon \mid \top] \mathbf{P}[\top] / \psi(k-j) \\ &\geq \mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\varepsilon \mid \top] \geq \mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\varepsilon] , \end{aligned}$$

where the last inequality relies on

$$\mathbf{P}[\mathcal{F}(\mathbf{g}') \in Y_\varepsilon \mid \tilde{\mathbf{g}}'_{k+1} \leq \beta \cap \top] = \mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\varepsilon \mid \tilde{\mathbf{g}}'_{k+1} \leq \beta] ,$$

which holds for any $\beta \in \mathbb{R}$. and $\mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\varepsilon \mid \tilde{\mathbf{g}}'_{k+1} \leq \beta] \geq \mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y_\varepsilon]$. For any β we also have that $\mathbf{P}[\top \mid \mathcal{F}(\mathbf{g}') \leq \beta] \geq \mathbf{P}[\top]$ which allows us to apply Bayes' theorem to find that $\mathbf{P}[\mathcal{F}(\mathbf{g}')_{k+1} \leq \beta \mid \top] \geq \mathbf{P}[\mathcal{G}(\mathbf{g}')_{k+1} \leq \beta]$. Integrating over all values of β gives the inequality.

Finally, we must show that Algorithm 16 is equivalent to post-processing $\mathcal{F}$. Let $\tilde{\mathbf{g}}$ denote the output of $\mathcal{F}(\mathbf{g})$. Then we post-process the entries by first setting $\tilde{\mathbf{h}}_i = \tilde{\mathbf{g}}_i + \tilde{\mathbf{g}}_i/k^{1/4}$ and then removing all values of $\tilde{\mathbf{h}}_i$ below $1 + \tau$. If the output was $\perp$ we repeatable sample $\tilde{H} \leftarrow \text{CSH}(H(\mathbf{x}'))$ until $\tilde{H}_i \neq i$ for at least one $j < i \leq k$. We then output the histogram $\tilde{H}$.

The proof is similar when $\gamma = \sqrt{j}$, but we use a different construction for $\mathbf{g}'$. We set $\mathbf{g}'_i = H(\mathbf{x}')_i$ for all $i \in [k]$ and $\mathbf{g}'_{k+1} = 0$.

The inequality $\mathbf{P}[\mathcal{G}(\mathbf{g}) \in Y] \leq e^{\hat{\varepsilon}(j)} \cdot \mathbf{P}[\mathcal{G}(\mathbf{g}') \in Y] + \hat{\delta}(j)$ holds where $\frac{1}{2}\sqrt{j + \sqrt{k}}$ is replaced with $\sqrt{j}$ in $\hat{\delta}(j)$ because

$$\|\mathbf{g} - \hat{\mathbf{g}}\|_2 = \left(\sum_{i \in [j]} (\mathbf{g}_i - \hat{\mathbf{g}}_i)^2 \right)^{1/2} \leq \left(\sum_{i \in [j]} 1^2 \right)^{1/2} = \sqrt{j} .$$

Since all possible elements in the sensitivity space falls in one of the categories above, we can compute the value of all inequalities above. The maximum of all parameters gives us an upper bound for δ . $\square$

3928

Part III

3929

3930

3931



Throughout the thesis, we use multiple probability distributions, which we will quickly define in this section.

Definition A.1 (Gaussian Distribution). *The zero-centered Gaussian (normal) distribution $\mathcal{N}(0, \sigma^2)$ with variance σ^2 has pdf $\phi(x, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{x^2}{2\sigma^2})$. We denote the cdf of the Gaussian distribution as $\Phi(x)$. We refer to Φ^{-1} as the inverse CDF.*

Definition A.2 (Laplace distribution). *The zero-centered Laplace distribution $\text{Lap}(0, b)$ with scale parameter $b > 0$ has probability density function $f_b(x) = \frac{1}{2b} \exp(-|x|/b)$ for all $x \in \mathbb{R}$.*

Definition A.3 ([Ros18] Beta Distribution). *The beta distribution $\text{Beta}(\alpha, \beta)$ is a distribution defined on $[0, 1]$ whose density is given by $p(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\text{B}(\alpha, \beta)}$, $\forall x \in [0, 1]$, where $\alpha, \beta > 0$ are shape parameters, $\text{B}(\alpha, \beta) \doteq \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$ is a normalization constant.*

Definition A.4 (Exponential Distribution). *Given $\lambda > 0$, a random variable X follows exponential distribution $\text{Exp}(\lambda)$, denoted by $X \sim \text{Exp}(\lambda)$, if it has density $p_X(x) \doteq \lambda e^{-\lambda x}$, $\forall x \geq 0$, and $p_X(x) \doteq 0$, $\forall x < 0$.*

Definition A.5 (Gumbel Distribution). *Given parameter $b \in \mathbb{R}$, the Gumbel distribution, $\text{Gumbel}(b)$, has pdf $p(z) = \frac{1}{b} \cdot \exp(-(\frac{z}{b} + \exp(-\frac{z}{b})))$, $\forall z \in \mathbb{R}$, and cumulative distribution function $F(z; b) = \exp(-e^{-z/b})$, $\forall z \in \mathbb{R}$.*

Definition A.6 (Discrete Gaussian Distribution). *Let $\sigma \in \mathbb{R}$ with $\sigma > 0$. The discrete Gaussian distribution with mean 0 and scale σ is denoted $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$. It is a probability distribution supported on the integers and defined by*

$$\forall x \in \mathbb{Z} : \mathbf{P}_{X \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)}[X = x] = \frac{e^{-x^2/(2\sigma^2)}}{\sum_{y \in \mathbb{Z}} e^{-y^2/(2\sigma^2)}}.$$

Definition A.7 (Geometric Distribution). We say X has distribution $\text{Geom}(p)$ with parameter $0 < p \leq 1$, if $\mathbf{P}[X = k] = (1 - p)^{k-1}p$ for all $k \in \mathbb{N}_{>0}$.

Definition A.8 (Discrete Laplace). The noise distribution for the discrete Laplace (two-sided geometric) $\text{DLap}(\alpha)$ is a two-sided geometric distribution with pmf $p(x) = \frac{1-\alpha}{1+\alpha} \cdot \alpha^{|x|}$ for $\alpha \in (0, 1)$ and $x \in \mathbb{Z}$.

3936 ABBREVIATIONS

- 3937 ANM Additive Noise Mechanism
- 3938 DP Differential Privacy
- 3939 IANM Independent Additive Noise Mechanism
- 3940 MST Minimum-Weight Spanning Tree
- 3941 MWPM Minimum-Weight (Perfect) Matching
- 3942 PLMR Private Lossless Multiple Release
- 3943 PPSACR Probability Proportional to Sizes with Adaptive Candidate Removal
- 3944 ρ -zCDP ρ -zero concentrated DP

- [ACM26] ACM SIGACT. *STOC Test of Time Award*. https://sigact.org/prizes/stoc_tot.html. Accessed: 2026-05-30. 2026 (cit. on pp. 14, 52).
- [Aam26] Anders Aamand. *Personal communication*. Communication with undisclosed researcher. 2026 (cit. on pp. 49–51, 68, 75).
- [Abo+22] John M. Abowd et al. “The 2020 Census Disclosure Avoidance System TopDown Algorithm.” In: *CoRR* abs/2204.08986 (2022). DOI: 10.48550/arXiv.2204.08986. arXiv: 2204.08986 (cit. on p. 5).
- [Ade+23] Temilola Adeleye et al. “Publishing Wikipedia usage data with strong privacy guarantees.” In: *CoRR* abs/2308.16298 (2023). DOI: 10.48550/arXiv.2308.16298. arXiv: 2308.16298 [cs.CR] (cit. on p. 5).
- [Aga+18] Naman Agarwal, Ananda Theertha Suresh, Felix X. Yu, Sanjiv Kumar, and Brendan McMahan. “cpSGD: Communication-Efficient and Differentially-Private Distributed SGD.” In: *Advances in Neural Information Processing Systems*. Vol. 31. 2018 (cit. on pp. 97, 98).
- [And+25] Joel Daniel Andersson, Lukas Retschmeier, Boel Nelson, and Rasmus Pagh. “Private Lossless Multiple Release.” In: *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13–19, 2025*. Vol. 267. Proceedings of Machine Learning Research. PMLR / OpenReview.net, 2025 (cit. on p. 15).
- [Arro6] Michael Arrington. *AOL Proudly Releases Massive Amounts of Private Data*. TechCrunch. 2006. URL: <https://techcrunch.com/2006/08/06/aol-proudly-releases-massive-amounts-of-user-search-data/> (visited on 04/27/2026) (cit. on p. 1).
- [Ash90] R.B. Ash. *Information Theory*. Dover Books on Advanced Mathematics. Dover Publications, 1990. ISBN: 978-0-486-66521-4 (cit. on p. 45).
- [BU17] Mitali Bafna and Jonathan R. Ullman. “The Price of Selection in Differential Privacy.” In: *Proceedings of the 30th Conference on Learning Theory, COLT 2017, Amsterdam, The Netherlands, 7–10 July 2017*. Vol. 65. Proceedings of Machine Learning Research. PMLR, 2017 (cit. on p. 123).
- [BV19] Victor Balcer and Salil P. Vadhan. “Differential Privacy on Finite Computers.” In: *J. Priv. Confidentiality* 9.2 (2019). DOI: 10.29012/jpc.679 (cit. on pp. 112, 116).
- [BW18] Borja Balle and Yu-Xiang Wang. “Improving the Gaussian Mechanism for Differential Privacy: Analytical Calibration and Optimal Denoising.” In: *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholmsmässan, Stockholm, Sweden, July 10–15, 2018*. Vol. 80. Proceedings of Machine Learning Research. PMLR, 2018 (cit. on pp. vii–ix, 9, 88).

- [Bat+17] MohammadHossein Bateni, Soheil Behnezhad, Mahsa Derakhshan, MohammadTaghi Hajiaghayi, Raimondas Kiveris, Silvio Lattanzi, and Vahab S. Mirrokni. "Affinity Clustering: Hierarchical Clustering at Scale." In: *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*. 2017 (cit. on p. 14).
- [BMS04] Peter S. Bearman, James Moody, and Katherine Stovel. "Chains of Affection: The Structure of Adolescent Romantic and Sexual Networks." In: *American Journal of Sociology* 110.1 (2004). Used with permission of University of Chicaco Press, permission conveyed through Copyright Clearance Center, Inc. (license 1730600-1, 16-May-2026). ISSN: 1537-5390. DOI: [10.1086/386272](https://doi.org/10.1086/386272) (cit. on p. 12).
- [BLP76] D. Elliott Bell and Leonard J. La Padula. *Secure Computer System: Unified Exposition and Multics Interpretation*: tech. rep. Fort Belvoir, VA: Defense Technical Information Center, 1976. DOI: [10.21236/ADA023588](https://doi.org/10.21236/ADA023588). (Visited on 01/23/2025) (cit. on p. 80).
- [Blo+13] Jeremiah Blocki, Avrim Blum, Anupam Datta, and Or Sheffet. "Differentially private data analysis of social networks via restricted sensitivity." In: *Innovations in Theoretical Computer Science, ITCS '13, Berkeley, CA, USA, January 9-12, 2013*. ITCS '13. ACM, 2013. DOI: [10.1145/2422436.2422449](https://doi.org/10.1145/2422436.2422449) (cit. on p. 13).
- [Bod+24] Greg Bodwin, Chengyuan Deng, Jie Gao, Gary Hoppenworth, Jalaj Upadhyay, and Chen Wang. "The Discrepancy of Shortest Paths." In: *51st International Colloquium on Automata, Languages, and Programming, ICALP 2024, Tallinn, Estonia, July 8-12, 2024*. Vol. 297. LIPIcs. Dagstuhl, Germany: Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2024. ISBN: 978-3-95977-322-5. DOI: [10.4230/LIPIcs.ICALP.2024.27](https://doi.org/10.4230/LIPIcs.ICALP.2024.27) (cit. on p. 52).
- [BMM24] Felipe T. Brito, André L. C. Mendonça, and Javam C. Machado. "A Differentially Private Guide for Graph Analytics." In: *Proceedings 27th International Conference on Extending Database Technology, EDBT 2024, Paestum, Italy, March 25 - March 28*. OpenProceedings.org, 2024. DOI: [10.48786/edbt.2024.86](https://doi.org/10.48786/edbt.2024.86) (cit. on p. 13).
- [Bro+92] Peter F. Brown, Vincent J. Della Pietra, Peter V. de Souza, Jennifer C. Lai, and Robert L. Mercer. "Class-Based n-gram Models of Natural Language." In: *Comput. Linguistics* 18.4 (1992) (cit. on p. 14).
- [BNS16] Mark Bun, Kobbi Nissim, and Uri Stemmer. "Simultaneous Private Learning of Multiple Concepts." In: *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science, Cambridge, MA, USA, January 14-16, 2016*. ACM, 2016. DOI: [10.1145/2840728.2840747](https://doi.org/10.1145/2840728.2840747) (cit. on pp. 112, 116).
- [BS16] Mark Bun and Thomas Steinke. "Concentrated Differential Privacy: Simplifications, Extensions, and Lower Bounds." In: *Theory of Cryptography - 14th International Conference, TCC 2016-B, Beijing, China, October 31 - November 3, 2016, Proceedings, Part I*. Vol. 9985. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2016. ISBN: 978-3-662-53641-4. DOI: [10.1007/978-3-662-53641-4_24](https://doi.org/10.1007/978-3-662-53641-4_24) (cit. on pp. 10, 11).

- 4033 [CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. “The
4034 Discrete Gaussian for Differential Privacy.” In: *Advances in Neural*
4035 *Information Processing Systems 33: Annual Conference on Neural Infor-*
4036 *mation Processing Systems 2020, NeurIPS 2020, December 6-12, 2020,*
4037 *virtual*. Vol. 33. Curran Associates, Inc., 2020 (cit. on pp. 90, 115, 126,
4038 127, 131).
- 4039 [Chao0] Bernard Chazelle. “A minimum spanning tree algorithm with Inverse-
4040 Ackermann type complexity.” In: *J. ACM* 47.6 (2000). ISSN: 0004-5411.
4041 DOI: 10.1145/355541.355562 (cit. on pp. 23, 38).
- 4042 [Che+23] Justin Y. Chen, Badih Ghazi, Ravi Kumar, Pasin Manurangsi, Shyam
4043 Narayanan, Jelani Nelson, and Yinzhan Xu. “Differentially Private
4044 All-Pairs Shortest Path Distances: Improved Algorithms and Lower
4045 Bounds.” In: *Proceedings of the 2023 ACM-SIAM Symposium on Discrete*
4046 *Algorithms, SODA 2023, Florence, Italy, January 22-25, 2023*. SIAM, 2023.
4047 ISBN: 9781611977554. DOI: 10.1137/1.9781611977554.ch184 (cit. on
4048 pp. 49, 52).
- 4049 [CL68] C. K. Chow and C. N. Liu. “Approximating discrete probability
4050 distributions with dependence trees.” In: *IEEE Trans. Inf. Theory* 14.3
4051 (1968). DOI: 10.1109/TIT.1968.1054142 (cit. on pp. 13, 22, 38).
- 4052 [Coh+19] Vincent Cohen-Addad, Varun Kanade, Frederik Mallmann-Trenn,
4053 and Claire Mathieu. “Hierarchical Clustering: Objective Functions
4054 and Algorithms.” In: *J. ACM* 66.4 (2019). ISSN: 1557-735X. DOI: 10.11
4055 45/3321386 (cit. on p. 52).
- 4056 [Coh97] Edith Cohen. “Size-Estimation Framework with Applications to Tran-
4057 sitive Closure and Reachability.” In: *J. Comput. Syst. Sci.* 55.3 (1997).
4058 DOI: 10.1006/jcss.1997.1534 (cit. on pp. 24, 36, 37).
- 4059 [Cor+12] Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and
4060 Thanh T. L. Tran. “Differentially Private Summaries for Sparse Data.”
4061 In: *15th International Conference on Database Theory*. Berlin, Germany:
4062 ACM, 2012. ISBN: 9781450307918. DOI: 10.1145/2274576.2274608
4063 (cit. on p. 89).
- 4064 [DNK26] Nicholas D’Silva, Surya Nepal, and Salil S. Kanhere. “SoK: Practical
4065 Aspects of Releasing Differentially Private Graphs.” In: *Proceedings*
4066 *of the ACM Asia Conference on Computer and Communications Security,*
4067 *ASIA CCS 2026, Bangalore, India, June 1-5, 2026*. ASIACCS ’26. To
4068 appear. Bangalore, India: ACM, 2026. DOI: 10.1145/3779208.380597
4069 2 (cit. on p. 13).
- 4070 [Das16] Sanjoy Dasgupta. “A cost function for similarity-based hierarchical
4071 clustering.” In: *Proceedings of the 48th Annual ACM SIGACT Symposium*
4072 *on Theory of Computing, STOC 2016, Cambridge, MA, USA, June 18-21,*
4073 *2016*. ACM, 2016. DOI: 10.1145/2897518.2897527 (cit. on pp. 14, 50,
4074 52–54, 64).
- 4075 [De12] Anindya De. “Lower bounds in differential privacy.” In: *Theory*
4076 *of Cryptography - 9th Theory of Cryptography Conference, TCC 2012,*
4077 *Taormina, Sicily, Italy, March 19-21, 2012. Proceedings*. Vol. 7194. Lecture
4078 Notes in Computer Science. Springer. Springer, 2012. DOI: 10.1007/9
4079 78-3-642-28914-9\18 (cit. on p. 52).

- [Den+25] Chengyuan Deng, Jie Gao, Jalaj Upadhyay, Chen Wang, and Samson Zhou. “On the Price of Differential Privacy for Hierarchical Clustering.” In: *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. DOI: [10.48550/arXiv.2504.15580](https://doi.org/10.48550/arXiv.2504.15580). arXiv: [2504.15580](https://arxiv.org/abs/2504.15580) [cs.DS] (cit. on pp. vii–ix, 18, 49, 50, 53).
- [Des20] Damien Desfontaines. *The Privacy Loss Random Variable*. <https://desfontain.es/blog/privacy-loss-random-variable.html>. Updated 2022-07-23. Accessed: 2026-06-23. 2020 (cit. on p. 6).
- [Des24] Damien Desfontaines. *Paper highlights: Utility-boosting geometric tricks*. Updated 2025-03-20. Sept. 29, 2024. URL: <https://desfontain.es/blog/geometric-tricks.html> (visited on 06/30/2026) (cit. on p. 19).
- [DP20] Damien Desfontaines and Balázs Pejó. “SoK: Differential privacies.” In: *Proc. Priv. Enhancing Technol.* 2020.2 (2020). ISSN: 2299-0984. DOI: [10.2478/popets-2020-0028](https://doi.org/10.2478/popets-2020-0028) (cit. on p. 9).
- [Die12] Reinhard Diestel. *Graph Theory, 4th Edition*. Vol. 173. Graduate texts in mathematics. Springer, 2012. ISBN: 978-3-642-14278-9. DOI: [10.1007/978-3-662-70107-2](https://doi.org/10.1007/978-3-662-70107-2) (cit. on p. 12).
- [Die+15] Ibai Diez, Paolo Bonifazi, Iñaki Escudero, Beatriz Mateos, Miguel A Muñoz, Sebastiano Stramaglia, and Jesus M Cortes. “A novel brain partition highlights the modular skeleton shared by structure and function.” In: *Scientific reports* 5.1 (2015), p. 10532 (cit. on p. 52).
- [Din+21] Zeyu Ding, Daniel Kifer, Sayed M. Saghaian N. E., Thomas Steinke, Yuxin Wang, Yingtai Xiao, and Danfeng Zhang. “The Permute-and-Flip Mechanism is Identical to Report-Noisy-Max with Exponential Noise.” In: *CoRR abs/2105.07260* (2021). DOI: [10.48550/arXiv.2105.07260](https://doi.org/10.48550/arXiv.2105.07260). arXiv: [2105.07260](https://arxiv.org/abs/2105.07260) (cit. on p. 103).
- [Din+25] Michael Dinitz, George Z. Li, Quanquan C. Liu, and Felix Zhou. “Differentially Private Matchings.” In: *CoRR abs/2501.00926* (2025). DOI: [10.48550/arXiv.2501.00926](https://doi.org/10.48550/arXiv.2501.00926). arXiv: [2501.00926](https://arxiv.org/abs/2501.00926) [cs.DS] (cit. on p. 52).
- [DNo3] Irit Dinur and Kobbi Nissim. “Revealing information while preserving privacy.” In: *Proceedings of the Twenty-Second ACM SIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems, June 9-12, 2003, San Diego, CA, USA*. ACM, 2003. DOI: [10.1145/773153.773173](https://doi.org/10.1145/773153.773173) (cit. on p. 52).
- [DRS19] Jinshuo Dong, Aaron Roth, and Weijie J. Su. “Gaussian Differential Privacy.” In: *CoRR abs/1905.02383.1* (2019). ISSN: 1467-9868. DOI: [10.48550/arXiv.1905.02383](https://doi.org/10.48550/arXiv.1905.02383). arXiv: [1905.02383](https://arxiv.org/abs/1905.02383) (cit. on pp. 5, 10).
- [DR19] David Durfee and Ryan M. Rogers. “Practical Differentially Private Top-k Selection with Pay-what-you-get Composition.” In: *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*. Vol. 32. Curran Associates, Inc., 2019 (cit. on pp. 17, 36, 123).

- [Dwo+06a] Cynthia Dwork, Krishnaram Kenthapadi, Frank McSherry, Ilya Mironov, and Moni Naor. "Our Data, Ourselves: Privacy Via Distributed Noise Generation." In: *Advances in Cryptology - EUROCRYPT 2006, 25th Annual International Conference on the Theory and Applications of Cryptographic Techniques, St. Petersburg, Russia, May 28 - June 1, 2006, Proceedings*. Vol. 4004. Lecture Notes in Computer Science. Springer, 2006. DOI: [10.1007/11761679_29](https://doi.org/10.1007/11761679_29) (cit. on p. 115).
- [Dwo+06b] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam D. Smith. "Calibrating Noise to Sensitivity in Private Data Analysis." In: *Theory of Cryptography, Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006, Proceedings*. Vol. 3876. Lecture Notes in Computer Science. New York, NY: Springer, 2006. ISBN: 3540327312. DOI: [10.1007/11681878_14](https://doi.org/10.1007/11681878_14) (cit. on pp. 2, 5, 7-9, 75, 114, 115).
- [DR14] Cynthia Dwork and Aaron Roth. "The Algorithmic Foundations of Differential Privacy." In: *Found. Trends Theor. Comput. Sci.* 9.3-4 (2014). DOI: [10.1561/0400000042](https://doi.org/10.1561/0400000042) (cit. on pp. 9, 11, 37, 90).
- [Dwo+15] Cynthia Dwork, Adam D. Smith, Thomas Steinke, Jonathan R. Ullman, and Salil P. Vadhan. "Robust traceability from trace amounts." In: *IEEE 56th Annual Symposium on Foundations of Computer Science, FOCS 2015, Berkeley, CA, USA, 17-20 October, 2015*. IEEE. IEEE Computer Society, 2015. DOI: [10.1109/FOCS.2015.46](https://doi.org/10.1109/FOCS.2015.46) (cit. on p. 52).
- [Ede+25] Talya Eden, Quanquan C. Liu, Sofya Raskhodnikova, and Adam D. Smith. "Triangle Counting With Local Edge Differential Privacy." In: *Random Struct. Algorithms* 66.4 (2025). DOI: [10.1002/rsa.70002](https://doi.org/10.1002/rsa.70002) (cit. on p. 52).
- [Edm71] Jack Edmonds. "Matroids and the greedy algorithm." In: *Math. Program.* 1.1 (1971). DOI: [10.1007/BF01584082](https://doi.org/10.1007/BF01584082) (cit. on p. 40).
- [Eis+98] Michael B Eisen, Paul T Spellman, Patrick O Brown, and David Botstein. "Cluster analysis and display of genome-wide expression patterns." In: *Proceedings of the National Academy of Sciences* 95.25 (1998), pp. 14863-14868 (cit. on p. 52).
- [Eli+20] Marek Eliás, Michael Kapralov, Janardhan Kulkarni, and Yin Tat Lee. "Differentially Private Release of Synthetic Graphs." In: *Proceedings of the 2020 ACM-SIAM Symposium on Discrete Algorithms, SODA 2020, Salt Lake City, UT, USA, January 5-8, 2020*. SIAM, 2020. ISBN: 9781611975994. DOI: [10.1137/1.9781611975994.34](https://doi.org/10.1137/1.9781611975994.34) (cit. on p. 52).
- [EC91] William H. R. Equitz and Thomas M. Cover. "Successive refinement of information." In: *IEEE Trans. Inf. Theory* 37.2 (1991). DOI: [10.1109/18.75242](https://doi.org/10.1109/18.75242) (cit. on pp. 77, 79, 99, 111).
- [ER59] P. Erdős and A. Rényi. "On Random Graphs I." In: *Publicationes Mathematicae Debrecen* 6 (1959) (cit. on pp. 24, 34, 44).
- [FLL22] Chenglin Fan, Ping Li, and Xiaoyun Li. "Private Graph All-Pairwise-Shortest-Path Distance Release with Improved Error Rate." In: *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*. Vol. 35. Curran Associates, Inc., 2022 (cit. on p. 52).

- 4173 [FT87] Michael L. Fredman and Robert Endre Tarjan. "Fibonacci heaps and
4174 their uses in improved network optimization algorithms." In: *J. ACM*
4175 34.3 (1987). ISSN: 0004-5411. DOI: [10.1145/28869.28874](https://doi.org/10.1145/28869.28874) (cit. on p. 38).
- 4176 [GAT20] Google Anonymization Team GAT. *Delta for thresholding*. [https://github.com/google/differential-privacy/blob/main/common_docs/](https://github.com/google/differential-privacy/blob/main/common_docs/Delta_For_Thresholding.pdf)
4177 [Delta_For_Thresholding.pdf](https://github.com/google/differential-privacy/blob/main/common_docs/Delta_For_Thresholding.pdf). [Online; accessed 8-December-2024].
4178 2020 (cit. on pp. 19, 87, 88, 112, 113, 116, 120, 124, 125).
- 4180 [Gei+26] Lukas Geis, Alexander Leonhardt, Johannes Meintrup, Ulrich Meyer,
4181 Manuel Penschuck, and Lukas Retschmeier. "Revisiting a Successful
4182 Reduction Rule for Dominating Set." In: *Proceedings of the 28th*
4183 *Symposium on Algorithm Engineering and Experiments, ALENEX 2026,*
4184 *Vancouver, BC, Canada, January 11-12, 2026*. SIAM, 2026. DOI: [10.1137/1.9781611978957.9](https://doi.org/10.1137/1.9781611978957.9) (cit. on p. 15).
- 4186 [GRS09] Arpita Ghosh, Tim Roughgarden, and Mukund Sundararajan. "Uni-
4187 versally utility-maximizing privacy mechanisms." In: *Proceedings of*
4188 *the 41st Annual ACM Symposium on Theory of Computing, STOC 2009,*
4189 *Bethesda, MD, USA, May 31 - June 2, 2009*. STOC '09. Bethesda, MD,
4190 USA: ACM, 2009. ISBN: 9781605585062. DOI: [10.1145/1536414.15364](https://doi.org/10.1145/1536414.1536464)
4191 [64](https://doi.org/10.1145/1536414.1536464) (cit. on p. 104).
- 4192 [Gir+20] Antonious M. Girgis, Deepesh Data, Kamalika Chaudhuri, Christina
4193 Fragouli, and Suhas N. Diggavi. "Successive Refinement of Privacy." In: *IEEE J. Sel. Areas Inf. Theory* 1.3 (2020). DOI: [10.1109/jsait.20](https://doi.org/10.1109/jsait.2020.3040403)
4194 [20.3040403](https://doi.org/10.1109/jsait.2020.3040403). URL: <https://doi.org/10.1109/jsait.2020.3040403>
4195 (cit. on p. 78).
- 4197 [Göt+12] Michaela Götz, Ashwin Machanavajjhala, Guozhang Wang, Xiaokui
4198 Xiao, and Johannes Gehrke. "Publishing Search Logs - A Comparative
4199 Study of Privacy Guarantees." In: *IEEE Trans. Knowl. Data Eng.* 24.3
4200 (2012). DOI: [10.1109/TKDE.2011.26](https://doi.org/10.1109/TKDE.2011.26) (cit. on pp. 112, 116).
- 4201 [Gra73] Mark S. Granovetter. "The Strength of Weak Ties." In: *American*
4202 *Journal of Sociology* 78.6 (1973). ISSN: 1537-5390. DOI: [10.1086/225469](https://doi.org/10.1086/225469)
4203 (cit. on p. 12).
- 4204 [GRU12] Anupam Gupta, Aaron Roth, and Jonathan R. Ullman. "Iterative
4205 Constructions and Private Data Release." In: *Theory of Cryptography -*
4206 *9th Theory of Cryptography Conference, TCC 2012, Taormina, Sicily, Italy,*
4207 *March 19-21, 2012. Proceedings*. Vol. 7194. Lecture Notes in Computer
4208 Science. Springer, 2012. ISBN: 9783642289149. DOI: [10.1007/978-3-64](https://doi.org/10.1007/978-3-642-28914-9_19)
4209 [2-28914-9_19](https://doi.org/10.1007/978-3-642-28914-9_19) (cit. on p. 52).
- 4210 [HSSo8] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. "Exploring
4211 Network Structure, Dynamics, and Function using NetworkX." In:
4212 *Proceedings of the 7th Python in Science Conference*. Pasadena, CA USA,
4213 2008 (cit. on p. 38).
- 4214 [Hay+09] Michael Hay, Chao Li, Gerome Miklau, and David D. Jensen. "Accu-
4215 rate Estimation of the Degree Distribution of Private Networks." In:
4216 *ICDM 2009, The Ninth IEEE International Conference on Data Mining,*
4217 *Miami, Florida, USA, 6-9 December 2009*. IEEE Computer Society, 2009.
4218 DOI: [10.1109/ICDM.2009.11](https://doi.org/10.1109/ICDM.2009.11) (cit. on p. 52).

- 4219 [Hay+08] Michael Hay, Gerome Miklau, David D. Jensen, Donald F. Towsley,
4220 and Philipp Weis. “Resisting structural re-identification in anonymized
4221 social networks.” In: *Proc. VLDB Endow.* 1.1 (2008). ISSN: 2150-8097.
4222 DOI: [10.14778/1453856.1453873](https://doi.org/10.14778/1453856.1453873) (cit. on p. 12).
- 4223 [HHE04] Johan Himberg, Aapo Hyvärinen, and Fabrizio Esposito. “Validating
4224 the independent components of neuroimaging time series via clus-
4225 tering and visualization.” In: *NeuroImage* 22.3 (2004). ISSN: 1053-8119.
4226 DOI: [10.1016/j.neuroimage.2004.03.027](https://doi.org/10.1016/j.neuroimage.2004.03.027) (cit. on p. 14).
- 4227 [HT24] Richard Hladík and Jakub Tetek. “Near-Universally-Optimal Differ-
4228 entially Private Minimum Spanning Trees.” In: *CoRR* abs/2404.15035
4229 (2024). DOI: [10.48550/arXiv.2404.15035](https://doi.org/10.48550/arXiv.2404.15035). arXiv: [2404.15035](https://arxiv.org/abs/2404.15035) [cs.CR]
4230 (cit. on pp. 17, 18, 24, 37, 38, 50, 52, 55, 60–62, 68).
- 4231 [HC25] Shlomi Hod and Ran Canetti. “Differentially Private Release of Is-
4232 rael’s National Registry of Live Births.” In: *IEEE Symposium on Se-
4233 curity and Privacy, SP 2025, San Francisco, CA, USA, May 12-15, 2025.*
4234 IEEE, 2025. DOI: [10.1109/SP61157.2025.00101](https://doi.org/10.1109/SP61157.2025.00101) (cit. on p. 5).
- 4235 [How25] Gary Howarth. *A Community-Driven Differential Privacy Deployment*
4236 *Registry*. 2025. DOI: [10.6028/nist.ir.8588.ipd](https://doi.org/10.6028/nist.ir.8588.ipd) (cit. on p. 5).
- 4237 [Hsu+14] Justin Hsu, Zhiyi Huang, Aaron Roth, Tim Roughgarden, and Zhiwei
4238 Steven Wu. “Private matchings and allocations.” In: *Symposium on*
4239 *Theory of Computing, STOC 2014, New York, NY, USA, May 31 - June*
4240 *03, 2014*. New York, New York: ACM, 2014. ISBN: 9781450327107. DOI:
4241 [10.1145/2591796.2591826](https://doi.org/10.1145/2591796.2591826) (cit. on p. 52).
- 4242 [Imo+23] Jacob Imola, Alessandro Epasto, Mohammad Mahdian, Vincent Cohen-
4243 Addad, and Vahab Mirrokni. “Differentially Private Hierarchical Clus-
4244 tering with Provable Approximation Guarantees.” In: *International*
4245 *Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu,*
4246 *Hawaii, USA*. Vol. 202. Proceedings of Machine Learning Research.
4247 PMLR, 2023 (cit. on pp. 50, 52, 65).
- 4248 [IPR27] Jacob Imola, Rasmus Pagh, and Lukas Retschmeier. “Lower Bounds
4249 for Private Graph Optimization Problems using Reconstruction At-
4250 tacks.” In: *Proc. ACM Manag. Data (PODS)* 3.2 (2027). DOI: [10.1145/3725240](https://doi.org/10.1145/3725240). URL: <https://doi.org/10.1145/3725240> (cit. on p. 15).
- 4252 [JS68] Nicholas Jardine and R Sibson. “A model for taxonomy.” In: *Mathe-*
4253 *matical Biosciences* 2.3-4 (1968), pp. 465–482 (cit. on p. 52).
- 4254 [Jar30] Vojtěch Jarník. “O jistém problému minimálním. (Z dopisu panu
4255 O. Borůvkovi) [On a certain problem of minimization].” Czech. In:
4256 *Acta Societatis Scientiarum Naturalium Moraviae* 6 (1930). (Visited on
4257 11/19/2024) (cit. on p. 37).
- 4258 [Jay+24] Rajesh Jayaram, Vahab Mirrokni, Shyam Narayanan, and Peilin
4259 Zhong. “Massively Parallel Algorithms for High-Dimensional Eu-
4260 clidean Minimum Spanning Tree.” In: *Proceedings of the 2024 ACM-*
4261 *SIAM Symposium on Discrete Algorithms, SODA 2024, Alexandria, VA,*
4262 *USA, January 7-10, 2024*. SIAM, 2024. DOI: [10.1137/1.9781611977912](https://doi.org/10.1137/1.9781611977912.139)
4263 [.139](https://doi.org/10.1137/1.9781611977912.139) (cit. on p. 14).
- 4264 [JM09] Carter Jernigan and Behram F. T. Mistree. “Gaydar: Facebook Friend-
4265 ships Expose Sexual Orientation.” In: *First Monday* 14.10 (2009). ISSN:
4266 1396-0466. DOI: [10.5210/fm.v14i10.2611](https://doi.org/10.5210/fm.v14i10.2611) (cit. on p. 12).

- [JY24] Matthew Joseph and Alexander Yu. “Some Constructions of Private, Efficient, and Optimal K-Norm and Elliptic Gaussian Noise.” In: *The Thirty Seventh Annual Conference on Learning Theory, June 30 - July 3, 2023, Edmonton, Canada*. Vol. 247. Proceedings of Machine Learning Research. PMLR, 2024 (cit. on pp. 118, 129).
- [Jum+21] John Jumper et al. “Highly accurate protein structure prediction with AlphaFold.” In: *Nature* 596.7873 (2021). ISSN: 1476-4687. DOI: 10.1038/s41586-021-03819-2 (cit. on p. 12).
- [KOV15] Peter Kairouz, Sewoong Oh, and Pramod Viswanath. “The Composition Theorem for Differential Privacy.” In: *Proceedings of the 32nd International Conference on Machine Learning, ICML 2015, Lille, France, 6-11 July 2015*. Vol. 37. JMLR Workshop and Conference Proceedings. Lille, France: JMLR.org, 2015 (cit. on pp. 5, 7).
- [Kar93] David R. Karger. “Global Min-cuts in RNC, and Other Ramifications of a Simple Min-Cut Algorithm.” In: *Proceedings of the Fourth Annual ACM/SIGACT-SIAM Symposium on Discrete Algorithms, 25-27 January 1993, Austin, Texas, USA*. Vol. 93. ACM/SIAM, 1993. URL: <http://dl.acm.org/citation.cfm?id=313559.313605> (cit. on pp. 57, 73).
- [KKT95] David R. Karger, Philip N. Klein, and Robert Endre Tarjan. “A Randomized Linear-Time Algorithm to Find Minimum Spanning Trees.” In: *J. ACM* 42.2 (1995). ISSN: 0004-5411. DOI: 10.1145/201019.201022. URL: <https://doi.org/10.1145/201019.201022> (cit. on pp. 23, 27, 38).
- [Kas+13] Shiva Prasad Kasiviswanathan, Kobbi Nissim, Sofya Raskhodnikova, and Adam D. Smith. “Analyzing Graphs with Node Differential Privacy.” In: *Theory of Cryptography - 10th Theory of Cryptography Conference, TCC 2013, Tokyo, Japan, March 3-6, 2013. Proceedings*. Vol. 7785. Lecture Notes in Computer Science. Tokyo, Japan: Springer, 2013. ISBN: 9783642365935. DOI: 10.1007/978-3-642-36594-2_26 (cit. on pp. 13, 52).
- [Kor+09] Aleksandra Korolova, Krishnaram Kenthapadi, Nina Mishra, and Alexandros Ntoulas. “Releasing search queries and clicks privately.” In: *Proceedings of the 18th International Conference on World Wide Web, WWW 2009, Madrid, Spain, April 20-24, 2009*. ACM, 2009. DOI: 10.1145/1526709.1526733 (cit. on pp. 88, 89, 112, 116).
- [KKPo1] Samuel Kotz, Tomaz J. Kozubowski, and Krzysztof Podgórski. *The Laplace Distribution and Generalizations*. Boston, MA: Birkhäuser Boston, 2001. ISBN: 978-1-4612-6646-4 978-1-4612-0173-1. DOI: 10.1007/978-1-4612-0173-1. (Visited on 04/07/2025) (cit. on pp. 99, 100).
- [KHP16] Fragkiskos Koufogiannis, Shuo Han, and George J. Pappas. “Gradual Release of Sensitive Data under Differential Privacy.” en. In: *J. Priv. Confidentiality* 7.2 (2016). Number: 2. ISSN: 2575-8527. DOI: 10.29012/jpc.v7i2.649. (Visited on 01/08/2025) (cit. on pp. 78, 81, 85, 90, 99).
- [Kru56] Joseph B. Kruskal. “On the Shortest Spanning Subtree of a Graph and the Traveling Salesman Problem.” In: *Proceedings of the American Mathematical Society* 7.1 (1956). ISSN: 00029939, 10886826. (Visited on 04/15/2024) (cit. on pp. 26, 27, 37, 40).

- 4315 [Kua25] Rebecca F. Kuang. *Katabasis*. Harper Collins Publishers, 2025. ISBN:
4316 9780063446243 (cit. on p. 5).
- 4317 [LRN09] Chih Lai, Taras Raza, and Dwight E. Nelson. “Approximate minimum
4318 spanning tree clustering in high-dimensional space.” In: *Intell. Data*
4319 *Anal.* 13.4 (2009). DOI: [10.3233/IDA-2009-0382](https://doi.org/10.3233/IDA-2009-0382) (cit. on p. 14).
- 4320 [Leb25] Christian Janos Lebeda. “Better Gaussian Mechanism using Corre-
4321 lated Noise.” In: *2025 Symposium on Simplicity in Algorithms, SOSA*
4322 *2025, New Orleans, LA, USA, January 13-15, 2025*. SIAM, 2025. DOI:
4323 [10.1137/1.9781611978315.9](https://doi.org/10.1137/1.9781611978315.9). eprint: [https://epubs.siam.org/doi](https://epubs.siam.org/doi/pdf/10.1137/1.9781611978315.9)
4324 [/pdf/10.1137/1.9781611978315.9](https://epubs.siam.org/doi/pdf/10.1137/1.9781611978315.9) (cit. on pp. vii–ix, 4, 19, 113, 114,
4325 117–119, 126, 130).
- 4326 [LAP22] Christian Janos Lebeda, Martin Aumüller, and Rasmus Pagh. “Repre-
4327 senting Sparse Vectors with Differential Privacy, Low Error, Optimal
4328 Space, and Fast Access.” In: *J. Priv. Confidentiality* 12.2 (2022). DOI:
4329 [10.29012/jpc.809](https://doi.org/10.29012/jpc.809) (cit. on pp. 112, 116).
- 4330 [LR24] Christian Janos Lebeda and Lukas Retschmeier. “The Correlated
4331 Gaussian Sparse Histogram Mechanism.” In: *CoRR abs/2412.10357*
4332 (2024). DOI: [10.48550/arXiv.2412.10357](https://doi.org/10.48550/arXiv.2412.10357). arXiv: [2412.10357](https://arxiv.org/abs/2412.10357) [cs.DS]
4333 (cit. on p. 122).
- 4334 [LR25] Christian Janos Lebeda and Lukas Retschmeier. “The Correlated
4335 Gaussian Sparse Histogram Mechanism.” In: *6th Symposium on Foun-*
4336 *ditions of Responsible Computing, FORC 2025, Stanford University, CA,*
4337 *USA, June 4-6, 2025*. Vol. 329. LIPIcs. Schloss Dagstuhl - Leibniz-
4338 Zentrum für Informatik, 2025. DOI: [10.4230/LIPIcs.FORC.2025.23](https://doi.org/10.4230/LIPIcs.FORC.2025.23)
4339 (cit. on p. 15).
- 4340 [LT23] Christian Janos Lebeda and Jakub Tetek. “Better Differentially Private
4341 Approximate Histograms and Heavy Hitters using the Misra-Gries
4342 Sketch.” In: *Proceedings of the 42nd ACM SIGMOD-SIGACT-SIGAI*
4343 *Symposium on Principles of Database Systems, PODS 2023, Seattle, WA,*
4344 *USA, June 18-23, 2023*. ACM, 2023. DOI: [10.1145/3584372.3588673](https://doi.org/10.1145/3584372.3588673)
4345 (cit. on pp. 113, 119).
- 4346 [Ler+26] Simon Lermen, Daniel Paleka, Joshua Swanson, Michael Aerni, Nicholas
4347 Carlini, and Florian Tramèr. “Large-scale online deanonymization
4348 with LLMs.” In: *CoRR. Proceedings of Machine Learning Research*
4349 *abs/2602.16800* (2026). DOI: [10.48550/arXiv.2602.16800](https://doi.org/10.48550/arXiv.2602.16800). arXiv:
4350 [2602.16800](https://arxiv.org/abs/2602.16800) (cit. on p. 1).
- 4351 [LRU14] Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman. *Mining of*
4352 *Massive Datasets, 2nd Ed.* Cambridge University Press, 2014. ISBN:
4353 978-1107077232. DOI: [10.1017/9781108684163](https://doi.org/10.1017/9781108684163) (cit. on p. 52).
- 4354 [Li+15] Chao Li, Gerome Miklau, Michael Hay, Andrew McGregor, and
4355 Vibhor Rastogi. “The matrix mechanism: optimizing linear counting
4356 queries under differential privacy.” In: *VLDB J.* 24.6 (2015). DOI:
4357 [10.1007/s00778-015-0398-x](https://doi.org/10.1007/s00778-015-0398-x) (cit. on p. 87).
- 4358 [Li26] Tianshi Li. “Agentic LLMs as Powerful Deanonymizers: Re-identification
4359 of Participants in the Anthropic Interviewer Dataset.” In: *CoRR*
4360 *abs/2601.05918* (2026). DOI: [10.48550/arXiv.2601.05918](https://doi.org/10.48550/arXiv.2601.05918). arXiv:
4361 [2601.05918](https://arxiv.org/abs/2601.05918) (cit. on p. 1).

- [Li+23] Yang Li, Michael Purcell, Thierry Rakotoarivelo, David Smith, Thilina Ranbaduge, and Kee Siong Ng. "Private Graph Data Release: A Survey." In: *ACM Comput. Surv.* 55.11 (2023). ISSN: 1557-7341. DOI: [10.1145/3569085](https://doi.org/10.1145/3569085) (cit. on pp. 13, 36).
- [Li+12] Yaping Li, Minghua Chen, Qiwei Li, and Wei Zhang. "Enabling Multilevel Trust in Privacy Preserving Data Mining." In: *IEEE Trans. Knowl. Data Eng.* 24.9 (2012). DOI: [10.1109/TKDE.2011.124](https://doi.org/10.1109/TKDE.2011.124) (cit. on pp. 79, 81).
- [Lig+17] Katrina Ligett, Seth Neel, Aaron Roth, Bo Waggoner, and Zhiwei Steven Wu. "Accuracy First: Selecting a Differential Privacy Level for Accuracy Constrained ERM." In: *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*. Vol. 30. 2017 (cit. on p. 78).
- [LUZ24] Jingcheng Liu, Jalaj Upadhyay, and Zongrui Zou. "Optimal Bounds on Private Graph Approximation." In: *Proceedings of the 2024 ACM-SIAM Symposium on Discrete Algorithms, SODA 2024, Alexandria, VA, USA, January 7-10, 2024*. SIAM, 2024. ISBN: 9781611977912. DOI: [10.1137/1.9781611977912.39](https://doi.org/10.1137/1.9781611977912.39) (cit. on p. 52).
- [McD89] Colin McDiarmid. "On the method of bounded differences." In: *Surveys in Combinatorics, 1989: Invited Papers at the Twelfth British Combinatorial Conference*. London Mathematical Society Lecture Note Series. Cambridge University Press, 1989 (cit. on p. 59).
- [MMS21] Ryan McKenna, Gerome Miklau, and Daniel Sheldon. "Winning the NIST Contest: A scalable and general approach to differentially private synthetic data." In: *J. Priv. Confidentiality* 11.3 (2021). DOI: [10.29012/jpc.778](https://doi.org/10.29012/jpc.778) (cit. on pp. 3, 13, 14, 22, 37).
- [MS20] Ryan McKenna and Daniel Sheldon. "Permute-and-Flip: A new mechanism for differentially private selection." In: *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*. Vol. 33. Curran Associates, Inc., 2020 (cit. on p. 37).
- [MT07] Frank McSherry and Kunal Talwar. "Mechanism Design via Differential Privacy." In: *48th Annual IEEE Symposium on Foundations of Computer Science, FOCS 2007, Providence, RI, USA, October 20-23, 2007, Proceedings*. IEEE. IEEE Computer Society, 2007. DOI: [10.1109/FOCS.2007.66](https://doi.org/10.1109/FOCS.2007.66) (cit. on pp. 25, 27, 36, 37, 90, 123).
- [Mei18] Sebastian Meiser. "Approximate and Probabilistic Differential Privacy Definitions." In: *IACR Cryptol. ePrint Arch.* (2018) (cit. on p. 7).
- [Mes+20] Solomon Messing, Christina DeGregorio, Bennett Hillenbrand, Gary King, Saurav Mahanti, Zagreb Mukerjee, Chaya Nayak, Nate Persily, Bogdan State, and Arjun Wilkins. *Facebook Privacy-Protected Full URLs Data Set*. Version V10. 2020. DOI: [10.7910/DVN/TDOAPG](https://doi.org/10.7910/DVN/TDOAPG) (cit. on p. 124).
- [MS17] Yuliya Mishura and Georgiy Shevchenko. "Markov and Diffusion Processes." In: *Theory and Statistical Applications of Stochastic Processes*. Hoboken, NJ / London: John Wiley & Sons / ISTE, 2017. Chap. 7. ISBN: 978-1-78630-050-8. DOI: [10.1002/9781119476597.ch7](https://doi.org/10.1002/9781119476597.ch7) (cit. on p. 108).

- 4410 [MG82] Jayadev Misra and David Gries. “Finding Repeated Elements.” In:
4411 *Sci. Comput. Program.* 2.2 (1982). ISSN: 0167-6423. DOI: [10.1016/0167-](https://doi.org/10.1016/0167-6423(82)90012-0)
4412 [6423\(82\)90012-0](https://doi.org/10.1016/0167-6423(82)90012-0) (cit. on p. 113).
- 4413 [Mue+22] Tamara T. Mueller, Dmitrii Usynin, Johannes C. Paetzold, Daniel
4414 Rueckert, and Georgios Kaissis. “SoK: Differential Privacy on Graph-
4415 Structured Data.” In: *CoRR abs/2203.09205* (2022). DOI: [10.48550/ar](https://doi.org/10.48550/arXiv.2203.09205)
4416 [Xiv.2203.09205](https://doi.org/10.48550/arXiv.2203.09205). arXiv: 2203.09205 [cs.CR] (cit. on pp. 13, 14, 36).
- 4417 [NSo8] Arvind Narayanan and Vitaly Shmatikov. “Robust De-anonymization
4418 of Large Sparse Datasets.” In: *2008 IEEE Symposium on Security and*
4419 *Privacy (SP 2008), 18-21 May 2008, Oakland, California, USA*. SP ’08.
4420 IEEE Computer Society, 2008. ISBN: 978-0-7695-3168-7. DOI: [10.1109](https://doi.org/10.1109/SP.2008.33)
4421 [/SP.2008.33](https://doi.org/10.1109/SP.2008.33) (cit. on p. 1).
- 4422 [NSo9] Arvind Narayanan and Vitaly Shmatikov. “De-anonymizing Social
4423 Networks.” In: *30th IEEE Symposium on Security and Privacy (SP 2009),*
4424 *17-20 May 2009, Oakland, California, USA*. SP ’09. USA: IEEE Computer
4425 Society, 2009. ISBN: 9780769536330. DOI: [10.1109/SP.2009.22](https://doi.org/10.1109/SP.2009.22) (cit. on
4426 p. 12).
- 4427 [Nat] National Longitudinal Study of Adolescent to Adult Health. *Add*
4428 *Health*. <https://addhealth.cpc.unc.edu/>. Accessed: 2026-05-22
4429 (cit. on p. 12).
- 4430 [NP33] Jerzy Neyman and Egon Sharpe Pearson. “IX. On the problem of
4431 the most efficient tests of statistical hypotheses.” In: *Philosophical*
4432 *Transactions of the Royal Society of London. Series A, Containing Papers of*
4433 *a Mathematical or Physical Character* 231.694-706 (1933). ISSN: 2053-9258.
4434 DOI: [10.1098/rsta.1933.0009](https://doi.org/10.1098/rsta.1933.0009) (cit. on p. 6).
- 4435 [NRS07] Kobbi Nissim, Sofya Raskhodnikova, and Adam D. Smith. “Smooth
4436 sensitivity and sampling in private data analysis.” In: *Proceedings of*
4437 *the 39th Annual ACM Symposium on Theory of Computing, San Diego,*
4438 *California, USA, June 11-13, 2007*. STOC07. ACM, 2007. DOI: [10.1145](https://doi.org/10.1145/1250790.1250803)
4439 [/1250790.1250803](https://doi.org/10.1145/1250790.1250803) (cit. on p. 13).
- 4440 [NSN24] Alex Nyffenegger, Matthias Stürmer, and Joel Niklaus. “Anonymity
4441 at Risk? Assessing Re-Identification Capabilities of Large Language
4442 Models in Court Decisions.” In: *Findings of the Association for Compu-*
4443 *tational Linguistics: NAACL 2024, Mexico City, Mexico, June 16-21, 2024*.
4444 Vol. NAACL 2024. Findings of ACL. Mexico City, Mexico: Association
4445 for Computational Linguistics, 2024. DOI: [10.18653/v1/2024.findin](https://doi.org/10.18653/v1/2024.findings-naacl.157)
4446 [gs-naacl.157](https://doi.org/10.18653/v1/2024.findings-naacl.157). URL: [https://doi.org/10.18653/v1/2024.findings-](https://doi.org/10.18653/v1/2024.findings-naacl.157)
4447 [naacl.157](https://doi.org/10.18653/v1/2024.findings-naacl.157) (cit. on p. 1).
- 4448 [Ohl90] Esbjörn Ohlsson. *Sequential poisson sampling from a business register*
4449 *and its application to the Swedish consumer price index*. Statistiska cen-
4450 tralbyrån, 1990 (cit. on pp. 24, 36).
- 4451 [PR24] Rasmus Pagh and Lukas Retschmeier. “Faster Private Minimum
4452 Spanning Trees.” In: *CoRR abs/2408.06997* (2024). DOI: [10.48550/a](https://doi.org/10.48550/arXiv.2408.06997)
4453 [rXiv.2408.06997](https://doi.org/10.48550/arXiv.2408.06997). arXiv: 2408.06997 [cs.DS] (cit. on pp. 3, 22, 23,
4454 38).

- [Pag+25] Rasmus Pagh, Lukas Retschmeier, Hao Wu, and Hanwen Zhang. "Optimal Bounds for Private Minimum Spanning Trees via Input Perturbation." In: *Proc. ACM Manag. Data (PODS)* 3.2 (June 2025). DOI: [10.1145/3725240](https://doi.org/10.1145/3725240). URL: <https://doi.org/10.1145/3725240> (cit. on pp. 3, 15, 44, 49, 68).
- [Pan24] Mingen Pan. "Randomized Response with Gradual Release of Privacy Budget." In: *CoRR* abs/2401.13952 (2024). DOI: [10.48550/arXiv.2401.13952](https://arxiv.org/abs/2401.13952). arXiv: [2401.13952](https://arxiv.org/abs/2401.13952) (cit. on pp. 19, 78, 107).
- [Pan14] Vijay Pandurangan. *On Taxis and Rainbows: Lessons from NYC's Improperly Anonymized Taxi Logs*. Medium. 2014. URL: <https://medium.com/vijay-pandurangan/of-taxis-and-rainbows-f6bc289679a1> (visited on 04/27/2026) (cit. on p. 1).
- [Pin18] Rafael Pinot. "Minimum spanning tree release under differential privacy constraints." In: *CoRR* abs/1801.06423 (2018). DOI: [10.48550/arXiv.1801.06423](https://arxiv.org/abs/1801.06423). arXiv: [1801.06423](https://arxiv.org/abs/1801.06423) [cs.CR] (cit. on pp. 14, 23, 37–39).
- [Pin+18] Rafael Pinot, Anne Morvan, Florian Yger, Cédric Gouy-Pailler, and Jamal Atif. "Graph-based Clustering under Differential Privacy." In: *Proceedings of the Thirty-Fourth Conference on Uncertainty in Artificial Intelligence, UAI 2018, Monterey, California, USA, August 6-10, 2018*. Monterey, California, United States: AUAI Press, 2018 (cit. on pp. 3, 14, 16, 17, 22, 37).
- [Pri57] Robert Clay Prim. "Shortest connection networks and some generalizations." In: *The Bell System Technical Journal* 36 (6 1957). ISSN: 0005-8580. DOI: [10.1002/j.1538-7305.1957.tb01515.x](https://doi.org/10.1002/j.1538-7305.1957.tb01515.x) (cit. on p. 37).
- [QSZ21] Gang Qiao, Weijie J. Su, and Li Zhang. "Oneshot Differentially Private Top-k Selection." In: *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*. Vol. 139. Proceedings of Machine Learning Research. PMLR, 2021 (cit. on pp. 37, 40, 123).
- [Ras+26] Sofya Raskhodnikova, Adam D. Smith, Connor Wagaman, and Anatoly Zavyalov. "Local Node Differential Privacy." In: *67th Annual IEEE Symposium on Foundations of Computer Science, FOCS 2026*. IEEE. IEEE Computer Society, 2026. DOI: [10.48550/arXiv.2602.15802](https://arxiv.org/abs/2602.15802). eprint: [2602.15802](https://arxiv.org/abs/2602.15802) (cit. on p. 13).
- [Rog+16] Ryan M. Rogers, Salil P. Vadhan, Aaron Roth, and Jonathan R. Ullman. "Privacy Odometers and Filters: Pay-as-you-Go Composition." In: *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*. Vol. 29. 2016 (cit. on p. 78).
- [Ros18] Sheldon Ross. *A First Course in Probability*. en. 10th ed. Upper Saddle River, NJ: Pearson, 2018 (cit. on p. 135).
- [Ros97] Bengt Rosén. "Asymptotic theory for order sampling." In: *Journal of Statistical Planning and Inference* 62.2 (1997). ISSN: 0378-3758. DOI: [https://doi.org/10.1016/S0378-3758\(96\)00185-1](https://doi.org/10.1016/S0378-3758(96)00185-1) (cit. on pp. 24, 36, 37, 123).

- 4502 [Sea16] Adam Sealfon. “Shortest Paths and Distances with Differential Pri-
4503 vacy.” In: *Proceedings of the 35th ACM SIGMOD-SIGACT-SIGAI Sym-*
4504 *posium on Principles of Database Systems, PODS 2016, San Francisco,*
4505 *CA, USA, June 26 - July 01, 2016.* PODS ’16. San Francisco, California,
4506 USA: ACM, 2016. ISBN: 9781450341912. DOI: [10.1145/2902251.29](https://doi.org/10.1145/2902251.2902291)
4507 [02291](https://doi.org/10.1145/2902251.2902291). URL: <https://doi.org/10.1145/2902251.2902291> (cit. on
4508 pp. vii–ix, 3, 13, 14, 18, 22, 23, 37, 38, 49, 50, 52, 54, 55, 68, 75).
- 4509 [SSo2] Jinwook Seo and Ben Shneiderman. “Interactively Exploring Hierar-
4510 chical Clustering Results.” In: *Computer* 35.7 (2002). ISSN: 0018-9162.
4511 DOI: [10.1109/MC.2002.1016905](https://doi.org/10.1109/MC.2002.1016905) (cit. on p. 14).
- 4512 [Sha+25] Ali Shahin Shamsabadi, Peter Snyder, Ralph Giles, Aurélien Bel-
4513 let, and Hamed Haddadi. “Nebula: Efficient, Private and Accurate
4514 Histogram Estimation.” In: *Proceedings of the 2025 ACM SIGSAC Con-*
4515 *ference on Computer and Communications Security, CCS 2025, Taipei,*
4516 *Taiwan, October 13-17, 2025.* CCS ’25. Taipei, Taiwan: ACM, 2025. ISBN:
4517 9798400715259. DOI: [10.1145/3719027.3744789](https://doi.org/10.1145/3719027.3744789) (cit. on p. 5).
- 4518 [Smiz0] Adam Smith. *Algorithms in Society: Lectures 9 and 10*. Lecture, BU
4519 CS591 S1, Boston University Computer Science. Spring 2020. 2020
4520 (cit. on p. 7).
- 4521 [Sne05] Peter HA Sneath. “Numerical taxonomy.” In: *Bergey’s manual® of*
4522 *systematic bacteriology*. Springer, 2005, pp. 39–42 (cit. on p. 52).
- 4523 [Sot+03] Christos Sotiriou, Soek-Ying Neo, Lisa M McShane, Edward L Korn,
4524 Philip M Long, Amir Jazaeri, Philippe Martiat, Steve B Fox, Adrian L
4525 Harris, and Edison T Liu. “Breast cancer classification and prognosis
4526 based on gene expression profiles from a population-based study.” In:
4527 *Proceedings of the National Academy of Sciences* 100.18 (2003), pp. 10393–
4528 10398 (cit. on p. 52).
- 4529 [SKK00] Michael Steinbach, George Karypis, and Vipin Kumar. “A Com-
4530 parison of Document Clustering Techniques.” In: *Proceedings of the*
4531 *International KDD Workshop on Text Mining* (June 2000) (cit. on p. 52).
- 4532 [Ste22] Thomas Steinke. “Composition of Differential Privacy & Privacy
4533 Amplification by Subsampling.” In: *CoRR* abs/2210.00597 (2022). DOI:
4534 [10.48550/arXiv.2210.00597](https://doi.org/10.48550/arXiv.2210.00597). arXiv: [2210.00597](https://arxiv.org/abs/2210.00597) (cit. on pp. xi, 5,
4535 11).
- 4536 [SU17] Thomas Steinke and Jonathan R. Ullman. “Tight Lower Bounds for
4537 Differentially Private Selection.” In: *58th IEEE Annual Symposium*
4538 *on Foundations of Computer Science, FOCS 2017, Berkeley, CA, USA,*
4539 *October 15-17, 2017.* Berkley, USA: IEEE Computer Society, 2017. DOI:
4540 [10.1109/FOCS.2017.57](https://doi.org/10.1109/FOCS.2017.57) (cit. on pp. 24, 33, 45).
- 4541 [Str65] V. Strassen. “The Existence of Probability Measures with Given
4542 Marginals.” In: *The Annals of Mathematical Statistics* 36.2 (1965). ISSN:
4543 0003-4851. DOI: [10.1214/aoms/1177700153](https://doi.org/10.1214/aoms/1177700153) (cit. on p. 74).
- 4544 [Tar75] Robert Endre Tarjan. “Efficiency of a Good But Not Linear Set Union
4545 Algorithm.” In: *J. ACM* 22.2 (1975). ISSN: 0004-5411. DOI: [10.1145/32](https://doi.org/10.1145/321879.321884)
4546 [1879.321884](https://doi.org/10.1145/321879.321884) (cit. on p. 44).

- [Toc14] Anthony Tockar. *Riding with the Stars: Passenger Privacy in the NYC Taxicab Dataset*. Neustar Research. 2014. URL: <https://web.archive.org/web/20200310042615/https://research.neustar.biz/2014/09/15/riding-with-the-stars-passenger-privacy-in-the-nyc-taxicab-dataset/> (visited on 04/29/2026) (cit. on p. 1).
- [Tro14] J. K. Trotter. *The Public NYC Taxicab Database That Accidentally Tracks Celebrities*. Gawker. 2014. URL: <https://www.gawkerarchives.com/the-public-nyc-taxicab-database-that-accidentally-track-1646724546> (visited on 04/29/2026) (cit. on p. 1).
- [Sto] “Univariate Stochastic Orders.” In: *Stochastic Orders*. New York, NY: Springer New York, 2007. ISBN: 978-0-387-34675-5. DOI: [10.1007/978-0-387-34675-5_1](https://doi.org/10.1007/978-0-387-34675-5_1) (cit. on p. 80).
- [Vad12] Salil P. Vadhan. “Pseudorandomness.” In: *Found. Trends Theor. Comput. Sci.* 7.1-3 (2012). ISSN: 1551-3068. DOI: [10.1561/04000000010](https://doi.org/10.1561/04000000010) (cit. on p. 63).
- [Vad17] Salil P. Vadhan. “The Complexity of Differential Privacy.” In: *Tutorials on the Foundations of Cryptography*. Springer International Publishing, 2017. DOI: [10.1007/978-3-319-57048-8_7](https://doi.org/10.1007/978-3-319-57048-8_7) (cit. on pp. 11, 54, 56, 61).
- [Ver18] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2018 (cit. on p. 75).
- [War63] Joe H. Ward. “Hierarchical Grouping to Optimize an Objective Function.” In: *Journal of the American Statistical Association* 58.301 (1963). ISSN: 1537-274X. DOI: [10.1080/01621459.1963.10500845](https://doi.org/10.1080/01621459.1963.10500845) (cit. on p. 52).
- [War65] Stanley L. Warner. “Randomized Response: A Survey Technique for Eliminating Evasive Answer Bias.” In: *Journal of the American Statistical Association* 60.309 (1965). ISSN: 1537-274X. DOI: [10.1080/01621459.1965.10480775](https://doi.org/10.1080/01621459.1965.10480775) (cit. on pp. 13, 107).
- [WZ10] Larry Wasserman and Shuheng Zhou. “A Statistical Framework for Differential Privacy.” In: *Journal of the American Statistical Association* 105.489 (2010). ISSN: 01621459. (Visited on 05/05/2026) (cit. on pp. 5, 7).
- [Web+18] Mark Weber, Jie Chen, Toyotaro Suzumura, Aldo Pareja, Tengfei Ma, Hiroki Kanezashi, Tim Kaler, Charles E. Leiserson, and Tao B. Schardl. “Scalable Graph Learning for Anti-Money Laundering: A First Look.” In: *CoRR abs/1812.00076* (2018). DOI: [10.48550/arXiv.1812.00076](https://doi.org/10.48550/arXiv.1812.00076). arXiv: [1812.00076](https://arxiv.org/abs/1812.00076) (cit. on p. 12).
- [Whi+22] Justin Whitehouse, Aaditya Ramdas, Zhiwei Steven Wu, and Ryan M. Rogers. “Brownian Noise Reduction: Maximizing Privacy Subject to Accuracy Constraints.” In: *Advances in Neural Information Processing Systems*. Vol. 35. New Orleans, LA, USA: Curran Associates Inc., 2022. ISBN: 9781713871088 (cit. on p. 78).
- [Who14] Chris Whong. *FOILing NYC’s Taxi Trip Data*. Chris Whong. 2014. URL: https://chriswhong.com/open-data/foil_nyc_taxi/ (visited on 04/27/2026) (cit. on p. 1).